

# PRIMITIVE GRAPH IDEALS AND HIGHER-CONTACT HILBERT LIMITS

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**ABSTRACT.** We determine the Hilbert graph and every embedded boundary curve of  $[t^a : bs^a : cst^{a-1}]$  for arbitrary  $a \geq 2$ . After removing its explicit divisorial content, the evaluation ideal factors into powers of  $(b, c^j)$ ,  $1 \leq j \leq a$ , and its Rees algebra is normal. The graph is a smooth surface obtained by  $a$  successive point blow-ups. Uniform monic equations give all  $2a + 1$  chamber and wall families, their associated points, and a sharp coefficient-jet bound independent of the Hilbert embedding degree. On the last wall a rational cuspidal tail acquires a punctual nilradical of length  $(a - 1)(a - 2)/2$  and nilpotence order  $a - 1$ : this is the exact arithmetic-genus correction required by flatness. We also glue the full polynomial-contact graphs at fixed degree under source and target changes, and compute the contraction of the marked boundary chain to one fixed-source Quot point. The all-order curve classification concerns the specified marked family, not every fibre over the full contact coefficient space. Its application to quadratic pencils retains the fixed-target comparison and the reconstruction theorem of the companion paper.

## 1. INTRODUCTION

A flat limit of embedded rational curves can remember arithmetic genus through a zero-dimensional nilpotent structure. Taking a cycle, taking a reduced support, or taking the scheme image of a stable map loses that information. This paper determines this phenomenon together with the parameter space that carries it in an all-order contact family.

Our central result is Theorem 3.1 and its embedded-curve description in Theorems 4.1 and 4.4. For

$$[s : t] \mapsto [t^a : bs^a : cst^{a-1}], \quad a \geq 2,$$

let  $T_a$  be the closure of the coefficient-retaining graph in the Hilbert scheme of  $\mathbb{P}^1 \times \mathbb{P}^2$ . We prove

$$T_a = \text{Bl}_{\prod_{j=1}^a (b, c^j)} \mathbb{A}^2.$$

The surface is smooth, with  $a$  exceptional curves. It carries an explicit universal curve described by at most  $a + 1$  equations per intermediate or terminal chart. The equations form monic Gröbner bases over the entire parameter charts. Consequently all boundary rings, including embedded associated points, follow from one uniform proof rather than separate saturation calculations for individual arcs.

For a degree- $m$  Hilbert evaluation matrix the exact ideal is

$$b^{(a-1)m(m+1)/2} \left( \prod_{j=1}^{a-1} (b, c^j)^{m-j+1} \right) (b, c^a)^{(m-a+1)(m-a+2)/2}.$$

Its principal factor has no effect on the graph. We compute the normal Rees algebra of the remaining ideal, its primitive exceptional valuations, the boundary intersection matrix, and its contractions. For a formal arc with  $\text{ord}(b) = p$ , the coefficient jet modulo  $\tau^{p+1}$  determines the embedded closed limit. This bound is uniformly optimal on that class of arcs and is independent of contact order and of the chosen Hilbert degree. It is different from the much larger raw maximal-minor order.

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The last wall is particularly instructive. Its reduced tail is  $R^a = \lambda F^{a-1}G$ , attached to the main component at a smooth point. At the cusp the actual Hilbert limit has nilradical

$$xk[x, F]/x(x, F)^{a-2}, \quad R\mathcal{N} = 0.$$

Its length is the arithmetic genus of the plane tail. For  $a = 3$  this is one square-zero embedded point; for larger  $a$  the nilpotence order grows. It cannot be removed while retaining the Hilbert polynomial. Theorem 4.3 also calculates the punctual excess and generic transverse algebras on every preceding wall and chamber.

**1.1. Intrinsic objects and their presentations.** A maximal-minor ideal is a Fitting ideal of a specified evaluation module. Its order and Smith factors are not numerical invariants of the underlying Hilbert map under arbitrary changes of embedding. Section 2 separates divisorial content, a primitive ideal, the finite-module length, and the exceptional Rees valuations. On the full monic coefficient chart the evaluation ideal already has no divisorial content, because its base locus has codimension two. Nonflat restriction can introduce content, as the displayed family shows. This distinction is necessary; merely dividing a restricted factor out of a matrix does not make all its numerical data intrinsic.

Theorem 5.1 then gives a single equivariant projective graph over the projective space of degree- $a$  triples. Every monic pivot chart is an explicit product with the full polynomial-contact graph  $\Gamma_a$ , with actual source and target undo maps and cocycles on overlaps. This is a fixed-degree rational-map compactification. It does not glue unaugmented graph functors with different generic Hilbert polynomials, nor does it classify all higher-corank or singular-pencil fibres. The complete classification proved here is the all-order marked family and its equivariant extensions, including moving contact centres.

**1.2. Comparison and an application independent of reconstruction.** The minimal fan of  $T_a$  is the Newton fan of its primitive monomial ideal. Its wall residues are genuine embedded parameters, even when their valuation vectors agree. General coefficient-support arrangements, by contrast, need not be minimal or invariant under polynomial coordinate changes. Section 5 compares these constructions with tropical Plücker data and comprehensive Gröbner systems.

The same coefficient family has a flat quotient of  $\mathcal{O}_{\mathbb{P}^1}^{\oplus 3}$  before any blow-up. At the origin that quotient is independent of the coefficient direction, so the map to the fixed-source Quot scheme contracts the whole exceptional chain. The Hilbert graph separates its points and retains the punctual genus correction. This is a concrete comparison of objects, not an identification based on a difference in terminology. Corollary 5.7 gives smooth rational graphs whose Hilbert limits have isolated nilradicals of arbitrarily large nilpotence order. It uses none of the inverse theory of Paper I.

**1.3. Proof route and retained theory.** The main proof proceeds from row ideals to the normal Rees algebra, then to monic universal equations and their local rings. The fixed-degree gluing and Quot comparison follow separately. Hilbert representability, regularity, Rees graph constructions, and toric normalization are classical inputs. The calculated all-order factorization, the uniform chart equations, the complete arc table, and the punctual genus-correction algebra are the explicit results proved here; no new general representability mechanism is claimed.

The previous determinantal graph, complete first-contact fibre, horizontal comparison, power ideals, complete quadrics, and singular pencil results are retained in the appendices, with their proofs and labels intact. They provide the full earlier framework and the fixed-target bridge to quadratic pencils without competing with the central proof route. The preservation master contains both paper bodies and is not a third submission. The sharp inverse is needed only when a specified effective failure family is first reconstructed and then inserted into this boundary construction. No internal boundary operator on an isolated raw finite algebra is asserted.

Historical claims are restricted to the explicitly stated theorems. No claim about priority over Ballico's 1993 work is made without a full-text comparison, which remains unavailable. The unchanged sharp-inverse proof is not represented as having received an external independent audit. Neither compilation nor finite symbolic checks can supply that audit or certify the originality of a general theorem.

## 2. EVALUATION IDEALS AND THEIR DIVISORIAL CONTENT

Throughout the new results the ground field  $k$  is algebraically closed of characteristic zero. The equations, flat families, and valuation formulas are defined over  $\mathbb{Z}$  and remain valid over a field whenever the displayed nonzero residues are retained. We distinguish a rational projective map, a choice of homogeneous coordinates for that map, and a choice of projective embedding of its target.

**2.1. A primitive ideal for a fixed projective map.** Let  $B$  be a regular integral scheme of finite type over  $k$ , and let  $I \subset \mathcal{O}_B$  be a nonzero coherent ideal. At a prime divisor  $D$  put  $c_D = \min_{f \in I} \text{ord}_D(f)$ , and set

$$C(I) = \sum_D c_D D, \quad I^{\text{pr}} = I \otimes \mathcal{O}_B(C(I)) \subset \mathcal{O}_B.$$

Only finitely many coefficients are nonzero locally. Regularity makes  $C(I)$  Cartier. Equivalently,  $I^{**} = \mathcal{O}_B(-C(I))$  and  $I^{\text{pr}} = I \otimes (I^{**})^{-1}$ . The inclusion follows by checking orders at prime divisors in the normal ring  $\mathcal{O}_B$ .

**Proposition 2.1** (Content and the graph). *Suppose  $h_0, \dots, h_N$  generate  $I$  and define a rational map  $B \dashrightarrow \mathbb{P}^N$ . Its schematic graph closure is  $\text{Bl}_I B$ , and*

$$\text{Bl}_I B \simeq \text{Bl}_{I^{\text{pr}}} B.$$

*Multiplication of the homogeneous coordinates by one nonzero rational function, and an invertible change of their basis, leave  $I^{\text{pr}}$  unchanged, whenever the resulting coordinates are regular. These assertions concern the same projective embedding; changing that embedding need not preserve the primitive ideal.*

*Proof.* On an affine factorial chart write  $I = gJ$ , where  $g$  is the greatest common divisor of the generators. Division by  $g$  identifies  $I^n$  with  $J^n$  compatibly with multiplication, hence identifies the two Rees algebras as graded algebras. The ratio chart with pivot  $h_i$  is  $A[h_0/h_i, \dots, h_N/h_i]$  inside the fraction field. These charts are exactly the closure of the generic graph. They also show the invariance under a common rational scalar. An invertible matrix changes the generating list but not its ideal. The local constructions agree under the unit changes of  $g$ , and therefore glue. This is the classical Rees graph construction, not a new representability assertion [Stacks Project].  $\square$

**Proposition 2.2** (Primitive finite determination). *On an affine factorial chart choose polynomial generators of  $I^{\text{pr}}$ . Let  $\gamma : \text{Spec } k[[\tau]] \rightarrow B$  have generic point in the domain of the rational map, and put  $\nu = \text{ord}_\gamma(I^{\text{pr}})$ . A second coefficient arc congruent to  $\gamma$  modulo  $\tau^{\nu+1}$  has the same closed projective limit. The bound is sufficient, and is not asserted to be optimal. It determines the closed graph point, not the whole formal family or a point of the normalization above it.*

*Proof.* Polynomial evaluation preserves this congruence. At least one primitive generator has order exactly  $\nu$ , so the second arc has the same minimum order. After division by  $\tau^\nu$ , the two residue vectors are equal and nonzero. The projective limits coincide. Changing a local trivialization multiplies this vector by a unit. A change of regular coordinates preserves the congruence relation on local-ring maps.  $\square$

The integer  $\nu$  is posterior: an oracle for a formal series need not reveal it in a predetermined number of queries. For polynomial or localized-polynomial arcs, polynomial division and exact linear algebra over  $k(\tau)$  compute it in finitely many operations. No efficiency claim follows from that termination statement.

**2.2. The evaluation module.** Write  $V = k^2$ ,  $W = k^3$ , and  $X = \mathbb{P}(V) \times \mathbb{P}(W)$ . A triple of degree  $a$  binary forms gives, by substitution and multiplication, the map

$$E_m : \text{Sym}^m V^* \otimes \text{Sym}^m W^* \longrightarrow \text{Sym}^{(a+1)m} V^*. \quad (2.1)$$

Our convention is a quotient matrix: columns are images of domain basis vectors and rows are coefficients in the target. Thus

$$q = (a+1)m + 1, \quad N = (m+1) \binom{m+2}{2}, \quad M_{a,m} \in A^{q \times N}.$$

Its maximal-minor ideal is  $\text{Fitt}_0(\text{coker } E_m)$ . Source and target changes act through symmetric powers in (2.1); in chosen bases the matrix changes by invertible left and right matrices, together with the corresponding base automorphism. The Fitting ideal is consequently covariant. Scalar changes of a representative of the triple instead multiply all columns by a common scalar to the  $m$ th power. This is a line-bundle twist on projective coefficient space, not an invariant numerical order.

For  $P_a(l) = (a+1)l + 1$  put

$$m_a = \frac{a(a+1)}{2} + 1. \quad (2.2)$$

Indeed, with  $d = a+1$ , the  $d$  linear terms  $\binom{l+1-i}{1}$ ,  $0 \leq i < d$ , sum to  $dl + d(3-d)/2$ . Adding  $(d-1)(d-2)/2$  constant terms gives  $dl + 1$ ; the number of terms is  $d(d-1)/2 + 1$ . Gotzmann regularity applies in the ordinary projective space of the Segre embedding. The restriction of its degree- $m$  section space to  $X$  is surjective: a monomial of bidegree  $(m, m)$  is a product of  $m$  Segre coordinates. The fixed kernel of this restriction belongs to every curve ideal in question. Thus the Hilbert immersion factors through the quotient space in (2.1). For  $m \geq m_a$  the degree- $m$  kernel generates the ideal *sheaf*; this is the truncation consequence of regularity, not a claim that the homogeneous ideal has no smaller minimal generators. These are the classical Hilbert-embedding inputs [Gotzmann(1978)].

**Proposition 2.3** (Rank, incidence, and content on the full chart). *Let  $B_a$  be the affine space of triples  $(f, g, r)$  with  $f$  monic of degree  $a$  and  $\deg g, \deg r < a$ . For  $m \geq m_a$ ,  $E_m$  has rank  $q$  exactly when the three homogenized forms have no common zero. The rank-drop support is irreducible of codimension two. In particular,  $I_q(M_{a,m})$  has no divisorial content on  $B_a$ .*

*Let  $C_f$  be the companion matrix for multiplication by  $x$  on  $A[x]/(f)$ . The smaller presentation*

$$A^{2a} \xrightarrow{[g(C_f) \quad r(C_f)]} A^a \longrightarrow A[x]/(f, g, r) \longrightarrow 0 \quad (2.3)$$

*defines a natural incidence Fitting ideal with the same radical as  $I_q(M_{a,m})$ . Equality of these two ideals is not asserted.*

*Proof.* On the base-point-free locus restriction to the graph is the quotient onto  $H^0(\mathbb{P}^1, \mathcal{O}((a+1)m))$  in a regularity degree. If a common form of positive degree divides the triple, every substituted degree- $m$  section is divisible by its  $m$ th power, so the quotient cannot have rank  $q$ . There is no common zero at infinity since  $f$  is monic and the other leading coefficients vanish. The incidence variety

$$\{(x, f, g, r) : f(x) = g(x) = r(x) = 0\} \subset \mathbb{A}^1 \times B_a$$

is affine space of dimension  $3a - 2$ : solve for the three constant coefficients. Its projection is finite, because  $x$  satisfies the monic equation  $f$ . Its image is therefore closed, irreducible, and of that dimension. An ideal with codimension-two support in a regular integral scheme has order zero at every prime divisor. Finally (2.3) is multiplication by  $g, r$  in the finite free algebra  $A[x]/(f)$ . Its cokernel is the stated quotient, whose support is exactly the same incidence image.  $\square$

Content removal need not commute with nonflat restriction. The full ideal in Proposition 2.3 is already primitive; its restriction to the two-dimensional families below has a large common factor. Thus the primitive construction does not turn the original minor order into an embedding-independent invariant.

**Example 2.4** (Degree changes without graph changes). For  $a = 1$ , translation reduces the triple to  $(x, v, w)$ . In every degree  $m \geq 1$  the maximal-minor ideal is

$$(v, w)^{m(m+1)/2}.$$

To see this, the row indexed by  $n$ ,  $0 \leq n \leq 2m$ , has ideal  $(v, w)^{\max(0, m-n)}$ , and a full minor chooses one entry in every row. The product gives the formula. All these ideals are primitive and define the same blow-up, although their orders along an arc are different. Their Smith lengths are lengths of different evaluation modules, not lengths attached to the rational Hilbert map alone.

**Proposition 2.5** (What survives a change of Hilbert degree). *For two degrees at least  $m_a$ , the graph modifications obtained from  $E_m$  are canonically isomorphic over  $B_a$ , with the same universal embedded curve and the same normalization. Their exceptional prime divisors therefore define the same divisorial valuations of  $k(B_a)$ . The orders of the evaluation ideals on those divisors, the Smith factors, and the primitive ideals themselves need not be equal. There is in general no inclusion-minimal nonzero ideal defining a given nontrivial blow-up: taking positive powers leaves its Proj unchanged.*

*Proof.* Both constructions are the closure of the same section in the same Hilbert scheme; the two Hilbert immersions only embed that scheme in different Grassmannians. The graph identification follows from Proposition 2.1. Normalization is intrinsic to the integral graph. On the normalization, an exceptional divisor is a prime divisor independently of its chosen relatively ample line bundle. Those line bundles can have different orders, as the preceding example demonstrates. Finally the Rees algebra of  $I^n$  is a Veronese subalgebra of that of  $I$ ; a proper nonzero ideal in a domain has strictly smaller powers at any point where it is proper.  $\square$

The worst-case presentation has  $m_a = \frac{1}{2}a^2 + O(a)$ ,  $q = \frac{1}{2}a^3 + O(a^2)$ , and  $N = \frac{1}{16}a^6 + O(a^5)$ . There are at most  $\binom{N}{q}$  candidate maximal minors, with

$$\log \binom{N}{q} = \frac{3}{2}a^3 \log a + O(a^3 + a^2 \log a).$$

This is a size bound, not a practical algorithm. The next theorem replaces this list by  $a$  two-generated ideals and at most  $a + 1$  curve equations on each chart of an all-order family.

### 3. A CHAIN MODEL FOR CONTACTS OF ARBITRARY ORDER

For  $a \geq 2$  consider the marked family over  $S_a = \text{Spec } k[b, c]$

$$[F_0 : F_1 : F_2] = [t^a : bs^a : cst^{a-1}]. \quad (3.1)$$

Let  $T_a$  be the closure of its Hilbert graph over  $b \neq 0$  in  $S_a \times \text{Hilb}^{P_a}(X)$ . This is a strict restriction of the coefficient graph. The full fibre product with  $\Gamma_a$  can have additional vertical components and is not the definition of  $T_a$ .

**Theorem 3.1** (The all-order chain model). *For  $m \geq m_a$  set*

$$C = \frac{(a-1)m(m+1)}{2}, \quad E_j = m - j + 1 \quad (1 \leq j < a), \quad E_a = \frac{(m-a+1)(m-a+2)}{2}.$$

*The restriction of the evaluation ideal is exactly*

$$I_q(M_{a,m})\mathcal{O}_{S_a} = b^C J_{a,m}, \quad J_{a,m} = \prod_{j=1}^a (b, c^j)^{E_j}. \quad (3.2)$$

*Its divisorial content is  $b^C$ . The ideals  $J_{a,m}$  and all their powers are integrally closed, and*

$$T_a = \text{Bl}_{J_{a,m}} S_a = \text{Bl}_{K_a} S_a, \quad K_a = \prod_{j=1}^a (b, c^j). \quad (3.3)$$

*This surface is smooth. It is obtained by blowing up the origin and then,  $a-1$  times, the intersection of the newest exceptional curve with the strict transform of  $b=0$ . Its exceptional divisors have valuations*

$$v_i(b) = i, \quad v_i(c) = 1 \quad (1 \leq i \leq a).$$

*The universal curve admits the explicit flat presentations below. The theorem classifies the marked family (3.1) for every  $a$ ; it does not identify the whole fibre of  $\Gamma_a \rightarrow B_a$  with the fibre of this surface.*

### 3.1. A row factorization, without enumerating minors.

*Proof of (3.2).* Every column has one nonzero entry. If  $h, l$  are the exponents of the second and third forms, that entry is  $b^h c^l$  in row  $n = i + a(m - h - l) + (a - 1)l$ ,  $0 \leq i \leq m$ . A nonzero full minor chooses one column in each row; columns belonging to distinct rows are necessarily distinct. The determinant ideal is therefore the product of the row ideals  $Q_n$ .

Put  $d = am - n$ . Rows with  $n \geq am$  have  $Q_n = (1)$ . For  $d \geq 0$  set

$$h_0 = \max \left( 0, \left\lceil \frac{d-m}{a-1} \right\rceil \right), \quad r = d - ah_0.$$

For a fixed  $h$ , the smallest feasible  $l$  is  $\max(0, d - ah)$ ; feasibility requires  $h \geq h_0$ . Removing dominated monomials gives

$$Q_n = \begin{cases} (b^{h_0}), & r \leq 0, \\ b^{h_0} (b, c^a)^u (b, c^\rho), & r = au + \rho > 0, \quad 0 \leq \rho < a, \end{cases} \quad (3.4)$$

where the last factor is omitted for  $\rho = 0$ . In fact the minimal list in the second case is  $b^{h_0+v} c^{\max(0, r-av)}$ ,  $0 \leq v \leq \lceil r/a \rceil$ . Expanding the right side of (3.4) and discarding dominated terms gives precisely this list.

The sum of the  $h_0$  is  $(a-1)(1 + \dots + m) = C$ . Among all rows, a positive residual value  $r \in \{1, \dots, m\}$  occurs

$$\nu(r) = \min(a, m - r + 1)$$

times. One occurrence comes from  $h_0 = 0$ ; for  $h_0 = h \geq 1$  it comes from  $m - a + 2 - h \leq r \leq m - h$ . Counting the residual factors gives

$$\sum_{\substack{r \equiv j \pmod{a} \\ 1 \leq r \leq m}} \nu(r) = m - j + 1 \quad (1 \leq j < a), \quad \sum_{r=1}^m \nu(r) \left\lfloor \frac{r}{a} \right\rfloor = \sum_{n=a}^m (n - a + 1) = E_a.$$

For the first equality write  $n = r + v$ ,  $0 \leq v < a$ , and count each  $n \in \{j, \dots, m\}$  once. For the second, at a fixed  $n$  sum  $\lfloor r/a \rfloor$  over  $\max(1, n - a + 1) \leq r \leq n$ ; the answer is  $\max(0, n - a + 1)$ . This proves the ideal identity. The argument in fact works for every  $m \geq a$ ; the regularity bound is used only to identify its graph with the Hilbert graph.  $\square$

### 3.2. The Rees algebra and its rays.

**Proposition 3.2** (The normal Rees algebra). *Put  $\eta_i = \sum_{j=1}^a E_j \min(i, j)$ . Then*

$$\mathcal{R}(J_{a,m}) = k[b^u c^v T^n : u, v, n \geq 0, \quad iu + v \geq n\eta_i \ (1 \leq i \leq a)]. \quad (3.5)$$

*The right side is already normal, not merely the normalization of the Rees algebra. For the raw ideal in (3.2) replace  $u \geq 0$  by  $u \geq nC$  and the last inequalities by  $iu + v \geq n(iC + \eta_i)$ . The exceptional Rees valuations of the primitive ideal are exactly  $v_1, \dots, v_a$ .*

*Proof.* For a prescribed integer exponent  $u$  of  $b$ , the least exponent of  $c$  in a product of the factors  $(b, c^j)^{E_j}$  is obtained by replacing  $c^j$  by  $b$  first for the largest  $j$ . Thus every integer abscissa on the lower Newton boundary has an integer ordinate achieved by an actual monomial generator. Its successive slopes are  $-a, \dots, -1$ , each with positive integral horizontal length  $E_a, \dots, E_1$ . The integral lattice points above this boundary are exactly the monomials of the ideal. Replacing  $E_j$  by  $nE_j$  proves the same assertion for every power. The half-space description of that boundary is  $iu + v \geq \eta_i$ ; adjoining the degree variable proves (3.5). A semigroup cut out by these integral linear inequalities is saturated in its group. Its semigroup algebra is normal. The compact facets have primitive inward normals  $(i, 1)$ , and their positive lengths show that none is redundant. They give exactly the exceptional prime divisors of the normalized blow-up. The shift by  $b^{nC}$  proves the raw formula. The raw ideal also records the nonexceptional strict transform of  $b = 0$ ; this is its content, not an additional exceptional ray.  $\square$

*Proof of the surface assertions in Theorem 3.1.* The graph Rees lemma and removal of  $b^C$  give the first equality in (3.3). The normal fan of the computed Newton polygon has, in order, the rays

$$(1, 0), (a, 1), (a-1, 1), \dots, (1, 1), (0, 1).$$

Every adjacent determinant is one. The same fan is obtained when all  $E_j$  are replaced by one, proving the second equality and smoothness. Successive subdivisions insert  $(1, 1), (2, 1), \dots, (a, 1)$  and give the stated sequence of point blow-ups. Equivalently, a product of positive powers of nonzero ideals is invertible on an integral test scheme exactly when every factor is invertible as a fractional ideal; the blow-up universal property gives the same identification.  $\square$

There are  $a + 1$  affine charts. The first and last are

$$T_0 = \operatorname{Spec} k[b, k], \quad c = bk; \quad T_a = \operatorname{Spec} k[c, z], \quad b = c^a z.$$

For  $1 \leq i < a$  the intermediate chart is

$$T_i = \operatorname{Spec} k[e, h], \quad c = eh, \quad b = e^{i+1} h^i, \quad e = b/c^i, \quad h = c^{i+1}/b. \quad (3.6)$$

Adjacent transition formulas are

$$\begin{aligned} k &= e^{-1}, \quad b = e^2 h \quad (T_0, T_1), \\ e_{i+1} &= h_i^{-1}, \quad h_{i+1} = e_i h_i^2 \quad (1 \leq i < a-1), \\ c &= e_{a-1} h_{a-1}, \quad z = h_{a-1}^{-1} \quad (T_{a-1}, T_a). \end{aligned} \quad (3.7)$$

All further overlaps are compositions of these formulas.

**3.3. Low-degree equations for the universal curve.** In the source chart  $s = 1$ , put  $x = t/s$ . Target coordinates  $F, G, R$  remain homogeneous. On  $T_0$  use

$$R - kx^{a-1}G, \quad x^a G - bF. \quad (3.8)$$

On  $T_i$ ,  $1 \leq i < a$ , use

$$\begin{aligned} A &= xR - ehF, \\ B_j &= x^{a-j}F^{j-1}G - e^{i+1-j}h^{i-j}R^j \quad (1 \leq j \leq i), \\ C_i &= R^{i+1} - hx^{a-i-1}F^iG. \end{aligned} \quad (3.9)$$

On  $T_a$  use

$$A = xR - cF, \quad B_j = x^{a-j}F^{j-1}G - zc^{a-j}R^j \quad (1 \leq j \leq a). \quad (3.10)$$

The ideal sheaves on  $X$  are obtained by bihomogenizing these equations and saturating by the source and target irrelevant ideals. In the first chart a convenient global list adds  $tR - bksF$  to the bihomogenizations of (3.8); this prevents a spurious component at  $s = 0$ . In the other charts the listed bihomogenizations already give the following two-source-chart cover.

**Proposition 3.3** (Flatness, exhaustion, and overlaps). *Equations (3.8)–(3.10) and (3.7) define the flat universal embedded curve over  $T_a$ . They cover all of that curve, including source infinity. Each intermediate chart has a monic Gröbner basis over  $k[e, h]$ , and the last chart has one over  $k[c, z]$ . Thus no coefficient stratum is excluded by a division of a leading coefficient.*

*Proof.* For the intermediate chart choose positive rational weights

$$\text{wt}(F) = \text{wt}(G) = 1, \quad \text{wt}(x) = \alpha, \quad \text{wt}(R) = 1 + \rho\alpha, \quad \frac{a-i-1}{i+1} < \rho < \frac{a-i}{i}, \quad \alpha > 0.$$

Refine by any monomial order. The leading monomials of (3.9) are respectively

$$xR, \quad x^{a-j}F^{j-1}G \quad (1 \leq j \leq i), \quad R^{i+1}.$$

Buchberger reductions are uniform in the coefficient ring. The  $S$ -polynomial of  $A, B_j$  is  $-ehB_{j+1}$  for  $j < i$ , and is  $eC_i$  for  $j = i$ . The one for  $A, C_i$  is  $hFB_i$ . For  $j < l$ , that of  $B_j, B_l$  is

$$e^{i+1-l}h^{i-l}R^j((xR)^{l-j} - (ehF)^{l-j}),$$

and reduces by  $A$ . The remaining pairs have relatively prime leading monomials. This proves the monic basis assertion and a free standard monomial basis over  $k[e, h]$ .

For the last chart use lexicographic order  $x > F > G > R$ . The same reductions replace  $eh$  by  $c$  and the coefficient of  $R^j$  by  $zc^{a-j}$ ; the pair  $A, B_a$  has relatively prime leading monomials. The first chart eliminates  $R$  and leaves the monic hypersurface  $x^aG - bF$ . These bases make the target-graded coordinate rings flat over their parameter rings. Localizing in a target coordinate and taking the degree-zero part preserves flatness.

For completeness put  $y = s/t$  on the other source chart. On every parameter chart the bihomogeneous equations reduce to

$$G = by^aF, \quad R = cyF. \quad (3.11)$$

For example  $A$  gives the second equation, and  $B_1$  gives the first; the other equations then vanish. Here  $F$  is a unit on the projective target. This is the entire family near source infinity and is flat. On  $s, t \neq 0$  the two displayed presentations agree by substitution.

Over the dense parameter torus all the equations are the graph of (3.1). Their flat coordinate algebras are torsion-free over the integral parameter ring, so each is the schematic closure of that generic graph. Substitution of (3.7) therefore identifies the ideal sheaves on each overlap, not merely their reduced supports: both are the same closure in the fixed  $X$ . The substitutions of  $b, c$  and the identity on  $[F : G : R]$  are explicit, so no target automorphism is hidden in this argument. The glued family is projective, flat, and has generic polynomial  $P_a$ ; connectedness gives that polynomial on every fibre. Its Hilbert map agrees with the defining map of  $T_a$  on a dense open. The graph identification proves that it is the universal curve.  $\square$



**Proposition 3.4** (Boundary geometry and contractions). *The scheme fibre of  $T_a \rightarrow S_a$  at the origin is the reduced chain  $E_1 \cup \cdots \cup E_a$ , with*

$$E_i^2 = -2 \ (i < a), \quad E_a^2 = -1, \quad E_i E_{i+1} = 1, \quad K_{T_a/S_a} = \sum_{i=1}^a i E_i.$$

If  $B', C'$  are the strict transforms of the two axes, then

$$\operatorname{div}(b) = B' + \sum i E_i, \quad \operatorname{div}(c) = C' + \sum E_i.$$

The Picard group is freely generated by the exceptional classes. A relative divisor  $D = \sum d_i E_i$  is nef precisely when

$$d_{i-1} - 2d_i + d_{i+1} \geq 0 \ (1 \leq i < a), \quad d_{a-1} - d_a \geq 0, \quad d_0 = 0.$$

Deleting rays gives the toric contraction morphisms. In particular, contracting the intermediate chain between  $(i, 1)$  and  $(j, 1)$  gives an  $A_{j-i-1}$  singularity when  $j - i > 1$ .

*Proof.* In an intermediate chart  $(b, c) = (e^{i+1}h^i, eh) = (eh)$ ; at the end charts its pullback is  $(b)$  or  $(c)$ . Thus the fibre is reduced, with ordinary crossings. The self-intersections follow either from the successive point blow-ups or the relation between adjacent primitive rays. The Jacobian in (3.6) is  $e^{i+1}h^i$ , and the axis orders are the stated valuations. Each point blow-up adds a free exceptional generator to the Picard group of the preceding smooth surface, starting with  $\operatorname{Pic}(\mathbb{A}^2) = 0$ . The only contracted complete curves are the  $E_i$ , so intersecting with them gives the nef inequalities. Coarsening a fan gives a proper birational toric morphism. The two rays  $(i, 1), (j, 1)$  have determinant  $j - i$  and lie at height one; the resulting affine surface is the cyclic quotient, equivalently the hypersurface  $uv = w^{j-i}$ .  $\square$

The unweighted ideal  $K_a$  is degree-independent and has exactly the required Newton rays, but is not an inclusion-minimal graph ideal. The powers  $K_a^n$  already preclude such a minimality assertion. Its boundary chain has a modular meaning for the marked family: it is the universal modification on integral tests on which every  $(b, c^j)$  becomes invertible. This is not a proposed intrinsic boundary complex for all pencils of all contact types.

#### 4. EMBEDDED LIMITS AND THE PUNCTUAL GENUS CORRECTION

The parameter fibre of  $T_a$  is a reduced chain; its universal curves need not be reduced or Cohen–Macaulay. These are different assertions about different schemes. We now determine both the one-dimensional parts and the embedded zero-dimensional structure of every curve over that chain.

**4.1. All formal arcs in the marked family.** Let  $b = \beta\tau^p$ ,  $c = \gamma\tau^q$ , with  $p, q > 0$  and  $\beta, \gamma \in k[[\tau]]^*$ . The units may be arbitrary formal series. The following ideals are written on  $s = 1$  with homogeneous target coordinates. At source infinity they are glued to  $G = R = 0$ ; equivalently, take the ideal sheaves specified by Proposition 3.3.

**Theorem 4.1** (The complete arc table). *There are  $a$  walls  $p = jq$ ,  $1 \leq j \leq a$ , and  $a + 1$  open chambers. The special embedded ideals are as follows:*

$$\mathcal{O}_0 \ (p < q) : \ (R, x^a G), \tag{4.1}$$

$$\mathcal{W}_1 \ (p = q) : \ (R - \kappa x^{a-1} G, x^a G), \quad \kappa = \bar{\gamma}/\bar{\beta}, \tag{4.2}$$

$$\mathcal{O}_i \ (iq < p < (i+1)q) : \ (xR, R^{i+1}, x^{a-j} F^{j-1} G \ (1 \leq j \leq i)), \quad 1 \leq i < a, \tag{4.3}$$

$$\mathcal{W}_j \ (p = jq) : \ (xR, x^{a-r} F^{r-1} G \ (1 \leq r < j), R^j - \lambda x^{a-j} F^{j-1} G), \quad 2 \leq j \leq a, \tag{4.4}$$

$$\mathcal{O}_a \ (p > aq) : \ (xR, x^{a-j} F^{j-1} G \ (1 \leq j \leq a)). \tag{4.5}$$

In (4.4),  $\lambda = \bar{\tau}^j / \bar{\beta}$ . Each wall is a one-dimensional family with nonzero residue parameter; each open chamber gives one fixed embedded curve. The closure of the  $j$ th wall family is  $E_j \simeq \mathbb{P}^1$ , with endpoints  $\mathcal{O}_{j-1}, \mathcal{O}_j$ . These are all components, points, and adjacencies of the parameter fibre of  $T_a$ . There is only one normalization branch at every point of this smooth parameter surface. The fan is minimal for these fixed-target embedded limits.

*Proof.* In the first chart  $k = c/b$  has nonnegative valuation exactly when  $p \leq q$ . Its residue is zero in the first chamber and is  $\kappa$  on the first wall. In  $T_i$  the two coordinates have valuations  $p - iq$  and  $(i + 1)q - p$ . Setting their positive-valuation coordinates to zero in (3.9) gives (4.3). Setting  $e = 0$  and retaining  $h = \lambda \neq 0$  gives the next wall; the lower wall is its expression in the adjacent chart. In the last chart  $z = b/c^a$  has nonnegative valuation exactly when  $p \geq aq$ . Its nonzero residue gives  $\mathcal{W}_a$  with  $\lambda = \bar{z}^{-1}$ ;  $z = 0$  gives (4.5). Properness and the chart cover exhaust all arcs. If  $c$  is identically zero, the first chart gives (4.1) as well.

Every nonzero wall residue is attained by an arc with  $p = jq$ . The transition formulas identify its two endpoints as stated. The Newton facets in Proposition 3.2 are all present; equivalently, the residue term in (4.2) or (4.4) changes the ideal sheaf as the residue varies. A wall cannot therefore be removed while retaining the embedded limit map. This minimal fan belongs to the marked two-parameter problem, not to an arbitrary coefficient presentation on  $B_a$ .  $\square$

**Theorem 4.2** (A sharp, degree-independent jet bound on the family). *Let a formal arc in  $S_a$  be centred at the origin and satisfy  $\text{ord}(b) = p < \infty$ . Its closed embedded Hilbert limit is determined by the coefficient arc modulo  $\tau^{p+1}$ . This bound is independent of  $a$  and of the Hilbert embedding degree. For every  $p \geq 1$  it is optimal as a uniform sufficient bound over arcs with that value of  $p$ .*

*Proof.* If  $\text{ord}(c) > p$ , including  $c = 0$ , any congruent arc is still in  $\mathcal{O}_0$ . Otherwise congruence preserves  $p, q$  and the leading coefficients of both  $b, c$ . It preserves every wall residue in Theorem 4.1, which proves sufficiency. For necessity take  $c = \tau^p$  and  $b = \beta\tau^p$  for two different nonzero constants  $\beta$ . The arcs agree modulo  $\tau^p$ , but the first-wall ideals differ: on  $G = 1$ , the nonzero class  $x^{a-1}$  has  $R = \beta^{-1}x^{a-1}$  and  $x^a = 0$ . Thus that jet does not determine the fixed-target limit.  $\square$

Here  $p$  is the intersection order with the marked base-point divisor  $b = 0$ . It is not the raw Smith length. In degree  $m$  the latter is

$$\mu = Cp + \sum_{j=1}^a E_j \min(p, jq), \quad (4.6)$$

whereas the primitive degree- $m$  ideal has order given by the sum without  $Cp$ , and the unweighted graph ideal has order  $\sum_{j=1}^a \min(p, jq)$ . These are three distinct quantities, none of which should be substituted for the optimal family bound  $p + 1$ . For  $a = 3, m = 7$ , arcs  $(b, c) = (\tau, \tau^2)$  and  $(\tau^2, \tau^3)$  have the same special Hilbert curve but raw Smith lengths 84 and 168. Ramifying an arc multiplies all ideal orders, while leaving its embedded closed limit unchanged.

**4.2. Local rings and associated points.** Let  $L_0$  denote the main component  $G = R = 0$ . Let  $L$  denote the tail  $x = R = 0$ , and put

$$P = (x = 0, [F : G : R] = [0 : 1 : 0]).$$

All local ideals below are ordinary ideals in the indicated affine rings, before passage to sheaves. Write  $H_P^0(\mathcal{O}_C)$  for the largest submodule supported at the closed point  $P$ , not for the whole nilradical of a curve with a generically thick tail.

**Theorem 4.3** (Pure tails and punctual excess). *For  $\mathcal{O}_0$  and  $\mathcal{W}_1$  the curve is Cohen–Macaulay, with components  $L_0, L$  of generic multiplicities 1,  $a$ . For  $1 \leq i < a$ , the curve  $\mathcal{O}_i$  has the same*

two minimal components and multiplicities. On  $F = 1$  its ideal is

$$(xR, x^{a-i}G, R^{i+1}) = (G, R) \cap (xR, x^{a-i}, R^{i+1}). \quad (4.7)$$

On  $G = 1$  it is

$$(xR, R^{i+1}, x^{a-i}(x, F)^{i-1}). \quad (4.8)$$

There is precisely one embedded associated point when  $i \geq 2$ , namely  $P$ , and none when  $i = 1$ . Its punctual module is

$$H_P^0(\mathcal{O}_{\mathcal{O}_i}) \simeq x^{a-i}k[x, F]/x^{a-i}(x, F)^{i-1}, \quad R \cdot H_P^0 = 0, \quad \text{length } H_P^0 = \binom{i}{2}. \quad (4.9)$$

For  $2 \leq j \leq a$  put  $d = a - j$ . The wall curve  $\mathcal{W}_j$  has on  $F = 1$  the complete-intersection ideal

$$(xR, R^j - \lambda x^d G). \quad (4.10)$$

On  $G = 1$  its ideal is

$$(xR, x^{d+1}(x, F)^{j-2}, R^j - \lambda x^d F^{j-1}). \quad (4.11)$$

Its pure quotient on this chart is obtained by adding  $x^{d+1}$ . The kernel is

$$H_P^0(\mathcal{O}_{\mathcal{W}_j}) \simeq x^{d+1}k[x, F]/x^{d+1}(x, F)^{j-2}, \quad R \cdot H_P^0 = 0, \quad \text{length } H_P^0 = \binom{j-1}{2}. \quad (4.12)$$

For  $j < a$  the minimal components are again  $L_0, L$  with multiplicities  $1, a$ . For  $j = a$  the tail is the reduced plane curve  $R^a = \lambda F^{a-1}G$ . In both cases the only possible embedded point is  $P$ , and it occurs exactly when  $j \geq 3$ .

*Proof.* Eliminating  $R$  in the first two cases leaves a hypersurface, and gives the multiplicities directly. For  $\mathcal{O}_i$ , localization at  $F$  reduces all the monomials in (4.3) to (4.7). The intersection identity follows by monomial divisibility. The second component is primary to  $(x, R)$  and is free of rank  $a$  over  $k[G]$ , with transverse basis

$$1, x, \dots, x^{a-i-1}, R, \dots, R^i.$$

The entire ring in (4.7) has a length-two Hilbert–Burch resolution; alternatively its three monomial generators have the two syzygies coming from  $xR^{i+1}$  and  $x^{a-i}GR$ . It is one-dimensional Cohen–Macaulay and has no embedded point.

On  $G = 1$  the ideal is (4.8). Adding  $x^{a-i}$  gives the pure primary ideal  $(xR, R^{i+1}, x^{a-i})$ . The kernel is generated by the image of  $x^{a-i}$ , with annihilator  $(R, (x, F)^{i-1})$  on that generator. This proves (4.9). For  $i \geq 2$  an explicit primary decomposition is

$$(xR, R^{i+1}, x^{a-i}) \cap (xR, R^{i+1}, (x, F)^{a-1}).$$

The second factor is primary to  $(x, F, R)$ . The displayed length is the number of monomials in two variables of total degree below  $i - 1$ .

For a wall, the equations reduce to (4.10) and (4.11). The former is a regular sequence of length two. In the latter, after adjoining  $x^{d+1}$ , there is a free  $k[F]$ -basis

$$1, x, \dots, x^d, R, \dots, R^{j-1};$$

the relation  $R^j = \lambda F^{j-1}x^d$  and  $xR = 0$  give its multiplication. Hence this pure ring is Cohen–Macaulay of rank  $a$  over  $k[F]$ . The kernel of the pure quotient is generated by  $x^{d+1}$  and has exactly the annihilator described in (4.12). No extra relation arises from multiplying  $R^j - \lambda x^d F^{j-1}$  by  $x$ : the resulting  $x^{d+1}F^{j-1}$  already lies in  $x^{d+1}(x, F)^{j-2}$ . A monomial reduction in  $x, F$ , followed by reduction in  $R$ , proves that these are all the relations. This also computes the length. For  $d > 0$  the pure ideal is primary to  $(x, R)$ ; for  $d = 0$  it is the reduced cuspidal tail ideal  $(x, R^a - \lambda F^{a-1})$ .

There are no missing associated points. Away from  $P$  the  $F$  chart or its overlap has just been treated. On  $R = 1$  the nonterminal cases have empty special fibre, while on the last wall  $F, G$  are units and (4.10) applies. Source infinity is (3.11). These charts cover the curve.  $\square$

**Theorem 4.4** (The cuspidal wall and its nilradical). *On the last wall,  $\mathcal{W}_a$  consists of its reduced curve  $L_0 \cup Q_a$  together with a punctual nilradical at  $P$ , where*

$$Q_a : x = 0, \quad R^a = \lambda F^{a-1}G.$$

The normalization of  $Q_a$  is  $\mathbb{P}^1$ , with map

$$[u : v] \longmapsto [u^a : \lambda^{-1}v^a : u^{a-1}v]. \quad (4.13)$$

The main component meets  $Q_a$  transversely at its smooth point  $[1 : 0 : 0]$ . Put  $g_a = (a-1)(a-2)/2$ . The nilradical  $\mathcal{N}$  has length  $g_a$  and, for  $a \geq 3$ , nilpotence order  $a-1$ . More precisely, on  $G = 1$  it is the ideal generated by  $x$  with

$$\mathcal{N} \simeq xk[x, F]/x(x, F)^{a-2}, \quad R\mathcal{N} = 0, \quad \text{length}(\mathcal{N}^r) = \binom{a-r}{2} \quad (1 \leq r \leq a-2). \quad (4.14)$$

For  $a = 2$  it is zero. The Hilbert polynomial of the reduced curve is  $(a+1)l + 1 - g_a$ ; this exact nilradical restores  $P_a(l) = (a+1)l + 1$ . Thus the punctual part is forced by arithmetic genus, not removable excess in a flat Hilbert limit.

*Proof.* Set  $j = a, d = 0$  in (4.11). The ring is

$$k[x, F, R]/(xR, x(x, F)^{a-2}, R^a - \lambda F^{a-1}).$$

Its reduction is the domain  $k[F, R]/(R^a - \lambda F^{a-1})$ ; the nilradical is exactly  $(x)$ . The formula for its module follows from Theorem 4.3. A basis of its  $r$ th power consists of  $x^u F^v$  with  $u \geq r$  and  $u + v \leq a-2$ . Counting these monomials gives (4.14) and proves the nilpotence statement, including nonvanishing of  $x^{a-2}$ . The parametrization (4.13) is birational; on the cusp chart its local semigroup is generated by  $a-1, a$ . Its missing nonnegative exponents number  $g_a$ . The only singularity of the plane tail is that cusp, while at the attaching point  $F = 1$  the full reduced local ring is  $k[x, R]/(xR)$  after eliminating  $G$ . The normalization of the whole reduction is consequently two copies of  $\mathbb{P}^1$ . A plane curve of degree  $a$  has arithmetic genus  $g_a$ ; attaching the main line at one point gives the stated reduced Hilbert polynomial. The exact sequence  $0 \rightarrow \mathcal{N} \rightarrow \mathcal{O}_{\mathcal{W}_a} \rightarrow \mathcal{O}_{(\mathcal{W}_a)_{\text{red}}} \rightarrow 0$  then proves the correction formula.  $\square$

**Proposition 4.5** (The terminal chamber). *The curve  $\mathcal{O}_a$  has three minimal components:  $L_0$  with multiplicity one, the tail  $x = G = 0$  with multiplicity one, and the tail  $x = F = 0$  with multiplicity  $a-1$ . Their reduced supports form a chain of three lines. On  $F = 1$  the ideal is  $(G, xR)$ ; on  $G = 1$  it is*

$$(xR, (x, F)^{a-1}).$$

The latter has pure quotient  $(x, F^{a-1})$  and punctual kernel  $xk[x, F]/x(x, F)^{a-2}$  of length  $g_a$ . On  $R = 1$  the ideal is  $(x, F^{a-1}G)$ , a hypersurface with no embedded point. Thus  $P$  is the only embedded associated point when  $a \geq 3$ , and there is none when  $a = 2$ . The full nilradical is not just this punctual module: the  $F = 0$  tail is generically nonreduced for  $a > 2$ .

*Proof.* Localize (4.5) in each target coordinate. On  $G = 1$  the monomials  $x^{a-j}F^{j-1}$  generate  $(x, F)^{a-1}$ . Adding  $x$  yields the pure ring, free of rank  $a-1$  over  $k[R]$ ; the kernel has the stated two-variable basis and is killed by  $R$ . On  $R = 1$ ,  $xR$  eliminates  $x$  and leaves  $F^{a-1}G$ . These ordinary affine calculations give the minimal multiplicities and all associated points. The normalization of the reduction is the disjoint union of its three lines. The pure plane tail  $F^{a-1}G = 0$  has arithmetic genus  $g_a$ , so the same length correction also recovers  $P_a$  in this terminal chamber.  $\square$

**4.3. The first higher-contact instance.** For  $a = 3$  the Gotzmann degree is 7, and

$$I_{29}(M_{3,7})\mathcal{O}_{S_3} = b^{56}(b, c)^7(b, c^2)^6(b, c^3)^{15}. \quad (4.15)$$

The parameter fibre has three components, with self-intersections  $-2, -2, -1$ . Every one is one-dimensional, their successive intersections are single reduced points, and there are no other intersections or normalization branches. The seven embedded curve families are collected here; each ideal is on  $s = 1$ .

| region        | ideal                                 | length $H_P^0$ |
|---------------|---------------------------------------|----------------|
| $p < q$       | $(R, x^3G)$                           | 0              |
| $p = q$       | $(R - \kappa x^2G, x^3G)$             | 0              |
| $q < p < 2q$  | $(xR, x^2G, R^2)$                     | 0              |
| $p = 2q$      | $(xR, x^2G, R^2 - \lambda xFG)$       | 0              |
| $2q < p < 3q$ | $(xR, x^2G, xFG, R^3)$                | 1              |
| $p = 3q$      | $(xR, x^2G, xFG, R^3 - \lambda F^2G)$ | 1              |
| $p > 3q$      | $(xR, x^2G, xFG, F^2G)$               | 1              |

At the last wall this length-one module is square-zero and is supported at the cusp of the cubic tail. At the preceding open chamber it is an embedded point on a generically triple line. These are different local algebras with the same required Euler-characteristic correction. For  $a \geq 4$  the last-wall nilradical is no longer square-zero, as (4.14) shows. This is why the first-contact extension algebra alone cannot describe higher contacts.

The entire  $S_3$  parameter fibre has been classified above, including its universal curves. The full  $B_3$  parameter fibre has more coefficient directions, and no equality with this three-component chain is asserted. The distinction remains important when the marked family is inserted into a local Smith normal form of a pencil.

## 5. COORDINATE DESCENT AND COMPARISONS OF COMPACTIFICATIONS

We first give an actual gluing statement at a fixed degree. It removes the dependence on a monic pivot chart without identifying constructions with different generic Hilbert polynomials.

**Theorem 5.1** (Equivariant gluing of the fixed-degree graphs). *Put*

$$\mathcal{P}_a = \mathbb{P}(H^0(\mathbb{P}^1, \mathcal{O}(a)) \otimes k^3), \quad H_a = \text{Hilb}^{P_a}(\mathbb{P}^1 \times \mathbb{P}^2),$$

and let  $\mathcal{U}_a \subset \mathcal{P}_a$  be the base-point-free locus. The closure

$$\mathcal{G}_a = \overline{\{(F, [\text{graph}(F)]) : F \in \mathcal{U}_a\}} \subset \mathcal{P}_a \times H_a$$

is an integral projective compactification of the graph of degree- $a$  morphisms with their coefficients retained. It carries a flat universal embedded curve and a compatible  $\text{PGL}_2 \times \text{PGL}_3$  action. On every monic pivot patch it is isomorphic, after the specified undoing of a target change, to  $\mathbb{A}^2 \times \Gamma_a$ . These patch isomorphisms satisfy the cocycle identity on all overlaps, including boundary points. The normalization also carries the action. The morphism

$$[\mathcal{G}_a / (\text{PGL}_2 \times \text{PGL}_3)] \longrightarrow [\mathcal{P}_a / (\text{PGL}_2 \times \text{PGL}_3)]$$

is representable and proper, with projective pullback to a scheme trivializing the group torsor.

*Proof.* The Hilbert scheme is projective and separated, so the closure is projective and integral and is an isomorphism over  $\mathcal{U}_a$ . Pullback of the universal Hilbert family is flat. A pair of source and target automorphisms takes a graph to the graph of the transformed map, preserves  $\mathcal{O}(1, 1)$ , and hence acts on its Hilbert scheme. It preserves the closure and its universal curve.

Here are the patch maps explicitly. Choose a source point where some coordinate form is nonzero, send it to  $[0 : 1]$ , and permute that target coordinate into position zero. In the projective

coefficient patch normalize its  $t^a$  coefficient to one and write

$$F_0 = f^h, \quad F_1 = \alpha_1 f^h + g^h, \quad F_2 = \alpha_2 f^h + r^h, \quad \deg g, \deg r < a.$$

This identifies the coefficient patch with  $\mathbb{A}^2 \times B_a$ . The target matrix and its inverse are

$$U(\alpha) = \begin{pmatrix} 1 & 0 & 0 \\ -\alpha_1 & 1 & 0 \\ -\alpha_2 & 0 & 1 \end{pmatrix}, \quad U(\alpha)^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ \alpha_1 & 1 & 0 \\ \alpha_2 & 0 & 1 \end{pmatrix}. \quad (5.1)$$

The first matrix identifies the graph with the polynomial-contact graph; the second returns it to the original fixed target. Since the product with  $\mathbb{A}^2$  is flat, taking the closure of the generic graph commutes with this product. Thus the closure on the patch is exactly the asserted product, with its scheme structure.

The patches cover  $\mathcal{P}_a$ : a nonzero form cannot vanish at  $a + 1$  distinct chosen source points, so finitely many such choices and the three target pivots suffice. On an overlap the change is the composition of the actual source matrices, the target permutations, and (5.1). Its inverse reverses that composition. The product of the matrices on a triple overlap is the identity as a projective transformation. The induced Hilbert maps therefore satisfy the same identity. No comparison based only on reduced closed points is used.

For normalization, the product of the smooth connected group with the normal variety  $\mathcal{G}_a^\nu$  is normal. The action composed with normalization is dominant and lifts uniquely to the normalization; uniqueness proves the action identities. Finally an equivariant morphism induces a representable map of the quotient stacks. After a torsor is trivialized its pullback is the projective morphism  $\mathcal{G}_a \rightarrow \mathcal{P}_a$ ; properness descends.  $\square$

This theorem glues *all* pivot charts of a fixed-degree rational-map problem. Its construction is classical graph compactification; the explicit patch maps state exactly how the contact graphs sit in it. It is not a gluing of every Smith and Kronecker stratum of every quadratic-pencil Hilbert problem. In particular, an adjugate tuple that vanishes identically supplies no point of  $\mathcal{P}_a$ .

**Corollary 5.2** (Moving contacts and mixed target coefficients). *Over  $\mathbb{A}_\alpha^1 \times S_a$  the family*

$$((x - \alpha)^a, b, c(x - \alpha)^{a-1})$$

*has strict Hilbert graph  $\mathbb{A}_\alpha^1 \times T_a$ , with the universal curve obtained by substituting  $x - \alpha$  for  $x$ . Over the further open parameter space  $1 - uv \neq 0$ , the same statement holds for*

$$((x - \alpha)^a, b + uc(x - \alpha)^{a-1}, vb + c(x - \alpha)^{a-1}), \quad (5.2)$$

*after undoing the target matrix  $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & u \\ 0 & v & 1 \end{pmatrix}$ . For  $a = 2$  both latter coordinate polynomials can have nonzero constant and linear terms. This family is an equivariant enlargement of the marked slice, not the unrestricted six-coefficient contact family.*

*Proof.* Translation is a source automorphism over  $\mathbb{A}_\alpha^1$ . The indicated target matrix is invertible precisely on the stated open, with inverse lower block  $(1 - uv)^{-1} \begin{pmatrix} 1 & -u \\ -v & 1 \end{pmatrix}$ . They identify the entire ambient families, not just their generic fibres. Applying these isomorphisms to the flat universal curve and its graph closure proves the result.  $\square$

**Proposition 5.3** (Conservation of the Hilbert polynomial). *A projective flat finitely presented family over a connected base cannot have, as its fibres, just the graphs of morphisms of two different degrees. A degree drop from  $a$  to  $a - d$  in a limit of  $\mathcal{G}_a$  must be accompanied by additional scheme structure whose Hilbert polynomial contributes the missing degree and Euler characteristic. The extra structure may include punctual nilpotents, as in Theorem 4.4.*

*Proof.* The Hilbert polynomial for  $\mathcal{O}(1, 1)$  is locally constant in a projective flat family, whereas the two graphs have polynomials  $(a + 1)l + 1$  and  $(a - d + 1)l + 1$ . These differ for  $d > 0$ . In a fixed-degree Hilbert compactification the polynomial is retained by the whole limiting scheme, not by deleting its tails or embedded associated points.  $\square$

Thus comparison across different generic polynomials cannot consist of an equality of the unaugmented flat graph functors. The horizontal comparison and common-refinement results in Section D continue to apply on a fixed dense open with a fixed generic embedded quotient. They are not changed into a false full-pullback statement by the equivariant gluing above.

### 5.1. Tropical data, residues, and coordinate changes.

**Proposition 5.4** (Comparison of coefficient presentations). *For a finite list of polynomial projective coordinates, the finite weight-layer and residue construction of Theorem B.1 extends to exact monomial profiles with nonnegative weights, including specified unit coordinates, and to polynomial coordinates translated about any fixed centre. It gives finitely many algebraic families of closed projective points, not finitely many embedded curves or abstract isomorphism types.*

*Under an invertible monomial change of coefficient torus coordinates, the weights transform by the exponent matrix and the residues by the same monomial substitution; the finite partitions have a common pullback refinement. Under a general polynomial coordinate change, the corresponding statement holds after composing the projective coordinate polynomials with that change. An exact monomial arc need not remain monomial in the new coordinates, and the two displayed fans need not be isomorphic.*

*Proof.* Expand each of the finitely many polynomials into its finite monomial support, after any specified translation or polynomial substitution. The equalities of weights partition the nonnegative orthant into a finite rational polyhedral arrangement. Within a cone, group terms of equal weight and partition the residue torus by the first nonzero coefficient sum in each polynomial. Zero polynomials are kept as zero; identically zero input coordinates are a separate designated pattern. Every choice is finite, and the residual vector is polynomial on each locally closed piece. Weight zero introduces constant residue terms but does not change the argument. For a monomial substitution the weight and residue formulas follow directly from substitution. For a polynomial substitution repeat the finite expansion on the original monomial arc; this compares the resulting points even when there is no transformation of fans by a linear weight map. For localized-polynomial changes work on an open where denominators are units and clear them by a common denominator, which does not change the projective map.  $\square$

Applied to maximal minors, valuations are realizable valuated-matroid data: the minors satisfy all Plücker equations, and the minimum valuation in each nonzero relation is attained at least twice. On the coordinate stratum of the nonzero minors these are points of the tropicalization of that Grassmannian stratum, as in Speyer–Sturmfels [Speyer–Sturmfels(2004), Theorem 2.1 and Section 3]. It is not enough to check only a selected set of tropical quadratic relations and then claim realizability. Valuations forget residues; our first wall already has the same valuations for every  $\kappa \neq 0$  but different embedded ideals. The primitive Hilbert vector retains these residues and determines the degree- $m$  kernel. It does not select among normalization branches over a graph point.

On  $S_a$  the individual nonzero minors are monomials. The normal fan of their Newton polygon is exactly the chain fan, after removal of content; all its walls are effective. This identifies the computed fan with the restriction to nonnegative weights of the state fan of this marked torus orbit, in the classical initial-ideal setting of [Bayer–Morrison(1988)]. No such minimality claim is made for the full coefficient-support arrangement. In comprehensive Gröbner terminology, which organizes specializations by leading-coefficient conditions [Manubens–Montes(2006)], Proposition 3.3 gives

a particularly simple system: a single monic basis on every affine parameter chart, with no further coefficient conditions. Its explicit standard monomials, rather than a Hilbert-scheme elimination, recover the curve ideals.

**Proposition 5.5** (Equality up to a target automorphism). *Let two embedded curves with polynomial  $P_a$  have degree- $m$  ideal kernels  $K, K' \subset H^0(X, \mathcal{O}(m, m))$ ,  $m \geq m_a$ . They differ by a target automorphism exactly when there is  $A \in \mathrm{PGL}_3$  for which*

$$(1 \otimes \mathrm{Sym}^m A^*)K' = K.$$

*This is an invariant, finite algebraic criterion: equality of the two fixed-rank subspaces is given by rank conditions, with  $\det A \neq 0$ . It does not classify abstract curve isomorphisms. On a fixed wall of the marked family, all nonzero residue values are related by a diagonal target automorphism, although their fixed-target Hilbert points differ.*

*Proof.* An automorphism acts on sections by pullback. The Hilbert immersion in degree  $m$  makes equality of the kernels equivalent to equality of ideal sheaves. This proves both implications and the algebraic criterion. Scaling  $F, G, R$  independently scales  $\kappa$ , or scales  $\lambda$  by the ratio between  $R^j$  and  $F^{j-1}G$ ; every nonzero ratio is available over  $k$ . No assertion about different walls is needed for this last observation.  $\square$

## 5.2. A concrete comparison with Quot and stable-map limits.

**Proposition 5.6** (A Quot contraction of the entire exceptional chain). *The marked family defines a flat quotient on the fixed source*

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^1}(-a) \xrightarrow{(t^a, bs^a, cst^{a-1})} \mathcal{O}_{\mathbb{P}^1}^{\oplus 3} \longrightarrow \mathcal{Q} \longrightarrow 0 \quad (5.3)$$

*over all of  $S_a$ . The resulting morphism from  $T_a$  to the fixed-source Quot scheme is constant on its entire exceptional chain. The common central quotient is*

$$\mathcal{Q}_0 \simeq \mathcal{O}_{a[1:0]} \oplus \mathcal{O}_{\mathbb{P}^1}^{\oplus 2}.$$

*After adding three fixed source markings away from  $[1:0]$ , this also gives stable quotients in the sense of Marian–Oprea–Pandharipande. In contrast, the Hilbert map distinguishes every point of the chain.*

*Proof.* The first coordinate  $t^a$  makes the sheaf map fibrewise injective, even when  $b = c = 0$ . Both source and target bundles are flat over  $S_a$ ; the fibrewise injectivity criterion for this finite-presentation sequence gives flatness of its cokernel over  $S_a$ . Thus it defines a morphism to the Quot scheme already before the blow-up. At the origin the sequence is multiplication by  $t^a$  in its first summand, which gives the stated quotient. Its pullback along  $T_a \rightarrow S_a$  therefore contracts the chain. The Hilbert map on that chain is a closed immersion because  $T_a$  is a closed graph in  $S_a \times H_a$ . For three markings the source is stable, the torsion is away from the markings, and  $\omega_{\mathbb{P}^1}(p_1 + p_2 + p_3) \otimes \mathcal{O}(a\epsilon)$  has positive degree for every  $\epsilon > 0$ . These verify the actual stability conditions of [Marian–Oprea–Pandharipande(2011), Section 2]; no unstable unmarked rational source is being called a stable quotient.  $\square$

**Corollary 5.7** (Non-Cohen–Macaulay limits of smooth rational graphs). *For each  $a \geq 3$  the Hilbert closure of smooth rational graphs in  $\mathbb{P}^1 \times \mathbb{P}^2$  with polynomial  $P_a$  contains a curve whose reduction is a line attached to a rational cuspidal plane curve of degree  $a$ , and whose punctual nilradical has length  $g_a$  and nilpotence order  $a - 1$ . This curve is invisible as a distinct punctual structure to the associated-cycle map. It is not the scheme-theoretic image of a map from a reduced nodal curve.*



*Proof.* Take  $b = \tau^a, c = \tau$ . The generic graph is smooth because its projection to the source is an isomorphism. Its limit is  $\mathcal{W}_a$  with  $\lambda = 1$ , so Theorem 4.4 proves the first assertions. An associated one-cycle records generic lengths at one-dimensional components and discards finite-length submodules. Finally the scheme-theoretic image of a morphism from a reduced scheme is reduced: if a local target function has a power mapping to zero, its pullback is nilpotent and hence zero. The asserted nonreduced curve cannot be such an image. This does not rule out comparison spaces between stable-map and Hilbert compactifications; it rules out identifying this Hilbert limit with the unmodified scheme image of a stable map.  $\square$

This application is independent of the finite failure algebra and of Paper I. General comparisons between stable maps and stable quotients already involve spaces dominating both constructions [Manolache(2014)]; the explicit contraction above is a different, fixed-source Hilbert-to-Quot calculation, not a replacement for those virtual-class comparison theorems.

**5.3. Return to the reconstructed pencil.** A specified effective failure family first reconstructs its pencil by Paper I. Wherever its fixed-target contact comparison lands in  $B_a$ , the graph in Theorem 5.1 supplies compatible models under source and target frame changes. If its coefficient map factors through (3.1) or (5.2), the chain, all local rings, and the punctual modules above pull back as strict horizontal families. The inherited complete-quadric comparison determines the target in that application; forgetting that target and its coefficient map would lose the meaning of the embedded limit. This statement does not identify every complete-quadric Hilbert component with  $\mathcal{G}_a$ , does not cover an identically zero adjugate tuple, and does not turn Kronecker data alone into a higher-corank fibre classification. No boundary operation on an isolated unframed multiplication table is inferred.

#### APPENDIX A. A DETERMINANTAL MODEL FOR EVERY CONTACT ORDER

The distinction between a curve in a Hilbert scheme and a fibre of its parameter space is essential. This section constructs the former Hilbert graph over the whole polynomial coefficient space. Later we return to flat limits of the fibres of that graph. The two operations have different Hilbert polynomials and different universal properties.

Fix an integer  $a \geq 1$  and set

$$B_a = \{(f, g, r) : f \text{ monic of degree } a, \quad \deg g, \deg r < a\}.$$

Write  $[s : t]$  for the source coordinates and  $x = t/s$ . Homogenization to degree  $a$  gives forms  $F_0, F_1, F_2$ . Let  $U_a \subset B_a$  be the open locus where these forms have no common zero. The graph  $C_b \subset X = \mathbb{P}^1 \times \mathbb{P}^2$  of  $[F_0 : F_1 : F_2]$  has, for the Segre polarization, polynomial

$$P_a(l) = (a + 1)l + 1.$$

The reduced closure of  $(b, [C_b])$ ,  $b \in U_a$ , is denoted by  $\Gamma_a \subset B_a \times \text{Hilb}^{P_a}(X)$ , as in Section H. It is an integral graph, not a quotient by projective changes of the target.

Put

$$d = a + 1, \quad m = \frac{d(d-1)}{2} + 1, \quad q = dm + 1, \quad N = (m+1) \binom{m+2}{2}. \quad (\text{A.1})$$

For  $0 \leq i \leq m$  and  $j_0 + j_1 + j_2 = m$ , let the corresponding column of the  $q \times N$  matrix  $M_a$  be the coefficient vector of

$$s^{m-i} t^i F_0^{j_0} F_1^{j_1} F_2^{j_2} \quad \text{in } \mathbb{C}[s, t]_{dm}. \quad (\text{A.2})$$

Thus its entries are explicit polynomials in the  $3a$  coefficients. Order rows by the exponent of  $t$ , and order columns lexicographically in  $(i, j_0, j_1, j_2)$ . For each  $q$ -element column set  $J$ , write

$\Delta_J = \det(M_a)_J$ ; identically zero determinants may be omitted. Let

$$\mathfrak{a}_a = (\Delta_J)_J \subset \mathcal{O}(B_a). \quad (\text{A.3})$$

The size of this presentation is not a complexity bound. In particular, enumerating every maximal minor is not the recommended way to specialize one matrix.

**Theorem A.1** (Determinantal Hilbert graph). *For every  $a \geq 1$  there is a canonical isomorphism over  $B_a$*

$$\Gamma_a \simeq \text{Bl}_{\mathfrak{a}_a}(B_a) = \text{Proj}_{B_a} \bigoplus_{n \geq 0} \mathfrak{a}_a^n. \quad (\text{A.4})$$

*Under this isomorphism the rational Plücker coordinates of the Hilbert map are  $[\Delta_J]_J$ . The universal curve on this blow-up is the flat embedded extension of the graph on  $U_a$ . The normalization of  $\Gamma_a$  is the normalization of this Rees model. This assertion does not normalize any scheme fibre.*

*Let  $R$  be a discrete valuation ring containing  $\mathbb{C}$ , with uniformizer  $\tau$ , fraction field  $K$ , and residue field  $k$ . For a coefficient arc  $\gamma : \text{Spec } R \rightarrow B_a$  with  $\gamma(\text{Spec } K) \subset U_a$ , set*

$$\mu(\gamma) = \min_J \text{ord}_\tau \Delta_J(\gamma), \quad \ell_J(\gamma) = \left( \tau^{-\mu(\gamma)} \Delta_J(\gamma) \right) \bmod \tau. \quad (\text{A.5})$$

*The special embedded curve is the unique Hilbert point whose Plücker vector is  $[\ell_J(\gamma)]_J$ . In particular this vector specifies its scheme structure, not merely its support or cycle. Two such arcs centred at the same coefficient point have the same point of  $\Gamma_a$  precisely when these nonzero vectors are proportional over  $k$ .*

*Proof.* We first check the degree at which the Hilbert point is taken. The Gotzmann representation of  $dl + 1$  has  $d$  terms of degree one and  $(d-1)(d-2)/2$  terms of degree zero, since

$$\sum_{i=0}^{d-1} \binom{l+1-i}{1} = dl + \frac{d(3-d)}{2}.$$

Its total number of terms is  $m$  in (A.1). Gotzmann regularity and the Grassmannian construction of the Hilbert scheme therefore give a closed immersion into the Grassmannian of rank- $q$  quotients in degree  $m$ ; see [Gotzmann(1978), Grothendieck(1961)]. We use the ordinary projective Hilbert scheme after the Segre embedding  $X \hookrightarrow \mathbb{P}^5$ , not an unproved multigraded regularity bound. The restriction

$$H^0(\mathbb{P}^5, \mathcal{O}(m)) \longrightarrow H^0(X, \mathcal{O}(m, m))$$

is onto. All kernels defining subschemes of  $X$  contain its fixed kernel. Hence this Hilbert immersion factors as a closed immersion into the Grassmannian of quotients of  $H^0(X, \mathcal{O}(m, m))$ , of dimension  $N$ .

For  $b \in U_a$ , evaluation on the graph is exactly

$$H^0(X, \mathcal{O}(m, m)) \longrightarrow H^0(C_b, \mathcal{O}(m, m)) = H^0(\mathbb{P}^1, \mathcal{O}(dm)).$$

This map is onto by the same regularity theorem. Its matrix is  $M_a(b)$  and its Plücker coordinates are its  $q$ -minors. This proves that  $\mathfrak{a}_a$  is nonzero and has no base point on  $U_a$ .

For an integral affine scheme  $\text{Spec } A$  and a nonzero ideal  $(h_0, \dots, h_n)$ , the graph closure of  $[h_0 : \dots : h_n]$  is  $\text{Proj } A[(h_0, \dots, h_n)T]$ . Indeed, on the chart indexed by a nonzero  $h_i$ , both schemes have ring  $A[h_0/h_i, \dots, h_n/h_i] \subset \text{Frac}(A)$ . These charts prove the assertion scheme-theoretically, with the Rees algebra rather than the symmetric algebra. Apply this to  $\mathfrak{a}_a$ . The Hilbert scheme is closed in the Plücker space, so the closure of its dense graph is this same blow-up. Pulling back the universal Hilbert family proves the flat-family assertion. Normalizing the identical integral schemes proves the normalization assertion.

Over  $R$ , at least one  $\Delta_J(\gamma)$  is nonzero. Dividing all coordinates by their common valuation gives a primitive vector in a finite free  $R$ -module. Its special projective point is  $[\ell_J]$ . The Grassmannian

and Hilbert equations hold over  $K$  and hence over  $R$ , since  $R$  has no  $\tau$ -torsion. The resulting  $R$ -point is the unique extension by separatedness and properness. The Hilbert immersion makes equality of its special projective coordinates equivalent to equality of the embedded special curves. Keeping the coefficient point supplies the last assertion for the retained-coefficient graph.  $\square$

*Remark A.2* (What the determinant construction supplies). The existence of a graph blow-up and the Hilbert immersion are classical. The explicit ideal (A.3) identifies their input for every  $B_a$  and makes the following valuation statements uniform in  $a$ . It is not a claim that arbitrary higher-contact fibres have only finitely many isomorphism classes. It does not identify points on the normalization solely from their Hilbert coordinates: distinct normalization points can lie over one graph point. A generically nonincident DVR arc has a unique lift to that normalization by properness and the generic isomorphism. Equality of closed Hilbert points is a statement on  $\Gamma_a$ , not an assertion that the lifts coincide.

**Proposition A.3** (A finite determinacy bound for each arc). *Keep the matrix and coordinates above fixed. If two  $R$ -valued coefficient arcs agree modulo  $\tau^{\mu(\gamma)+1}$  and the first is generically in  $U_a$ , the second is also generically in  $U_a$  and their embedded special curves agree. Thus a jet of length  $\mu(\gamma) + 1$  suffices for this arc. The assertion concerns a Hilbert point, not the whole formal family.*

*Proof.* Polynomial evaluation preserves congruences. All maximal minors are therefore congruent modulo  $\tau^{\mu+1}$ , and at least one has the same nonzero coefficient in degree  $\mu$ . The primitive vectors in (A.5) agree. To justify the generic-locus assertion, a triple with common homogeneous factor of positive degree has all images in (A.2) divisible by its  $m$ th power and cannot span  $\mathbb{C}[s, t]_{dm}$ . Thus a nonzero maximal minor forces absence of a common zero over an algebraic closure of  $K$ .  $\square$

**Proposition A.4** (Smith data and specialization without enumerating minors). *Let the nonzero invariant factors of  $M_a(\gamma)$  over  $R$  have valuations  $e_1 \leq \dots \leq e_q$ . Then*

$$\mu(\gamma) = \sum_{i=1}^q e_i.$$

*The cokernel of this degree- $m$  evaluation map is  $\bigoplus_i R/(\tau^{e_i})$ . Its least annihilating exponent is  $e_q$ , and its length is  $\mu$ . The saturated row lattice of  $M_a(\gamma)$  in  $R^N$  has special exterior line  $[\ell_J]$ . The kernel of the corresponding rank- $q$  quotient over  $k$ , viewed in the Segre coordinate ring, is the degree- $m$  part of the special ideal. It generates its ideal sheaf.*

*Proof.* Smith normal form follows by choosing an entry of least valuation, using it to clear its row and column, and repeating. Elementary invertible operations preserve the ideal of maximal minors. In normal form that ideal is generated by  $\tau^{e_1+\dots+e_q}$ , which proves the formula and the cokernel statements. Over a DVR a saturated submodule of a finite free module is a direct summand. Saturating the row lattice divides its exterior line by exactly this common factor. Reduction gives the stated quotient. Gotzmann regularity, applied to the Hilbert point already constructed, says that its saturated homogeneous ideal is generated in degrees at most  $m$ , and its sheaf is generated by the degree- $m$  part.  $\square$

The exponent  $e_q$  here is an exponent for a finite module in one fixed projective degree. It is not a bound for saturation of every affine presentation of a pulled-back parameter graph.

## APPENDIX B. FINITE COEFFICIENT-STATE CHAMBERS

A valuation without its leading coefficient loses information. The following finite statement keeps that information, including cancellation. Its use of coefficient weights is different from

a one-parameter torus acting on the fixed target. Classical state polytopes organize initial ideals under the latter action [Bayer–Morrison(1988)]; here the weighted variables are the input coefficients of (A.2).

Write these coefficients as  $z_1, \dots, z_{3a}$  and expand the finite set of nonzero minors as

$$\Delta_J(z) = \sum_{\alpha \in E} c_{J\alpha} z^\alpha,$$

where  $E$  is the union of their finite monomial supports. Allow a specified subset of the  $z_i$  to be identically zero, and omit the corresponding monomials first. For this theorem take  $R = k[[\tau]]$  (or  $k[\tau]_{(\tau)}$ ), where  $k/\mathbb{C}$  is any field extension, so the residue coefficients are actual constants. No coefficient-field choice was required for the preceding general DVR statements. On the remaining coordinates consider exact monomial arcs

$$z_i = \xi_i \tau^{w_i}, \quad \xi_i \in k^*, \quad w_i \in \mathbb{Z}_{>0}. \quad (\text{B.1})$$

The constant monic coefficient of  $f$  is not among these variables.

**Theorem B.1** (Finite decorated chamber classification). *For each fixed  $a$  and each choice of zero coordinates there is an explicit finite rational polyhedral fan in the nonnegative weight orthant, and for each relatively open cone a finite locally closed partition of the residue torus, with the following properties. The pairs for which (B.1) is generically in  $U_a$  are specified by these partitions. On each such pair of a cone and a residue stratum, the embedded Hilbert limit is given by a single displayed vector of polynomial functions of  $\xi$ , independent of  $w$  in that relatively open cone. These vectors classify the special embedded curves by projective equality. They include nonreduced curves and embedded scheme structure.*

*One such fan is the hyperplane arrangement*

$$(\alpha - \beta) \cdot w = 0 \quad (\alpha, \beta \in E) \quad (\text{B.2})$$

*intersected with the coordinate orthant and all its faces. The residue partition and the vectors require only finite polynomial operations on the coefficients  $c_{J\alpha}$ .*

*Proof.* On the relative interior of a cone, the weak order of all numbers  $\alpha \cdot w$  is fixed. Let  $E_1, \dots, E_t$  be its ordered equality classes. For each  $J$  and  $h$  form

$$b_{Jh}(\xi) = \sum_{\alpha \in E_h} c_{J\alpha} \xi^\alpha.$$

Partition the residue torus by the first index  $h(J)$  at which  $b_{Jh}$  is nonzero, allowing  $h(J) = \infty$  when all are zero. Each condition is a finite list of polynomial equalities and one inequality, so there are finitely many locally closed pieces. Discard the piece on which all  $h(J)$  are infinite. On the others let  $h_* = \min_J h(J)$ . The answer is the projective vector

$$L_J(\xi) = \begin{cases} b_{J, h_*}(\xi), & h(J) = h_*, \\ 0, & h(J) > h_*. \end{cases} \quad (\text{B.3})$$

Indeed substitution into every minor is a finite sum of powers of  $\tau$ , with its terms grouped exactly as above. Equalities in a residue stratum account for every possible cancellation, not just a generic noncancellation assumption. The least surviving exponent is the common valuation in (A.5). Theorem A.1 proves both classification and retention of scheme structure. The same rank argument as in Proposition A.3 identifies the discarded set with the triples having a generic common zero.  $\square$

*Remark B.2* (Finite classification and its scope). For fixed residue coefficients, the theorem gives finitely many embedded limits as the positive integral weights vary. As the residue coefficients vary, it gives finitely many algebraic families, not finitely many curves. The arrangement need

not be minimal and does not assert that every one of its walls changes the curve. The degree bound is uniform for each  $a$ , but no claim of efficient enumeration of its potentially enormous minors is made. Arbitrary formal arcs are treated by the pointwise jet bound, not by declaring their leading coefficient monomials to be exact arcs. In particular, repeated cancellation in a formal arc cannot be discarded by this theorem.

The construction gives a practical distinction between two sorts of audit. An individual polynomial arc can be evaluated and its row lattice saturated, without computing the whole arrangement. The finite arrangement is a proof of finite classification of exact monomial profiles. The next section computes the entire answer on a nontrivial two-dimensional slice, including its minimal walls and the geometry on those walls.

### APPENDIX C. A TWO-WALL CLASSIFICATION ON A NONSYMMETRIC SLICE

The determinant construction can be evaluated without listing its maximal minors. We do so on a slice on which both a jet direction and a conic direction vary. Put

$$S = \text{Spec } \mathbb{C}[b, c] \longrightarrow B_2, \quad (f, g, r) = (x^2, b, cx).$$

Let  $T$  be the schematic closure of  $S_{b \neq 0}$  in  $\Gamma_2 \times_{B_2} S$ . Thus  $T$  is the horizontal main component of this base change, not its full pullback. All target coordinates  $[F : G : R]$  in this section are fixed.

**Theorem C.1** (The two-wall surface). *The surface  $T$  is the blow-up of  $S$  at  $(b, c) = (0, 0)$  followed by the blow-up of the point  $e = c = 0$  in the chart  $b = ec$ . Equivalently,*

$$T \simeq \text{Bl}_{(b,c)(b,c^2)} S.$$

*The maximal-minor ideal in Theorem A.1 restricts to the explicitly factored ideal*

$$\mathfrak{a}_2 \mathcal{O}_S = b^{10}(b, c)^4(b, c^2)^6. \quad (\text{C.1})$$

*The surface is smooth and has the following three affine charts:*

| <i>chart</i> | <i>coordinates</i> | $(b, c)$      |
|--------------|--------------------|---------------|
| $T_0$        | $(b, k)$           | $(b, bk)$     |
| $T_1$        | $(e, h)$           | $(e^2h, eh)$  |
| $T_2$        | $(c, z)$           | $(c^2z, c)$ . |

(C.2)

*On  $T_1$  and  $T_2$  the universal curve has respectively the bihomogeneous ideals*

$$I_1 = (R^2 - hFG, tG - esR, tR - ehsF), \quad (\text{C.3})$$

$$I_2 = (FG - zR^2, tG - czsR, tR - csF). \quad (\text{C.4})$$

*On  $T_0$  its ideal sheaf is generated by*

$$t^2G - bs^2F, \quad sR - ktG, \quad tR - bksF, \quad R^2 - bk^2FG. \quad (\text{C.5})$$

*These equations define flat families of polynomial  $3l + 1$ . The fibre  $T_{(0,0)}$  is a reduced chain  $E_{11} \cup E_{21}$  of two projective lines. Their self-intersections are  $-2$  and  $-1$ , respectively. The first line parametrizes primitive tails, and the second the conic pencil through the attaching point; their intersection parametrizes the doubled conic  $R^2 = 0$ .*

*Proof.* Perform the stated blow-ups. The first chart of the first blow-up is  $c = bk$  and is unchanged by the second blow-up. In the other chart write  $b = ec$  and blow up  $(e, c)$ . Its charts  $c = eh$  and  $e = cz$  give exactly (C.2). The transition maps are

$$\begin{aligned} k &= e^{-1}, & b &= e^2h, & e &\neq 0 \text{ on } T_1, & k &\neq 0 \text{ on } T_0, \\ z &= h^{-1}, & c &= eh, & h &\neq 0 \text{ on } T_1, & z &\neq 0 \text{ on } T_2. \end{aligned}$$

In the first overlap the inverse is  $e = k^{-1}$ ,  $h = bk^2$ ; in the second it is  $h = z^{-1}$ ,  $e = cz$ . The third overlap is the composite, so the cocycle identity holds as an identity of rings. To check the alternative blow-up description, first blow up  $(b, c)$ ; on the chart  $b = ec$  the transform of  $(b, c^2)$  is  $c(e, c)$ , whereas on  $c = bk$  it is the invertible ideal  $(b)$ . Removing the invertible factors and applying the universal property of a blow-up proves the product-ideal assertion.

We next construct the morphism to the Hilbert scheme. On  $T_1$  the Hilbert–Burch chart of Section I has centre and parameters  $\lambda = B = C = 0$ ,  $A = -h$ , and retained coefficient  $e$ . Its equations are (C.3). On  $T_2$  a Hilbert–Burch matrix is

$$\begin{pmatrix} F & R \\ -zR & -G \\ -t & -cs \end{pmatrix}.$$

Its minors are  $-(FG - zR^2)$ ,  $tR - csF$ , and  $-(tG - czsR)$ . On every geometric fibre the ideal has height two: if  $c \neq 0$ , the equation  $tR = csF$  and the other equations give a one-dimensional graph or its single-contact limit; if  $c = 0$ , they give the main section and the conic  $FG - zR^2$  at  $t = 0$ . This conic is nonzero for every  $z$ . There is no common divisor of the three minors. The fibrewise Hilbert–Burch resolution

$$0 \longrightarrow \mathcal{O}(-1, -2)^2 \longrightarrow \mathcal{O}(-1, -1)^2 \oplus \mathcal{O}(0, -2) \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}_C \longrightarrow 0$$

is therefore exact, and its constant Euler characteristic gives flatness and the polynomial  $3l + 1$ . Equivalently one may apply the fibrewise flatness criterion successively to this complex of relatively locally free sheaves.

For  $T_0$ , on  $s \neq 0$  eliminate  $R$  from (C.5). The remaining equation in the relative  $\mathbb{P}^1$  with coordinates  $[F : G]$  is

$$x^2G = bF, \quad R = kxG.$$

It is a relative effective Cartier divisor, since  $x^2G - bF$  is nonzero on every fibre. On  $t \neq 0$ , with  $y = s/t$ , the equations instead give  $G = by^2F$  and  $R = bkyF$ , the graph of a section. These two descriptions prove flatness and exclude any excess component at infinity. This is also the primitive division family of Theorem H.1, with  $w = kx$ . On the overlaps substitution of the displayed transitions makes the ideal sheaves identical; on  $T_1 \cap T_2$  the quadratic equations differ by the invertible factor  $-h$ . Thus the three flat families glue in the fixed target. Their generic graph is  $[t^2 : bs^2 : cst]$ .

It remains to prove that the constructed surface is the whole horizontal graph, rather than a resolution mapping to it. For  $a = 2$ , the matrix  $M_2$  has 13 rows and 75 columns. After restricting to  $F_0 = t^2$ ,  $F_1 = bs^2$ ,  $F_2 = cst$ , every column has exactly one nonzero entry. Its row index is  $i + 2j_0 + j_2$  and its entry is  $b^{j_1}c^{j_2}$ . A nonzero maximal minor chooses exactly one column from each row. Consequently its ideal is the product of the ideals generated by the entries in the individual rows. With  $J_1 = (b, c)$  and  $J_2 = (b, c^2)$  those row ideals, in order from 0 to 12, are

$$b^4, \quad b^3J_1, \quad b^2J_2, \quad bJ_1J_2, \quad J_2^2, \quad J_1J_2, \quad J_2, \quad J_1, \quad \mathcal{O}_S, \mathcal{O}_S, \mathcal{O}_S, \mathcal{O}_S, \mathcal{O}_S.$$

They follow by taking all  $0 \leq i \leq 4$  and  $j_0 + j_1 + j_2 = 4$  with the indicated row index; divisibility removes redundant monomials. Their product proves (C.1).

The graph closure of the restricted rational Plücker map is therefore  $\mathrm{Bl}_{b^{10}J_1^4J_2^6}S$ , by the graph argument in Theorem A.1. This is not an appeal to unrestricted base change for blow-ups. A principal nonzero factor does not change the Rees blow-up. Moreover blowing up  $J_1^4J_2^6$  is the same as blowing up  $J_1J_2$ : on either integral blow-up, invertibility of the product makes each factor invertible as a fractional ideal, and conversely. For completeness, if two nonzero finitely generated ideals have invertible product, then locally, after dividing their product by its generator, they are inverse fractional ideals. Thus each is invertible; apply this repeatedly to the positive

powers. The universal properties supply inverse morphisms between the two blow-ups, identical on the dense open. This proves that the twice blown-up surface is exactly  $T$ . The flat curve constructed above equals its Hilbert universal curve since the two Hilbert morphisms agree on the dense open of an integral surface and the Hilbert scheme is separated. In particular no unproved normality of a graph image is used in the argument.

On  $T_1$  the ideal of the origin pulls back to  $(eh)$ ; on  $T_0$  it is  $(b)$ , and on  $T_2$  it is  $(c)$ . Hence the exceptional fibre is reduced with a transverse intersection. The strict transform of the first exceptional curve loses one from its self-intersection when the second point is blown up. This gives  $E_{11}^2 = -2$ ,  $E_{21}^2 = -1$ , and  $E_{11}E_{21} = 1$ . Substitution  $h = 0$  and  $e = 0$  in (C.3) gives its two modular descriptions and their common doubled conic.  $\square$

**Corollary C.2** (Complete valued-arc limits on the slice). *Let  $R$  be a DVR containing  $\mathbb{C}$ , with uniformizer  $\tau$  and residue field  $k$ . Let  $b = \beta\tau^p$ ,  $c = \gamma\tau^q$ , where  $p, q > 0$  and  $\beta, \gamma \in R^*$ ; write bars for their residues. The special embedded curve has the following ideal sheaf. The displayed bihomogeneous ideals are understood with irrelevant saturation.*

| weights      | residue parameter                | special ideal         |       |
|--------------|----------------------------------|-----------------------|-------|
| $p < q$      | $k = 0$                          | $(R, t^2G)$           | (C.6) |
| $p = q$      | $k = \bar{\gamma}/\bar{\beta}$   | $(sR - ktG, tR, R^2)$ |       |
| $q < p < 2q$ |                                  | $(tG, tR, R^2)$       |       |
| $p = 2q$     | $h = \bar{\gamma}^2/\bar{\beta}$ | $(tG, tR, R^2 - hFG)$ |       |
| $p > 2q$     |                                  | $(tG, tR, FG)$        |       |

These five cases exhaust all generically nonincident DVR arcs centred at the origin with  $bc \neq 0$  in the fraction field, including units with arbitrary higher coefficients. If  $c = 0$  identically, the first row applies. Moreover the determinantal order on this slice is

$$\mu = 10p + 4 \min(p, q) + 6 \min(p, 2q), \quad (\text{C.7})$$

with  $\mu = 20p$  when  $c = 0$ . The two walls  $p = q$  and  $p = 2q$  are actual walls for embedded limits; coefficients on the walls cannot be omitted.

*Proof.* For  $p \leq q$ , use  $k = c/b$  on  $T_0$ . For  $q < p \leq 2q$  use  $e = b/c$  and  $h = c^2/b$  on  $T_1$ . For  $p > 2q$  use  $z = b/c^2$  on  $T_2$ . In each case these coordinates belong to  $R$ , and their reductions give the table. The formula for  $\mu$  follows by taking valuations in (C.1); an ideal generated by two elements has valuation equal to the lesser of their valuations. It is unaffected by higher unit coefficients. For  $k \neq 0$  the first generator  $t^2G$  of the primitive central fibre follows from  $sR - ktG$  and  $tR$  after taking ideal sheaves. For  $k = 0$  it must be retained. This explains why setting  $k = 0$  in just the last three generators would give the wrong scheme.  $\square$

**Proposition C.3** (Components, multiplicities, and the parameter boundary). *Every curve in (C.6) is Cohen–Macaulay and has no embedded associated points. In the first three rows the minimal primes are  $(G, R)$  and  $(t, R)$ ; the main section has multiplicity one and the tail multiplicity two. In the fourth row the primes are  $(G, R)$  and  $(t, R^2 - hFG)$ , each of multiplicity one. In the fifth they are  $(G, R)$ ,  $(t, F)$ , and  $(t, G)$ , each of multiplicity one. The fourth curve is a main section attached to a smooth conic; the fifth is a chain of three reduced lines. The normalization of each reduced curve is the disjoint union of its smooth components. None of these attached curves is normal as a whole.*

The parameter surface  $T$  is normal and smooth. Its exceptional boundary has simple normal crossings, with

$$\begin{array}{c|cc} & E_{11} & E_{21} \\ \hline \text{ord}(b) & 1 & 2 \\ \text{ord}(c) & 1 & 1 \end{array}, \quad K_{T/S} = E_{11} + 2E_{21}.$$

*These are statements about a parameter-space modification, not a stable reduction theorem for curves.*

*Proof.* For the primitive rows, the affine presentation  $R = kxG$ ,  $x^2G = 0$  is a hypersurface in a smooth relative surface. Near infinity it is the smooth main section. The two generic lengths are read from  $x^2G$ ; the residue  $k$  changes the embedding but not the lengths. For the conic rows the Hilbert–Burch resolution above makes every local ring on the curve Cohen–Macaulay of dimension one. At the attaching point use  $F = s = 1$ : the ideals are  $(xG, xR, R^2 - hG)$ , with the analogous  $FG$  presentation in the last row. The ideal decompositions

$$(tG, tR, Q) = (G, R) \cap (t, Q), \quad Q \in \{R^2, R^2 - hFG, FG\},$$

as ideal sheaves give the claimed minimal components and lengths. A one-dimensional Cohen–Macaulay scheme has no embedded points. When  $h \neq 0$  the conic is smooth. When  $Q = FG$ , its two lines meet at  $[0 : 0 : 1]$  and only the line  $G = 0$  contains the attaching point  $[1 : 0 : 0]$ . This gives the chain description. The assertions on the normalization of the reductions follow.

The valuations follow directly from  $b = e^2h$ ,  $c = eh$ :  $E_{11}$  is  $h = 0$  and  $E_{21}$  is  $e = 0$ . For a blow-up of a smooth surface at a point the relative canonical divisor is its exceptional curve, as the Jacobian in a chart shows. Pulling the first exceptional divisor through the second blow-up adds one more copy of  $E_{21}$ . This proves the discrepancy formula.  $\square$

**Corollary C.4** (Pointwise, but not uniform, finite determinacy). *Even on  $B_2$  no integer  $n$  has the property that the coefficient  $n$ -jet determines every embedded graph limit. In contrast, Proposition A.3 gives an explicit finite bound for each generically nonincident arc.*

*Proof.* Given  $n$ , choose  $M > n$  and two distinct nonzero constants  $\beta_1, \beta_2$ . The arcs

$$f = x^2, \quad g = \beta_i \tau^{2M}, \quad r = \tau^M x \quad (i = 1, 2)$$

agree modulo  $\tau^{n+1}$  and are generically base-point free. Their limits are the distinct fixed-target conics  $R^2 - \beta_i^{-1}FG$  attached to the same main section, by the fourth row of (C.6). This proves the claim.  $\square$

**Proposition C.5** (No uniform saturation exponent for full pullbacks). *The uniformizer saturation exponents of the full pulled-back parameter graph at contact order two are unbounded. This is compatible with the finite degree-module statement of Proposition A.4.*

*Proof.* In the symmetric family of Theorem L.3, take  $\delta = \tau^m$ . Its full vertical torsion is a nonzero ideal module tensored with  $\mathbb{C}[\tau]/(\tau^m)$ . On a chart where that ideal is nonzero it is killed by  $\tau^m$  and not by  $\tau^{m-1}$ . Thus the least exponent at which uniformizer saturation stabilizes is at least  $m$ . Since  $m$  is arbitrary, there is no uniform bound with contact order fixed at two. This assertion is about the full parameter-family pullback; it does not replace the embedded-curve jet theorem with a negative statement.  $\square$

#### APPENDIX D. BASE CHANGE AND COMPARISON OF HORIZONTAL MODELS

We now distinguish the Hilbert graph of embedded curves from the horizontal modification of a projective parameter family. In the notation of Section E, let  $W \rightarrow B$  be projective and of finite presentation, with  $B$  integral of finite type over  $\mathbb{C}$ . Fix a dense open  $U \subset B$  over which  $W$  is flat with polynomial  $P$ . The admissible tests are locally Noetherian morphisms  $Y \rightarrow B$  for which  $U_Y$  is schematically dense. The horizontal object  $W_Y^h$  is the schematic closure of  $W_{U_Y}$  in  $W_Y$ . Only tests for which this closure is flat belong to the represented horizontal-flat subfunctor. Throughout,

$$H_B = \overline{s(U)} \subset \text{Hilb}^P(W/B)$$



means the *horizontal Hilbert-main-component modification*. Its terminal property refers to this admissible category and this prescribed generic embedded quotient.

**Lemma D.1** (Flat schematic density). *Let  $V \subset Y$  be a quasi-compact schematically dense open in a locally Noetherian scheme. If  $Z \rightarrow Y$  is flat, then  $Z_V$  is schematically dense in  $Z$ .*

*Proof.* The assertion is local on  $Y$  and  $Z$ . For  $Y = \text{Spec } A$ , write  $V = \bigcup_{i=1}^r D(f_i)$ . Schematic density is the injectivity of  $A \rightarrow \prod_i A_{f_i}$ . Tensoring with any flat  $A$ -module preserves this injection; finiteness of the product identifies the tensor product with the product of the localizations. Apply this to the rings of affine opens of  $Z$ . A section vanishing on  $Z_V$  is zero. This is precisely the required schematic density.  $\square$

**Theorem D.2** (Strict base change and coherent comparison). *Let  $B' \rightarrow B$  be an integral finite-type base change such that  $U' = U \times_B B'$  is nonempty. Use the pullback polarization and write  $H_{B'}$  for the horizontal Hilbert-main-component modification of  $W_{B'}$  relative to  $U'$ . There is a canonical closed immersion*

$$H_{B'} = \overline{U'}^{H_B \times_B B'} \hookrightarrow H_B \times_B B'. \quad (\text{D.1})$$

*It is an equality if and only if  $U'$  is schematically dense in the full fibre product. In particular it is an equality for flat base change. For composable base changes meeting the indicated generic opens, these comparison maps compose to the comparison for the composite. They obey the corresponding cocycle identity.*

*If  $B^\nu \rightarrow B$  is the normalization, then  $H_{B^\nu} \rightarrow H_B$  is finite and birational, and the normalizations of these two integral schemes are canonically isomorphic over  $B$ .*

*Proof.* The relative Hilbert scheme and its universal family commute with base change. In the resulting Hilbert scheme over  $B'$ , the closure of  $s(U')$  is contained in the closed subscheme  $H_B \times_B B'$ , since the ideal of  $H_B$  vanishes on  $s(U')$ . It is therefore exactly the schematic closure on the right of (D.1). This proves the first assertion and the density criterion. Flat base change preserves the schematic density of  $U \subset H_B$  by Lemma D.1; note that the projection of the fibre product to  $H_B$  is flat.

For a further base change  $B'' \rightarrow B'$ , both routes give the schematic closure of  $U''$  in the same relative Hilbert scheme. The closed immersions consequently agree, including their scheme structures. This proves transitivity and the cocycle, not merely agreement on closed points. Finally, normalization of a finite-type integral  $\mathbb{C}$ -scheme is finite. The map  $H_B \times_B B^\nu \rightarrow H_B$  is finite; its closed integral main component is finite and birational because it agrees on a dense normal open. Finite birational integral extensions have the same integral closure in their common function field. This proves the last assertion.  $\square$

**Example D.3** (The full pullback need not commute). Take  $B = \mathbb{A}_{x,y}^2$ ,  $W = \text{Bl}_0 B$ , and  $U = B \setminus \{0\}$ . The generic polynomial is  $P = 1$ . A flat family of length-one embedded subschemes of  $W/B$  is a section, so its relative Hilbert scheme is  $W$  itself and  $H_B = W$ . Restrict to  $B' = \{y = 0\}$ . With homogeneous coordinates  $[u : v] = [x : y]$ , the full pullback has equation  $xv = 0$  in  $\mathbb{A}_x^1 \times \mathbb{P}^1$ . It contains a vertical projective line. The closure of  $x \neq 0$  is instead  $v = 0$ , isomorphic to  $B'$ . Thus (D.1) is a strict inclusion. The parameter  $y = 0$  is a nonflat base change; replacing the full pullback by its reduction would not remove its extra component.

**Proposition D.4** (Common refinements on one generic open). *Fix  $B, U$ . For two integral proper  $U$ -modifications  $Y_1, Y_2$ , let  $Y_1 \star_U Y_2$  be the schematic closure of the diagonal  $U$  in  $Y_1 \times_B Y_2$ . The projections are proper birational maps and this object is terminal among integral  $U$ -modifications equipped with maps to both  $Y_1$  and  $Y_2$  restricting to the identity on  $U$ . Repeated products are canonically associative and symmetric, and their comparison maps satisfy all product cocycles. If*

*a refinement of a stratum has the same reduced closure and a smaller dense generic-flat open, its horizontal model is canonically unchanged.*

*Proof.* A pair of compatible maps gives a map into the product. Since its integral source has dense  $U$ , the map factors through the schematic closure of that open. The factorization is unique because the closure is a closed subscheme. Properness follows from properness of the product, and birationality from the common dense open. For three or more factors every iterated object is the closure of the same diagonal in the full product; this proves associativity, symmetry, and their compatibility. Shrinking a dense open of an integral section does not change its schematic closure, proving the last assertion.  $\square$

The theorem supplies comparison for dominant restrictions and strict base changes, and the proposition supplies common refinements on a fixed generic problem. A stratum whose closure lies entirely outside  $U$  does not meet these hypotheses. In particular these results do not identify Hilbert components with different generic polynomials. For such strata the original Noetherian-induction construction remains a choice of strata, not a canonical incidence poset. If one asks for a flat family whose fibres are the entire original fibres on two strata in one connected base, equality of their Hilbert polynomials is necessary. If  $W$  is already flat over all of  $B$ , the horizontal model is  $B$  itself. These observations specify both the full-base flat case and the precise limitation of the comparison theorem.

**D.1. Comparison with full flattening.** For an admissible  $Y \rightarrow B$  on which  $W_Y$  is flat, Lemma D.1 gives  $W_Y = W_Y^h$ . There is therefore a natural inclusion of functors and a commutative diagram

$$\begin{array}{ccc} \{Y : W_Y \text{ is flat}\} & \longrightarrow & \{Y : W_Y^h \text{ is flat}\} \\ \downarrow & & \downarrow \\ \mathrm{Hom}_B(Y, B) & \xlongequal{\quad} & \mathrm{Hom}_B(Y, B). \end{array}$$

The top arrow is not in general surjective: the extra projective line in Example D.3 is exactly an obstruction to full flattening. The usual flattening functor of a finitely presented sheaf concerns its full tensor pullback and, when represented, is represented by a monomorphism; see [Stacks Project], Tag 052F. By contrast, a horizontal modification may have a positive-dimensional exceptional fibre. Flattening by blow-up uses a pure transform and must likewise not be confused with flatness of the full pullback. Theorem A.1 computes a Rees ideal for the *embedded-curve graph*; it does not assert that the same ideal flattens every fibre of that parameter graph.

The horizontal main component is integral, hence reduced. Neither normality nor Cohen–Macaulayness follows from the construction: for  $W = B$  the construction returns  $B$ , which may be a nonnormal integral curve or a non-Cohen–Macaulay integral higher-dimensional scheme. The explicit surface in Theorem C.1 has stronger properties because its charts have been computed. Independence of polarization identifies representing objects for the same horizontal functor; it does not identify the ambient Hilbert schemes or their other components.

## APPENDIX E. UNIVERSAL HORIZONTAL MODELS OF THE HILBERT BOUNDARY

A boundary fibre and a limit of boundary fibres are different objects. The first is obtained by imposing the coefficient equations; the second also removes the torsion created by the chosen specialization. We begin with a construction that organizes this distinction over a whole base. The representability input is the classical projective Hilbert scheme, not a new representability theorem. Its application below supplies a universal property for the horizontal models of the pencil boundary.

**E.1. The flat horizontal functor.** Let  $p : W \rightarrow B$  be a projective morphism of finite presentation, with  $B$  an integral scheme of finite type over  $\mathbb{C}$ . Fix a relatively ample line bundle and a nonempty open  $U \subset B$  on which  $p$  is flat with constant Hilbert polynomial  $P$ . The scheme  $W$  need not be reduced, equidimensional, or flat over the boundary of  $U$ .

A locally Noetherian  $B$ -scheme  $T$  is called  $U$ -admissible if  $U_T = U \times_B T$  is schematically dense in  $T$ . Write

$$W_T^h = \overline{W_{U_T}}^{W_T}, \quad \mathcal{J}_T = \ker(\mathcal{O}_{W_T} \rightarrow j_* \mathcal{O}_{W_{U_T}}). \quad (\text{E.1})$$

The bar denotes schematic closure, and  $j$  is the indicated open immersion. On these test schemes define  $\mathfrak{F}_{W,U}(T)$  to be the singleton if  $W_T^h \rightarrow T$  is flat, and the empty set otherwise. Morphisms are  $B$ -morphisms between admissible test schemes. Whenever an object exists, its pullback is the horizontal closure and is again flat: a flat scheme over  $T$  has schematically dense inverse image of  $U_T$ . Thus this is a functor on the stated category. This last assertion does not concern arbitrary nonflat pullback of  $W$ .

**Theorem E.1** (Universal horizontal modification). *There are an integral projective birational  $B$ -scheme  $F_{W,U}$  and a closed subscheme*

$$\mathcal{W} \subset W \times_B F_{W,U}$$

*with the following properties. The morphism  $F_{W,U} \rightarrow B$  is an isomorphism over  $U$ . The family  $\mathcal{W} \rightarrow F_{W,U}$  is flat, and is the schematic closure of  $W_U$ . For every  $U$ -admissible  $T$ , composition with  $F_{W,U} \rightarrow B$  gives a natural bijection*

$$\text{Hom}_B(T, F_{W,U}) \simeq \mathfrak{F}_{W,U}(T). \quad (\text{E.2})$$

*Under this bijection  $\mathcal{W}_T = W_T^h$ . Consequently  $F_{W,U}$  is terminal among  $U$ -admissible modifications on which the horizontal closure is flat. It is independent of the chosen relatively ample line bundle and of replacing  $U$  by a smaller nonempty open on which  $p$  is flat.*

*Proof.* Let  $H = \text{Hilb}^P(W/B)$ , which is projective over  $B$  by the projective Hilbert-scheme theorem [Grothendieck(1961)]. The entire fibre  $W_u$  defines a section  $s : U \rightarrow H$ . Put  $F_{W,U} = \overline{s(U)} \subset H$ , with its schematic closure structure, and restrict the universal quotient on  $H$  to obtain  $\mathcal{W}$ . Since  $U$  is integral, its schematic closure is integral. The section is a closed immersion over  $U$ , so the restriction of  $F_{W,U}$  over  $U$  is exactly  $s(U)$ . This proves projectivity, birationality, and the assertion over  $U$ .

Flatness of the universal Hilbert quotient proves flatness of  $\mathcal{W}$ . A flat module over an integral base injects into its localization at the generic point. Hence the open  $W_U$  is schematically dense in  $\mathcal{W}$ , and  $\mathcal{W}$  is its closure inside  $W \times_B F_{W,U}$ .

Suppose first that  $W_T^h$  is flat. This closed subscheme is of finite presentation and projective over  $T$ . Its Hilbert polynomial is locally constant and equals  $P$  on  $U_T$ . Every nonempty open-and-closed piece of  $T$  meets  $U_T$ , so the polynomial is  $P$  everywhere. Representability gives  $h : T \rightarrow H$ . On  $U_T$  this is the section  $s$ . The pullback of the ideal of  $F_{W,U}$  therefore vanishes on  $U_T$  and hence on  $T$ , by schematic density. Thus  $h$  factors through  $F_{W,U}$ .

Conversely, pull the universal flat quotient back by any  $h : T \rightarrow F_{W,U}$ . Flat base change preserves schematic density of the inverse image of  $U_T$ : this can be checked on affine charts by tensoring the injection into the finite product of localizations associated to an affine cover of  $U_T$ . The pulled-back quotient is therefore precisely the closure in (E.1). This also proves uniqueness. In particular two possible lifts give the same embedded quotient, not merely isomorphic abstract fibres. All constructions commute with morphisms of admissible tests.

The same universal property identifies the constructions from two polarizations. For a smaller open  $U'$ , its section is schematically dense in  $s(U)$ ; their closures in the Hilbert scheme coincide. Alternatively apply the universal property in both directions: the universal flat closure over the

integral modification is already determined by the generic fibre. The two resulting morphisms are inverse because they agree over the dense open and the targets are separated.  $\square$

The theorem modifies the base to flatten a *horizontal closed subscheme*. It does not assert that the full pullback  $W \times_B F$  is flat. Nor is  $F \rightarrow B$  the monomorphism representing flatness of the full pulled-back sheaf in the usual flattening stratification; compare [Stacks Project, Tag 05PS]. This distinction accounts for the vertical component in Theorem K.1.

**E.2. Finite saturation and base torsion.** Compatibility with completion reduces to the following finite kernel calculation.

**Lemma E.2** (Flat change of rings and finite saturation). *Let  $A$  be Noetherian, let  $I, J \subset A$  be ideals, and let  $A \rightarrow A'$  be flat with  $A'$  Noetherian. Choose  $N$  such that  $(I : J^N) = (I : J^{N+1})$ . Then*

$$(I : J^\infty)A' = (IA' : (JA')^\infty) = (IA' : (JA')^N). \quad (\text{E.3})$$

*This applies to Noetherian completion. If the map is faithfully flat, the least stabilization exponent is unchanged.*

*Proof.* For a finite set  $j_1, \dots, j_m$  of generators of  $J^n$ , the ideal quotient modulo  $I$  is the kernel of  $A/I \rightarrow (A/I)^m$ ,  $a \mapsto (j_1 a, \dots, j_m a)$ . Tensoring this kernel sequence with  $A'$  proves  $(I : J^n)A' = (IA' : (JA')^n)$  for every  $n$ . The increasing sequence of ideal quotients stabilizes by Noetherianity. Equality of consecutive terms implies equality thereafter, since colon by  $J$  sends one term to the next. The same equality at  $N$  holds after flat change of rings. Taking the stabilized terms proves the formula. Faithful flatness detects equality and inequality of the finite terms, which gives the last assertion. There is no assertion of a uniform exponent for all ideals, or for all arcs in a coefficient space.  $\square$

**Proposition E.3** (The horizontal quotient over a discrete valuation ring). *Let  $R$  be a discrete valuation ring with uniformizer  $\tau$ , residue field  $k$ , and fraction field  $K$ . Let  $A$  be a Noetherian  $R$ -algebra. Set*

$$T(A) = \{a \in A : \tau^n a = 0 \text{ for some } n\}, \quad A^h = A/T(A).$$

*Then  $T(A)$  is an ideal killed by some  $\tau^N$ , and  $A^h$  is the unique flat quotient of  $A$  with generic fibre  $A_K$ . For a presentation  $A = D/I$  over an  $R$ -algebra  $D$  with no  $\tau$ -torsion,*

$$A^h = D/(I : \tau^\infty), \quad T(A) = (I : \tau^\infty)/I. \quad (\text{E.4})$$

*Moreover*

$$\text{Tor}_1^R(A, k) = \text{Ann}_A(\tau), \quad \text{Tor}_i^R(A, k) = 0 \quad (i \geq 2). \quad (\text{E.5})$$

*Writing  $T_n = \text{Ann}_A(\tau^n)$  gives  $T_n/T_{n-1} \simeq \text{Ann}_A(\tau) \cap \tau^{n-1}A$ . These statements glue for schemes of finite type over  $R$ .*

*Proof.* The annihilator ideals stabilize in  $A$ . Their union is an ideal, so stabilization supplies  $N$ . Multiplication by  $\tau$  on  $A/T(A)$  is injective. A torsion-free module over a valuation ring is flat [Stacks Project, Tag 0539]; this proves flatness. If  $A/J$  is another flat quotient with the same generic fibre, its  $\tau$ -torsion is zero, so  $T(A) \subset J$ . The submodule  $J/T(A)$  of the torsion-free module  $A^h$  is generically zero, hence zero. This proves uniqueness and the presentation.

Resolve  $k$  by  $0 \rightarrow R \xrightarrow{\tau} R \rightarrow k \rightarrow 0$  to obtain all of (E.5). Multiplication by  $\tau^{n-1}$  on  $T_n$  has kernel  $T_{n-1}$  and image exactly the stated intersection. Localization commutes with this finite filtration, so the formulas glue. Flat change of discrete valuation rings commutes with the horizontal quotient by Lemma E.2. In general  $T(A)$  is larger than  $\text{Ann}_A(\tau)$ .  $\square$

For polynomial coefficient arcs the saturation is effective: in an auxiliary variable  $y$ , it is the elimination ideal

$$(I : \tau^\infty) = (I, 1 - y\tau)D[y] \cap D.$$

This gives a finite elimination procedure when the coefficients are provided by polynomial or localized polynomial data. For an arbitrary formal power series arc the formula remains exact, but it is not an algorithm requiring only a fixed finite jet of unspecified series.

**E.3. A stratumwise compactification for every pencil specialization.** Let  $G$  be the coefficient space of pencils under consideration and let  $\tilde{\Gamma} \rightarrow G$  be the normalization of its integral projective Hilbert graph. More generally the next assertion applies to any fixed projective graph model of finite type. Normalization is performed on the total graph before any specialization; it is finite over a complex finite-type scheme, hence the map remains projective. For every integral closed  $B \subset G$  put  $W_B = \tilde{\Gamma} \times_G B$ , *without* reducing this pullback, and choose its dense generic-flat open  $U_B$ .

**Theorem E.4** (Stratumwise universal boundary models). *The modification  $F_{W_B, U_B} \rightarrow B$  and its universal family give a canonical horizontal model on every such  $B$ . Every discrete valuation ring arc  $\gamma : \text{Spec } R \rightarrow B$  whose generic point lies in  $U_B$  has a unique lift to this modification, and*

$$\gamma^* \mathcal{W} = (\tilde{\Gamma} \times_G \text{Spec } R)^h. \quad (\text{E.6})$$

*There exists a finite stratification of  $G$  by integral locally closed strata  $S_i$ , with projective modifications over their reduced closures  $\overline{S}_i$ , such that every discrete valuation ring arc in  $G$  is covered by one of these formulas, determined by the stratum containing its generic point.*

*Proof.* The first construction is Theorem E.1. An arc whose generic point belongs to  $U_B$  is admissible. Its horizontal quotient is flat by Proposition E.3, so the universal property gives the unique lift and the equality of embedded quotients. Equivalently, extend the generic lift by properness and apply uniqueness of a flat quotient with that generic fibre.

To obtain finitely many strata, take dense generic-flat opens in each of the finitely many irreducible components of the reduced base, shrinking them to be disjoint and to have constant Hilbert polynomial. Apply the same procedure to their closed complement. Its dimension strictly decreases, so the procedure terminates. Each stratum is a dense open in its integral reduced closure after the previous closed pieces are removed. Apply the first construction to that closure and its generic-flat open. If the generic point of an arc lies in  $S_i$ , the whole arc factors through  $\overline{S}_i$ : the pullback of its defining ideal vanishes generically in the domain  $R$  and hence vanishes in  $R$ . The preceding argument applies. The stratification is not claimed unique; the horizontal modification attached to a specified integral closure is canonical.  $\square$

This supplies universal horizontal models for arbitrary contact order where a projective Hilbert graph has been defined. It does not use an a priori classification of its components. In particular it does not identify higher-corank Hilbert fibres from Kronecker invariants. The complete component formulas proved in Theorems I.2 and M.1 retain their hypotheses. A single modification of the full base cannot be substituted without argument for the stratumwise assertion: the generic boundary fibre on a multiple-incidence stratum has a different Hilbert polynomial from the point fibre on the nonincident open.

For the coefficient spaces  $B_a$ , this theorem turns an arcwise saturation into the pullback of a projective universal family. For  $B_2$ , Section J gives finite equations for *every* arc through the first nonreduced contact. Thus the universal property is accompanied by an explicit specialization calculation, rather than being used as a substitute for it.

## APPENDIX F. VERSAL CONTACT COORDINATES FOR REGULAR SYMMETRIC PENCILS

The contact order does not determine a neighbourhood of a pencil. We describe the missing parameters and then use them to calculate open charts of the ambient Hilbert boundary. The normal-form input belongs to the classical deformation theory of symmetric pencils [Dmytryshyn(2011/2018), Theorem 2.1, Corollary 2.1 and Lemma 3.1]. We give the tangent calculation and the formal reduction needed here, because a normal form for a pencil and a normal form for its embedded graph are different assertions.

All completions in this section are over  $\mathbb{C}$ . The pencil parameter is fixed; reparametrizing it is not part of the local congruence action. A source-and-parameter framing is a smooth atlas of the Grassmannian of genuine pencils. Statements on this atlas descend by change of frame, rather than by identifying different embedded Hilbert points.

**Lemma F.1** (The contact deformation module). *Put  $D = \text{diag}(x^{d_1}, \dots, x^{d_c})$ , where  $d_i \geq 1$ . For deformations of this symmetric matrix germ under congruences which reduce to the identity, the tangent quotient is*

$$\mathcal{T}_D = \bigoplus_i \mathbb{C}[x]_{< d_i} \oplus \bigoplus_{i < j} \mathbb{C}[x]_{< \min(d_i, d_j)}. \quad (\text{F.1})$$

A formally miniversal family is the symmetric polynomial matrix

$$S_{ii} = x^{d_i} + p_{ii}(x), \quad S_{ij} = p_{ij}(x), \quad \deg p_{ii} < d_i, \quad \deg p_{ij} < \min(d_i, d_j). \quad (\text{F.2})$$

Its coefficients vanish at the origin. Miniversality means existence of a formal classifying map and a congruence after every complete local base change, with an isomorphism on the tangent quotient. It does not assert uniqueness of that map in the presence of stabilizers.

*Proof.* An infinitesimal change  $1 + \epsilon H$  adds  $H^t D + DH$ . Its diagonal entries generate  $(x^{d_i})$ , and its  $(i, j)$  entry ranges over  $(x^{d_i}, x^{d_j}) = (x^{\min(d_i, d_j)})$ . Different unordered entries can be prescribed independently. This proves (F.1), including the factor two on the diagonal, which is invertible.

For completeness let  $(R, \mathfrak{m})$  be complete local and let  $S(x) \equiv D \pmod{\mathfrak{m}}$ . Work successively modulo  $\mathfrak{m}^{q+1}$ . Divide each coefficient of the degree- $q$  error in an off-diagonal position by  $x^{\min(d_i, d_j)}$ , and each diagonal error by  $x^{d_i}$ . Put the remainders into the indicated polynomials. Remove the quotients with  $H$  of order  $q$ . On an off-diagonal entry use the side with the smaller exponent; on the diagonal divide by two. Terms involving an already deformed coefficient and  $H$  have order at least  $q + 1$ . The compatible products of these changes converge  $\mathfrak{m}$ -adically, as do the finitely many polynomial coefficients. This constructs the formal reduction. The tangent quotient proves minimality of the number of parameters. The choices made by this algorithm are coordinates, not invariants.  $\square$

**Theorem F.2** (Ambient realization of the contact parameters). *Let  $R_0$  be a regular complex symmetric pencil. At each spectral point  $\lambda$ , let  $d_{\lambda,1}, \dots, d_{\lambda,c_\lambda}$  be its positive Smith exponents. On a source-and-parameter frame atlas, the completed pencil base has a formally smooth map to the product of the coefficient spaces (F.2), with independent coordinates in all factors. After a congruence on each completed parameter disc, its residual symmetric matrix is exactly (F.2); the complementary block is a unit. The number of essential congruence parameters is*

$$N = \sum_{\lambda, i} d_{\lambda, i} + \sum_{\lambda} \sum_{i < j} \min(d_{\lambda, i}, d_{\lambda, j}). \quad (\text{F.3})$$

The remaining parameters are smooth frame-orbit directions. Thus these models describe deformations in the full regular-pencil neighbourhood, not only a selected two-parameter slice.

For one corank-two point with exponents  $a \leq b$ , the residual matrix can be written

$$S(x) = \begin{pmatrix} f(x) & g(x) \\ g(x) & f(x)k(x) + r(x) \end{pmatrix}, \quad (\text{F.4})$$

where  $f$  is monic of degree  $a$ ,  $g, r$  have degree less than  $a$ , and  $k$  is monic of degree  $b - a$ . For  $a = b$ ,  $k = 1$ . The central values are  $f = x^a$ ,  $g = r = 0$ , and  $k = x^{b-a}$ . These are  $2a + b$  independent formal coordinates. With  $A_f = \mathcal{O}[x]/(f)$  and multiplication matrices  $M_g, M_r$  on its basis  $1, x, \dots, x^{a-1}$ , the exact incidence module and ideal are

$$q_*\mathcal{O}_D = \text{coker}[M_g \ M_r], \quad \text{Fitt}_0(q_*\mathcal{O}_D) = I_a([M_g \ M_r]). \quad (\text{F.5})$$

These identities hold over arbitrary base algebras in this polynomial model. They do not identify its nonreduced Hilbert modification with the blow-up of that Fitting ideal.

*Proof.* Choose a nonsingular pencil member. Fixing it by congruence identifies the other member with a self-adjoint endomorphism  $T$  for a fixed nondegenerate symmetric form. Its primary subspaces at distinct eigenvalues are a direct sum, even when  $T$  is not semisimple. Within one primary subspace the symmetric Jordan presentation is  $\bigoplus_i L_{d_i}(x)$ , where  $L_d(x) = E_d(xI - N_d)$  as in Lemma V.1. If  $v_d(x) = (1, x, \dots, x^{d-1})^t$ , then

$$L_d(x)v_d(x) = x^d e_1. \quad (\text{F.6})$$

The complement to  $e_1$  is a unit block. Differentiating its Schur complement in the direction of a constant symmetric perturbation  $H$  gives, on an  $(i, j)$  block,  $v_{d_i}^t H_{ij} v_{d_j}$ . For  $0 \leq l < \min(d_i, d_j)$  the perturbation  $H_{ij} = e_1 e_{l+1}^t$  gives  $x^l$ ; insert its transpose in the opposite block. On a diagonal block use  $(e_1 e_{l+1}^t + e_{l+1} e_1^t)/2$  for  $l > 0$  and  $e_1 e_1^t$  for  $l = 0$ . They give every coefficient in (F.1). Block-diagonal perturbations prescribe these coefficients independently at distinct eigenvalues. This proves surjectivity from actual pencil deformations, not merely from arbitrary matrix-valued power series.

Here is also the dimension check against the congruence orbit. The centralizer between Jordan blocks of sizes  $d_i, d_j$  at the same eigenvalue has dimension  $\min(d_i, d_j)$ , by solving  $N_{d_i} X = X N_{d_j}$  along its diagonals. A commuting endomorphism on a single block is polynomial in  $N_d$ , hence self-adjoint. A skew-adjoint commuting endomorphism has no diagonal contribution; its two opposite off-diagonal blocks determine one another and contribute  $\min(d_i, d_j)$ . Distinct primary eigenvalues contribute zero, since the corresponding Sylvester map is invertible. Therefore the orthogonal stabilizer has the second sum in (F.3) as its dimension. Subtracting its orbit dimension from  $n(n+1)/2$  gives  $n$  plus that sum, namely  $N$ . Congruence directions are killed by the contact quotient. Surjectivity and equality of dimensions identify their kernel exactly.

Apply the formal reduction of Lemma F.1 to the universal pencil. Its classifying coefficient map has the surjective differential just computed. Complete its coefficients by coordinates in the kernel. The formal inverse function theorem identifies the completed smooth base with these coefficients and the complementary power-series variables. This is the claimed formal smoothness. Equivalently a transverse affine pencil family is miniversal by the congruence tangent-space criterion in [Dmytryshyn(2011/2018), Lemma 3.1]. Normalizing the chosen nonsingular member and then undoing that normalization restores the smooth source-and-parameter framing directions.

For two blocks the first normal form is  $\begin{pmatrix} f & g \\ g & h \end{pmatrix}$  with  $h$  monic of degree  $b$ . Euclidean division  $h = fk + r$  is a polynomial change of coefficient coordinates with polynomial inverse. This proves (F.4) and its dimension. The submaximal-minor ideal on the residual chart is  $(f, g, h) = (f, g, r)$ . Quotienting first by the monic polynomial  $f$  makes a free module of rank  $a$ , and quotienting further by  $g, r$  gives the displayed cokernel. The Fitting formula follows directly from this presentation. No reduced-fibre argument or extraction of an ideal root is involved.  $\square$

The theorem includes higher corank in its normal-coordinate statement. The explicit Hilbert charts below concern corank two. The all-pencil inverse and the separate singular-pencil invariant theorem are not restricted by this choice. In particular the balanced two-parameter Jordan slice

has only two directions inside a  $3a$ -parameter contact base; it is not a versal neighbourhood when  $a > 1$ .

## APPENDIX G. FIXED-TARGET COMPARISON OF CONTACT GRAPHS

This section makes the change of category in the contact construction explicit. A formal congruence depending on the pencil parameter acts on an incidence bundle; it is not an automorphism of the fixed target whose Hilbert scheme is under consideration. We retain the incidence projection and undo every such change before taking a Hilbert point.

**G.1. Objects, targets, and the gluing statement.** Work on a finite-type framed atlas  $S$  of regular pencils avoiding corank at least three, with central pencil  $R_0$ . A frame here means a lift of the projective source, an ordered pencil basis, and a local parameter at each of the finitely many central corank-two points. These choices are not quotiented in an embedded Hilbert fibre. Let  $\mathcal{L} \subset \mathbb{P}(\mathrm{Sym}^2 V) \times S$  be the universal pencil line,  $X = \mathrm{CQ}(V)$ , and

$$\mathcal{Y} = \mathcal{L} \times_{\mathbb{P}(\mathrm{Sym}^2 V)} X.$$

The targets and their maps are

$$\begin{array}{ccccc} \mathcal{Z} & \hookrightarrow & \mathcal{Y} & \hookrightarrow & X \times S \\ & & \downarrow & & \downarrow \\ & & \mathcal{L} & \longrightarrow & S. \end{array} \tag{G.1}$$

Here  $\mathcal{Z}$  denotes an embedded flat family, and the last vertical arrow is projection. The middle map is a closed immersion because  $\mathcal{L}$  is the universal family of actual lines. An object in the graph retains its point of  $S$ , not just the curve in  $X$ . On the complement of the central contact supports, the complete-quadric lift is the forced section of this incidence target.

Write  $\widehat{S} = \mathrm{Spf} \widehat{\mathcal{O}}_{S, R_0}$ . For a local Artin complex algebra  $T$  with a specified map to  $\widehat{S}$ , define  $\mathcal{D}_X(T)$  to be the set of embedded  $T$ -flat closed subschemes of (G.1), with the fixed central curve, the specified Hilbert polynomial, and the forced lift off the contacts. Isomorphism means equality as closed subschemes, with the identity on the retained base. Restrict this functor, when needed, to the completed reduced closure of the nonincident graph. We do not use the abstract deformation functor of the curve or identify objects by an ambient automorphism.

**Lemma G.1** (Patching closed ideals with their quotients). *Let  $T$  be a local Artin complex algebra, let  $W$  be a Noetherian ambient scheme over a neighbourhood of a point of  $\mathcal{L}_T$ , and complete along that point's inverse image. Let  $W^\circ$  be its complement. Compatible coherent ideal quotients on  $W^\circ$  and the completion, with their identification on the punctured completion, glue uniquely to a coherent ideal quotient on  $W$ . The assertion applies to quotients with torsion supported at the contact and to their flat families over  $T$ . Flatness, equality of ideals and the specified closed incidence equations can be checked on this cover. The same statement holds for finitely many disjoint contact supports and after any Artin base change.*

*Proof.* Use the ring-theoretic completion on affine neighbourhoods and its pullback on projective charts. The map from this completion is flat, Noetherian, and is an isomorphism over the closed support. The formal patching equivalence for quasi-coherent sheaves is therefore [Bhatt(2014), Proposition 5.6(4), Example 5.8]; its hypotheses are also covered by Theorem 1.4 and Lemma 5.12(2) there. Flatness of the completion is essential: the unrestricted module assertion for a nonflat completion is not the statement being invoked. In particular no torsion-freeness at the contact is imposed.

Apply this symmetric monoidal equivalence to the quotient with its unit and multiplication. The ambient structure map and the ideal kernel are retained as part of the data. Surjectivity and the ordinary degree-zero algebra property are detected on the jointly surjective flat cover



consisting of the completion and the open complement. Thus the glued object is an ordinary ideal quotient. Finite presentation descends on this faithfully flat cover; the ambient rings are Noetherian. The same cover detects base flatness by the local criterion for flatness, or by vanishing of the relevant Tor modules. It detects all the additional incidence equations. Patching on the projective affine cover then glues the coherent quotients. Iterating over disjoint supports gives the last assertion. This argument supplies descent data on the punctured completions; faithfulness of completion alone would only give uniqueness, not existence.  $\square$

At the  $i$ th contact split a nonsingular block of size  $n - 2$ . The remaining symmetric Schur complement has entries  $a_i, b_i, c_i$ . On this rank open the only noninvertible minor ideal below the determinant is

$$J_i = (a_i, b_i, c_i) = I_{n-1}.$$

The determinant factor is an invertible ideal on the integral total space and does not change the simultaneous graph. A congruence  $M_i(x_i)$  and the Schur splitting replace this generating triple by an invertible linear transformation over the completed base ring. Consequently they give an isomorphism between the corresponding projective graph targets over the formal parameter disc. Denote it by  $\Theta_i$ ; it preserves  $x_i$  and is part of the comparison data. In normal coordinates the triple is  $(f_i, g_i, r_i)$ , the fourth polynomial  $k_i$  having been removed by the invertible row operation  $c_i \mapsto c_i - k_i a_i$ .

**Proposition G.2** (Completed embedded comparison). *Fix the framed atlas, formal Schur splittings and normalizing congruences just described. Let  $\mathcal{D}_{\text{nor}}$  be the functor of embedded normal-graph quotients on the formal contact discs, together with their identification with the forced lift on the punctured discs. Then there is an equivalence of functors*

$$\mathcal{D}_X \simeq \mathcal{D}_{\text{nor}} \tag{G.2}$$

obtained by restriction and  $\Theta_i$ , with inverse given by  $\Theta_i^{-1}$  and ideal patching. No congruence quotient is taken on either side. Changing the chosen frames or congruences conjugates these functors and their inverse maps; it does not identify distinct fixed-target Hilbert points.

On the reduced graph component, the complete local rings are the complete local rings of the corresponding polynomial graph charts, with the smooth parameters of the contact atlas adjoined. In particular this assertion holds both at the division charts of Theorem H.1 and at every determinantal conic chart of Lemma I.1. The comparison is compatible with restriction to fibres. Where these total charts are regular, normalization of the total graph changes neither them nor their scheme fibres.

*Proof.* The completed normal form of Theorem F.2 is used over  $T[[x_i]]$ , with the base maximal-ideal filtration for its inductive construction. The parameter  $x_i$  is fixed; its powers specify the remainder degrees, not the topology in that induction. Since  $T$  is Artin the base induction is finite. For the complete universal base it is the inverse limit of these finite inductions, so no interchange of two unrelated convergence assertions is required.

An embedded family restricts to a coherent quotient on each completed contact neighbourhood and to the forced quotient on the complement. Apply  $\Theta_i$  on the first and the induced identification on the punctured neighbourhood. Conversely undo  $\Theta_i$ , impose the original incidence equations, and apply Lemma G.1. These two operations are inverse on quotients and their structure maps, proving (G.2) for every  $T$ , including nonreduced bases. The use of the image Rees algebra can be recorded explicitly: a homomorphism  $R \rightarrow R'$  gives

$$\mathcal{R}_R(J) \otimes_R R' \twoheadrightarrow \mathcal{R}_{R'}(JR').$$

Proj of this surjection embeds the actual image graph into the base-changed incidence target. We do not assert that this surjection is always an isomorphism. The closed ideals compared in the proposition are these actual embedded ideals, rather than an arbitrary base change of a blow-up.

For precision about the graph component, work first on the reduced finite-type Hilbert graph over the atlas. The closure ideal of its nonincident open can be computed in each finite presentation by saturating with the ideal of the excluded base locus. Noetherianity makes the increasing sequence of ideal quotients stationary; flat completion commutes with each finite ideal quotient. It therefore commutes with this closure computation. The rings are excellent complex algebras, and their reduced completions stay reduced. The patching equivalence preserves the nonincident section and the retained base, hence also this specified reduced closure. It is not a statement that unrelated components of an abstract curve Hilbert scheme have disappeared.

The contact map on the completed smooth framed base is formally smooth, with the bounded coefficient parameters as independent coordinates and the remaining directions as power-series variables. The scalar  $k_i$  is one of these independent smooth factors for the graph problem: the row operation above removes it without changing the embedded ideal when the operation is undone. Different contact discs patch independently. The polynomial graph charts give algebraic representatives for their contact quotients; their recovery maps retain the base coefficients. Applying the functor equivalence gives the asserted isomorphism of completed local rings with the smooth factors adjoined.

This is not yet an algebraization by formal assertion. In each application below the morphism on the special fibre is constructed algebraically first: a primitive line over the finite contact algebra, or a conic together with the fixed main curve, supplies its explicit coherent ideal. The finitely many coefficients of the central gauge on that contact algebra are algebraic. Thus these are finite-presentation morphisms into the actual Hilbert graph. The completed comparison proves their local isomorphism on the specified fibre charts. The residue fields at closed points are  $\mathbb{C}$  and the local rings are Noetherian; the formal criterion for etaleness, [Stacks Project, Lemma 41.11.3], applies. The coefficient or ideal recovery gives a monomorphism. Such a morphism is an open immersion on these neighbourhoods. This orders the argument as algebraic construction, completed comparison, then etale monomorphism, rather than treating formal normal coordinates as an algebraic family without descent.

An alternative normalizing congruence acts through a transformation of the same relative incidence target. Both directions of the comparison compose with that transformation and its inverse. The composition in the fixed target remains the identity. The same identity is the cocycle on overlaps of the source-frame atlas, so the comparison descends. Finally a regular total chart is normal; the finite normalization of the total graph is its identity there. Taking its fibre retains the original base equations, possibly with nilpotents. No commutation of fibre and normalization is used.  $\square$

## G.2. Cohomological recovery along the primitive locus.

**Lemma G.3** (The union sequence and the retained coefficient base). *Let  $D = \text{Spec } \mathbb{C}[x]/(x^a)$  and let  $Z_w = C_0 \cup T$ , with  $T = \mathbb{P}_D^1$  the primitive tail and intersection the attaching section  $D$ . For  $L = \mathcal{O}(a-1, 1)|_{Z_w}$  there is an exact sequence*

$$0 \rightarrow L \rightarrow L|_{C_0} \oplus L|_T \xrightarrow{\text{res}_C - \text{res}_T} L|_D \rightarrow 0. \quad (\text{G.3})$$

*The restriction from  $T$  is onto,  $H^1(Z_w, L) = 0$ , and  $h^0(Z_w, L) = 2a$ . For each such  $w$  these assertions persist on a neighbourhood in the integral graph  $\Gamma_a$  over  $B_a$ . On that neighbourhood the recovery map has a line-bundle cokernel, and vanishing of the retained third coordinate in that cokernel is a scheme-theoretic equality, not a fibrewise assertion.*

*Proof.* The ideals of a scheme-theoretic union give  $0 \rightarrow \mathcal{O}_{Z_w} \rightarrow \mathcal{O}_C \oplus \mathcal{O}_T \rightarrow \mathcal{O}_D \rightarrow 0$ ; tensor by  $L$ . A line bundle on the local Artin scheme  $D$  is free, and the evaluation  $H^0(\mathbb{P}_D^1, \mathcal{O}(1)) = \mathcal{O}_D^2 \rightarrow \mathcal{O}_D$  at its section has a unit coefficient, hence is surjective. On the main component  $L = \mathcal{O}(a-1)$  and on the tail it is  $\mathcal{O}(1)$  up to an invertible Artin factor. The cohomology sequence gives  $a + 2a - a = 2a$  and the claimed vanishing, over the Artin ring rather than only its residue field.

To specify homogenization, write  $W$  for degree- $(a-1)$  homogenization of  $w$ , and  $Q_0$  for degree- $(a-2)$  homogenization of  $q$ ; when  $a = 1$  the latter space is zero. The map is

$$(W, Q_0) \mapsto WG - sQ_0F \quad \text{in } H^0(Z, \mathcal{O}(a-1, 1)).$$

The section to be recovered is  $s^{a-1}R$ . There are  $a + (a-1)$  unknown coefficients. Restricting to the main component, then to the tail, proves injectivity at every central  $w$ . Upper semicontinuity and cohomology and base change give a pointwise open neighbourhood on which the map is a split injection of ranks  $2a-1$  and  $2a$ . These neighbourhoods cover the whole central locus; no single uniform affine shrink is asserted. Its cokernel is a line bundle. The class of  $s^{a-1}R$  is zero on the dense nonincident graph. A section of a line bundle on an integral reduced neighbourhood that vanishes densely is zero. The recovered coefficients are thus regular and satisfy the division identity. The projection to  $B_a$  is indispensable: it already supplies  $f, g, r$ . All comparisons retain it.  $\square$

## APPENDIX H. EUCLIDEAN CHARTS OF THE AMBIENT HILBERT BOUNDARY

We first calculate a universal polynomial graph in a fixed target. This separates the Hilbert calculation from the choice of pencil normal coordinates. Write  $\mathbb{C}[x]_{<a}$  for polynomials of degree less than  $a$ , including zero. Let

$$B_a = \{(f, g, r) : f \text{ monic of degree } a, \quad g, r \in \mathbb{C}[x]_{<a}\} \simeq \mathbb{A}^{3a}.$$

Homogenize the three polynomials to degree  $a$ . They have no common zero at infinity, where their value is  $[1 : 0 : 0]$ . Over the dense open where they have no common zero, their graph in  $\mathbb{P}^1 \times \mathbb{P}^2$  has Hilbert polynomial  $(a+1)l + 1$  for  $\mathcal{O}(1, 1)$ . Denote its reduced Hilbert graph closure over  $B_a$  by  $\Gamma_a$ . It is integral, since it is the reduced closure of the graph over an integral dense open. We never quotient its embedded curves by target automorphisms.

**Theorem H.1** (An open division chart). *Let  $W_a = \{(f, g, w) : f \text{ monic of degree } a, \quad g, w \in \mathbb{C}[x]_{<a}\} \simeq \mathbb{A}^{3a}$  and define*

$$\mu : W_a \longrightarrow B_a, \quad (f, g, w) \mapsto (f, g, r = \text{rem}_f(gw)). \quad (\text{H.1})$$

*Put  $q = (gw - r)/f$ . The graph modification on the universal parameter line over  $W_a$  is the blow-up of  $(f, g)$ , and its two-generator Rees equation is*

$$gU - fV = 0. \quad (\text{H.2})$$

*It is a projective flat family embedded in  $\mathbb{P}^1 \times \mathbb{P}^2 \times W_a$ , by the third graph coordinate*

$$[F : G : R] = [U : V : wV - qU] \quad (\text{H.3})$$

*on the finite parameter chart. At infinity it is the unique original graph. There is an open neighbourhood of*

$$\{f = x^a, \quad g = 0, \quad w \in \mathbb{C}[x]_{<a}\} \subset W_a$$

*on which this family identifies  $W_a$  with an open subscheme of  $\Gamma_a$ . In particular  $\Gamma_a$  is smooth and already normal there. The assertion is an open chart of the Hilbert graph, not just a generically injective parametrization of some limits.*

For any point of this chart, let  $d = \gcd(f, g)$  be monic, let  $r_0 = \deg d$ , and put  $f = df_1$ ,  $g = dg_1$ . Its embedded fibre has the scheme decomposition

$$C \cup T_d, \quad T_d = \mathbb{P}_{\mathbb{C}[x]/(d)}^1, \quad C \cap T_d = \operatorname{Spec} \mathbb{C}[x]/(d). \quad (\text{H.4})$$

The attaching section on  $T_d$  is  $[U : V] = [f_1 : g_1]$ . It has no embedded associated points. If  $d = \prod_i (x - \alpha_i)^{m_i}$ , this is one main component and thick tails of multiplicities  $m_i$  at the distinct  $\alpha_i$ . Its graph degree on the main component is  $a - r_0$ , and its tail cycle degree is  $r_0$ . Thus the same open chart describes splitting, collision, and simultaneous unequal thicknesses by an exact gcd law.

*Proof.* In the polynomial ring of the universal line and  $W_a$ , the elements  $f, g$  form a regular sequence:  $f$  is monic, and modulo  $f$  the coefficient  $g_0$  still occurs freely and linearly in  $g$ . The Rees algebra is consequently the symmetric algebra modulo the single syzygy  $gU - fV$ . One can alternatively compare coefficients in its injection into the polynomial ring in a Rees variable. On every geometric fibre the homogeneous equation is nonzero, because  $f$  is monic. The fibrewise Cartier criterion therefore proves flatness [Stacks Project, Tag 062Y, part (3)]. The original ideal  $(f, g, r)$  is exactly  $(f, g)$ , since  $r = gw - fq$ . Formula (H.3) is a rank-two linear embedding on every finite parameter chart; the first two coordinates are unchanged. Thus the Rees graph is closed in the three-coordinate graph. Infinity is disjoint from the base ideal and needs no blow-up. This constructs the claimed flat embedded family and its morphism to the Hilbert scheme.

Its image lies in  $\Gamma_a$ : over  $\operatorname{Res}(f, g) \neq 0$  it is the original graph, that open is dense in the integral scheme  $W_a$ , and the reduced graph is closed. To prove that the resulting map is an open chart we construct a regular inverse. This step cannot be replaced by pointwise injectivity alone.

Let  $Z_w$  be a curve with  $f = x^a, g = 0$  in the family, and let  $L = \mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^2}(a - 1, 1)|_{Z_w}$ . On its main component  $L$  has degree  $a - 1$ . On its thick tail it is  $\mathcal{O}(1)$  over  $A = \mathbb{C}[x]/(x^a)$ , and their intersection has length  $a$ . The restriction map from tail sections onto the intersection is surjective. The union exact sequence therefore gives

$$H^1(Z_w, L) = 0, \quad h^0(Z_w, L) = 2a. \quad (\text{H.5})$$

These statements hold in a Hilbert neighbourhood by cohomology and base change. On that neighbourhood consider the linear map to the rank- $2a$  bundle of sections

$$\Psi : \mathbb{C}[x]_{<a} \oplus \mathbb{C}[x]_{<a-1} \longrightarrow H^0(L), \quad (w, q) \longmapsto wG - qF. \quad (\text{H.6})$$

The expressions are homogenized to degree  $a - 1$  in the first factor; in particular the second polynomial has zero leading coefficient. For  $a = 1$  its second summand is zero. At  $Z_w$  this map is injective. Indeed restriction to the main component makes  $q$  zero. Restriction to the tail then makes  $w$  zero modulo  $x^a$ , hence zero as a polynomial of degree less than  $a$ . After shrinking,  $\Psi$  is a split injection with line-bundle cokernel.

The section to be represented is  $R$ , multiplied by the homogeneous coordinate vanishing at infinity to the power  $a - 1$ . On the dense open  $\operatorname{Res}(f, g) \neq 0$ , Euclidean inversion of  $g$  in  $\mathbb{C}[x]/(f)$  gives its unique representation  $R = wG - qF$  with the bounds in (H.6). Its class in the line-bundle cokernel therefore vanishes on a dense open of the integral reduced Hilbert graph, and hence vanishes identically. The split injection recovers regular coefficients  $w, q$ . The polynomial equality  $r = gw - fq$  follows on the dense open and therefore everywhere. This defines the inverse of (H.1). Conversely the reconstructed curves agree with the universal Hilbert curve on a dense open; the two maps to the separated Hilbert scheme agree on the entire reduced neighbourhood. This proves the open-chart assertion, including scheme structure and normality.

For the fibre calculation, factor (H.2) as  $d(g_1U - f_1V)$ . The two factors are relatively prime on each affine chart:  $f_1, g_1$  are coprime. Their principal ideals intersect in their product. The second factor is the graph of the coprime pair, and the first is  $\mathbb{P}^1$  over  $\mathbb{C}[x]/(d)$ . Since  $f_1, g_1$

generate the unit ideal modulo  $d$ , their intersection is exactly the displayed section over that Artin ring. Each chart is a hypersurface in a regular surface, hence Cohen–Macaulay; its associated primes are precisely its component primes. This also proves absence of embedded points. The Chinese remainder theorem separates the distinct roots of  $d$  with their full multiplicities. Finally,  $\chi(\mathcal{O}_{T_d}(l)) = r_0(l + 1)$  and the attaching scheme has Euler characteristic  $r_0$ . Adding the main component gives  $(a + 1)l + 1$  for  $\mathcal{O}(1, 1)$ , as required.  $\square$

**H.1. From the universal division chart to actual pencils.** For a corank-two incidence  $p$  of length  $a$ , let  $D_p$  be its scheme-theoretic intersection with the pencil. The strict transform of the pencil meets the exceptional projective plane bundle over  $D_p$  in a section  $s_p$ . A *primitive line tail* means a line subbundle in that plane bundle containing  $s_p$ , over all of  $D_p$ , not merely a line on its reduced support. Its union with the strict transform is taken with their common attaching subscheme  $D_p$ . This definition concerns a fixed embedded curve in complete quadrics.

**Theorem H.2** (Ambient jet-tail charts and their collision law). *Let  $R_0$  be a regular pencil with no corank at least three. Suppose its corank-two points have Smith exponents  $(a_i, b_i)$ ,  $a_i \leq b_i$ ; points of corank one are unrestricted. In the full normalized pencil Hilbert graph, the locus over  $R_0$  with a primitive line tail at each corank-two point is open and is*

$$\prod_i \text{Res}_{\mathbb{C}[x_i]/(x_i^{a_i})/\mathbb{C}} \mathbb{P}^1 = \prod_i J_{a_i-1}(\mathbb{P}^1). \quad (\text{H.7})$$

*It is smooth, irreducible, and has dimension  $\sum_i a_i$ . For  $a_i = 1$  it recovers the reduced-incidence fibre factor  $\mathbb{P}^1$ ; for  $a_i > 1$  constant directions form only its zero-jet section. All points of (H.7) are limits of genuine pencils. No extra framing or congruence quotient is taken in this scheme fibre.*

*More precisely, after a smooth frame atlas, the completed local map at any such lift is the product of the division maps (H.1), the  $k_i$  coordinates in (F.4), and complementary smooth coordinates. Thus, in each chart, its nonincident open is*

$$\prod_i \text{Res}_{x_i}(f_i, g_i) \neq 0. \quad (\text{H.8})$$

*The modification is smooth in these charts, but the resultant boundary need not have normal crossings. Its relative curve has local equation  $g_i U_i - f_i V_i = 0$ . Formula (H.4) classifies all fibres in the chart, including mixed nonreduced contacts and collisions, with multiplicities and attaching lengths. On the actual complete-quadric curve the exterior multidegrees are*

$$\deg_C H_q = q - \left( \sum_i \deg \gcd(f_i, g_i) \right) \mathbf{1}_{q=n-1}, \quad \deg_{(T_i)_{\text{red}}} H_q = \mathbf{1}_{q=n-1}. \quad (\text{H.9})$$

*Each local thick factor has its corresponding cycle multiplicity. The polarization is  $\bigotimes_{q=1}^{n-1} H_q$  and the Hilbert polynomial is  $\binom{n}{2}l + 1$ .*

*Proof.* We first specify the formal comparison which makes this an ambient statement. Work on the smooth atlas of framed pencils and choose a parameter disc at each point in question. Theorem F.2 supplies independent formal coordinates  $(f_i, g_i, r_i, k_i)$  and smooth complementary coordinates. On such a disc a unit  $(n - 2)$ -block reduces the submaximal-minor ideal to  $(f_i, g_i, r_i)$ . The rank locus is smooth of codimension three, and its complete-quadric modification is its blow-up. Equivariant Gaussian elimination identifies its graph with the three-coordinate normal graph; the extra unit-block and target coordinates are fixed functions of the pencil and of the disc parameter.

This comparison is also a comparison of the embedded Hilbert problems, for the following reason. The parameter  $x_i$  is retained: we work in the incidence target over the universal pencil line, not in an abstract unparametrized normal plane. The first complete-quadric projection

recovers that line coordinate, so adding it does not change any embedded Hilbert point in the pencil graph. A source congruence depending on  $x_i$  is an automorphism of this incidence target over the disc. It need not, and is not claimed to, be one constant automorphism of the fixed projective target. Away from the finitely many contact supports the submaximal ideal is the unit ideal; the curve there is forced to be the original lift, with no additional embedded choices. On the punctured discs the two descriptions are this same forced lift. The local ideals therefore glue uniquely. For Artin base extensions this can be checked in affine charts: equality and flatness of their coherent ideal quotients are detected after the faithfully flat completion at each remaining support. Thus the completed embedded deformation functors on the graph closure agree with the polynomial normal-graph functors and the complementary smooth factors. This uses the actual Rees image, not a general assertion that blow-up commutes with nonflat base change. The same argument applies to families specializing the dense nonincident graph, so their reduced closures and their completed local rings agree. Excellence over  $\mathbb{C}$  and normality of the smooth charts identify the normalization there.

Apply Theorem H.1 to each factor. In the central fibre,  $f_i = x_i^{a_i}$  and  $g_i = r_i = 0$ , and  $w_i$  is an arbitrary polynomial of degree less than  $a_i$ . The opposite chart interchanges  $g_i$  and  $r_i$ . On the overlap its slope is  $w'_i = w_i^{-1}$  in  $\mathbb{C}[x_i]/(x_i^{a_i})$ . These are exactly the two standard charts for lines through the section  $s_p$  over that local Artin algebra. They represent  $\text{Res}_{D_p/\mathbb{C}} \mathbb{P}^1$ , equivalently the indicated jet scheme.

There is an algebraic morphism from this jet space to the actual Hilbert fibre, not only a formal map. Pull back the universal line subbundle over  $D_p$ , take its projectivization in the exceptional plane bundle, and unite it with the fixed strict transform along its attaching section  $D_p$ . The union exact sequence has flat outer terms over the jet space and a surjective restriction to the attaching scheme, so the union is flat. Its line subbundle is recovered from the embedded tail, also after base change; hence this morphism is a monomorphism onto its image. The completed local calculation above and the inverse in (H.6) identify its completed local rings with those of the Hilbert fibre. A finite-type morphism with these completed local isomorphisms and equal residue fields is étale there; an étale monomorphism is an open immersion. This proves the claimed openness algebraically. The charts of the total Hilbert graph are regular, so its normalization changes neither their rings nor this open fibre. The construction descends from the frame atlas and preserves distinctions among tails in the fixed target, rather than quotienting by automorphisms of  $R_0$ .

Every such point specializes from the dense open in its smooth chart, hence occurs from a formal one-parameter family of pencils; properness of the Hilbert graph gives the same embedded limit. Alternatively choose  $g_i = t$  and lift its prescribed slope by  $r_i = \text{rem}_{f_i}(g_i w_i)$ , then use the formally smooth pencil-coordinate map. Its local ideal is  $(x_i^{a_i}, t)$ . The two jet charts are affine spaces of dimension  $a_i$ , with the usual open overlap. They are smooth and irreducible. Products at distinct supports follow from the independent contact coordinates.

In a chart  $r_i$  is a multiple of  $g_i$  modulo  $f_i$ , so a common zero of the triple is exactly a common zero of  $f_i, g_i$ . This gives (H.8). The fibre and multiplicity statements follow from the scheme calculation in (H.4). Only the last exterior system has a base ideal on the corank-two open. Its base length is the degree of the indicated gcd; the smaller systems are units. This proves (H.9) and the Hilbert polynomial, with no assumption that the gcd is squarefree.  $\square$

The theorem determines a full neighbourhood of every primitive line-tail lift. It does not identify this open with the whole proper fibre. In fact the first nonreduced case already has another kind of component, as the next result shows.

## APPENDIX I. THE COMPLETE FIRST NONREDUCED CONTACT FIBRE

We determine the entire fibre, including its nilpotents, at the first nonreduced contact. The calculation also determines the common boundary of the two components found in Theorem T.1. It takes place in the graph over the coefficient base: the coefficients are part of a point throughout.

Put  $B_2 = \mathbb{A}^6$ , with coordinates the coefficients of a monic quadratic  $f$  and of two polynomials  $g, r$  of degree at most one. Let  $\Gamma_2$  be the reduced closure in  $B_2 \times \text{Hilb}_{3l+1}(\mathbb{P}^1 \times \mathbb{P}^2)$  of the graphs of  $[f : g : r]$ , on their common nonvanishing locus. The polarization is  $\mathcal{O}(1, 1)$ . Write  $\nu : \tilde{\Gamma}_2 \rightarrow \Gamma_2$  for the normalization of the total graph, and set

$$o = (x^2, 0, 0), \quad X_2 = (\tilde{\Gamma}_2)_o.$$

The scheme  $X_2$  is not defined by normalizing a fibre. In the target use coordinates  $[F : G : R]$  and put  $P = [1 : 0 : 0]$ . The main curve at  $o$  is  $C_0 = \mathbb{P}^1 \times \{P\}$ .

**I.1. An embedded chart containing all conic tails.** Let  $[s : t]$  be coordinates on the first projective line, with  $x = t/s$ . Consider the affine space with coordinates  $(c, e, \lambda, A, B, C)$ , and put  $z = t - cs$ ,  $H = R - \lambda G$ . In the product with  $\mathbb{P}^1 \times \mathbb{P}^2$  form the matrix

$$\mathcal{M} = \begin{pmatrix} H & AF + CG \\ -G & H + BF \\ es & -z \end{pmatrix}. \quad (\text{I.1})$$

Its minors generate the bihomogeneous ideal

$$\begin{aligned} Q &= H^2 + BFH + AFG + CG^2, \\ L_1 &= zG - es(H + BF), \\ L_2 &= zH + es(AF + CG). \end{aligned} \quad (\text{I.2})$$

The associated map to  $B_2$ , written on  $s = 1$  with  $z = x - c$ , is

$$\begin{aligned} f &= z^2 + e^2C, & g &= eBz - e^2A, \\ r &= \lambda g - eAz - e^2BC. \end{aligned} \quad (\text{I.3})$$

**Lemma I.1** (The determinantal family and its recovery maps). *The minors in (I.2) define a flat family of embedded curves of Hilbert polynomial  $3l + 1$ . It defines a morphism to  $\Gamma_2$ . Near every point with  $e = 0$ , this morphism is an open immersion. Its image is regular and is unchanged by normalization of the total graph. At  $e = 0$  the curve is the scheme-theoretic union of  $C_0$  and the plane conic  $Q = 0$  over  $x = c$ , with the attachment placed at that base point. The intersection is the reduced point  $(c, P)$ , even when  $Q$  is a doubled line. Constant changes of target coordinates fixing  $P$  provide such charts around every plane conic through  $P$ .*

*Proof.* The minors have height two on every geometric fibre. When  $e \neq 0$ , one eliminates  $H$  locally from  $L_1, L_2$ ; equivalently the rank-one condition is the two-generator graph of  $f, g$ , with the indicated third coordinate. At  $e = 0$  the ideal is

$$(zG, zH, Q) = (G, H) \cap (z, Q),$$

where  $Q \in (G, H)$  is nonzero. The equality follows by reducing modulo  $z$  and then using that  $z$  is regular on the polynomial ring in  $F, G, H$  modulo  $Q$ . This gives the asserted union and intersection. In particular the height is two also for a singular conic. The Hilbert–Burch complex of (I.1) is consequently exact on every fibre. Its sheaf form is

$$0 \longrightarrow \mathcal{O}(-1, -2)^2 \longrightarrow \mathcal{O}(-1, -1)^2 \oplus \mathcal{O}(0, -2) \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}_C \longrightarrow 0. \quad (\text{I.4})$$

The fibrewise exactness criterion for a complex of base-flat locally free sheaves gives base flatness of the cokernel. Computing the Euler polynomial from this complex gives  $3l + 1$ . This is an

application of the classical determinantal resolution, not an assertion that the resolution theorem is new.

Substitution of (I.3) into the three minors gives zero. On the dense open

$$e(A^2 + CB^2) \neq 0$$

the three binary quadratics  $f, g, r$  are a basis, the curve is their graph, and the image is a smooth conic. Thus the flat family gives a morphism into the reduced graph closure. The assertion at infinity is direct:  $s = 0$  implies  $z = t \neq 0$ , so  $L_1, L_2$  give  $G = H = 0$ . There is no additional component or choice at infinity.

We prove the open-immersion assertion by regular recovery, not by counting parameters. At a central curve  $C_0 \cup Q$ , the union exact sequence and its twists give

$$H^1(\mathcal{O}(0, 2)|_{C_0 \cup Q}) = 0, \quad h^0 = 5, \quad H^1(\mathcal{O}(1, 1)|_{C_0 \cup Q}) = 0, \quad h^0 = 4.$$

Indeed  $H^0(Q, \mathcal{O}_Q(2))$  has dimension five and  $H^0(Q, \mathcal{O}_Q(1))$  dimension three for every nonzero plane conic, including a doubled line, by  $0 \rightarrow \mathcal{O}_{\mathbb{P}^2}(j-2) \rightarrow \mathcal{O}_{\mathbb{P}^2}(j) \rightarrow \mathcal{O}_Q(j) \rightarrow 0$ . The restriction to  $P$  is surjective. Combining it with the main component proves both equalities and the surjectivity of the ambient restriction maps. Cohomology and base change therefore give, in a neighbourhood of this point in the integral graph  $\Gamma_2$ , a line of quadratic equations and a rank-two bundle of equations of bidegree  $(1, 1)$ .

Normalize the coefficient of  $R^2$  in the quadratic equation to one. Its  $F^2$  coefficient is zero on the dense graph, because infinity maps to  $P$ , and hence is zero throughout this neighbourhood. Completing the square in  $R, G$  uniquely writes the equation as  $Q$  in (I.2); it recovers  $\lambda, A, B, C$  regularly. The projection to  $B_2$  supplies  $f$ , so  $c$  is minus half its linear coefficient. Normalize the two bilinear equations by their coefficients at  $tG, tH$ ; the relevant minor is a unit at the central curve. The negative coefficient of  $sH$  in the equation with  $tG$  coefficient one recovers  $e$ .

Here is an explicit verification on the dense graph, which also checks the normalizations. Write  $f = (x - c)^2 + d$ ,  $g = eB(x - c) + u$ , and  $r - \lambda g = -eA(x - c) + v$ . The coefficient of  $x^3$  in  $Q(f, g, r - \lambda g) = 0$  gives the displayed opposite linear coefficients. When  $A^2 + CB^2 \neq 0$ , the remaining coefficients give

$$u = -e^2A, \quad v = -e^2BC, \quad d = e^2C.$$

The normalized bilinear equations are then precisely  $L_1, L_2$ . All these are equalities of regular functions or sections on an integral neighbourhood, so they extend from its dense graph open. The recovered parameters satisfy (I.3), and the universal curve equals (I.2), first densely and then by separatedness of the Hilbert scheme. Recovery is also the identity on the determinantal parameter space. These two morphisms are inverse after restricting to their common open neighbourhoods. The parameter space is smooth; its image is therefore regular and normalization does not change it.

Finally any nonzero quadratic form vanishing at  $P$  is nonzero on some vector not proportional to  $P$ . Choose that vector as the  $R$  direction. This is a constant target change fixing  $P$ , followed by rescaling. The preceding argument applies. Such changes are changes of chart, not an equivalence relation on Hilbert points.  $\square$

**I.2. The full fibre and its nilpotents.** Let  $\mathbb{F}_2 = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(2))$  be the compactification of  $\text{Tot}(\mathcal{O}(2))$  whose section at infinity has self-intersection  $-2$ . Denote that section by  $D_\infty$ . Let  $\mathcal{Q}_P \simeq \mathbb{P}^4$  be the projective space of plane conics through  $P$ , and let  $D_{\text{dbl}} \simeq \mathbb{P}^1 \subset \mathcal{Q}_P$  be its locus of doubled lines through  $P$ .

**Theorem I.2** (Complete first nonreduced fibre). *The scheme  $X_2$  has exactly two irreducible components, with reduced structures  $\mathcal{Q}_P \simeq \mathbb{P}^4$  and  $\mathbb{F}_2$ . Their scheme-theoretic intersection, when*



each is given its reduced closed structure, is the reduced projective line

$$D_{\text{dbl}} = D_{\infty}.$$

The open complement of  $D_{\infty}$  in the second component is  $J_1(\mathbb{P}^1) = \text{Tot}(T\mathbb{P}^1)$ , and the first component parametrizes exactly the curves  $C_0 \cup Q$ ,  $Q \in \mathcal{Q}_P$ . Thus the second component has dimension exactly two and the conic component has dimension exactly four.

The fibre is connected and generically reduced on both components, but is not reduced. In the charts of Lemma I.1 its scheme structure is

$$\mathbb{C}[\lambda, e, A, B, C]/(eA, eB, e^2C). \quad (\text{I.5})$$

At a doubled line with slope  $\lambda_0$  the completed local ring is obtained by replacing the polynomial ring by  $\mathbb{C}[[\lambda - \lambda_0, e, A, B, C]]$  in this formula. The nilradical is generated by  $eC$ , has square zero, and is supported on the locus of conics singular at  $P$ . The fibre is not Cohen–Macaulay at that locus. Its reduced support is the union of the two indicated smooth components, and the normalization of that reduced support is their disjoint union. This last normalization is different from the normalization of the total Hilbert graph used to define  $X_2$ .

*Proof.* First take the fibre of (I.3) at  $o$ . Its coefficient equations give  $c = 0$ ,  $e^2C = 0$ ,  $eB = 0$ ,  $e^2A = 0$  and  $e(\lambda B - A) = 0$ , together with a redundant constant equation. The resulting ideal is exactly

$$(c, eA, eB, e^2C).$$

No radical has been taken. In particular the charts compute the scheme fibre of a regular total graph and therefore of its normalization as well.

The conic family is algebraic over all of  $\mathcal{Q}_P$ . Its ideal is  $I_{C_0} \cap I_Q$ , with  $Q$  placed at  $x = 0$ ; the intersection with  $C_0$  is the constant reduced point  $P$ . The union exact sequence proves flatness and the Hilbert polynomial. The family is a closed immersion into the Hilbert fibre: its inverse on its image recovers the quadratic equation as the kernel of the ambient restriction in degree  $(0, 2)$ . This also gives the relative residual construction; locally

$$(I_{C_0} \cap I_Q) : I_{C_0} = I_Q.$$

For the displayed ideals this follows by dividing by  $G$  and  $H$ : no nonzero element in the hypersurface ring of a conic is annihilated by both  $G$  and  $H$ . This remains true in the universal conic family by its same presentation. Alternatively the kernel-line construction alone proves the asserted relative closed immersion. The regular charts of Lemma I.1 show that this family lifts uniquely to the normalization near every one of its points; we do not use a general lifting assertion for maps from normal schemes.

The primitive open of the other component is the established jet chart. Write a jet as  $w = w_0 + w_1x$  modulo  $x^2$ . Near its direction at infinity use

$$\lambda = w_0, \quad e = 1/w_1.$$

In (I.2) put  $c = A = B = C = 0$ . This gives

$$(xG - eH, xH, H^2), \quad H = R - \lambda G. \quad (\text{I.6})$$

For  $e \neq 0$  it is the primitive thick-line curve with slope  $w = \lambda + x/e$ . At  $e = 0$  it is

$$(xG, xH, H^2) = (G, H) \cap (x, H^2),$$

the main curve together with the planar doubled line  $H^2 = 0$  at  $x = 0$ . Under the other affine coordinate of the direction line,  $w'_0 = 1/w_0$  and  $w'_1 = -w_1/w_0^2$ ; hence  $e' = -w_0^2e$  on the boundary chart. These are exactly the transition functions of the compactification of  $T\mathbb{P}^1 = \mathcal{O}(2)$  described above. Formula (I.6) and the jet families glue to a morphism from  $\mathbb{F}_2$  into  $X_2$ . The local inverse recovers  $\lambda, e$ ; the original jet chart supplies the inverse on its complement. Thus this

is a closed immersion, since its source is proper. Its infinity section is precisely  $D_{\text{dbl}}$ . It has self-intersection  $-2$  by the transition functions. The sum of the two reduced component ideals in (I.5) is  $(e, A, B, C)$ , giving the stated reduced intersection.

It remains to prove that these two families exhaust the fibre. Their union  $K$  is closed and proper. It is also open in the underlying fibre: at a point of the conic family the local fibre (I.5) has support

$$V(e) \cup V(A, B, C),$$

which is exactly the union just constructed; at a jet point the original open jet chart has no other branch. The scheme  $\tilde{\Gamma}_2$  is integral and normal, and its projection to the normal integral base  $B_2$  is proper and birational. Consequently all its geometric fibres are connected by Stein factorization [Stacks Project, Lemma 37.53.6]. The nonempty subset  $K$  is open and closed in  $X_2$ , so it is all of its support. This is an exhaustion argument for the proper fibre, not an inference from the existence of two families of curves.

For clarity, the local primary calculation is

$$(eA, eB, e^2C) = (e) \cap (A, B, C) \cap (A, B, e^2). \quad (\text{I.7})$$

It is irredundant wherever all the displayed coordinate functions vanish. It follows either by divisibility of monomials, or by intersecting first  $(e)$  with  $(A, B, C)$  and then with  $(A, B, e^2)$ . The two minimal primes are  $(e)$  and  $(A, B, C)$ ;  $(e, A, B)$  is the embedded associated prime when present. The radical is  $(eA, eB, eC)$ , so the nilradical is generated by  $eC$  and

$$\text{Ann}(eC) = (e, A, B), \quad (eC)^2 = 0.$$

In conic coordinates the equations  $A = B = 0$  say exactly that  $Q$  is singular at  $P$ . These calculations also apply after localization and completion and identify the claimed support intrinsically. They prove generic reducedness, nonreducedness and failure of the Cohen–Macaulay property. The two reduced components are smooth and are distinct; the final assertion about their normalization follows.  $\square$

**Corollary I.3** (Explicit adjacency and a numerical consequence). *Every doubled line through the attaching point is the common limit of primitive thick-line jets and of smooth conic tails. At such a point there is no extra component beyond the two in Theorem I.2. The Euler characteristic of the underlying complex fibre is seven.*

*Proof.* The first specialization is (I.6). For the second, set  $e = 0$ ,  $\lambda$  constant,  $B = C = 0$ ,  $A = \tau$ ; then  $Q = H^2 + \tau FG$  is nonsingular for  $\tau \neq 0$  and specializes to the doubled line. These are algebraic one-parameter families. Exhaustion was proved in the theorem. Euler characteristic is additive for the closed union, giving  $5 + 4 - 2 = 7$ .  $\square$

## APPENDIX J. ALL ARCS THROUGH A DOUBLE CONTACT

The six coefficient directions of  $B_2$  need not vanish in pairs, and a collision need not have a chosen square root. In this section we give a finite presentation of the full pullback, its horizontal closure, and its vertical torsion for every discrete valuation ring arc through  $o = (x^2, 0, 0)$ . The presentation is by a fixed collection of coordinate systems with their recovery opens. It is not a list of irreducible components indexed by the valuation of a single chosen coefficient.

Let  $R$  be a discrete valuation ring which is a  $\mathbb{C}$ -algebra, with uniformizer  $\tau$  and residue field  $\mathbb{C}$ . The same statements hold after extending that residue field. Write an arbitrary arc centred at  $o$  uniquely as

$$f = (x - c)^2 + d, \quad g = u(x - c) + v, \quad r = w(x - c) + z_0, \quad (\text{J.1})$$

where  $c, d, u, v, w, z_0 \in \tau R$ . No relation between these six elements is assumed. In particular neither  $g = r = 0$  nor a prescribed ramification order is imposed. Set

$$H_R = \tilde{\Gamma}_2 \times_{B_2} \text{Spec } R, \quad Y_R = \overline{(H_R)_K}^{H_R},$$

where  $K = \text{Frac } R$  and total-graph normalization precedes the base change.

**J.1. Five conic coordinates and two division coordinates.** The vector space of conic equations through  $P = [1 : 0 : 0]$  has basis  $FG, FR, G^2, GR, R^2$ . Evaluation at the following five vectors gives linearly independent functionals:

$$(0, 0, 1), \quad (0, 1, 0), \quad (0, 1, 1), \quad (1, 1, 0), \quad (1, 0, 1). \quad (\text{J.2})$$

Here  $R$  in a target coordinate denotes the third homogeneous coordinate, not the discrete valuation ring. To avoid confusion below it will only occur in homogeneous equations. Choose matrices  $M_i$  whose first column is  $(1, 0, 0)$  and whose third columns are the vectors in (J.2):

$$\begin{aligned} M_0 &= I_3, & M_1 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, & M_2 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \\ M_3 &= \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, & M_4 &= \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (\text{J.3})$$

For coordinate system  $i$  replace the coefficient column by  $M_i^{-1}(f, g, r)^t$ , and recompute the six coefficients in (J.1). The first polynomial stays monic of degree two and the other two have degree at most one. The central point remains  $o$ . These are changes of target coordinates, always undone when the embedded curve is returned to the fixed target; they are not quotient identifications of Hilbert points.

For any one of these transformed six-tuples put

$$\begin{aligned} D_i &= R[\lambda, e, A, B, C], \\ K_i &= (e^2 C - d, eB - u, e^2 A + v, \lambda u - eA - w, \lambda v - e^2 BC - z_0) \subset D_i. \end{aligned} \quad (\text{J.4})$$

There are also two division coordinate systems. The first has

$$D_+ = R[\alpha, \beta], \quad K_+ = (\alpha u + \beta v - w, \alpha v - \beta u d - z_0). \quad (\text{J.5})$$

The second is obtained by interchanging the pairs  $(u, v)$  and  $(w, z_0)$  and is denoted  $(D_-, K_-)$ .

**Theorem J.1** (Finite-chart miniversal specialization). *For every arc (J.1), the seven coordinate systems (J.4) and (J.5), restricted to their Hilbert recovery opens, cover  $H_R$ . On each such open its ring is the corresponding  $D_\bullet/K_\bullet$ , its horizontal ring is*

$$D_\bullet/(K_\bullet : \tau^\infty), \quad (\text{J.6})$$

*and its vertical torsion ideal is  $(K_\bullet : \tau^\infty)/K_\bullet$ . These formulas retain the scheme structures, including all embedded primes and nilpotents. They commute with localization, completion, and flat extensions of discrete valuation rings. They determine the entire specialization, not merely a neighbourhood of one chosen conic or primitive tail.*

*The recovery opens are the opens of the total coefficient graph, not extra conditions on the six coefficients of the arc. Thus the theorem permits nonzero  $g$  and  $r$  jets, arbitrary relative orders of vanishing, unequal residual directions, and arbitrary ramification. It includes arcs entirely contained in an incidence stratum.*

*Proof.* In the Hilbert–Burch coordinates of Lemma I.1, put  $z = x - c$  and  $H = R - \lambda G$ . The base map reads

$$f = z^2 + e^2 C, \quad g = eBz - e^2 A, \quad r = \lambda g - eAz - e^2 BC.$$

Equating its coefficients with those of the retained arc gives exactly the five equations in (J.4); the sixth coefficient eliminates  $c$ . In particular the constant equation  $\lambda v - e^2 BC = z_0$  must not be discarded for a general arc. No radical has been taken. The total chart is regular on its recovery open, so the same computation applies to the normalization of the total graph.

In the primitive chart write the recovered direction as  $\alpha + \beta z$ . Division by  $f = z^2 + d$  gives

$$(\alpha + \beta z)(uz + v) - \beta u(z^2 + d) = (\alpha u + \beta v)z + \alpha v - \beta ud.$$

The condition that this equal  $r$  proves (J.5). Interchanging  $g$  and  $r$  gives the other primitive chart. These are the degree-two division charts of Theorem H.1, after the invertible change between  $x$  and  $z$  coordinates. Their equations also retain the coefficient projection.

We specify the opens used here. On a conic chart the restriction maps in degrees  $(0, 2)$  and  $(1, 1)$  have ranks five and four. Restrict to the opens defined by these ranks, by the nonzero selected conic coefficient, and by the  $tG, tH$  pivot for the two bilinear equations. The normalized inverse is precisely the recovery map in Lemma I.1; the matrices in (J.3) transport these opens. They contain every central conic on which the corresponding evaluation functional is nonzero. On a division chart use the split-injection open in Lemma G.3. These opens contain every central primitive tail for which the selected homogeneous direction coordinate is a unit. Localizing the displayed rings to these opens is understood; no global affine-isomorphism claim outside the recovery domains is needed.

A nonzero conic through  $P$  cannot vanish on all five vectors in (J.2). Indeed their evaluations recover, in order, the coefficients of  $R^2, G^2, GR, FG$ , and  $FR$ . Thus the five conic coordinate systems cover the complete conic component. A primitive jet over  $\mathbb{C}[x]/(x^2)$  is represented by a pair with at least one unit, and therefore belongs to one of the two division coordinate systems. The complete-fibre theorem I.2 shows that these are all central points, including the doubled-line intersection. The same charts compute their scheme structures, not only their closed points.

Their union is an open neighbourhood of the entire special fibre in  $H_R$ . Its complement is closed in the proper  $R$ -scheme  $H_R$ . A nonempty closed subset of a proper scheme over a local discrete valuation ring has image meeting its closed point. Since the complement misses that fibre, it is empty. This proves exhaustion of  $H_R$ , including its generic fibre. Now apply Proposition E.3 on the seven coordinate domains. Its uniqueness assertion glues the horizontal quotients. Lemma E.2 proves compatibility on overlaps and with the stated flat changes of rings. The torsion formula follows from the same proposition.  $\square$

**Corollary J.2** (Universal interpretation of the seven presentations). *Choose the stratum containing the generic point of the arc in Theorem J.1. Its seven horizontal presentations are the pullback of the universal family on the corresponding modification in Theorem E.4. Hence specialization does not require an additional choice of a component, a normalization of the special fibre, or a stable-map replacement.*

*Proof.* Both constructions are flat embedded quotients with the original generic fibre. Uniqueness in Proposition E.3 identifies them on every coordinate domain and hence globally.  $\square$

The adjective miniversal here refers to the full six-dimensional coefficient base and its framed fixed-target comparison. It does not assert a finite list of valuation chambers, nor a classification of the full fibre for  $a \geq 3$ . The theorem gives the exact ideals and their saturation for every arc through the double contact. Its specialization to  $g = r = 0$  recovers the previously computed collision; it is not restricted to that subbase.

**J.2. Which conics are horizontal?** The vertical component in the symmetric collision has a concrete modular explanation: it records limits approached in coefficient directions other than the double-incidence subbase. Let

$$\mathcal{Q}_P = \mathbb{P}\langle FG, FR, G^2, GR, R^2 \rangle, \quad S = V(q_{FG}, q_{FR}) \subset \mathcal{Q}_P.$$

Thus  $S \simeq \mathbb{P}^2$  is the linear space of conics singular at  $P$ . Its ideal is generated by the two nontrivial first derivatives at  $P$ ; this is an intrinsic first-jet condition on the conic equation.

**Theorem J.3** (Arc realization and the meaning of vertical excess). *Every conic  $Q \in \mathcal{Q}_P$ , including a doubled line, occurs as the tail of the special Hilbert curve of an arc in  $B_2$  whose generic point is nonincident. Among arcs in the double-incidence subbase  $g = r = 0$  with  $f$  generically squarefree, the conic tails reached horizontally are exactly  $S$ .*

*Proof.* Choose one of the five coordinates so that  $Q$  is

$$(R - \lambda_0 G)^2 + B_0 F(R - \lambda_0 G) + A_0 FG + C_0 G^2 = 0.$$

Take  $e = \tau$ ,  $c = 0$ ,  $\lambda = \lambda_0$ , and choose polynomial series  $A(\tau), B(\tau), C(\tau)$  with prescribed constant terms  $A_0, B_0, C_0$  and  $A(\tau)^2 + C(\tau)B(\tau)^2 \neq 0$  in  $K$ . Such a choice always exists because this polynomial is not identically zero on the three-dimensional coefficient space; a generic line through the specified point is not contained in its zero locus. The Hilbert–Burch base map produces an arc centred at  $o$ . On its generic point  $e(A^2 + CB^2) \neq 0$ , so the three quadratics span the space of binary quadratics and define the nonincident graph. The flat determinantal family has special curve  $C_0 \cup Q$  by Lemma I.1. Undoing the constant coordinate change gives the required embedded arc in the original target.

For the second assertion use any conic coordinate system for a horizontal limit; the transformations in (J.3) preserve the condition  $g = r = 0$ . On the generic fibre the coefficient equations give  $eB = eA = 0$ . Also  $e \neq 0$  there, since  $e^2 C = d$  and the centred quadratic is generically squarefree, so  $d \neq 0$ . Thus  $A = B = 0$  on the horizontal closure. Its conic limit lies in  $S$ . Conversely, in a chart for any  $Q \in S$ , set  $A = B = 0$ ,  $e = \tau$ , and take  $C(\tau)$  with the specified constant term and nonzero generic value. Then  $f = x^2 + \tau^2 C(\tau)$  is generically squarefree and  $g = r = 0$ . The displayed flat Hilbert family gives a point of the horizontal closure: it agrees with that closure generically and factors through its saturated ideal over the torsion-free ring  $R$ . Its conic limit is  $Q$ . This includes doubled lines by taking  $C(0) = 0$  and  $C(\tau) \neq 0$ .  $\square$

The first assertion concerns closures of individual generic Hilbert points on generically nonincident arcs. The second concerns points of the horizontal family of complete incidence fibres. The two notions agree on their common incidence interpretation but should not be conflated. In particular the vertical  $\mathbb{P}^4$  is not a collection of spurious curves: its complement of  $S$  consists of valid limits that require different coefficient directions. No variation of a stability parameter is involved.

## APPENDIX K. COLLISION OF TWO INCIDENCE POINTS

The complete fibre of Theorem I.2 does not by itself describe which of its points arise by colliding two reduced incidences. We determine this specialization as a family, including its vertical excess and its scheme multiplicities. All spaces in this section are parameter spaces for embedded curves. A semistable model of such a parameter space is not a semistable model of every curve parametrized by it.

Retain the coefficient graph  $\tilde{\Gamma}_2 \rightarrow B_2$  and the point  $o$  of Section I. Set

$$T = \operatorname{Spec} \mathbb{C}[\delta], \quad \gamma : T \longrightarrow B_2, \quad \delta \longmapsto (x^2 - \delta, 0, 0), \quad \mathcal{H} = \tilde{\Gamma}_2 \times_{B_2} T. \quad (\text{K.1})$$

Let  $\mathcal{Y}$  be the schematic closure of  $\mathcal{H}|_{\delta \neq 0}$  in  $\mathcal{H}$ . Thus the closure is formed *after* this base change. Write  $\Lambda \simeq \mathbb{P}^1$  for the space of lines through  $P = [1 : 0 : 0]$  in the residual plane, and  $\mathcal{Q}_P \simeq \mathbb{P}^4$  for its conics through  $P$ . Its linear subspace

$$S = \{Q \in \mathcal{Q}_P : Q \text{ is singular at } P\} \simeq \mathbb{P}^2 = \text{Sym}^2 \Lambda$$

parametrizes unordered pairs of lines through  $P$ , with repetitions. We distinguish this plane from the entire conic component.

**Theorem K.1** (The horizontal collision space and its excess). *The scheme  $\mathcal{Y}$  is a smooth integral threefold, projective and flat over  $T$ . Define*

$$Z = \text{Bl}_{\Delta_\Lambda \times \{0\}}(\Lambda \times \Lambda \times \mathbb{A}_t^1), \quad \iota(\ell_+, \ell_-, t) = (\ell_-, \ell_+, -t). \quad (\text{K.2})$$

*The involution lifts to  $Z$ , and there is an isomorphism over  $T$*

$$\mathcal{Y} \simeq Z / \langle \iota \rangle, \quad \delta = t^2. \quad (\text{K.3})$$

*For  $\delta \neq 0$  the geometric fibre is  $\Lambda \times \Lambda$ , with the two factors attached at  $x = \sqrt{\delta}$  and  $x = -\sqrt{\delta}$ . The identification with an ordered product requires the displayed root cover; it is not a global ordering over  $T \setminus \{0\}$ .*

*There are smooth divisors  $E \simeq \mathbb{F}_2$  and  $S \simeq \mathbb{P}^2$  in  $\mathcal{Y}$  such that*

$$\mathcal{Y}_0 = E + 2S, \quad D = E \cap S \simeq \mathbb{P}^1, \quad \mathcal{O}_S(S) = \mathcal{O}_{\mathbb{P}^2}(-1). \quad (\text{K.4})$$

*The equality is an equality of effective Cartier divisors in the smooth threefold. The reduced support has normal crossings. The curve  $D$  is the negative section in  $E$  and the doubled-line conic in  $S = \text{Sym}^2 \Lambda$ .*

*The full pullback  $\mathcal{H}$  is reduced but is not flat over  $T$ . It has precisely the horizontal component  $\mathcal{Y}$  and the vertical component  $\mathcal{Q}_P$  over 0. Their scheme intersection is the reduced plane  $S$ , and their union is scheme-theoretic:*

$$0 \longrightarrow \mathcal{O}_{\mathcal{H}} \longrightarrow \mathcal{O}_{\mathcal{Y}} \oplus \mathcal{O}_{\mathcal{Q}_P} \longrightarrow \mathcal{O}_S \longrightarrow 0. \quad (\text{K.5})$$

*Closed-immersion direct images are implicit in this formula. The entire  $\mathbb{C}[\delta]$ -torsion is killed by  $\delta$  and is*

$$\ker(\mathcal{O}_{\mathcal{H}} \longrightarrow \mathcal{O}_{\mathcal{Y}}) = i_* \mathcal{I}_{S/\mathcal{Q}_P} = \text{Tor}_1^{\mathbb{C}[\delta]}(\mathcal{O}_{\mathcal{H}}, \mathbb{C}), \quad (\text{K.6})$$

*where  $i : \mathcal{Q}_P \hookrightarrow \mathcal{H}$ . Consequently colliding two reduced incidences reaches exactly  $E \cup S$  in the support of  $X_2 = \mathcal{H}_0$ , not every conic in  $\mathcal{Q}_P$ .*

*Proof.* The target and the coefficients are retained throughout. In a conic chart of Lemma I.1, taking the base change (K.1) gives exactly

$$\mathcal{H} : \quad \mathbb{C}[\lambda, e, A, B, C, \delta] / (eA, eB, e^2C + \delta). \quad (\text{K.7})$$

Indeed the linear coefficient of  $f$  gives  $c = 0$ , its constant coefficient gives  $e^2C = -\delta$ , the linear coefficient of  $g$  gives  $eB = 0$ , and that of  $r$  gives  $eA = 0$ . The two remaining constant equations are then redundant. These are equations in a regular chart of the normalized total graph; no fibre normalization or fibrewise radical is being taken.

After eliminating  $\delta$ , the ideal is  $(eA, eB)$ . Saturation by  $\delta = -e^2C$  gives  $(A, B)$ . The horizontal chart is therefore

$$\mathbb{C}[\lambda, e, C], \quad \delta = -e^2C. \quad (\text{K.8})$$

The other reduced component is  $e = 0$ , and their intersection is  $e = A = B = 0$ . On the conic component this is exactly the singular-at- $P$  plane. The identity

$$(eA, eB) = (e) \cap (A, B)$$

proves reducedness of the pullback on this chart. It also gives (K.5) locally. Its annihilator calculation is

$$((eA, eB) : e^2C) = (A, B),$$

so every base-torsion section is already killed by  $\delta$ .

For completeness we construct the projective horizontal model, rather than identify it from its closed points. Choose an affine coordinate  $w$  on  $\Lambda$  containing both directions and put

$$w_+ = \lambda + d, \quad w_- = \lambda - d.$$

Near the diagonal the ordered centre has ideal  $(d, t)$ . The two blow-up charts and their involutions are

$$\begin{aligned} t = ed : & & (\lambda, d, e) &\longmapsto (\lambda, -d, e), \\ d = ut : & & (\lambda, t, u) &\longmapsto (\lambda, -t, u). \end{aligned}$$

Their invariant rings, respectively, are

$$\mathbb{C}[\lambda, e, C], \quad C = -d^2, \quad \delta = -e^2C; \quad \mathbb{C}[\lambda, u, \delta], \quad \delta = t^2. \quad (\text{K.9})$$

On the overlap  $e = 1/u$  and  $C = -\delta u^2$ . These are regular rings. Common affine coordinates cover all pairs in  $\Lambda \times \Lambda$ : a point of  $\Lambda$  outside either of two specified directions may be chosen as infinity. The invariant charts and their target-coordinate translates glue, giving a scheme quotient. This also verifies smoothness at the fixed locus; no theorem about quotients by a *free* action is applied there. Projectivity follows, for example, by tensoring the translates of a relatively ample line bundle on the blow-up, taking an even power to trivialize stabilizer actions on fibres, and descending it to the finite quotient. Equivalently its invariant section algebra gives this projective quotient. The affine calculations exhibit the same quotient directly.

We next map it to the retained-coefficient Hilbert graph. On the first invariant chart use (I.2) with  $c = A = B = 0$ :

$$(H^2 + CG^2, xG - eH, xH + eCG), \quad H = R - \lambda G. \quad (\text{K.10})$$

The fixed coefficients are  $(x^2 - \delta, 0, 0)$  because  $\delta = -e^2C$ . On the second chart use the division family with  $f = x^2 - \delta$ ,  $g = r = 0$ , and  $w = \lambda + ux$ ; its equations include  $fG = 0$  and  $R = wG$ . The original flat-family results apply after these base changes. On  $u \neq 0$  these ideals agree with (K.10), since  $e = 1/u$ . Under changes of the direction coordinate they give the same embedded family: this holds on the dense locus  $t \neq 0$  and hence everywhere by separatedness of the Hilbert scheme on each integral overlap. Their maps to the coefficient graph therefore glue. They land in its regular division and conic charts, where total normalization is an isomorphism.

For  $t \neq 0$ , the first ideal places the direction  $w_+$  at  $x = t$  and  $w_-$  at  $x = -t$ . It records all choices of the two lines, including equality of their directions, which is covered by the second chart. The common affine-coordinate argument covers the whole generic fibre. The recovery maps in the conic and division charts identify the quotient with the horizontal charts of  $\mathcal{H}$ , not just with their supports. The two resulting morphisms are inverse on these charts, proving (K.3). This constructs the isomorphism algebraically; formal normal forms are not used to manufacture it.

The strict transform of  $t = 0$  in  $Z$  is  $\Lambda \times \Lambda$ . Its quotient is  $\text{Sym}^2 \Lambda = S$ . The exceptional divisor is

$$\mathbb{P}(N_{\Delta_\Lambda/(\Lambda \times \Lambda)} \oplus \mathcal{O}) = \mathbb{P}(\mathcal{O}(2) \oplus \mathcal{O}) = \mathbb{F}_2.$$

The involution acts trivially on this divisor, since it is  $-1$  on both normal directions before projectivization. Their common curve is the diagonal, whose image in  $S$  is the doubled-line conic. On the chart (K.8),  $S = (e = 0)$ ,  $E = (C = 0)$ , and  $\delta = -e^2C$ . This proves the divisor multiplicities and transverse reduced intersection. On  $E$  the coordinate  $u = 1/e$  is the affine tangent-jet coordinate, so  $e = 0$  is its negative section, as in (I.6). Since  $2S + E$  is principal and

$E|_S$  is the degree-two conic,  $2S|_S = -2H_S$ . The Picard group of  $S = \mathbb{P}^2$  has no torsion, giving  $\mathcal{O}_S(S) = \mathcal{O}_S(-1)$ .

The quotient is integral and hence torsion-free as a module over  $\mathbb{C}[\delta]$ . Localize this ring at each prime: its nonfield localizations are discrete valuation rings. The valuation-ring flatness criterion [Stacks Project, Tag 0539] proves flatness at every base point, and hence over  $T$ . The complete first-contact theorem covers every point of  $\mathcal{H}_0$  by its conic charts or primitive charts. On a primitive chart the pullback is  $\mathbb{C}[\lambda, u, \delta]$  and has no vertical part. Away from 0 all fibres are the reduced-incidence product. Thus the local descriptions just proved exhaust  $\mathcal{H}$ . They glue (K.5); no unexamined component can be supported over a different point of  $T$ . The kernel of the projection to  $\mathcal{Y}$  is, on the vertical component, exactly its ideal of  $S$ . Resolving  $\mathbb{C}$  over  $\mathbb{C}[\delta]$  by multiplication by  $\delta$  identifies this kernel with the stated Tor sheaf. It is nonzero, so  $\mathcal{H}$  is not flat. The last assertion follows from the proper flat horizontal model and (K.4).  $\square$

### K.1. The ramified model and monodromy.

**Theorem K.2** (Ordered resolution of the collision). *The normalization of  $\mathcal{Y} \times_{\mathbb{A}_\delta^1} \mathbb{A}_t^1$ , with  $\delta = t^2$ , is  $Z$  in (K.2). It is smooth and its fibre over  $t = 0$  is the reduced normal-crossing union of  $\Lambda \times \Lambda$  and  $\mathbb{F}_2$ , meeting along the diagonal section. In particular the multiplicity two in (K.4) is caused by forgetting the order of the colliding supports.*

*On the punctured complex base the monodromy on  $H^2(\mathcal{Y}_\delta, \mathbb{Q})$  interchanges the two ruling classes. Its invariant and anti-invariant spaces both have dimension one. These assertions concern the degeneration of the parameter space. They do not replace thick embedded Hilbert tails by reduced stable-map tails.*

*Proof.* The first invariant chart after base change is

$$\mathbb{C}[\lambda, e, C, t]/(t^2 + e^2C).$$

Its normalization is obtained by adjoining  $d = t/e$ , with  $d^2 = -C$ , and is  $\mathbb{C}[\lambda, e, d]$ ,  $t = ed$ . This extension is finite because  $d$  satisfies a monic equation, and it has the same fraction field; the polynomial ring is normal. The second chart becomes  $\mathbb{C}[\lambda, t, u]$  with  $d = tu$ . These finite normal charts glue by  $e = 1/u$ , giving precisely the ordered blow-up. They prove normalization of the base change itself, not of its special fibre. The central equations  $t = ed$  and  $t = 0$  in the two smooth charts give the reduced normal crossings and the stated components.

Over  $t \neq 0$  the blow-up is the constant product  $\Lambda \times \Lambda \times \mathbb{G}_m$ . Descent identifies  $(\ell_+, \ell_-, t)$  with  $(\ell_-, \ell_+, -t)$ . A loop around  $\delta = 0$  exchanges the roots and hence the ruling classes. Their sum and difference give the two claimed eigenspaces. This calculation uses the ordering cover, not a claim that the whole collision family is topologically locally trivial at 0.  $\square$

### K.2. The nilpotent sheaf of the full contact fibre.

**Theorem K.3** (Global nilradical and coherent cohomology). *Let  $j : S = \mathbb{P}^2 \hookrightarrow X_2$  be the singular-conic plane. The nilradical  $\mathcal{N}$  of the complete scheme fibre satisfies*

$$\mathcal{N} \simeq j_*\mathcal{O}_{\mathbb{P}^2}(-1), \quad \mathcal{N}^2 = 0. \quad (\text{K.11})$$

*The isomorphism specifies the module; it does not assert that the square-zero algebra extension is split. Moreover*

$$H^0(X_2, \mathcal{O}_{X_2}) = \mathbb{C}, \quad H^q(X_2, \mathcal{O}_{X_2}) = 0 \quad (q > 0).$$

*The same coherent-cohomology statement holds for every complete fibre  $X_2^r \times (\mathbb{P}^1)^s$  of Theorem M.1, although for  $r > 0$  that fibre is nonreduced and is not Cohen–Macaulay on its nilpotent support.*



*Proof.* Restrict the quotient  $\mathcal{O}_{X_2} \rightarrow \mathcal{O}_{Y_0}$  in the conic charts. Its source and target have presentations

$$\frac{\mathbb{C}[\lambda, e, A, B, C]}{(eA, eB, e^2C)} \longrightarrow \frac{\mathbb{C}[\lambda, e, C]}{(e^2C)}.$$

Both nilradicals are generated by  $eC$ , and the map identifies them as modules on  $S$ . Off  $S$  both vanish. This is consequently a global isomorphism of nilradical sheaves. In the smooth threefold  $\mathcal{Y}$ , the nilradical of  $E + 2S$  is

$$\mathcal{O}_{\mathcal{Y}}(-E - S)/\mathcal{O}_{\mathcal{Y}}(-E - 2S) \simeq \mathcal{O}_S(-E - S) \simeq \mathcal{O}_S(S).$$

The last isomorphism uses the principal divisor  $E + 2S$ ; (K.4) now gives (K.11). Thus the local nilpotent generator has a nontrivial global twist; its support alone would not determine it.

The reduced union has the exact sequence

$$0 \longrightarrow \mathcal{O}_{(X_2)_{\text{red}}} \longrightarrow \mathcal{O}_{\mathbb{P}^4} \oplus \mathcal{O}_{\mathbb{F}_2} \longrightarrow \mathcal{O}_{\mathbb{P}^1} \longrightarrow 0.$$

Each of the three smooth terms has constants as its only coherent cohomology: for  $\mathbb{F}_2$  this follows from its projective-line bundle over  $\mathbb{P}^1$ . The map on constants is the difference and is surjective. Also  $H^q(\mathbb{P}^2, \mathcal{O}(-1)) = 0$  for every  $q$ . Apply the cohomology sequences first to this union and then to  $0 \rightarrow \mathcal{N} \rightarrow \mathcal{O}_{X_2} \rightarrow \mathcal{O}_{(X_2)_{\text{red}}} \rightarrow 0$ . They give the assertions. The product statement follows from Kunneth over  $\mathbb{C}$  and the already proved scheme-product theorem. It is not an inference of reducedness or flatness from constant coherent cohomology.  $\square$

### K.3. Several simultaneous collisions.

**Corollary K.4** (Independent collision multiplicities). *For  $r$  independent copies of (K.1) and  $s$  additional simple contacts, the horizontal collision space is  $\mathcal{Y}^r \times (\mathbb{P}^1)^s$  over  $\mathbb{A}_{\delta}^r$ . Its central fibre has  $2^r$  reduced components*

$$S^{|J|} \times E^{r-|J|} \times (\mathbb{P}^1)^s, \quad J \subseteq \{1, \dots, r\},$$

*each of dimension  $2r + s$  and of cycle multiplicity  $2^{|J|}$ . Intersections replace differing factors by  $D$ . After  $\delta_i = t_i^2$  and normalization the ordered model is  $Z^r \times (\mathbb{P}^1)^s$ . Its total space is smooth, and the inverse image of the union of the base coordinate hyperplanes is a reduced simple-normal-crossing divisor. The fibre over the origin instead has the product local equations  $u_i v_i = 0$  in disjoint coordinate pairs, omitting one factor at a smooth point of a component. The full central pullback is  $X_2^r \times (\mathbb{P}^1)^s$ , whose conic factors are  $\mathbb{P}^4$ , not  $S$ . Thus the horizontal and full fibres have different component dimensions even though both keep all choices at separated supports.*

*Proof.* Form the product of the explicit flat horizontal models. The open where all  $\delta_i$  are nonzero is dense in this integral product. On each chart its closure is obtained by saturating with  $\prod_i \delta_i$ , and removes precisely the vertical ideals in the individual factors. Hence it is the horizontal closure of the full product pullback. At the generic point of a component indexed by  $J$ , the central equations have the forms  $e_i^2$  for  $i \in J$  and  $C_i$  for the other indices, up to units. Their independent local lengths multiply to  $2^{|J|}$ . The intersection descriptions follow from the two smooth divisors in each factor. The finite normalization charts used in Theorem K.2 tensor over the independent complex variables, giving the stated smooth ordered model. Locally  $t_i = u_i v_i$  at the two-component locus and  $t_i = u_i$  away from it. The equation of the full inverse image of the coordinate boundary is therefore the product of the appearing distinct coordinates, whereas the central fibre is cut out by the individual  $t_i$ . The last assertion is the complete-fibre product theorem, not a normalization-of-fibre claim.  $\square$

## APPENDIX L. THE EXTENSION CLASS AND RAMIFIED COLLISION MODELS

The nilradical as a module does not determine a nonreduced scheme. We compute the missing algebra extension, and then determine how arbitrary ramification changes the horizontal normalization and the vertical torsion of the collision family. The two calculations separate intrinsic nonreducedness of a fibre from torsion introduced by a nonflat pullback.

**L.1. The square-zero algebra is not locally split along the doubled lines.** Write  $X = X_2$ ,  $X_{\text{red}} = \mathcal{Q}_P \cup \mathbb{F}_2$ ,  $S \simeq \mathbb{P}^2 \subset \mathcal{Q}_P$  for the singular-conic plane, and  $D \simeq \mathbb{P}^1$  for the doubled-line locus. The closed immersions are denoted  $j : S \hookrightarrow X_{\text{red}}$  and  $i : D \hookrightarrow X_{\text{red}}$ . By Theorem K.3, the nilradical is  $N = j_* \mathcal{O}_S(-1)$ . The square-zero extension of sheaves of algebras

$$0 \longrightarrow N \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{X_{\text{red}}} \longrightarrow 0 \quad (\text{L.1})$$

defines a class  $\xi \in \text{Ext}_{X_{\text{red}}}^1(L_{X_{\text{red}}/\mathbb{C}}, N)$ . Here  $L$  is the cotangent complex; the degree-one group can equally be computed using the naive cotangent complex. It is not the ordinary extension group of the sheaf of differentials.

**Theorem L.1** (The nonsplitting locus and its obstruction sheaf). *The algebra extension (L.1) splits locally on  $X_{\text{red}} \setminus D$  and does not split locally at any point of  $D$ . Moreover*

$$\mathcal{E}xt_{X_{\text{red}}}^1(L_{X_{\text{red}}/\mathbb{C}}, N) \simeq i_* \mathcal{O}_D, \quad (\text{L.2})$$

where the image of  $\xi$  is the section 1. In a conic chart the class is represented by the conormal homomorphism

$$eA \longmapsto 0, \quad eB \longmapsto 0, \quad eC \longmapsto eC. \quad (\text{L.3})$$

In particular the nilpotent algebra is not the trivial square-zero extension of its reduced support, even Zariski locally along  $D$ .

*Proof.* Work first in the polynomial chart

$$P_0 = \mathbb{C}[\lambda, e, A, B, C], \quad I = (eA, eB, e^2C), \quad J = (eA, eB, eC), \quad R_0 = P_0/J.$$

Then  $\mathcal{O}_X = P_0/I$  and  $N = J/I$  is generated by  $n = eC$  with annihilator  $(e, A, B)$ . Hence, as a module,  $N = \mathbb{C}[\lambda, C]n$ . Its extension class is the homomorphism  $J/J^2 \rightarrow N$  induced by lifting the polynomial generators to  $P_0/I$ , namely (L.3). This is the presentation description of square-zero extensions [Stacks Project, Section 91.2, especially Lemma 91.2.3 and Remark 91.2.4].

We compute the whole local degree-one group. The module of relations on  $eA, eB, eC$  over  $P_0$  is generated by the three Koszul relations on  $A, B, C$ , since multiplication by  $e$  is injective in  $P_0$ . For a homomorphism to  $N$ , the relations  $C(eA) - A(eC) = 0$  and  $C(eB) - B(eC) = 0$  force the images of  $eA$  and  $eB$  to vanish:  $A$  and  $B$  annihilate  $N$ , whereas multiplication by  $C$  on  $N$  is injective. The image of  $eC$  is unrestricted. Consequently  $\text{Hom}_{R_0}(J/J^2, N) = N$ .

The image of a derivation from the polynomial presentation is  $\delta(eC) = C\delta(e) + e\delta(C) = C\delta(e)$  in  $N$ ; its values on  $eA, eB$  are zero. All multiples of  $C$  occur. Therefore

$$\text{Ext}_{R_0}^1(L_{R_0/\mathbb{C}}, N) = \frac{\text{Hom}_{R_0}(J/J^2, N)}{\text{im } \text{Hom}_{R_0}(\Omega_{P_0/\mathbb{C}} \otimes R_0, N)} = N/CN \simeq \mathbb{C}[\lambda]. \quad (\text{L.4})$$

The class of the given algebra maps to  $n \bmod CN$ , hence to 1. Only the truncation in degrees  $-1, 0$  is used in this computation; we have not asserted that the full cotangent complex has length two, or that  $J$  is a complete intersection.

The annihilator of the last module is the ideal of  $D = V(e, A, B, C)$  in this chart. At every point of  $D$  the class is a nonzero generator, so an algebra section is impossible. Where  $C$  is invertible,  $R_0$  is the polynomial algebra  $\mathbb{C}[\lambda, A, B, C, C^{-1}]$  and its variables give an explicit section into  $(P_0/I)[C^{-1}]$ . Outside  $S$ , the module  $N$  is zero. These opens, together with the primitive charts away from  $D$ , prove the asserted local splitting elsewhere.

The complete-fibre charts cover  $X$ . Their extension classes are restrictions of the same global algebra extension. Thus their local generators glue to a nowhere vanishing section of the obstruction sheaf supported on  $D$ . The map  $\mathcal{O}_D$  sending 1 to this section is an isomorphism on every displayed chart, and is zero off  $D$  on both sides. This proves (L.2). We identify the image of the global class in the local obstruction sheaf; no assertion that the entire global hyperextension group is one-dimensional is required.  $\square$

**Corollary L.2** (Associated subvarieties of the contact fibre). *The associated subvarieties of  $X_2$  are its two reduced components  $\mathcal{Q}_P$  and  $\mathbb{F}_2$ , and the embedded subvariety  $S$ . The embedded associated prime disappears precisely off  $S$ . The support of the nilradical is  $S$ , whereas the locus where its algebra extension fails to split locally is the strictly smaller curve  $D$ .*

*Proof.* The primary decomposition

$$(eA, eB, e^2C) = (e) \cap (A, B, C) \cap (A, B, e^2)$$

gives the associated primes  $(e)$ ,  $(A, B, C)$ , and  $(e, A, B)$  on conic charts. The last is the generic point of  $S$ ; the first two are those of the two components. The primitive charts away from their boundary are reduced and smooth. Since these charts cover all of  $X_2$ , there are no additional associated subvarieties. The final distinction follows from Theorem L.1 and  $N = j_*\mathcal{O}_S(-1)$ .  $\square$

**L.2. Arbitrary ramification of the ordered collision.** For  $m \geq 1$  replace the coefficient parameter in Theorem K.1 by  $\delta = \tau^m$ . Let  $H_m$  be the full pullback and  $Y_m$  its horizontal closure. The following statements are over  $\mathbb{C}[\tau]$ ; they also hold after localization or completion at  $\tau = 0$ .

**Theorem L.3** (Ramified torsion and toric normalization). *The horizontal space is the flat base change of the previous horizontal model. Its conic coordinate ring is*

$$\mathbb{C}[\lambda, e, C, \tau]/(e^2C + \tau^m). \quad (\text{L.5})$$

*The normalization of this chart is*

$$\mathbb{C}[\lambda, s, r]^{\mu_m}, \quad \zeta(s, r) = (\zeta s, \zeta^{-2}r), \quad e = s^m, \quad C = -r^m, \quad \tau = s^2r. \quad (\text{L.6})$$

*The two central divisors in this normalization have multiplicities  $2/\gcd(m, 2)$  and 1, respectively. If  $m = 2q$ , the normalized chart can equivalently be written*

$$\mathbb{C}[\lambda, e, h, \tau]/(eh - \tau^q), \quad C = -h^2. \quad (\text{L.7})$$

*It is smooth for  $q = 1$ ; for  $q > 1$  its transverse surface singularity has equation  $eh = \tau^q$ .*

*The full pullback has the scheme-theoretic union sequence*

$$0 \rightarrow \mathcal{O}_{H_m} \rightarrow \mathcal{O}_{Y_m} \oplus \mathcal{O}_{\mathcal{Q}_P} \otimes_{\mathbb{C}} \mathbb{C}[\tau]/(\tau^m) \rightarrow \mathcal{O}_S \otimes_{\mathbb{C}} \mathbb{C}[\tau]/(\tau^m) \rightarrow 0. \quad (\text{L.8})$$

*Its base torsion, with closed-immersion direct images understood, is*

$$T_m = \mathcal{I}_{S/\mathcal{Q}_P} \otimes_{\mathbb{C}} \mathbb{C}[\tau]/(\tau^m). \quad (\text{L.9})$$

*It has exact torsion order  $m$ . In particular  $\text{Tor}_1^{\mathbb{C}[\tau]}(\mathcal{O}_{H_m}, \mathbb{C}) = \tau^{m-1}T_m$ , and all higher Tor sheaves vanish. For  $m > 1$  the full pullback is not reduced along the generic point of its vertical component.*

*Proof.* The map  $\mathbb{C}[\delta] \rightarrow \mathbb{C}[\tau]$ ,  $\delta \mapsto \tau^m$ , is finite free. The union sequence and flat horizontal quotient in Theorem K.1 therefore survive this base change. Its conic chart is

$$\mathbb{C}[\lambda, e, A, B, C, \tau]/(eA, eB, e^2C + \tau^m).$$

The horizontal ring is (L.5), either by flat saturation or by uniqueness of the flat quotient with the same generic fibre. Pulling back the original torsion ideal proves (L.9). Multiplication by  $\tau^{m-1}$  is nonzero on this sheaf and multiplication by  $\tau^m$  is zero. Its annihilator by  $\tau$  is its last

nonzero layer. The two-term resolution of the residue field proves the Tor statements. At the generic vertical point  $e = 0$  and one of  $A, B$  is a unit, leaving the nonreduced equation  $\tau^m = 0$  when  $m > 1$ .

For normalization, omit the independent coordinate  $\lambda$ . The ring in (L.5) is a domain: its relation is linear in  $C$ , with relatively prime coefficients  $e^2$  and  $\tau^m$ . Embed it in  $\mathbb{C}[s, r]$  by the displayed monomials. Its exponent lattice is generated by  $(m, 0), (0, m), (2, 1)$ . This lattice has index  $m$  in  $\mathbb{Z}^2$  and is exactly

$$\{(a, b) \in \mathbb{Z}^2 : a - 2b \equiv 0 \pmod{m}\},$$

the character lattice of the invariant field in (L.6). Thus the invariant ring and the original ring have the same fraction field. The invariant ring is normal: an element of its fraction field integral over it is integral over the normal polynomial ring, belongs to that ring, and is fixed by the group. It is finite over the original ring, because  $s$  and  $r$  satisfy the monic equations  $s^m = e$  and  $r^m = -C$ , and the invariant submodule of this finite module is finite. These facts prove that it is exactly the normalization, not a cover with a larger fraction field.

The inverse image of  $\tau = 0$  is supported on  $s = 0$  and  $r = 0$ . At the generic point of  $s = 0$  the stabilizer has order  $\gcd(m, 2)$ , so the quotient has this ramification index there. Upstairs  $\tau = s^2 r$  vanishes to order two. Downstairs its order is  $2/\gcd(m, 2)$ . At the generic point of  $r = 0$  the stabilizer is trivial and the order is one. There are no other central prime divisors.

If  $m = 2q$ , first take invariants by the order-two subgroup, giving  $\mathbb{C}[s^2, r]$ . The remaining action has order  $q$  and opposite weights on  $s^2, r$ . Its invariant ring is generated by  $e = s^{2q}$ ,  $h = r^q$ , and  $\tau = s^2 r$ , with the single relation  $eh = \tau^q$ . Every invariant monomial is a product of these three, as is seen by subtracting the smaller of its two exponents. This proves (L.7). The derivative criterion gives smoothness at  $q = 1$  and the stated surface singularity for  $q > 1$ .  $\square$

These local normalizations glue uniquely because they are integral closures in the common function field of  $Y_m$ . On the primitive chart the pullback is already  $\mathbb{C}[\lambda, u, \tau]$ , so no additional normalization is needed. For  $m = 2$  the result recovers the ordered normal-crossing parameter space. For larger even  $m$  reducedness of the normalized central divisor does not imply smoothness of the total space. For odd  $m$  the doubled divisor remains. None of these operations normalizes the individual embedded curve or asserts its stable reduction as a map.

#### APPENDIX M. PRODUCTS AND ADJACENCY FOR THE FIRST NONREDUCED BOUNDARY

The local fibre calculation applies to arbitrary larger Smith exponents and to any number of distinct contacts. This gives a complete boundary class, rather than a primitive open in a class whose complement has not been identified.

Let  $R_0$  be a regular symmetric pencil whose line avoids corank at least three. At its distinct corank-two points write the positive Smith exponents as  $(a_i, b_i)$ ,  $a_i \leq b_i$ . Assume

$$a_i \in \{1, 2\} \quad \text{for all } i. \tag{M.1}$$

There is no bound on  $b_i$  except the dimension constraint on the pencil. Let  $r$  count the points with  $a_i = 2$  and  $s$  those with  $a_i = 1$ . Denote by  $\widehat{G}$  the normalization of the total reduced Hilbert graph of lifted pencil lines and by  $\mathfrak{X}_{R_0}$  its scheme fibre. The polarization on complete quadrics is  $\bigotimes_{q=1}^{n-1} H_q$ , with Hilbert polynomial  $\binom{n}{2}l + 1$ .

**Theorem M.1** (Complete fibres and their component intersections). *After choosing parameters and residual frames at the distinct contact points, there is an isomorphism of schemes*

$$\mathfrak{X}_{R_0} \simeq X_2^r \times (\mathbb{P}^1)^s. \tag{M.2}$$

Frame changes act by the corresponding transition isomorphisms; the construction is intrinsic without an ordering of the supports. In particular the reduced fibre has exactly  $2^r$  irreducible components. For a subset  $S$  of the  $r$  length-two contacts, the component is

$$(\mathbb{P}^4)^{|S|} \times (\mathbb{F}_2)^{r-|S|} \times (\mathbb{P}^1)^s, \quad \dim = s + 2r + 2|S|. \quad (\text{M.3})$$

The intersection of two reduced components is obtained by replacing each factor where their choices differ by the doubled-line  $\mathbb{P}^1$ , and leaving the common factors unchanged. These intersections are reduced. This describes all multiple intersections as well.

The fibre is connected and is reduced exactly when  $r = 0$ . For  $r > 0$  its nilradical has nilpotence order exactly  $r + 1$ : its  $(r + 1)$ st power is zero, and its  $r$ th power is nonzero. Locally its equations are tensor products of (I.5) and smooth jet charts. In particular the reduced support is not a replacement for its Hilbert fibre scheme. The Euler characteristic of the underlying complex space is  $7^r 2^s$ .

*Proof.* We construct the morphism before using completed rings. At a length-two support, the universal curve over  $X_2$  contains its fixed main curve. In the charts (I.2), the fibre relations  $eA = eB = e^2C = 0$  give

$$xG = eH, \quad xH = -eCG, \quad x^2G = x^2H = 0.$$

Consequently the quotient of the fixed main ideal by the curve ideal is annihilated by  $x^2$ . Thus the entire modification, including the nilpotent directions in its parameter scheme, is specified by a coherent ideal quotient on the first infinitesimal neighbourhood of the support. At a length-one contact the corresponding quotient is annihilated by  $x$ . These are finite algebraic neighbourhoods, not unbounded formal data.

Use the central Schur splitting and undo its congruence on these finite quotients. Only finitely many coefficients of the central gauge are involved. Include the fixed main lift, and extend by its forced ideal off the supports. This gives algebraic closed ideal families in the fixed complete-quadric target. Multiplicative stability of the ideals is preserved by the target coordinate isomorphisms. Distinct supports are disjoint, so the constructions combine independently. Flatness follows from the local flat families and the patching lemma; their Hilbert polynomial is the one stated above. Indeed each contact of length  $a_i$  removes  $a_i$  from the last exterior degree of the main curve and puts that degree on the tail. The attachment lengths give Euler characteristic one for the total curve. The smaller exterior systems have no base point on this rank open.

This constructs a finite-presentation morphism from the right side of (M.2) to the fibre of the reduced Hilbert graph. Each point is a specialization of nonincident graph points: for primitive tails this is the division family, and for conic tails it is the dense open  $e(A^2 + CB^2) \neq 0$  in (I.3). A polynomial line through a specified point with generic direction meets that open; normalization of its closure gives an algebraic one-parameter specialization. The nilpotent part of the constructed map also factors through the reduced total graph, by the completed comparison in Proposition G.2, not merely because its closed points do. That proposition identifies its completed total neighbourhoods with the regular polynomial graph charts and the smooth contact factors. Hence the total graph is regular there, the morphism lifts uniquely to  $\hat{G}$ , and the completed local maps on the fibre are isomorphisms.

Recovery of the embedded ideal on each disjoint finite neighbourhood makes this morphism a monomorphism. The formal etaleness criterion therefore makes it an open immersion. Its source is proper, so it is also a closed immersion. The fibre of  $\hat{G} \rightarrow G$  is connected: this is a proper birational morphism from an integral normal variety to the smooth Grassmannian, and Stein factorization applies. The nonempty open-and-closed image is the entire fibre. This proves the scheme isomorphism, not just an equality of reduced supports. It also explains why the arbitrary values of  $b_i$  make no difference:  $k_i$  is removed by a generating-row transformation which is undone

in the fixed incidence target, while the contact algebra and the resulting finite quotient have length  $a_i$ .

The reduced components and intersections follow from Theorem I.2 by products. Tensor products over  $\mathbb{C}$  preserve the displayed reduced smooth components and their reduced intersections. The nilradical is the sum of the pullbacks  $N_i$  of the square-zero nilradicals of the  $r$  factors. The quotient by this sum is reduced, so there is no additional nilradical. Every monomial of degree  $r + 1$  repeats a factor and vanishes. The tensor product of nonzero sections of all  $N_i$  is nonzero on a product of their nonzero affine charts, so the  $r$ th power is nonzero. The Euler characteristic follows by multiplicativity and Corollary I.3.

Finally the construction uses the actual support schemes and the fixed embedded ideals. Permuting the support labels permutes the factors; changing local source frames applies their functorial transition maps. These satisfy the cocycle identity because the coordinate changes are undone in the fixed target. This gives the asserted unlabelled interpretation, rather than a preferred global ordering of contacts.  $\square$

**M.1. Resultant strata on the higher-contact open.** For larger  $a_i$  the division chart remains useful even though the whole proper fibre is not classified here. The following records exactly its boundary stratification. It is also a check that repeated roots are not being mistaken for multiple reduced components.

**Proposition M.2** (Gcd partitions, codimensions and multiplicity). *In  $W_a = \{(f, g, w)\}$ , with  $f$  monic of degree  $a$  and  $\deg g, \deg w < a$ , fix a partition  $\mu = (\mu_1, \dots, \mu_k)$  of  $d \leq a$ . The locally closed locus where the monic gcd of  $f, g$  is  $\prod_{j=1}^k (x - \xi_j)^{\mu_j}$  with distinct  $\xi_j$  has dimension*

$$3a - 2d + k$$

*and codimension  $2d - k$ . Its finite attaching scheme is the disjoint union of the primary schemes  $\text{Spec } \mathbb{C}[x]/((x - \xi_j)^{\mu_j})$ , and its tails are projective lines over those schemes. A multiple root is one primary thickening, not several components. At a point with squarefree  $f$  and precisely  $d$  common roots, the resultant divisor is locally a product of  $d$  independent parameters and has multiplicity  $d$ .*

*Proof.* Parametrize the gcd by its distinct roots, giving  $k$  parameters, and write  $f = hf_1$ ,  $g = hg_1$ . The monic polynomial  $f_1$  has  $a - d$  coefficients and  $g_1$  has  $a - d$  coefficients; impose the open coprimality condition on this residual pair and the conditions specifying the exact gcd partition. Recovering the monic gcd is regular on the constant-gcd-degree stratum by the fixed-rank Euclidean linear equations. Up to the finite permutations of equal parts, this gives dimension  $k + 2(a - d)$ ;  $w$  contributes  $a$ . The statement includes  $d = a$ , where  $g = 0$  and  $f = h$ . The primary decomposition is the Chinese remainder theorem for distinct roots and the gcd fibre equation of Theorem H.1. At a squarefree  $f$ , its roots are etale local coordinates and  $\text{Res}(f, g) = \prod_{f(\xi)=0} g(\xi)$ . The evaluations of a polynomial of degree  $< a$  at these roots are independent by the Vandermonde matrix. The nonzero factors are units, proving the final assertion. No normal-crossing statement is made at a multiple root of  $f$ .  $\square$

**M.2. A testable extension beyond length two.** The following conjecture concerns the reduced components of a local fibre, not the miniversal congruence quotient. It distinguishes two questions: component geometry and a possible nilpotent structure on their union.

*Remark M.3* (Conjecture: rational components of the corank-two contact fibre). For every  $a \geq 1$ , let  $X_a$  be the fibre over  $(x^a, 0, 0)$  of the normalization of the total reduced graph  $\Gamma_a \rightarrow B_a$  defined in Theorem H.1. Every irreducible component of  $(X_a)_{\text{red}}$  has rational normalization. The closure of the primitive jet open is an irreducible component of dimension  $a$ , and its intersections with the other components contain a locus of curves whose tail cycle has a repeated line through the attaching point. For  $a \geq 2$  the scheme  $X_a$  is not reduced.

For  $a = 1$  the fibre is  $\mathbb{P}^1$ . For  $a = 2$  the conjecture follows from the complete description above, including the doubled-line intersection and the square-zero ideal. For  $a \geq 3$  it is a conjecture, not a component classification. The statement is falsifiable separately by the birational type of a component, its adjacency to the primitive component, or reducedness of the fibre. An eventual classification should refine it by the degrees and primary thicknesses of tail cycles of total degree  $a$ . The present product theorem makes no assertion about higher-corank contacts or singular pencils in the Hilbert problem; the independent inverse and singular-pencil invariant theorems continue to apply on their original domains.

#### APPENDIX N. EXTENSION CLASSES AND NORMALIZED RAMIFICATION CHARTS

We record two local consequences that separate properties of the embedded curve from properties of its parameter space.

**N.1. The presentation complex and explicit splitting.** For a polynomial presentation  $R = P/J$  and an  $R$ -module  $N$ , square-zero  $\mathbb{C}$ -algebra extensions with specified kernel  $N$  and specified quotient  $R$  are classified by  $\text{Ext}_R^1(L_{R/\mathbb{C}}, N)$ . We use cohomological degrees:  $J/J^2$  is in degree  $-1$  and  $\Omega_{P/\mathbb{C}} \otimes_P R$  in degree  $0$ . The group is

$$\text{coker}(\text{Hom}_R(\Omega_{P/\mathbb{C}} \otimes_P R, N) \longrightarrow \text{Hom}_R(J/J^2, N)).$$

Indeed a lift of polynomial generators gives a conormal map; changing lifts changes it by a derivation. The part of the cotangent complex in degrees at most  $-2$  contributes only in  $\text{Ext}$  degrees at least two when the target is in degree zero. This justifies truncation for this particular group, not a claim that the whole cotangent complex is two-term.

In the chart of Theorem L.1, invert  $C$ . Then the extension algebra is explicitly

$$\mathbb{C}[\lambda, A, B, C, C^{-1}, n]/(n^2, An, Bn), \quad e = n/C.$$

Writing its elements as  $r + an$  gives multiplication  $(r + an)(s + bn) = rs + (rb + sa)n$ , where the coefficients of  $n$  are restricted modulo  $(A, B)$ . This is the actual split algebra on this chart. On an overlap of two obstruction charts, let  $\xi_i$  be the conormal representative obtained from the same algebra extension. A change of polynomial lifts changes  $\xi_i$  by a derivation. Therefore  $[\xi_i] = [\xi_j]$  under the induced overlap map on the  $\text{Ext}$  sheaves. Taking the trivialization that sends  $1$  to  $[\xi_i]$  makes the overlap function exactly  $1$ , rather than an unspecified unit. This proves the asserted trivialization of the local obstruction sheaf. It does not trivialize the nilradical line bundle on the singular-conic plane, and it does not identify the entire global hyper- $\text{Ext}$  group with its sheaf of local degree-one groups. The singular-conic plane is the locus where the linear part of the conic equation at the attaching point vanishes; the doubled-line curve is its discriminant-zero locus. Thus the support of the embedded associated subvariety and the obstruction to an algebra retraction are different loci.

**Proposition N.1** (Singularities and divisor classes after ramification). *Let  $m \geq 1$  and put  $g = \gcd(m, 2)$ . For the normalized transverse surface in Theorem L.3, the character lattice and its saturated semigroup are*

$$M_m = \{(a, b) \in \mathbb{Z}^2 : a - 2b \equiv 0 \pmod{m}\}, \quad M_m \cap \mathbb{R}_{\geq 0}^2.$$

*The primitive rays of the dual cone in  $N_m = \text{Hom}(M_m, \mathbb{Z})$  are  $(1/g, 0)$  and  $(0, 1)$ . Its divisor class group is  $\mathbb{Z}/(m/g)$ . For odd  $m$  it is the cyclic quotient of type  $\frac{1}{m}(1, -2)$ ; it is smooth for  $m = 1$  and has a unique singular point for every odd  $m > 1$ . For even  $m = 2q$  it is of type  $\frac{1}{q}(1, -1)$ , with equation  $eh = \tau^q$ ; it is smooth for  $q = 1$  and has a unique singular point otherwise. In the full chart the additional coordinate  $\lambda$  gives the product with an affine line. The singular locus is correspondingly empty or that line.*

*Proof.* The exponent lattice and its saturation were computed in Theorem L.3. Their dual lattice is  $N_m = \mathbb{Z}^2 + \mathbb{Z}(1/m, -2/m)$ . On the first coordinate axis its primitive nonzero vector is  $(1/g, 0)$ ; on the second it is  $(0, 1)$ . This follows by solving the congruence in the unused coordinate. The index of the lattice generated by these two rays in  $N_m$  is  $m/g$ . The torus-invariant divisor sequence is the cokernel of

$$M_m \longrightarrow \mathbb{Z}^2, \quad (a, b) \longmapsto (a/g, b).$$

For odd  $m$  its image is  $a - 2b \equiv 0 \pmod{m}$ . For even  $m = 2q$  write  $a = 2a'$ ; its image is  $a' - b \equiv 0 \pmod{q}$ . In both cases the cokernel is the stated cyclic group. This computes the class group of the normal affine toric surface, including the smooth cases.

For odd  $m$  neither weight has a nontrivial stabilizer on a coordinate axis away from the origin. The quotient is therefore smooth away from the origin. Its cone is unimodular exactly when  $m = 1$ , proving singularity at the origin otherwise. For even  $m$  taking invariants under the order-two pseudoreflection first gives coordinates  $s^2, r$  with residual opposite weights of order  $q$ . The displayed hypersurface and its derivatives prove the remaining assertions. Formation of the product with the independent coordinate does not change this transverse analysis.  $\square$

**Lemma N.2** (Gluing integral closures). *Let an integral Noetherian scheme be covered by affine charts  $\text{Spec } A_i$  in a fixed function field. Their integral closures in that field glue to its normalization. In particular the ramified invariant charts and the unchanged primitive charts in Theorem L.3 glue without a choice of a cyclic cover or an ordering of branches.*

*Proof.* Integral closure commutes with localization. One implication is immediate by localizing a monic equation. For the other, clear the finitely many denominators in a monic equation for an element integral over a localization: a suitable multiple of that element by an element of the multiplicative set is integral over the original ring. Thus both chart normalizations restrict to the same ring on every principal overlap. Affine refinements and the cocycle identity of localization give the gluing. All rings here are finite-type over  $\mathbb{C}$ , so these normalizations are finite.  $\square$

## APPENDIX O. POWER COEFFICIENTS AND THEIR TWISTS

For a vector space  $V$ , put  $M(V) = \text{Sym}^2 V$ , with congruence action  $A \mapsto gAg^\dagger$ , and define

$$B_h(V) = \text{Sym}(M(V)) / \langle (v^2)^{h+1} : v \in V \rangle. \quad (\text{O.1})$$

The degree of  $M(V)$  is one. The generating space of the ideal is the Cartan inclusion  $\text{Sym}^{2h+2} V \hookrightarrow \text{Sym}^{h+1}(\text{Sym}^2 V)$ , normalized by  $v^{2h+2} \mapsto (v^2)^{h+1}$ . This definition requires neither a volume form nor a complementary space. It agrees with  $B_{n,h}$  in Theorem S.1 when  $W = V^*$ .

**Proposition O.1** (The coefficient map and its twists). *Let  $n = \dim V$  and  $N = nh$ . The quotient multiplication  $\mu_p$  gives an equivariant injection*

$$\mu_p^* : B_h(V)_p^* \hookrightarrow \text{Sym}^p(M(V)^*), \quad (\mu_p^* \varphi)(A) = \varphi(A^p). \quad (\text{O.2})$$

*Its image is the full coefficient space defining  $J_p$ , without a choice of apolar coordinates. The socle and perfect pairings are*

$$B_h(V)_N = (\det V)^{2h}, \quad B_h(V)_p^* \simeq B_h(V)_{N-p} \otimes (\det V^*)^{2h}. \quad (\text{O.3})$$

*For a line  $L$ ,  $B_h(V \otimes L)_p = B_h(V)_p \otimes L^{2p}$ , compatibly with all products. On  $\mathbb{P}(M(V))$ , the  $p$ th power is a section of  $B_h(V)_p \otimes \mathcal{O}(p)$ , and its base ideal is the image of  $B_h(V)_p^* \otimes \mathcal{O}(-p) \rightarrow \mathcal{O}$ .*

*Proof.* A homogeneous polynomial of degree  $p$  is identified with its symmetric  $p$ -linear polarization, normalized so that its value on  $A$  is its value on  $(A, \dots, A)$ . Dualizing the surjection  $\mu_p$  gives exactly (O.2). Multinomial coefficients appearing in a monomial basis express this normalization and are nonzero; no dual representation is silently replaced by the original one. Theorem S.1



identifies the quotient with determinant apolarity. The determinant of  $A : V^* \rightarrow V$  takes values in  $(\det V)^2$ . Choosing a trivialization gives a scalar polynomial whose inverse-system line has character  $(\det V^*)^{2h}$ ; its dual socle therefore has character  $(\det V)^{2h}$ . The Gorenstein pairing gives the second formula. Each generator in  $M(V)$  has scalar weight two, proving the line-twist assertion directly from (O.1). The tautological line  $\mathcal{O}(-1) \rightarrow M(V) \otimes \mathcal{O}$  has  $p$ th product  $\mathcal{O}(-p) \rightarrow B_h(V)_p \otimes \mathcal{O}$ , which proves the last assertion.  $\square$

**O.1. Graph schemes and classical inputs.** The projective graph argument uses the following precise statement. It applies to the ambient space and also to the two-dimensional total base of the degeneration computed below.

**Lemma O.2** (Simultaneous graph with invertible twists). *Let  $X$  be an integral noetherian complex scheme and let  $W_i \otimes L_i^{-1} \rightarrow \mathcal{O}_X$  be nonzero coherent systems of sections, with finite-dimensional  $W_i$  and invertible  $L_i$ . Write  $K_i$  for their image ideals, and let  $U = X \setminus \bigcup_i V(K_i)$ . The scheme-theoretic graph closure of  $U \rightarrow \prod_i \mathbb{P}(W_i^*)$  is  $\text{Bl}_{\prod_i K_i} X$  with its specified graph embedding. This statement identifies the scheme and the maps, not only their normalizations. Replacing the product ideal by an invertible twist or a positive power does not change its blow-up over  $X$ .*

*Proof.* The open  $U$  is dense because all  $K_i$  are nonzero and  $X$  is integral. After the Segre embedding the simultaneous map is given by all products of one section from each system. On a trivializing affine open, these products generate  $K = \prod_i K_i$ . The homogeneous coordinate algebra of the graph is the image of the polynomial algebra on these products in  $\mathcal{O}_X[t]$ , namely the Rees algebra  $\bigoplus_{m \geq 0} K^m t^m$ . It is a subalgebra of the function-field polynomial ring. Thus its Proj is exactly the schematic closure of the graph, with no extra component. The Segre equations hold on the dense graph and hence on this integral closure scheme, so the graph lands in the product, not merely in its ambient projective space. Local trivializations glue. An invertible ideal changes the degree- $m$  piece by its  $m$ th tensor power and does not change relative Proj; a positive power replaces the Rees algebra by a Veronese subalgebra.  $\square$

For the power system,  $L_i = \mathcal{O}(p)$  and  $W_i = B_h(V)_p^*$ ; for the exterior system of order  $q$ ,  $L_i = \mathcal{O}(q)$  and its sections are the  $q$ -minors. The determinant system is invertible on the integral ambient space. It may be omitted from the blow-up product, although its zero divisor must still be retained in contact formulas. Likewise the first-coordinate identity map has unit base ideal. These observations specify the projective twists in the proof of Theorem Q.1.

The precise classical scheme-level complete-quadric construction is [33, Theorem 6.3]; the graph of exterior-power maps and its successive symmetric-rank blow-ups are stated in [34, Remark 2.5 and Construction 2.6]. The relative rank version is recalled in [35, Definition 2.5, Remark 2.6 and Theorem 2.14]. Consequently smoothness, the flag description, and the orbit stratification in Corollary Q.2 are classical consequences of the graph identification, not new classifications.

There is also a precise invariant-ideal predecessor to the universal identity. Henriques–Varbaro [32, §2.4, Theorems 2.6–2.8] recall Abeasis’s classification for the symmetric coordinate ring, its products of determinantal ideals and their integral closures, and the passage between invariant generators and products indexed by diagrams. Our coefficient-space comparison uses this same multiplicity-free setting. Theorem P.1 identifies a particular multiplication coefficient map with the balanced determinantal product as an ordinary ideal; it is not an independent classification of invariant ideals. The exact coefficient identification in Proposition O.1 explains which map is being specialized on nonreduced bases. We assert no exhaustive historical nonanticipation of the displayed identity in all invariant-ideal terminology. Similarly the multiplier calculation in Corollary AD.1 is the substitution of its exponents in the symmetric determinantal formula [32, Theorem 4.8], not an independent multiplier-ideal theorem.

## APPENDIX P. UNIVERSAL POWER IDEALS

The power-zero construction admits an equality of ideals before any restriction to a pencil or to a valuation ring. This is the point at which the reciprocal section acquires a global boundary geometry. Let  $V$  have dimension  $n \geq 2$ , put  $M = \text{Sym}^2 V$ , and use the algebra  $B_{n,h}$  of Theorem S.1 with  $W = V^*$ . Thus  $(B_{n,h})_1 = M$ . If  $A$  is a symmetric matrix with entries in a commutative  $\mathbb{C}$ -algebra  $T$ , denote its element of  $M \otimes T$  by  $\ell_A$ , and set

$$J_p(A) = \text{im}((B_{n,h})_p^* \otimes T \longrightarrow T, \quad \varphi \longmapsto \varphi(\ell_A^p)). \quad (\text{P.1})$$

Write  $I_q(A)$  for the ideal of all  $q$ -minors, with  $I_0(A) = T$ . In this section powers of ideals are ordinary powers unless parentheses explicitly indicate symbolic powers.

**Theorem P.1** (Universal determinantal formula). *For every commutative  $\mathbb{C}$ -algebra  $T$ , every symmetric  $n \times n$  matrix  $A$  over  $T$ , and  $0 \leq p \leq nh$ , write  $p = hq + s$  with  $0 \leq s < h$ . Then*

$$J_p(A) = I_q(A)^{h-s} I_{q+1}(A)^s. \quad (\text{P.2})$$

When  $s = 0$  the second factor is omitted. For  $p > nh$ ,  $J_p(A) = 0$ . These are equalities of ideals, not of radicals or integral closures, and commute with every homomorphism of  $\mathbb{C}$ -algebras. In particular, for every symmetric map  $A : T^n \rightarrow T^n$ ,

$$J_{h(n-j)}(A) = \text{Fitt}_j(\text{coker } A)^h, \quad 0 \leq j \leq n. \quad (\text{P.3})$$

The formulas also hold locally, and hence as ideal sheaves, for symmetric maps of vector bundles with a line-bundle twist on an arbitrary  $\mathbb{C}$ -scheme.

We separate the classical determinantal input. For an arbitrary matrix  $X$  over a characteristic-zero polynomial ring, the balancing inclusion

$$I_u(X)I_v(X) \subseteq I_{u+1}(X)I_{v-1}(X), \quad 0 \leq u \leq v-2, \quad (\text{P.4})$$

is the determinantal symmetrization lemma. See [31, Lemma 2.5 and Theorem 2.4], which attributes the characteristic-zero product theorem to De Concini–Eisenbud–Procesi. Because this is a polynomial ideal inclusion, it remains true after specialization to a symmetric matrix; no flatness of that specialization is needed. This classical lemma is not a claim of the present paper.

**Lemma P.2** (The bounded-row generating space). *Let  $S = \text{Sym}(\text{Sym}^2 V^*)$  and let  $A$  be its universal symmetric matrix. For  $p = hq + s$ , the degree- $p$  part generating  $I_q(A)^{h-s} I_{q+1}(A)^s$  is*

$$\bigoplus_{\substack{|\lambda|=p, \ell(\lambda) \leq n \\ \lambda_1 \leq h}} \mathbb{S}_{2\lambda}(V^*) \subseteq S_p. \quad (\text{P.5})$$

*Proof.* The symmetric Cauchy decomposition of  $S_p$  is multiplicity free, with summands  $\mathbb{S}_{2\lambda}(V^*)$ ,  $|\lambda| = p$ ; see [32, §2.4]. We may use the dual general linear group to regard these as polynomial representations. Every minor has torus weight at most two in any one coordinate: its row set and column set each use that coordinate at most once. A product of  $h$  minors therefore has no weight with a coordinate exceeding  $2h$ . Its invariant span cannot contain a highest weight  $2\lambda$  with  $\lambda_1 > h$ . This proves one inclusion.

For the converse, let  $c_1 \geq \cdots \geq c_h \geq 0$  be the column lengths of a partition  $\lambda$  occurring on the right, padded with zeros. The product of the leading principal minors of orders  $c_i$  is a nonzero highest-weight vector of weight  $2\lambda$  in  $S_p$ . It belongs initially to  $\prod_{i=1}^h I_{c_i}(A)$ . Whenever two column lengths differ by at least two, apply (P.4) to increase the smaller length by one and decrease the larger by one. The sum of the squares of the lengths strictly decreases, so the procedure terminates. At termination the lengths are  $q+1$  repeated  $s$  times and  $q$  repeated  $h-s$  times, since their sum is  $p$ . Thus that highest-weight vector belongs to  $I_q^{h-s} I_{q+1}^s$ . The latter ideal

is invariant; its degree- $p$  part contains the full irreducible summand. Multiplicity freeness proves the equality. The case  $q = 0$  includes zero-length minors, which are the constant 1.  $\square$

*Proof of Theorem P.1.* First work over  $S$  with its universal matrix. The coefficient map of  $\ell_A^p$  is the dual of the multiplication quotient  $\text{Sym}^p M \rightarrow (B_{n,h})_p$ , up to nonzero multinomial constants. Consequently its degree- $p$  coefficient span is exactly  $(B_{n,h})_p^* \subseteq S_p$ . The character formula (S.3) identifies that span with (P.5). Both ideals in (P.2) are generated in degree  $p$ , so the lemma proves their equality in  $S$ . This step uses the full coefficient space, not just evaluation on diagonal matrices or closed points.

For a matrix over  $T$ , substitute its entries in this polynomial identity. Taking the image of the fixed finite-dimensional coefficient map commutes with scalar extension, as does taking minors and products of ideals. This proves the assertion even when  $T$  is nonreduced, non-Noetherian, or not a domain. The Fitting formula follows from its minor presentation. Changing a vector-bundle frame applies an invertible congruence to  $A$  and an invertible linear transformation to each coefficient space; changing a local frame of the twisting line multiplies generators by a unit. Therefore the local ideal identities glue. Finally,  $(B_{n,h})_p = 0$  for  $p > nh$ .  $\square$

**Example P.3** (A nonreduced test base). Take  $T = \mathbb{C}[\epsilon]/(\epsilon^3)$ ,  $n = h = 2$ , and  $A = \text{diag}(\epsilon, 0)$ . Then  $J_1(A) = (\epsilon)$  and  $J_2(A) = (\epsilon^2) \neq 0$ , whereas  $J_3(A) = J_4(A) = 0$ . Every closed-point matrix is zero. Thus equality on geometric fibres would not establish the theorem. Nor does (P.3) license taking unique ideal roots: for example, in  $\mathbb{C}[\epsilon]/(\epsilon^2)$  the distinct ideals  $(\epsilon)$  and  $0$  have the same square. At  $h = 1$  the power ideals are the Fitting ideals themselves over every base.

#### APPENDIX Q. THE RECIPROCAL POWER GRAPH AND COMPLETE QUADRICS

Let  $P = \mathbb{P}(M)$ , let  $P^{\text{nd}}$  be the nonsingular-quadric open, and define  $\mathcal{G}_h$  as the scheme-theoretic closure of the graph

$$P^{\text{nd}} \longrightarrow P \times \prod_{p=1}^{nh-1} \mathbb{P}((B_{n,h})_p), \quad [A] \longmapsto ([A], [\ell_A], \dots, [\ell_A^{nh-1}]). \quad (\text{Q.1})$$

The first factor records the base point. We use the reduced, integral graph closure, equivalently its scheme-theoretic image; no additional normalization is inserted in this definition. Let  $\text{CQ}(V)$  be the classical space of complete quadrics: the graph closure of  $[A] \mapsto ([\wedge^2 A], \dots, [\wedge^{n-1} A])$  over  $P$ . For  $n = 2$  the empty product gives  $\text{CQ}(V) = P$ .

**Theorem Q.1** (A compactification determined by multiplication). *For every  $h \geq 1$  there is a canonical  $\text{GL}(V)$ -equivariant isomorphism over  $P$*

$$\mathcal{G}_h \simeq \text{CQ}(V). \quad (\text{Q.2})$$

*In particular the graph is smooth and projective, is independent of  $h$  as a scheme over  $P$ , and has a simple-normal-crossing boundary. Write  $\pi : \text{CQ}(V) \rightarrow P$ , and let  $E_r$  be its boundary divisor of rank break  $r$ ,  $1 \leq r < n$ ; for  $r < n - 1$  it is exceptional over rank at most  $r$ , and  $E_{n-1}$  is the strict transform of the determinant divisor. Then every power ideal has the exact total transform*

$$J_p \mathcal{O}_{\text{CQ}(V)} = \mathcal{O}_{\text{CQ}(V)} \left( - \sum_{r=1}^{n-1} (p - hr)_+ E_r \right), \quad 0 \leq p \leq nh, \quad (\text{Q.3})$$

*where  $(a)_+ = \max(a, 0)$ . Removing this common divisor from the universal  $p$ th power gives a nowhere-zero projective section on the whole compactification.*

The smoothness, boundary and flag interpretation of complete quadrics are classical, due to the complete-form construction; see [33, 34]. The assertion here identifies that particular compactification, with all of its scheme structure, from the multiplication in one reciprocal

normalization-fibre section. It does not assert that complete quadrics is a newly constructed variety.

*Proof.* We first recall the graph calculation with its scheme structure. If rational maps from an integral scheme are given by generators of nonzero ideals  $K_1, \dots, K_a$ , their simultaneous graph closure is  $\text{Bl}_{K_1 \dots K_a}$  of the base. Indeed, after the Segre embedding the coordinates are all products of one generator from each ideal. The closure of this graph is the Proj of the Rees algebra of that product: on a generator chart its coordinate ring is the subring of the function field obtained by adjoining the generator ratios. These are exactly the affine charts of both constructions. Tensoring an ideal by an invertible ideal, or replacing it by a positive power, does not change the blow-up, the latter by the Veronese description of Proj.

Apply this calculation on  $P$ . The sheaf  $I_1 \mathcal{O}_P$  is the unit ideal, and  $I_n \mathcal{O}_P$  is invertible. For each  $2 \leq q \leq n-1$ , the exponent of  $I_q$  in  $\prod_{p=1}^{nh-1} J_p$  is

$$\sum_{s=0}^{h-1} (h-s) + \sum_{s=1}^{h-1} s = h^2.$$

Theorem P.1 therefore gives, up to an invertible ideal,

$$\prod_{p=1}^{nh-1} J_p = \left( \prod_{q=2}^{n-1} I_q \right)^{h^2}.$$

The same graph calculation for the exterior-power maps identifies  $\text{CQ}(V)$  with the blow-up of  $\prod_{q=2}^{n-1} I_q$ . This proves (Q.2) as an isomorphism of schemes, not only after normalization. Both isomorphisms are the identity on the common dense open, hence are canonical and equivariant.

For completeness, the local calculation behind the boundary formula is as follows. The complete-quadric construction blows up the strict rank loci successively. On its Gaussian charts, after a congruence and projective normalization, its matrix has the form

$$U \text{diag}(1, \tau_1, \tau_1 \tau_2, \dots, \tau_1 \cdots \tau_{n-1}) U^t, \quad (\text{Q.4})$$

with  $U$  lower unitriangular and  $E_r = (\tau_r = 0)$ . One obtains these charts inductively by making the first pivot one, taking a Schur complement, blowing up the zero residual matrix, and repeating. At each step the next nonzero residual quadratic form has a nonisotropic vector, so congruence translates of these charts cover the construction. This also explains the smooth charts and the normal crossings stated in the classical construction.

Cauchy–Binet, applied to (Q.4), shows that every  $q$ -minor is divisible by  $\tau_1^{q-1} \tau_2^{q-2} \cdots \tau_{q-1}$ . The leading principal  $q$ -minor is exactly that product. Thus, on these charts and therefore globally,

$$I_q \mathcal{O}_{\text{CQ}(V)} = \mathcal{O} \left( - \sum_{r < q} (q-r) E_r \right). \quad (\text{Q.5})$$

Using (P.2), its exponent along  $E_r$  is  $(h-s)(q-r)_+ + s(q+1-r)_+ = (p-hr)_+$ . This proves (Q.3). The coefficient sections generate the transformed ideal; division by its local generator makes them generate the unit ideal, proving the last assertion.  $\square$

**Corollary Q.2** (Native boundary strata). *The power graph has one congruence orbit for every subset  $I \subseteq \{1, \dots, n-1\}$ . It is the stratum  $E_I^\circ = (\bigcap_{r \in I} E_r) \setminus \bigcup_{r \notin I} E_r$ , of codimension  $|I|$ , and its closure is  $\bigcap_{r \in I} E_r$ . Its points are flags with nondegenerate projective quadratic forms on successive quotients; the quotient dimensions are the successive gaps in  $\{0\} \cup I \cup \{n\}$ . The simultaneous normalized power values recover this complete-quadric datum. They are not the one common limit for the full cotangent-coordinate group.*

*Proof.* The flag description and boundary stratification are the classical complete-quadric description. The general linear group acts transitively on flags of fixed dimensions, and over  $\mathbb{C}$  every nondegenerate quadratic form on a fixed-dimensional quotient is congruent to the identity. Independent diagonal scalings remove the projective scalars. The chart (Q.4) realizes each subset of vanishing  $\tau_r$  and permits further coordinates to vanish, giving the stated codimensions and closures. The graph is a closed subscheme of the product in (Q.1); its projections therefore retain the point, including its flag and quotient forms. The group here is the native congruence group of  $V$ .  $\square$

*Remark Q.3* (Relative construction and base change). For a vector bundle  $\mathcal{V}$  on any  $\mathbb{C}$ -scheme  $S$ , the associated bundle with fibre  $\text{CQ}(\mathbb{C}^n)$  is a relative smooth projective scheme over  $S$ . Equivariance glues the power maps and the boundary formula. Formation of this associated bundle and the polynomial ideal identity commute with arbitrary base change. This does not say that taking the graph closure of an arbitrary restricted family commutes with nonflat base change. The relative ambient resolution and the closure of an individual family's graph are different constructions.

## APPENDIX R. DESCENT AND THE SCHEME STRUCTURE OF THE CONTACT CHARTS

This section records the descent, exactness, and overlap arguments used by the contact theorems. In particular, formal coordinates are never used as a substitute for an algebraic family in the fixed target. The coefficient projection is retained in every graph point.

**R.1. The category of embedded deformations.** Fix a framed coefficient atlas  $S$ , a central point  $s$ , and a central embedded curve. The test category has objects  $(A, \beta)$ , where  $A$  is a local Artin  $\mathbb{C}$ -algebra with residue field  $\mathbb{C}$  and  $\beta : \hat{\mathcal{O}}_{S,s} \rightarrow A$  is a local map. Its morphisms are local algebra maps commuting with  $\beta$ . The Hilbert deformation functor is covariant on this algebra category, or contravariant on the opposite category of test schemes. It is *set-valued*: an object is an embedded ideal quotient, and two objects are equal only when their ideals in the fixed target coincide. No automorphism group of the target is divided out.

Let  $X = \text{CQ}(V)$ , let  $\mathcal{L}$  be the universal pencil line, and let  $\mathcal{Y} = \mathcal{L} \times_{\mathbb{P}(\text{Sym}^2 V)} X$ . For each completed contact neighbourhood the comparison diagram is

$$\begin{array}{ccccc} \hat{\mathcal{Y}}_{i,A} & \xrightarrow{\Theta_i} & \hat{\mathcal{N}}_{i,A} & \xrightarrow{\Theta_i^{-1}} & \hat{\mathcal{Y}}_{i,A} \\ \downarrow & & \downarrow & & \downarrow \\ \hat{\mathcal{L}}_{i,A} & = & \hat{\mathcal{L}}_{i,A} & = & \hat{\mathcal{L}}_{i,A}. \end{array} \quad (\text{R.1})$$

The last upper object maps to  $X \times \text{Spec } A$  by the original incidence embedding. The middle upper object is the normal-coordinate graph target, not a new fixed target. The operation producing a Hilbert quotient always includes the last arrow. On a common domain of two choices define  $\Phi_{ji} = \Theta_j \Theta_i^{-1}$ . Then

$$\Phi_{kj} \Phi_{ji} = \Phi_{ki}, \quad \Theta_j^{-1} \Phi_{ji} \Theta_i = 1. \quad (\text{R.2})$$

These identities hold on the ambient targets and hence on their ideal quotients and on every Artin base change. They give the triple-overlap cocycle and show why a parameter-dependent congruence does not identify distinct curves in the fixed target.

**Proposition R.1** (Precise patching and algebraic comparison). *On an affine contact chart let  $D$  be a Noetherian ring and let  $J$  cut out the closed contact support. Use the ring-theoretic completion  $\hat{D}$ , the open  $U = \text{Spec } D \setminus V(J)$ , and the overlap  $V = \text{Spec } \hat{D} \setminus V(J\hat{D})$ . Finitely presented ideal quotients on  $U$  and  $\text{Spec } \hat{D}$ , with compatible quotient algebra structures on  $V$ , patch uniquely to an ideal quotient on  $\text{Spec } D$ . Quotients with contact-supported torsion are allowed. Base flatness, equality, and the closed incidence equations are detected by this patching cover.*

*Consequently, suppose an algebraic family of finite presentation has first supplied a morphism  $h : T \rightarrow H$  into a Hilbert scheme or its fixed coefficient graph. If recovery makes  $h$  a monomorphism and the embedded comparison identifies completed local rings at all closed points, then  $h$  is an open immersion. This conclusion uses the algebraic family, not only an isomorphism of formal functors.*

*Proof.* Completion of a Noetherian ring is flat and induces an isomorphism over  $D/J$ . Bhatt's flat formal-gluing statement [Bhatt(2014), Proposition 5.6(4), Example 5.8] therefore applies to quasi-coherent sheaves. Apply its monoidal equivalence to the quotient, its unit, multiplication, and its structure map from the ambient ring. Surjectivity and finite presentation descend on the jointly surjective flat cover given by the completion and the open complement. The glued algebra is thus an ordinary finitely presented quotient, not a derived replacement. The same cover detects base flatness and the vanishing of the prescribed incidence equations. The gluing theorem supplies existence; faithful flatness alone would not supply the missing overlap descent datum. Iteration over disjoint supports and over projective affine charts gives the claimed form for the incidence target.

For the last assertion, use the completed local criterion for etaleness at closed points, whose residue fields are  $\mathbb{C}$  [Stacks Project, Lemma 41.11.3]. The morphism is already of finite presentation. Its etale locus is open; its closed complement in a finite-type complex chart would contain a closed point if nonempty. Thus it is etale everywhere under consideration. An etale monomorphism is an open immersion. Completed comparison here means comparison on all Artin quotients, including nilpotent test rings; an agreement of tangent dimensions or closed reduced points is insufficient.  $\square$

The reduced graph closure is compatible with completion by Lemma E.2, using the finite stabilized colon ideal, not a general assertion that closure commutes with every base change. Its completed rings remain reduced for the following precise reason. A reduced local complex finite-type ring is excellent, so its formal fibres are geometrically reduced; see [Stacks Project, Tag 0BJ0]. The injection into the finite product of its minimal-prime fraction fields remains injective after flat completion. Each resulting generic formal fibre is reduced. The completion, as a subring of their product, is therefore reduced. Normalization is unchanged on the regular total charts; this says nothing about normalizing their possibly nonreduced fibres.

**Lemma R.2** (The larger Smith exponent is a smooth graph factor). *In the formal corank-two model  $\begin{pmatrix} f & g \\ g & fk+r \end{pmatrix}$ , the coefficient parameters of  $k$  are independent smooth factors for the relative embedded graph comparison. They are not forgotten in the retained coefficient point.*

*Proof.* The generating triples  $(f, g, fk + r)$  and  $(f, g, r)$  are related over the completed parameter ring by

$$T_k = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -k & 0 & 1 \end{pmatrix}, \quad T_k^{-1} = T_{-k}, \quad \det T_k = 1.$$

Thus their projective graph targets are isomorphic over the parameter disc. Apply  $T_k$  on that target and  $T_k^{-1}$  before patching into the fixed incidence target. This gives inverse operations on embedded ideals over every Artin coefficient algebra. The independence of the coefficients of  $k$  is the coefficient normal form of Theorem F.2. Therefore the completed graph rings acquire these parameters as power-series variables. This is an isomorphism of relative graph presentations with a specified inverse, not a constant ambient automorphism of  $X$  or an unlabelled Hilbert-point inversion.  $\square$

## R.2. Exactness and recovery at every conic.

**Lemma R.3** (Height, flatness, and the recovery minors). *The three minors in (I.2) have codimension two on every geometric fibre of the six-parameter family. Their Hilbert–Burch complex is fibrewise exact, and its cokernel is flat over that parameter space. At a conic-tail point the ambient restriction maps in degrees  $(0, 2)$  and  $(1, 1)$  have respectively ranks five and four, from ambient spaces of dimension six. Thus recovery uses kernels of ranks one and two, including at doubled lines.*

*Proof.* At  $e = 0$  the ideal is  $(zG, zH, Q) = (G, H) \cap (z, Q)$ , where  $Q \neq 0$  and is independent of  $z$ . There is no common codimension-one factor: any factor common to  $zG, zH$  divides  $z$ , which does not divide  $Q$ . The displayed union has pure support of codimension two. At  $e \neq 0$ , on  $s = 1$  eliminate  $H$  from  $zG - e(H + BF) = 0$ . The other linear minor becomes

$$(z^2 + e^2C)G + (e^2A - eBz)F = 0.$$

It is nonzero, with coefficient of  $z^2G$  equal to one. Together with the first linear equation it is a regular sequence in the affine polynomial ring; the remaining quadratic follows from the syzygies. At  $s = 0$  the two bilinear equations force  $G = H = 0$ . There is no component at infinity and no intermediate parameter value at which the height drops. The perfect-height-two criterion therefore gives the Hilbert–Burch resolution on every fibre. The fibrewise exactness criterion for a complex of base-flat locally free modules gives exactness and flatness of the cokernel [Stacks Project, Lemma 10.99.5].

On  $C_0 \cup Q$  use its union sequence and the plane-hypersurface sequence  $0 \rightarrow \mathcal{O}_{\mathbb{P}^2}(j-2) \rightarrow \mathcal{O}_{\mathbb{P}^2}(j) \rightarrow \mathcal{O}_Q(j) \rightarrow 0$ . They give  $h^0 = 5, 4$  and  $H^1 = 0$  in the respective degrees, and surjectivity of restriction from the ambient six-dimensional spaces. This argument uses only that the quadratic equation is nonzero, so includes reducible conics and doubled lines. The nonzero 5 by 5 and 4 by 4 restriction minors define the constant-rank opens. Normalize the selected  $R^2$  coefficient of the kernel quadratic to one, and normalize the two bilinear equations by their  $tG, tH$  coefficient matrix. Its 2 by 2 pivot is invertible at the conic-tail point. These are the open minors used by the recovery maps. The coefficient projection supplies  $f$  and hence  $c$ ; recovery of  $e$  from the normalized bilinear equation would not replace that projection.  $\square$

**Lemma R.4** (Connectedness, scheme exhaustion, and unlabelled descent). *Let  $p : Z \rightarrow B$  be proper and birational, with  $Z$  integral and normal and  $B$  integral and normal of finite type over  $\mathbb{C}$ . Its geometric fibres are connected. A nonempty proper union of constructed closed families which is open in the support of a fibre therefore exhausts that support. If algebraic charts cover those families and compute the fibre ideals, the same charts determine the complete scheme fibre.*

*For the disjoint-contact product theorem, the map from the product of local families to the Hilbert fibre is a representable algebraic morphism of finite presentation. Completed comparison on all Artin rings and ideal recovery make it an open immersion. Its properness and the preceding connectedness then give the asserted isomorphism. The construction descends without global spectral labels.*

*Proof.* Stein factorization gives connected geometric fibres followed by a finite birational map to  $B$ . Normality of  $B$  makes the latter an isomorphism; equivalently use [Stacks Project, Lemma 37.53.6]. The stated union is closed by properness and open by the local chart descriptions, hence is the whole nonempty connected support. Connectedness alone does not compute nilpotents: those are supplied by the fibre ideals on the covering charts.

For the product assertion, the contact-supported quotients are finite-presentation algebraic quotients on disjoint finite contact neighbourhoods. Patching them to the fixed main curve produces a flat embedded family with the required polynomial, so Hilbert representability supplies

the morphism before formal comparison is applied. Restriction to the supported quotients recovers each factor on every test scheme, giving a monomorphism. Proposition R.1 now applies to its completed local maps, including nonreduced Artin tests. A proper open immersion has open-and-closed image; connectedness supplies surjectivity. On the finite étale cover ordering the distinct supports, changes of order permute the corresponding factors and the supported quotients. These permutation maps satisfy the cocycle identity, so descent gives the unlabelled object. This does not impose an ordering through a collision of supports.  $\square$

Our convention for  $\mathbb{F}_2 = \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(2))$  is the projectivization of one-dimensional subspaces. The chart  $[1 : v]$  is  $\text{Tot } \mathcal{O}(2)$ ; its section at infinity is the subline  $\mathcal{O}(2)$  and has normal bundle  $\text{Hom}(\mathcal{O}(2), \mathcal{O}) = \mathcal{O}(-2)$ . This fixes the negative section and the sign used in the component-intersection formulas.

**R.3. The quotient overlap maps and divisor classes.** Write the ordered directions as  $w_{\pm} = \lambda \pm h$ , and the ordered root parameter as  $t$ . On the blow-up of  $(h, t)$  use  $t = eh$  or  $h = ut$ . The involution sends  $h$  to  $-h$  in the first chart and  $t$  to  $-t$  in the second. Its fixed locus is the exceptional divisor. Its invariant coordinates are respectively  $(\lambda, e, C = -h^2)$  and  $(\lambda, u, \delta = t^2)$ ; their rings are polynomial even at the fixed locus.

For completeness, a change of direction coordinate  $\phi(w) = (aw + b)/(cw + d)$ , with  $\Delta_{\phi} = ad - bc \neq 0$ , gives on the first invariant chart

$$\begin{aligned} K &= (c\lambda + d)^2 + c^2C, \\ \lambda' &= ((a\lambda + b)(c\lambda + d) + acC)/K, \\ C' &= \Delta_{\phi}^2 C / K^2, \quad e' = Ke / \Delta_{\phi}. \end{aligned} \tag{R.3}$$

On the second chart it gives

$$\begin{aligned} K &= (c\lambda + d)^2 - c^2\delta u^2, \\ \lambda' &= ((a\lambda + b)(c\lambda + d) - ac\delta u^2)/K, \\ u' &= \Delta_{\phi} u / K, \quad \delta' = \delta. \end{aligned} \tag{R.4}$$

These formulas are used where both directions are in both affine charts, so  $K$  is invertible. They follow by taking the mean and half-difference of  $\phi(w_+), \phi(w_-)$ . The overlap between the two blow-up charts is  $e = 1/u$ ,  $C = -\delta u^2$ . These identities supply every overlap, since any two directions admit a common point at infinity outside them. The triple-overlap identities follow from composition of the fractional linear maps, before taking invariants.

**Lemma R.5** (Projectivity and ramification of the collision quotient). *The finite involution quotient of the ordered blow-up is projective over the unordered coefficient line. If  $T_0$  is the strict transform of  $t = 0$  and  $F_0$  its exceptional divisor, and if  $S, E$  are their respective images, then for the quotient map  $q$*

$$q^*S = T_0, \quad q^*E = 2F_0, \quad \text{div}(t) = T_0 + F_0, \quad \text{div}(\delta) = 2S + E.$$

*In particular  $\mathcal{O}_S(S) = \mathcal{O}_{\mathbb{P}^2}(-1)$ .*

*Proof.* Choose a relatively ample line bundle on the blow-up and tensor its two translates. It has the permutation linearization. Squaring kills the characters of every stabilizer on its fibres. It thus descends to the finite quotient; equivalently, products of the translates of sufficiently high-degree sections give invariant sections and their affine projective charts. The descended line bundle is ample, since its finite pullback is ample. This proves projectivity without assuming that the action is free. In these charts its overlap maps are those of the tensor product linearization, followed by their squares, so they satisfy descent on triple overlaps.



The invariant charts show ramification index two at the generic exceptional divisor and index one at the generic strict transform. They prove the first two identities. The blow-up of  $(h, t)$  gives  $\text{div}(t) = T_0 + F_0$ , and  $\delta = t^2$  gives the last identity after pullback by  $q$ . On  $S = \text{Sym}^2 \mathbb{P}^1$  the divisor  $E|_S$  is the doubled-line conic, of class  $2H$ . Since  $E + 2S$  is principal,  $2[S|_S] = -2H$  in  $\text{Pic}(\mathbb{P}^2) = \mathbb{Z}H$ . This group has no two-torsion, so  $[S|_S] = -H$  as claimed.  $\square$

In the full conic pullback the finite colon calculation is  $((eA, eB) : e^2C) = (A, B)$  and further colons have the same value. Thus for the original unramified coefficient line all torsion is already annihilated by  $\delta$ . The horizontal ring is integral and dominates  $\mathbb{C}[\delta]$ , hence is torsion-free over that base. The discrete valuation rings used to infer flatness are the nonfield localizations of  $\mathbb{C}[\delta]$ , not localizations of the threefold coordinate ring. The higher Tor groups vanish by (E.5); for ramified arcs their first Tor is only the last layer of the larger torsion ideal.

Finally choose  $\delta_0 > 0$  and the positive root at that base point. On the punctured analytic disc the local system  $R^2 p_* \mathbb{Q}$  for the horizontal parameter space has basis the two ruling classes of  $\mathbb{P}^1 \times \mathbb{P}^1$ . The loop  $\delta(\theta) = \delta_0 \exp(2\pi\sqrt{-1}\theta)$  lifts to an exchange of the two roots, and acts by exchanging that basis. Its invariant line is the sum and its anti-invariant line the difference. In the normalization after  $\delta = t^2$ , the element  $h = t/e$  is taken in the fraction field of the integral base-changed horizontal ring; adjoining it is finite and birational. This is normalization of the base-changed total space. It is neither normalization of the original total graph nor normalization of its central fibre. The resulting normal crossings belong to the parameter space, not to a stable-map model of every embedded curve.

#### APPENDIX S. QUADRATIC SECTIONS OF RECIPROCAL NORMALIZATION FIBRES

The reciprocal presentation has a distinguished linear coefficient section on which the entire algebra, rather than only its minimal relations, can be determined. Throughout this section a decomposition  $E = \mathbb{C}a \oplus W$  is fixed. The section is equivariant for  $\text{GL}(W)$ ; it is not asserted to be independent of an arbitrary change of that decomposition.

Put  $d = \dim W$ ,  $g \geq k \geq 2$ , and  $h = \lfloor g/2 \rfloor$ . In the algebra  $F_{g,k}(W)$  of (AM.1), let  $\mathfrak{l}$  be the ideal generated by the coefficients of all  $Q_i$  with  $i \neq 2$ . Write  $Q = Q_2$  and

$$B_{d,h} = \text{Sym}((\text{Sym}^2 W)^*) / (\text{coeff } Q^{h+1}). \quad (\text{S.1})$$

Here the ordinary degree of a coefficient of  $Q$  is one; its reciprocal weight is two. These two gradings will always be specified separately. A partition  $\lambda \subseteq (h^d)$  has at most  $d$  rows, each of length at most  $h$ .

**Theorem S.1** (The entire quadratic coefficient section). *There is an isomorphism of  $\text{GL}(W)$ -equivariant graded algebras*

$$F_{g,k}(W)/\mathfrak{l} \simeq B_{d,h}. \quad (\text{S.2})$$

*Under the coefficient/differential pairing,  $B_{d,h}$  is the apolar algebra of the  $h$ th power of the determinant of a generic symmetric  $d \times d$  matrix. Its ordinary-degree decomposition is*

$$(B_{d,h})_p \simeq \bigoplus_{\substack{\lambda \subseteq (h^d) \\ |\lambda|=p}} \mathbb{S}_{2\lambda}(W^*). \quad (\text{S.3})$$

In particular it is Gorenstein of socle degree  $dh$ , with one-dimensional socle representation  $(\det W^*)^{2h}$ . Its Hilbert function and length are

$$H_{d,h}(p) = \sum_{\substack{\lambda \subseteq (h^d) \\ |\lambda|=p}} \prod_{1 \leq i < j \leq d} \frac{2\lambda_i - 2\lambda_j + j - i}{j - i}, \quad (\text{S.4})$$

$$L_{d,h} = \prod_{i=1}^d \frac{h + d - i + 1}{d - i + 1} \prod_{1 \leq i < j \leq d} \frac{2h + 2d - i - j + 2}{2d - i - j + 2}. \quad (\text{S.5})$$

The Hilbert function is symmetric. A degree-one element is strong Lefschetz precisely when its associated symmetric matrix has full rank. More generally, an element associated with a rank- $r$  matrix has nilpotence index  $hr + 1$  (index one for the zero element). Consequently

$$\text{length } F_{g,k}(W) \geq L_{d,h}, \quad \mathbf{m}^{dh} \neq 0 \text{ in } F_{g,k}(W). \quad (\text{S.6})$$

The Gorenstein and Lefschetz assertions concern the section  $B_{d,h}$ , not the full fibre  $F_{g,k}(W)$ .

*Proof.* After setting the indicated coefficients to zero, the reciprocal series is  $(1 + Qz^2)^{-1}$ . The first even integer in  $\{g + 1, \dots, g + k\}$  is  $2(h + 1)$ ; it belongs to this interval because  $k \geq 2$ . Every odd coefficient vanishes, and the even ones are signed powers of  $Q$ . Thus the specialized ideal is exactly  $(\text{coeff } Q^{h+1})$ : all its higher powers have coefficients in this ideal. This proves (S.2) as an equality of ideals, not only an equality of radicals.

For clarity about the pairing, write  $Q(x) = \sum_i y_{ii} x_i^2 + \sum_{i < j} y_{ij} x_i x_j$  and let  $X$  have independent symmetric entries  $X_{ij}$ ,  $i \leq j$ . Send  $y_{ij}$  to  $\partial/\partial X_{ij}$ . Then  $Q(x)$  acts by differentiation in the rank-one direction  $xx^t$ . Since  $\det(X + zxx^t)^h$  has degree at most  $h$  in  $z$ , the coefficients of  $Q(x)^{h+1}$  annihilate  $(\det X)^h$ . The reverse containment is the symmetric-matrix case of [30, Remark 11, pp. 8–9]: that apolar ideal is generated by the full general-linear orbit of the  $(h + 1)$ st power of a rank-one differentiation direction. Powers of linear forms span their symmetric power, so this orbit span is precisely the coefficient space just displayed. No assertion that an arbitrary quotient of an apolar algebra is Gorenstein is used.

We record explicitly how the classical representation formulas translate here. In [30, Example 9 and Proposition 10], the surviving highest weights are indexed by  $k_1, \dots, k_d \geq 0$  with  $\sum k_i \leq h$ , and their degrees are  $\sum ik_i$ . Set  $\lambda_i = \sum_{j \geq i} k_j$ . These are exactly the partitions in  $(h^d)$ , of size  $\sum ik_i$ , and their polynomial representations are  $\mathbb{S}_{2\lambda}(W^*)$ . This proves (S.3); the ordinary Weyl dimension formula gives (S.4). The same apolar module is the finite-dimensional highest-weight  $\mathfrak{sp}_{2d}$ -module of weight  $h\omega_d$ , with its Levi character shifted by  $\det^h$ ; see [30, Proposition 16 and Lemma 13]. For the positive roots  $e_i - e_j$ ,  $e_i + e_j$ , and  $2e_i$ , its Weyl dimension factors are respectively 1,  $(2h + 2d - i - j + 2)/(2d - i - j + 2)$ , and  $(h + d - i + 1)/(d - i + 1)$ . Their product is (S.5). Apolarity to one homogeneous polynomial of degree  $dh$  gives the perfect pairings and the socle. Equivalently, the complementary partition  $(h - \lambda_d, \dots, h - \lambda_1)$  gives the dual summand with the indicated determinant twist.

The strong Lefschetz assertion is the symmetric-matrix case of [30, Theorem 4 and Example 5]. There is a direct argument for the asserted nilpotence index. If  $\ell_A = \sum_{i \leq j} A_{ij} y_{ij}$ , then  $\ell_A^p = 0$  in the apolar algebra exactly when  $\partial_z^p \det(X + zA)^h = 0$  as a polynomial in  $X$ . Congruence reduces a rank- $r$  symmetric matrix to  $\text{diag}(1, \dots, 1, 0, \dots, 0)$ . The last determinant power has degree exactly  $hr$  in  $z$ , as is seen by taking  $X$  diagonal with independent entries. This proves the index. Finally a surjective local algebra map surjects on every power of its maximal ideal. The section has nonzero degree  $dh$ , proving both assertions in (S.6).  $\square$

**Corollary S.2** (Size and Loewy depth of the actual common boundary fibre). *In the notation of Theorem AT.1, put  $d = n^2 - 1$  and  $h = n(n - 1)/2$ . The entire normalization fibre at  $K_\infty$*

has length at least  $L_{d,h}$  and its maximal ideal has nonzero  $dh$ th power. For the ternary pencil boundary,

$$\text{length } F_{6,2}(\mathbb{C}^8) \geq 922268360, \quad \mathfrak{m}^{24} \neq 0. \quad (\text{S.7})$$

Its quadratic coefficient section has exactly that length, ordinary socle degree 24, and reciprocal-weight socle 48.

*Proof.* The common boundary fibre is  $F_{g,k_n}(\mathbb{C}^{n^2-1})$ , with  $g = n(n-1) \geq k_n \geq 2$ . Apply the theorem. For  $n = 3$ , substitute  $(d, h) = (8, 3)$  into (S.5). This computes a whole closed section, not the length of the original fibre. The previously obtained 44 coefficient variables and 9867 minimal equations remain counts of different invariants.  $\square$

#### APPENDIX T. A NON-EQUIDIMENSIONAL AMBIENT HILBERT FIBRE

**Theorem T.1** (Two different ambient boundary components). *Let  $n \geq 4$  and let  $R_0$  be a regular symmetric pencil with one corank-two point having Smith exponents  $(2, 2)$  and otherwise only corank-one points. Its normalized Hilbert fibre has an irreducible component of dimension two, with open part  $J_1(\mathbb{P}^1)$ . It also has a distinct component of dimension at least four. In particular the fibre is reducible and not equidimensional.*

*The second assertion is witnessed by every smooth conic through the attaching point in the exceptional  $\mathbb{P}^2$ : union with the fixed main strict transform gives a four-dimensional locally closed family of embedded Hilbert limits. These are reduced conic tails, not the thick line tails parametrized by the jet open. The assertion is about the actual fibre over  $R_0$ , not the fibre of a chosen parameter slice. It does not assert that these exhaust its components.*

*Proof.* Theorem H.2 gives the smooth irreducible open  $J_1(\mathbb{P}^1)$  of dimension two in the normalized fibre. Its closure is an irreducible component of dimension two. For the other family use the full normal coordinates with  $a = b = 2$ ,  $k = 1$ , and take

$$\begin{aligned} f &= x^2 + t\alpha_1x + t^2\alpha_0, \\ g &= t\beta_1x + t^2\beta_0, \quad r = t\gamma_1x + t^2\gamma_0, \quad \beta_1\gamma_0 - \beta_0\gamma_1 \neq 0. \end{aligned} \quad (\text{T.1})$$

The three coefficient quadrics span all quadratic forms in  $x, t$ . Their ideal is exactly  $(x, t)^2$ . The simultaneous graph on the parameter surface is therefore  $\text{Bl}_{(x,t)}$ , and its exceptional line maps as

$$[X : T] \longmapsto [X^2 + \alpha_1XT + \alpha_0T^2 : \beta_1XT + \beta_0T^2 : \gamma_1XT + \gamma_0T^2]. \quad (\text{T.2})$$

This is a complete Veronese embedding, giving a smooth conic through  $[1 : 0 : 0]$ . The graph is a closed embedding in the incidence target, since these three generators define its Rees graph. Its flat special curve is the fixed main component and this reduced conic, meeting at the point corresponding to  $[X : T] = [1 : 0]$ . The intersection has length one. Their last-exterior degrees are  $n - 3$  and two, respectively, and their Euler characteristic is one. The other exterior degrees are unchanged. Thus these are limits with the required Hilbert polynomial.

Conversely every smooth conic through  $[1 : 0 : 0]$  has a Veronese parametrization of the displayed form: choose its preimage of that point as  $[1 : 0]$ , and normalize the first leading coefficient to one. Nonsingularity gives independence of the three quadrics. The normal-coordinate realization theorem lifts these arcs to actual pencil arcs. The congruence on the parameter disc specializes on the exceptional divisor to its fixed value at the contact point; it consequently gives the fixed exceptional plane identification, not a quotient identifying distinct conics.

Smooth conics through a fixed point form an open of a hyperplane in  $\mathbb{P}^5$ , of dimension four. Their universal conic is flat and its attachment to the fixed main component is one reduced point; the union construction is therefore a flat family. It is a locally closed immersion into the Hilbert scheme, as the conic is recovered as the residual component to the fixed main curve. Equivalently

its homogeneous quadratic ideal in the exceptional plane determines it uniquely, in families. Since all its closed points are limits from (T.1), this reduced family factors through the reduced Hilbert graph fibre. The normalization of that graph is finite and surjective, and its inverse image over this family is finite and surjective. Some component of this inverse image has dimension four. Hence the normalized fibre has a component of dimension at least four. We use this finite inverse image, not a purported automatic lifting of a nondominant normal family to a normalization. This component is distinct from the dimension-two component already obtained. The asserted non-equidimensionality follows.  $\square$

The distinction has a concrete source. For a thick line the length-two parameter algebra acts nontrivially on the tail. For a conic tail the parameter acts by its closed-point value, and degree two is carried by a reduced curve in the exceptional plane. Smith data of the closed pencil do not choose between these Hilbert objects. The full versal neighbourhood contains both, whereas the two constant transverse parameters of the original Jordan slice see only a section of the jet-tail locus.

#### APPENDIX U. MULTIPLE REDUCED INCIDENCE AND ITS COMPLETE HILBERT BOUNDARY

We remove the restriction to a single incidence point. The independence of the incidence conditions is proved from the radicals of distinct members of a regular pencil; it is not added as a transversality hypothesis. The resulting modification has a global Fitting-ideal description even though the incidence points need not be globally labelled.

Put  $M = \text{Sym}^2 V$ ,  $P = \mathbb{P}(M)$  and  $G = \text{Gr}(2, M)$ , with  $n = \dim V \geq 3$ . Let  $Z_j \subset P$  have its symmetric determinantal scheme structure. When  $n = 3$  put  $Z_0 = \emptyset$ . Let  $G_{\text{reg}}$  be the open of regular pencils and let  $X_G = \mathbb{P}(\mathcal{R})$  be the universal line. The image of  $X_G \times_P Z_{n-3}$  is closed, since the projection from this incidence is proper. Remove it and call the resulting open  $G_* \subset G_{\text{reg}}$ . Every line over  $G_*$  meets  $Z_{n-2}$  in a finite scheme: a positive-dimensional intersection would contain the line and contradict regularity. Consequently

$$q_*^{\text{inc}} : D_* := X_{G_*} \times_P Z_{n-2} \longrightarrow G_*$$

is proper and quasi-finite, hence finite. Define

$$U_{\text{red}} = G_* \setminus q_*^{\text{inc}}(\text{Supp } \Omega_{D_*/G_*}), \quad D = D_* \times_{G_*} U_{\text{red}}, \quad q : D \rightarrow U_{\text{red}}. \quad (\text{U.1})$$

This is open because  $q_*^{\text{inc}}$  is finite. Its geometric incidences are precisely the finite reduced intersections: for a finite algebra over  $\mathbb{C}$ , vanishing of relative differentials is equivalent to being a product of copies of  $\mathbb{C}$ . Unlike the single-incidence open, it allows every possible number of such points. Set

$$\mathcal{Q} = q_* \mathcal{O}_D, \quad \mathcal{I} = \text{Fitt}_0(\mathcal{Q}), \quad B_{\text{red}} = \text{Bl}_{\mathcal{I}} U_{\text{red}}. \quad (\text{U.2})$$

The zeroth Fitting ideal of the zero module is the unit ideal.

##### U.1. The general incidence calculation.

**Lemma U.1** (Transverse incidence of a family of curves). *Let  $S, Y$  be smooth complex varieties, let  $Z \subset Y$  be smooth of codimension  $c \geq 2$ , and let  $X \rightarrow S$  be a smooth family of embedded curves in  $Y$ . Suppose  $D = X \times_Y Z \rightarrow S$  is finite and unramified. Fix  $s \in S$  with incidence points  $p_1, \dots, p_k$ . Write  $a_i \subset N_{Z/Y, p_i}$  for the normal image of  $T_{p_i} X_s$ . Assume  $a_i \neq 0$  and that the joint normal evaluation map*

$$T_s S \longrightarrow \bigoplus_{i=1}^k N_{Z/Y, p_i} / a_i$$

is surjective. Etale locally at  $s$ , the incidence splits into closed immersions  $D_i \simeq \Sigma_i \hookrightarrow S$ , with independent smooth centres of codimension  $c - 1$ . Near  $p_i$  there are coordinates in which

$$I_D = (x_i, u_{i1}, \dots, u_{i,c-1}), \quad I_{\Sigma_i} = (u_{i1}, \dots, u_{i,c-1}), \quad (\text{U.3})$$

where all base parameters  $u_{ij}$  are independent. If  $\mathcal{Q}_S = q_* \mathcal{O}_D$ , then

$$\text{Fitt}_0(\mathcal{Q}_S) = \prod_i I_{\Sigma_i} = \bigcap_i I_{\Sigma_i}.$$

The blow-up of this ideal is the fibre product of the individual blow-ups. It is smooth, with etale normal-crossing exceptional boundary and fibre  $\prod_i \mathbb{P}^{c-2}$  at  $s$ .

*Proof.* Over the strict henselization at  $s$ , the finite algebra splits into factors indexed by its geometric fibre. These factors and their idempotents descend to an etale neighbourhood. For each factor the unit map from the base ring has cokernel whose fibre at  $s$  is zero: finite base change identifies that fibre with the corresponding one-dimensional residue algebra. Nakayama's lemma, followed by shrinking, makes the unit map surjective. Each factor is therefore a closed immersion. This argument uses the finite algebra, not just the set of its geometric points.

The map of the curve tangent into the normal space is nonzero, so one local equation of  $Z$  has nonzero relative derivative. Its zero set is etale over the base. After an etale base change choose this as a section and use that equation as the relative coordinate  $x_i$ . Modulo  $x_i$ , the other equations become base functions  $u_{ij}$ . Their differentials give exactly the joint normal evaluation in the statement. Surjectivity makes them independent regular parameters, proving the normal form and smoothness of all intersections of the centres. In particular the length-one factors in this argument are reduced on complex geometric fibres; reducedness has not been inferred from an arbitrary higher fibre length.

Now  $\mathcal{Q}_S = \bigoplus_i \mathcal{O}_S / I_{\Sigma_i}$ . Fitting ideals of a direct sum multiply. In the displayed independent coordinates, intersection equals product: in a polynomial coordinate ring a monomial in every ideal uses a variable from each disjoint group, and the equality persists under etale flat extension. Equivalently one may verify it in the faithfully flat completion. The simultaneous blow-up is covered by charts in which, for each  $i$ , one  $u_{ij}$  is retained and the others are its multiples by new ratios. These charts are smooth and identify the fibre product with the blow-up of the product ideal. The latter equality also follows from the simultaneous Rees graph lemma. There are  $k$  independent exceptional parameters, and their zero sets have normal crossings. The fibre consists of independent choices of a line in each normal space of dimension  $c - 1$ , proving the last assertion.  $\square$

The arrangement blow-up mechanism is classical. Li's wonderful-model Theorems 1.2 and 1.3 [Li(2009)] give the normal-crossing and product-ideal description for a building set. Here we have proved the independent-centre case explicitly. The pencil-specific assertion needed to apply it at every point of  $U_{\text{red}}$  is the following.

**Lemma U.2** (Independence at distinct spectral points). *For every  $R \in U_{\text{red}}$ , with incidence points  $p_1, \dots, p_k$ , the joint normal evaluation of Lemma U.1 is surjective. In particular the local incidence centres have codimension two and meet transversely, without any additional genericity assumption on the other elementary divisors of  $R$ .*

*Proof.* Choose a nonsingular member  $Q_0$  of the pencil and write its other members as  $Q_1 - \lambda Q_0$ . Put  $T = Q_0^{-1} Q_1$  on  $V^*$ . The radicals  $K_i = \ker(T - \lambda_i)$  at distinct spectral points lie in distinct generalized eigenspaces. Their sum is direct. Restriction therefore gives a surjection

$$\text{Sym}^2 V \longrightarrow \bigoplus_i \text{Sym}^2(K_i^*) :$$

extend a basis of the direct sum of the  $K_i$  to a basis of  $V^*$  and prescribe the indicated diagonal blocks of a symmetric form. Varying  $Q_1$  with  $Q_0$  fixed realizes every one of these restrictions. At a corank-two point the normal space to the rank locus is the three-dimensional space of forms on  $K_i$ , with its harmless local projective twist. The tangent of the pencil maps to the nonzero line  $a_i$ , because the incidence is reduced. Quotienting by  $a_i$  gives the joint evaluation in the statement. A change of basis of the pencil contributes only multiples of its tangent image in each summand, so the map factors through  $T_R G = \text{Hom}(R, M/R)$ . It remains surjective after this quotient. This proves the claim also when the tangent form on  $K_i$  has rank one rather than rank two.  $\square$

**U.2. Identification of the full Hilbert graph.** Let  $G^\circ \subset U_{\text{red}}$  be the open with no corank-two incidence. A pencil over it has a unique lifted embedded curve in  $Q = \text{CQ}(V)$ . Use the polarization  $H = \bigotimes_{q=1}^{n-1} H_q$ , where  $H_q$  is the resolved exterior hyperplane bundle, and put  $P_H(l) = \binom{n}{2}l + 1$ . Let  $\Gamma_{\text{red}}$  be the reduced closure of this graph in  $U_{\text{red}} \times \text{Hilb}^{P_H}(Q)$ , and let  $\hat{\Gamma}_{\text{red}}$  be its normalization. Thus  $\hat{\Gamma}_{\text{red}} = \hat{G} \times_G U_{\text{red}}$  with the conventions of Proposition Z.1. This equality uses localization of integral closure [Stacks Project, Section 29.55, Lemmas 29.55.3–4]; it is an open restriction, not an arbitrary base-change assertion.

**Theorem U.3** (The complete multiple-incidence modification). *There is a canonical congruence-equivariant isomorphism*

$$\hat{\Gamma}_{\text{red}} \simeq B_{\text{red}} = \text{Bl}_{\text{Fitt}_0(q_* \mathcal{O}_D)} U_{\text{red}}. \quad (\text{U.4})$$

*The space is smooth, and its exceptional boundary has normal crossings etale locally. Over a pencil with  $k$  distinct reduced corank-two incidences the whole scheme fibre is  $(\mathbb{P}^1)^k$ . A point of that fibre specifies independently, at each  $p_i$ , a line through  $[a_i]$  in*

$$\pi^{-1}(p_i) = \mathbb{P}(N_{Z_{n-2}/P, p_i}) \simeq \mathbb{P}^2.$$

*Its Hilbert curve is the strict transform  $C_R$  of the pencil line, with these  $k$  embedded tails attached at their respective points. It is reduced and nodal, with no embedded components. For each tail  $F_i$  and each  $1 \leq q < n$ ,*

$$\deg_{F_i} H_q = \mathbf{1}_{q=n-1}, \quad \deg_{C_R} H_q = q - k \mathbf{1}_{q=n-1}. \quad (\text{U.5})$$

*The incidence scheme satisfies  $k \leq \lfloor n/2 \rfloor$ .*

*On a labelled etale neighbourhood the exceptional divisors  $E_i$  meet like coordinate hyperplanes; every subset occurs locally, and*

$$K_{B_{\text{red}}} = \beta^* K_{U_{\text{red}}} + \sum_i E_i.$$

*No global labelling, or global simplicity of each irreducible boundary component, is asserted. The universal curve is a flat nodal family; near its  $i$ th node its equation is  $x_i y_i = t_i$ , with  $t_i = 0$  the corresponding exceptional divisor. The base blow-up has its usual effective-Cartier image-ideal universal property. These statements classify all fibres over  $U_{\text{red}}$ , not only the fibres of selected smoothing arcs.*

*Proof.* Apply the preceding two lemmas with  $c = 3$ . Locally write  $I_i = (u_i, v_i)$ . In the simultaneous blow-up chart put  $u_i = t_i$ ,  $v_i = t_i w_i$ . Near the  $i$ th incidence the image of the ambient rank-locus ideal on the pulled-back universal line is  $(x_i, t_i)$ . Its blow-up has charts

$$\mathcal{O}_B[x_i, y_i]/(x_i y_i - t_i), \quad \mathcal{O}_B[z_i], \quad x_i = t_i z_i.$$

The incidence points have disjoint neighbourhoods on the curve, so these modifications do not interfere. Outside them the ideal is the unit ideal. The local hypersurfaces are flat over  $B$ : they are relative Cartier divisors in a smooth relative plane and their nonzero defining equations remain nonzerodivisors on every fibre. The fibrewise Cartier flatness criterion applies. These are

the standard nodal local models, so all the stated fibre properties follow. Their total spaces are smooth in the displayed independent coordinates.

Here is the scheme-level embedding needed for the Hilbert argument. If  $\mathcal{K}$  is the rank-locus ideal on the open ambient space and  $f : X_B \rightarrow P$  is evaluation, there is a surjection of graded algebras

$$f^* \left( \bigoplus_{d \geq 0} \mathcal{K}^d \right) \twoheadrightarrow \bigoplus_{d \geq 0} (\mathcal{K} \mathcal{O}_{X_B})^d. \quad (\text{U.6})$$

Its kernel removes precisely the relations annihilated on passing to the image ideal. The target is the image Rees algebra inside  $\mathcal{O}_{X_B}[T]$ ; it is not an assumed tensor-product identification. Taking Proj gives a closed immersion into the base-changed ambient blow-up, which is  $Q$  over rank at least  $n - 2$ . The universal line is already embedded in  $P \times B$ . We obtain a flat closed family of curves in  $Q \times B$ . This agrees with the lifted pencil on the dense nonincident open.

In normal coordinates its exceptional map at  $p_i$  is  $[x_i : t_i] \mapsto [x_i a_i + t_i b_i]$ . The vectors  $a_i, b_i$  are independent, and their span determines an embedded line, whose homogeneous ideal is its linear annihilator. Distinct directions modulo  $a_i$  therefore give distinct closed subschemes of the exceptional plane, not merely different parametrizations of one curve. Tails at different  $p_i$  cannot be confused. The induced proper map  $B \rightarrow \Gamma_{\text{red}}$  has finite geometric fibres and is birational. It is consequently finite, and the smooth source is the normalization of the reduced integral graph. This proves (U.4), including its scheme fibres. Both maps restrict to the identity on the dense graph, giving canonicity and equivariance and gluing the etale calculations.

The smaller minor systems have no base point, while the last minor system has one simple base point at each  $p_i$ . Subtracting these base points and adding one unit of degree on each tail gives (U.5). The determinant has vanishing order at least two at each distinct corank-two point, so  $2k \leq n$ . The nodal tree has Euler characteristic one and total  $H$ -degree  $\binom{n}{2}$ . This proves the Hilbert polynomial used above. Finally the product blow-up charts give independent parameters  $t_i$ ; the Jacobian for each codimension-two blow-up contributes one  $E_i$  to the canonical divisor. These computations also prove the local intersection statement.  $\square$

**Corollary U.4** (Multiplication systems at every node). *The ideal  $\mathcal{I}$  in (U.2) is also obtained by taking the zeroth Fitting ideal of the pushforward of the zero scheme of  $J_{h(n-2)+1}$  on the universal line, for every  $h \geq 1$ . At  $h = 1$  this is the distinguished construction without an exponent choice. On every tail the resolved  $j$ th power is the complete system of degree*

$$\delta_j = (j - h(n - 2))_+ - 2(j - h(n - 1))_+.$$

*The exceptional parameter space at a  $k$ -incidence pencil has Chow ring  $\mathbb{Z}[\xi_1, \dots, \xi_k]/(\xi_1^2, \dots, \xi_k^2)$ . In particular at a double incidence the full fibre is a smooth quadric surface  $\mathbb{P}^1 \times \mathbb{P}^1$ , not merely two independent one-parameter examples.*

*Proof.* On this rank open the universal formula gives  $J_{h(n-2)+1} = I_{n-1}$  because  $I_{n-2} = \mathcal{O}$ . Its zero scheme is exactly  $D$ . Finite pushforward and Fitting ideals give the specified centre. The power assertion is Proposition AA.3 at each disjoint tail. For clarity, a coefficient ideal generated in degree  $b$  has degree- $b$  part equal to the linear span of its actual generators. In the residual binary ring this part is  $f^{(b-h)+} \text{Sym}^{b-2(b-h)+}(\text{Sym}^2 W^*)$ ; division by the common factor therefore leaves all forms, not merely a subspace with the same base locus. The Chow ring is the projective bundle formula applied to the identified scheme fibre  $(\mathbb{P}^1)^k$ .  $\square$

## APPENDIX V. HIGHER CONTACT, THICK TAILS, AND SINGULAR MULTIPLICATION GRAPHS

Nonreduced incidence has a different Hilbert boundary from reduced multiple incidence. We give a two-parameter family of genuine symmetric pencils for every contact order  $m \geq 2$ , determine

its entire normalized Hilbert graph, and compute the nonreduced fibres and the singularities of the universal curve. The result is a slice theorem, not an assertion that the whole ambient Hilbert modification commutes with restriction to that slice.

**V.1. Symmetric linear pencils of every contact order.** Let  $E_m$  be the reversal matrix,  $N_m$  the nilpotent matrix with ones on its superdiagonal, and put

$$L_m(x) = E_m(xI_m - N_m), \quad e = e_1.$$

This is symmetric. In dimension  $n = 2m$ , consider the affine family of pencils, with  $x$  the pencil parameter and  $(u, v) \in S = \mathbb{A}^2$ ,

$$A_{u,v}(x) = \begin{pmatrix} L_m(x) + uee^t & vee^t \\ vee^t & L_m(x) - uee^t \end{pmatrix}. \quad (\text{V.1})$$

The projective pencil is obtained by homogenizing every entry to degree one. Its leading matrix is  $E_m \oplus E_m$ , so infinity is nonsingular. For  $n > 2m$  one may adjoin fixed scalar blocks  $x - \lambda_j$  with distinct nonzero  $\lambda_j$  and restrict the parameter base to a neighbourhood where these additional simple roots stay disjoint from the cluster. All assertions below then hold near the same cluster; we state the global slice formula first for  $n = 2m$ .

**Lemma V.1** (Exact contact slice). *For the family (V.1), the complement of the two  $e_1$  directions is a nonsingular block over  $\mathbb{C}[x]$ , of constant nonzero determinant. Its Schur complement is exactly*

$$\begin{pmatrix} x^m + u & v \\ v & x^m - u \end{pmatrix}. \quad (\text{V.2})$$

Consequently, on the affine pencil chart,

$$I_{n-1}(A) = (x^m, u, v), \quad \det A = c_m(x^{2m} - u^2 - v^2), \quad c_m \in \mathbb{C}^*.$$

All pencils are regular and have rank at least  $n - 2$  everywhere. Only the origin of  $S$  has a corank-two incidence; its intersection scheme has length  $m$ . Its two positive local Smith exponents are  $m, m$ . The finite incidence algebra pushed to  $S$  is  $(\mathcal{O}_S/(u, v))^{\oplus m}$ , so its zeroth Fitting ideal is  $(u, v)^m$ .

*Proof.* Delete row and column one from  $L_m(x)$ . The remaining matrix is anti-triangular with  $-1$  on its antidiagonal, hence has constant nonzero determinant. Since  $\det L_m(x) = \det(E_m)x^m$ , its scalar Schur complement is  $x^m$ ; the ratio of the two constant determinants is one. This can also be read directly by eliminating along the superdiagonal. The two kernel directions receive exactly the displayed perturbation, proving (V.2). The unit-block minor identity then gives  $I_{n-1} = I_1(\text{Schur complement}) = (x^m, u, v)$  in characteristic zero. All smaller required minors are units. The determinant and rank claims follow from the same block decomposition, including infinity. At  $u = v = 0$  the residual matrix is  $x^m I_2$ , which proves both the intersection length and the Smith exponents. The incidence algebra is  $\mathbb{C}[x, u, v]/(x^m, u, v)$ ; its  $\mathbb{C}[u, v]$ -module basis is  $1, x, \dots, x^{m-1}$  with  $(u, v)$  annihilating every basis vector. The direct-sum formula for Fitting ideals proves the last assertion.  $\square$

Let  $\Gamma_m$  be the reduced closure of the lifted-pencil graph over  $S \setminus \{0\}$  in  $S \times \text{Hilb}^{P_H}(\text{CQ}(V))$ , with  $P_H(l) = \binom{n}{2}l + 1$ , and let  $\widehat{\Gamma}_m$  be its normalization. This definition is independent of whether normalization of the global pencil graph commutes with this particular slice; no such commutation is assumed.

**Theorem V.2** (The full contact-slice Hilbert modification). *For every  $m \geq 2$  the slice graph is*

$$\widehat{\Gamma}_m \simeq \text{Bl}_0 \mathbb{A}^2 = \text{Bl}_{\text{Fitt}_0(q_* \mathcal{O}_D)} S. \quad (\text{V.3})$$



Its whole exceptional fibre is  $\mathbb{P}^1$ ; the Hilbert curves over that exceptional fibre are nonreduced. In a base chart  $u = t$ ,  $v = tw$ , the total multiplication graph is the blow-up of  $(x^m, t)$  on the pulled-back pencil surface. With  $R = \mathbb{C}[x, t, w]$ , its Rees presentation and its two projective charts are

$$\begin{aligned} \mathcal{R}(x^m, t) &= R[U, V]/(tU - x^mV), \\ D_+(U) : \quad t &= x^m y, \quad D_+(V) : \quad x^m = tz. \end{aligned} \tag{V.4}$$

Here  $U$  represents  $x^m T$  and  $V$  represents  $tT$  in  $R[T]$ . The charts glue by  $y = z^{-1}$ . The universal curve is flat over the base blow-up, is normal, and has a transverse  $A_{m-1}$  surface singularity along the section  $x = t = z = 0$  of its exceptional divisor.

For every direction  $[\alpha : \beta] \in \mathbb{P}^1$ , its special Hilbert curve is

$$C_0 \cup T_m, \quad T_m \simeq \mathbb{P}_{\mathbb{C}[x]/(x^m)}^1, \quad C_0 \cap T_m \simeq \text{Spec } \mathbb{C}[x]/(x^m). \tag{V.5}$$

The intersection is at one section of  $T_m$ ; the support of  $T_m$  is the residual line through  $[I_2]$  with second member  $\begin{pmatrix} \alpha & \beta \\ \beta & -\alpha \end{pmatrix}$  in the exceptional plane. The fibre has no embedded associated points. Its cycle is  $[C_0] + m[F]$ , where  $F = (T_m)_{\text{red}}$ . Every direction occurs, and these embedded thickened curves are pairwise distinct. The multidegrees are

$$\deg_{C_0} H_q = q - m \mathbf{1}_{q=n-1}, \quad \deg_F H_q = \mathbf{1}_{q=n-1}.$$

For  $\delta_j = (j - h(n-2))_+ - 2(j - h(n-1))_+$ , the  $j$ th resolved multiplication system on  $T_m$  is, over  $\mathbb{C}[x]/(x^m)$ , the complete system  $\mathcal{O}_{T_m}(\delta_j)$ . In particular its reduced degree is  $\delta_j$  and its degree as a cycle is  $m\delta_j$ .

*Proof.* All rank loci below corank two are avoided, so the relevant ambient complete-quadric map is the blow-up of the rank- $(n-2)$  locus. Pull the exact ideal in Lemma V.1 to  $\text{Bl}_0 S$ . In the stated chart it is  $(x^m, t)$ . These two elements are a regular sequence. The only homogeneous Rees relation is  $tU - x^mV$ : successive comparison of powers of  $T$ , using that  $x^m$  and  $t$  are relatively prime, expresses every relation as a multiple of this one. Thus the relation is saturated in the image subalgebra of  $R[T]$ , and the stated Proj charts follow. The other base chart is identical with the roles of  $u, v$  exchanged.

Both chart equations remain nonzerodivisors on every fibre of their smooth relative affine planes, so the fibrewise Cartier flatness criterion proves flatness. The first chart eliminates  $t$  and is smooth. The second is a hypersurface with singular locus  $x = t = z = 0$ , of codimension two in its total space. It is a domain and Cohen–Macaulay, so regularity in codimension one proves normality. Its transverse completed equation is  $x^m = tz$ , the  $A_{m-1}$  surface singularity. This describes a singular universal curve over a smooth parameter blow-up; the two smoothness assertions are not conflated.

At  $t = 0$  the first chart has ring  $\mathbb{C}[x, y]/(x^m y)$  and the second has ring  $\mathbb{C}[x, z]/(x^m)$ . In the first ring the ideals  $(y)$  and  $(x^m)$  intersect in  $(x^m y)$ . These pieces glue to the strict transform  $C_0$  and to the thick tail whose two rings are  $(\mathbb{C}[x]/(x^m))[y]$  and  $(\mathbb{C}[x]/(x^m))[z]$ , with  $y = z^{-1}$  and  $x$  unchanged. This proves (V.5), including its length- $m$  intersection. The local hypersurfaces of the fibre are Cohen–Macaulay of dimension one and have only their component primes as associated primes. There are no embedded points.

The Rees quotient (U.6) embeds this flat family into the fixed complete-quadric space times the base blow-up. Its generic curve is the prescribed lift. The support of each tail is the indicated residual line. Distinct  $[\alpha : \beta]$  give distinct residual lines, hence distinct embedded closed subschemes. The resulting proper map to  $\Gamma_m$  is quasi-finite and birational, so the smooth base blow-up is its normalization. This proves the whole fibre statement in (V.3). The Fitting formula follows from  $(u, v)^m$  and invariance of Proj under a positive Veronese power.

The last minor system loses a base divisor of length  $m$  on  $C_0$ , while the smaller systems lose none. The graph line bundle restricts to  $\mathcal{O}(1)$  on the residual projective line over  $\mathbb{C}[x]/(x^m)$ . It

follows that  $\chi(\mathcal{O}_{T_m} \otimes H^l) = m(l+1)$  and the intersection has Euler characteristic  $m$ . Adding to the main component gives  $\binom{n}{2}l+1$ , as required. More generally the unit-block congruence is invertible over  $\mathbb{C}[x]$  and thus over  $\mathbb{C}[x]/(x^m)$ . The binary multiplication coefficient calculation of Corollary U.4, extended to this Artin ring, spans every degree- $\delta_j$  binary form. Multiplication by the unit-block socle is a split injection on these coefficient modules: it has a unit coordinate in a trivializing frame. This proves the relative complete-system assertion, not just its reduced-fibre version. The degree- $j$  coordinates originally have the twist  $\mathcal{O}(j)$  on the pencil line; it is trivial on this local thickening. Removing their Cartier common factors produces the stated graph line bundle.  $\square$

## V.2. The singularity and the stable-map comparison.

**Proposition V.3** (Local deformation module of the multiplication graph). *At a singular point of a one-parameter direction in Theorem V.2, let  $A_m = \mathbb{C}[[x, t, z]]/(x^m - tz)$ . The abstract local cotangent deformation modules are*

$$T^1(A_m) = \mathbb{C}[[x]]/(x^{m-1}), \quad T^i(A_m) = 0 \quad (i \geq 2). \quad (\text{V.6})$$

The first equality has dimension  $m-1$ . A formally versal family of hypersurface deformations is obtained by adding  $\sum_{a=0}^{m-2} c_a x^a$  to  $x^m - tz$ . These are deformation modules of the total graph singularity, not of its fixed-base relative curve or of the raw failure algebra.

In a transverse surface the reduced tail is  $\mathbb{Q}$ -Cartier with  $F^2 = -1/m$ . Its minimal resolution introduces the  $A_{m-1}$  chain; the total fibre has multiplicities  $m$  on the strict transform of  $F$  and  $m-1, m-2, \dots, 1$  along the consecutive chain, and multiplicity one on  $C_0$ .

*Proof.* The cotangent complex of a hypersurface is represented by the conormal free module mapping to the three ambient differentials. Dualizing gives cokernel by the partial derivatives  $mx^{m-1}, -z, -t$ , hence (V.6); there are no higher terms. Lifting an equation across a small extension has no obstruction, and the displayed monomials form a basis of the cokernel, giving the formal versality assertion. This is the classical hypersurface calculation applied to the exact Rees equation.

In the first Rees chart the exceptional Cartier ideal is generated by  $x^m$ , so its Cartier divisor is  $mF$ . Its associated ideal bundle restricts to  $\mathcal{O}_F(1)$ ; thus  $mF^2 = -1$ . The standard cyclic quotient presentation can be checked directly:

$$x = ab, \quad t = a^m, \quad z = b^m, \quad A_m = \mathbb{C}[[a, b]]^{\mu_m}, \quad (a, b) \mapsto (\zeta a, \zeta^{-1} b).$$

Subdivide its two-dimensional cone by the primitive rays  $((m-i)/m, i/m)$ ,  $1 \leq i < m$ , in the quotient lattice. Consecutive cones are smooth. The internal curves form the chain with self-intersection  $-2$ , and the orders of  $t = a^m$  are  $m-i$ . The end attached to the tail has multiplicity  $m-1$ ; the next ones decrease by one. The main component meets the other, already smooth, end of the tail, where  $t = x^m y$  gives multiplicities  $m, 1$ . These charts prove the stated fibre description.  $\square$

**Theorem V.4** (A thick Hilbert tail and a covered stable-map tail). *Fix a nonzero direction  $(u, v) = \tau(\alpha, \beta)$  in the slice, with contact order  $m$ . After the base change  $\tau = s^m$ , the normalization of its total Hilbert graph is the blow-up of the smooth pencil surface at  $(x, s) = (0, 0)$ . Its special fibre is a reduced nodal union  $C_0 \cup F'$ . The induced map to complete quadrics is an embedding on  $C_0$  and a degree- $m$  map on  $F'$  onto  $F$ ,*

$$[X : S] \longmapsto [X^m : S^m] \quad (\text{V.7})$$

in the residual line coordinates. It is a stable map with tail deck group  $\mu_m$ . It has the same cycle  $[C_0] + m[F]$  as the Hilbert limit, but the latter has the nonreduced scheme structure (V.5), not the reduced image of this map.

*Proof.* The base-changed graph is the blow-up of  $(x^m, s^m)$ . This particular base change is flat, and the Rees presentation also verifies it directly. On the chart  $x^m = s^m z$  adjoining  $w = x/s$  gives the integral element  $w^m = z$  and the normal chart  $\mathbb{C}[s, w]$ , with  $x = sw$ . On the other chart  $s^m = x^m y$  adjoining  $r = s/x$  gives  $r^m = y$  and the normal chart  $\mathbb{C}[x, r]$ , with  $s = xr$ . These charts are finite and birational over the original charts and glue by  $w = r^{-1}$ . They are exactly the point blow-up charts. They also show a reduced nodal fibre after normalization.

The old graph coordinates  $[x^m : s^m]$  restrict to (V.7) on the exceptional line of the point blow-up. Away from the contact point normalization does not alter the original lifted line. Both components are nonconstant; the main component is embedded and the nonconstant tail has precisely the finite deck group  $\mu_m$ , which fixes the attachment point. Thus the map is stable. The image of  $F'$  is reduced while the flat Hilbert fibre has the explicit nilpotent  $x$  of order  $m$ , so the two objects differ scheme-theoretically even though pushforward cycles and degrees agree. This comparison is calculated here; it is not deduced from a general identification of Hilbert and stable-map compactifications.  $\square$

For the reduced-incidence local model ( $m = 1$ ), the two local descriptions coincide with the nodal model. For every  $m \geq 2$ , the length of the contact, the tail multiplicity, the transverse graph singularity  $A_{m-1}$ , and the stable tail covering degree record the same integer in four different objects. The assertions concern all orders in the specified symmetric Jordan slices. They do not classify arbitrary higher-corank contacts or the entire ambient Hilbert fibre over a nonreduced-incidence pencil.

#### APPENDIX W. POWER GRAPHS OF EVERY RANK

For  $1 \leq r \leq n$ , let  $P^{(r)} \subset \mathbb{P}(\text{Sym}^2 V)$  be the rank-exactly- $r$  locus. Define  $\mathcal{G}_{r,h}$  as the scheme-theoretic closure of

$$P^{(r)} \longrightarrow \mathbb{P}(\text{Sym}^2 V) \times \prod_{p=1}^{hr} \mathbb{P}(B_h(V)_p), \quad [A] \longmapsto ([A], [A], \dots, [A^{hr}]). \quad (\text{W.1})$$

Keeping the final power is essential when  $r < n$ . Write  $\mathcal{U}$  for the tautological rank- $r$  bundle on  $\text{Gr}(r, V)$  and  $\text{CQ}(\mathcal{U})$  for its relative complete-quadric space. For rank one this means  $\mathbb{P}(\text{Sym}^2 \mathcal{U}) = \text{Gr}(1, V)$ .

**Theorem W.1** (The rank-stratified power graph). *There is a canonical isomorphism*

$$\mathcal{G}_{r,h} \simeq \text{CQ}(\mathcal{U}) \longrightarrow \text{Gr}(r, V). \quad (\text{W.2})$$

*The final power in (W.1) recovers the image  $r$ -plane by the degree- $2h$  Pluecker embedding. In particular the graph is smooth and projective, independently of  $h$ . For its relative boundary divisors  $E_i$ ,  $1 \leq i < r$ , its power base ideals have total transforms*

$$J_p \mathcal{O}_{\mathcal{G}_{r,h}} = \mathcal{O}_{\mathcal{G}_{r,h}} \left( - \sum_{i=1}^{r-1} (p - hi)_+ E_i \right), \quad 0 \leq p \leq hr. \quad (\text{W.3})$$

*The projective power maps in this formula have their common divisors removed. All powers  $p > hr$  vanish on the rank- $r$  locus.*

*Proof.* For  $U \subset V$ , the map  $B_h(U) \rightarrow B_h(V)$  from the presentation (O.1) is injective: any linear projection  $V \rightarrow U$  splitting the inclusion induces a retraction on the quotient algebras. The injection itself is independent of that projection. Locally this gives subbundles  $B_h(\mathcal{U})_p \subset B_h(V)_p$ . If  $A$  has image  $U$  and rank  $r$ , it is a nonsingular element of  $\text{Sym}^2 U$ , and  $A^{hr}$  generates the socle  $B_h(U)_{hr} = (\det U)^{2h}$ . As  $U$  varies these lines give the homogeneous embedding

$$\text{Gr}(r, V) \hookrightarrow \mathbb{P}(\mathbb{S}_{(2h)r} V) \hookrightarrow \mathbb{P}(B_h(V)_{hr}).$$

Indeed the line at the coordinate  $r$ -plane is a highest-weight line of weight  $(2h)^r$ ; its orbit map is the Pluecker embedding followed by its degree- $2h$  linear system. In particular this is a closed immersion, not just a recovery of the image plane on closed points.

Over the recovered Grassmannian, the remaining graph is precisely the relative simultaneous power graph for  $B_h(\mathcal{U})$ . By Theorem Q.1 on each frame chart, it is  $\text{CQ}(\mathcal{U})$ . The relative graph embedding, the subbundle injections above, and the final-power Grassmannian embedding together give a closed immersion into the product in (W.1). Its dense open is the graph of  $P^{(r)}$ . Both its source and that graph closure are integral, so equality of their scheme-theoretic images proves (W.2) without a normalization step. Applying (Q.3) in rank  $r$  proves the transform formula. Local split inclusion of the coefficient bundles ensures that ambient and relative power ideals agree. The vanishing assertion follows from the socle degree of  $B_h(U)$ .  $\square$

The varieties of complete singular quadrics are classical: see [35, Definition 2.5, Remark 2.6 and Theorem 2.14]. The point of the theorem is the exact recovery of their image-plane projection and entire scheme structure from the power maps in one fixed algebra  $B_h(V)$ , simultaneously for every  $r$ . The classical orbit stratification is background, not an additional novelty claim.

#### APPENDIX X. CONTACTS AND MINIMAL INDICES OF ARBITRARY PENCILS

Let  $R = \langle A_0, A_1 \rangle \subset \text{Sym}^2 V$  be a genuine pencil,  $\Lambda = \mathbb{P}(R)$ , and  $r$  its normal rank. Its tautological symmetric map is  $A : V^* \otimes \mathcal{O}_\Lambda \rightarrow V \otimes \mathcal{O}_\Lambda(1)$ . Write

$$K = \ker A, \quad U = K^\perp \subset V \otimes \mathcal{O}_\Lambda.$$

The kernel is saturated and hence a subbundle on the smooth curve;  $U$  is the saturation of the generic image. Thus  $A$  is a generically nonsingular section of  $\text{Sym}^2 U \otimes \mathcal{O}_\Lambda(1)$ . The map on the rank- $r$  open lifts uniquely to  $\tilde{\Lambda} \rightarrow \mathcal{G}_{r,h}$  by properness. Its Grassmannian projection is the map defined by  $U$ , and its source is still  $\Lambda$ . Let  $D_i$  be its contact divisor with  $E_i$ .

**Lemma X.1** (Symmetric diagonalization with a kernel). *Over  $\mathbb{C}[[t]]$ , a symmetric matrix of rank  $r$  over the fraction field is congruent to  $\text{diag}(t^{e_1}, \dots, t^{e_r}, 0, \dots, 0)$ , with  $e_1 \leq \dots \leq e_r$ . For the local matrix of a genuine pencil,  $e_1 = 0$ . The exponents are its nonzero Smith exponents.*

*Proof.* Choose a matrix entry of smallest valuation. If a diagonal entry has that valuation, it is a pivot. Otherwise an off-diagonal entry does, and the value of the quadratic form on  $e_i + ce_j$  has the same valuation for some  $c \in \mathbb{C}$ : its leading coefficient is a nonzero polynomial in  $c$ , since 2 is invertible. Extend this primitive vector to a basis. The pivot divides every entry in its row, so symmetric row-and-column elimination clears the row and column over the ring. All remaining entries have valuation at least that of the pivot. Repeat until the remaining block is zero. Units have square roots in  $\mathbb{C}[[t]]$  and can be absorbed into the basis. Invertible congruence preserves determinantal ideals, which identify the resulting exponents with Smith exponents. A genuine pencil has a nonzero matrix at every projective parameter, giving a unit entry and  $e_1 = 0$ . This proof supplies the local congruence step rather than relying on left-right Smith form alone; compare the classical pencil congruence treatment in [36].  $\square$

For  $j \geq 0$  define the finite multiplication-syzygy map

$$C_j(R) : V^* \otimes \text{Sym}^j R^* \longrightarrow V \otimes \text{Sym}^{j+1} R^*, \quad q \longmapsto Aq, \quad \kappa_j = \dim \ker C_j(R). \quad (\text{X.1})$$

Set  $\kappa_{-1} = \kappa_{-2} = 0$ . Under a basis of  $R$  this is the block Toeplitz map with rows  $A_0 q_0$ ,  $A_0 q_i + A_1 q_{i-1}$ , and  $A_1 q_j$ ; its definition is independent of that basis.

**Theorem X.2** (Complete power-and-syzygy data). *For every complex symmetric pencil, including every singular pencil, the following formulas recover its complete congruence data. If  $\varepsilon_1, \dots, \varepsilon_c$  are its minimal indices, where  $c = n - r$ , then*

$$K \simeq \bigoplus_{a=1}^c \mathcal{O}_\Lambda(-\varepsilon_a), \quad \kappa_j = \sum_{a=1}^c (j - \varepsilon_a + 1)_+, \quad (\text{X.2})$$

$$\#\{a : \varepsilon_a = j\} = \kappa_j - 2\kappa_{j-1} + \kappa_{j-2}. \quad (\text{X.3})$$

*It is enough to use  $0 \leq j \leq \lfloor (n-1)/2 \rfloor$ . At  $x \in \Lambda$ , let  $0 = e_1(x) \leq \dots \leq e_r(x)$  be the nonzero local Smith exponents. With  $v_0 = 0$  and  $v_p(x) = \text{length}(\mathcal{O}_{\Lambda, x}/J_p)$  for  $p \leq hr$ , one has*

$$v_{hq+s}(x) = h \sum_{i=1}^q e_i(x) + se_{q+1}(x), \quad (\text{X.4})$$

$$e_i(x) = \frac{v_{hi}(x) - v_{h(i-1)}(x)}{h}, \quad \text{mult}_x D_i = e_{i+1}(x) - e_i(x). \quad (\text{X.5})$$

*In the first formula  $s = 0$  at  $q = r$  and the last term is omitted. The positive exponents at each supported point, together with the minimal indices, determine the pencil up to congruence and projective reparametrization of  $\Lambda$ . The spectral datum includes the positions of the points, not only their local partitions.*

*There is also a global conservation law:*

$$\sum_{i=1}^{r-1} (r-i) \deg D_i + 2 \deg(\Lambda \longrightarrow \text{Gr}(r, V)) = r, \quad \deg(\Lambda \longrightarrow \text{Gr}(r, V)) = \sum_a \varepsilon_a. \quad (\text{X.6})$$

*The Grassmannian degree uses its Pluecker polarization. The first summand is the dimension of the regular summand in the Kronecker congruence form. Every object and number in the theorem is invariant under the ungraded failure-algebra isomorphisms of Theorem I.B.1.*

*Proof.* We use the classical complex symmetric Kronecker congruence theorem [36]. A singular block of minimal index  $\varepsilon$  has size  $2\varepsilon + 1$  and form

$$S_\varepsilon(s, t) = \begin{pmatrix} 0 & L_\varepsilon(s, t)^t \\ L_\varepsilon(s, t) & 0 \end{pmatrix}, \quad L_\varepsilon = \begin{pmatrix} s & t & & 0 \\ & s & t & \\ & & \ddots & t \end{pmatrix},$$

where  $L_\varepsilon$  has  $\varepsilon$  rows and  $\varepsilon + 1$  columns;  $S_0$  is a zero  $1 \times 1$  block. Its kernel is generated by  $((-t)^\varepsilon, s(-t)^{\varepsilon-1}, \dots, s^\varepsilon)$  in the first block. These components have no common projective zero, so it gives  $\mathcal{O}(-\varepsilon)$  as a kernel subbundle. A regular block has zero kernel as a sheaf. Direct sums prove the first formula in (X.2). Taking sections after twist by  $\mathcal{O}(j)$  identifies them with (X.1), proving the second formula. The second finite difference of  $(j - \varepsilon + 1)_+$  is one exactly at  $j = \varepsilon$  and zero otherwise. Each singular block uses  $2\varepsilon + 1 \leq n$  dimensions, proving the stated finite bound. The canonical-form classification is used here as a classical input, not reproved or claimed as new.

Locally, symmetric diagonalization factors the matrix through the saturated  $r$ -plane  $U$  with an  $r \times r$  symmetric block. Theorem P.1, or its rank- $r$  version via the split inclusion  $B_h(U) \hookrightarrow B_h(V)$ , yields (X.4). Finite differences recover the  $e_i$ . In the relative complete-quadric chart the successive ratios of diagonal entries have valuations  $e_{i+1} - e_i$ , proving the contact formula. The torsion of the pencil cokernel has precisely the positive Smith exponents. The regular blocks of the classical congruence form are specified by these elementary divisors, and its singular blocks by the minimal indices; over  $\mathbb{C}$  there are no additional signature parameters. Retaining the supported points on  $\Lambda$  makes the statement compatible with arbitrary projective changes of the pencil basis.

The exact sequence  $0 \rightarrow U \rightarrow V \otimes \mathcal{O} \rightarrow K^* \rightarrow 0$  gives  $\deg U = -\sum \varepsilon_a$ , so the Grassmannian degree is  $\sum \varepsilon_a$ . The determinant of the induced symmetric map  $U^* \rightarrow U(1)$  is a nonzero section of  $(\det U)^2 \otimes \mathcal{O}(r)$ ; its zero divisor has degree  $r - 2\sum \varepsilon_a$ . At each point its order is  $\sum e_i = \sum_{i < r} (r - i) \text{mult}_x D_i$ . This proves (X.6). The block-size identity  $n = n_{\text{reg}} + \sum (2\varepsilon_a + 1)$  identifies the first summand with  $n_{\text{reg}}$ . Finally the intrinsic first relation extracts  $\Lambda_A$  on its source and preserves all the maps involved; changes of source lift or pencil basis give isomorphic kernel maps and equal ideals, hence the asserted invariance.  $\square$

**Example X.3** (Power contacts alone miss a singular invariant). The size-six pencils  $S_0 \oplus S_2$  and  $S_1 \oplus S_1$  both have constant rank four and no spectral torsion. All their power ideals are the unit ideal for  $p \leq 4h$  and zero for  $p > 4h$ , and every contact divisor is empty. Their kernel splittings are respectively  $\mathcal{O} \oplus \mathcal{O}(-2)$  and  $\mathcal{O}(-1)^2$ . Thus their first three syzygy nullities are  $(1, 2, 4)$  and  $(0, 2, 4)$ , distinguishing the two congruence types. Both have Grassmannian degree two, as required by (X.6); the full splitting, rather than just its degree, is necessary.

**Proposition X.4** (Finite closed tests in families). *For  $j \geq 0$  and  $0 \leq a \leq n(j+1)$ , on the pencil Grassmannian each condition  $\kappa_j \geq a$  is the closed determinantal condition on the bundle map  $C_j$  given by minors of size  $n(j+1) - a + 1$ . These conditions commute with arbitrary base change as schemes. Their fibrewise ranks need not be constant, so no unqualified assertion that the kernel bundles commute with every base change is made. Along a specialization over an integral curve,  $\kappa_j$  can only increase.*

*Proof.* Use the tautological rank-two subbundle in (X.1). Its source has rank  $n(j+1)$  and its target rank  $n(j+2)$ . Nullity at least  $a$  is equivalent to rank at most  $n(j+1) - a$ , giving the stated minors. The ideals are polynomial in the bundle map and therefore commute with arbitrary substitution. Upper semicontinuity of nullity, or specialization of these minors, gives the last assertion.  $\square$

## APPENDIX Y. BOUNDARY CONTACTS AND SPECTRAL SCHEMES

Let  $R \subset M$  be a regular pencil and  $\Lambda = \mathbb{P}(R)$ . Its map to  $P$  lifts uniquely to a morphism  $\tilde{\iota}_R : \Lambda \rightarrow \text{CQ}(V)$ : it lifts on the nonsingular open, and properness extends it uniquely across the missing points of the smooth curve. Put  $D_r(R) = \tilde{\iota}_R^* E_r$ . These are effective divisors because the generic member is nonsingular.

**Theorem Y.1** (Contact-divisor reconstruction of spectral data). *For every  $h \geq 1$ , every regular pencil, and  $0 \leq p \leq nh$ ,*

$$J_p|_{\Lambda} = \mathcal{O}_{\Lambda} \left( - \sum_{r=1}^{n-1} (p - hr)_+ D_r(R) \right). \quad (\text{Y.1})$$

*At a point  $x \in \Lambda$ , let  $0 = e_1 \leq \dots \leq e_n$  be the symmetric Smith exponents of the local pencil matrix. Then*

$$\text{mult}_x D_r(R) = e_{r+1} - e_r \quad (1 \leq r < n). \quad (\text{Y.2})$$

*Consequently the supported contact divisors determine every elementary divisor, hence the full Segre symbol. If  $v_p(x) = \text{length}(\mathcal{O}_{\Lambda, x}/J_p)$  and  $v_0(x) = 0$ , then*

$$\text{mult}_x D_r(R) = \frac{v_{h(r+1)}(x) - 2v_{hr}(x) + v_{h(r-1)}(x)}{h}. \quad (\text{Y.3})$$

*Proof.* Pull back (Q.3); equality of extended ideals under composition gives (Y.1). The pencil matrix is nonzero at every point, since  $R$  is a genuine two-dimensional subspace; hence its entries generate the unit ideal locally and  $e_1 = 0$ . Over  $\mathbb{C}[[t]]$  symmetric elimination gives a congruence to  $\text{diag}(t^{e_1}, \dots, t^{e_n})$ , absorbing unit factors by square roots. Its unique lift has, in

(Q.4),  $\tau_r = t^{e_{r+1}-e_r}$  up to a unit. This proves (Y.2); faithful flatness of completion gives the local divisor statement. Taking valuations in (Y.1) and second differences at multiples of  $h$  proves (Y.3). Recover  $e_i$  by summing the first  $i - 1$  contact multiplicities. The positive exponents, with their points on  $\Lambda$ , are precisely the elementary-divisor partitions of the spectral cokernel.  $\square$

**Example Y.2** (A boundary point and its contact order are distinct data). For a symmetric germ with exponents  $(0, 1, 3)$  and  $h = 2$ , the valuations of  $J_1, \dots, J_6$  are  $(0, 0, 1, 2, 5, 8)$ . Its two boundary contacts are 1 and 2. The germs  $\text{diag}(1, t, t^3)$  and  $\text{diag}(1, t^2, t^5)$  reach the same full-flag complete-quadric point but have contacts  $(1, 2)$  and  $(2, 3)$ . The compactification retains the flag; the arc with its divisorial contacts retains the elementary exponents. These assertions must not be conflated. For singular pencils the universal ideal theorem still holds and detects all Fitting schemes and the normal rank; Fitting ideals alone do not determine Kronecker minimal indices.

#### APPENDIX Z. A PROJECTIVE INCIDENCE COMPACTIFICATION FOR PENCILS

The complete-quadric graph concerns members of pencils. There is also a projective parameter space carrying both the original finite failure algebra and flat limits of their resolved member curves. This construction keeps the pencil and does not identify distinct pencils at a common full-coordinate-change limit.

**Proposition Z.1** (Resolved pencil curves and the failure family). *Let  $n \geq 3$ ,  $G = \text{Gr}(2, M)$ , and let  $G^\circ$  be the open set of regular pencils whose lines avoid the rank-at-most- $(n - 2)$  locus in  $P$ . There is a normal projective  $\text{GL}(V)$ -equivariant scheme  $\hat{G}$  with a projective birational morphism  $\rho : \hat{G} \rightarrow G$ , an isomorphism over  $G^\circ$ , and a flat family of subschemes  $\mathcal{C} \subseteq \hat{G} \times \text{CQ}(V)$  with the following properties. Over  $G^\circ$  it is exactly the lifted pencil line. Every fibre over  $R$  lies scheme-theoretically in  $\pi^{-1}(\mathbb{P}(R))$ . All the resolved power maps restrict to  $\mathcal{C}$ , for every  $h$ . The sharp finite failure algebra on  $G$  pulls back to a finite locally free algebra on  $\hat{G}$ , with its original multiplication.*

*Proof.* The rank-at-most- $(n - 2)$  locus has codimension three in  $P$ , so a general line avoids it; a general pencil is regular. Thus  $G^\circ$  is nonempty and dense. Over that open,  $\pi$  is an isomorphism along every pencil line and these lifts form a flat family of embedded projective lines.

For  $1 \leq q < n$ , let  $H_q$  be the pullback to  $\text{CQ}(V)$  of the hyperplane bundle for its  $q$ th exterior-power projection, taking  $q = 1$  to mean  $P$ . The bundle  $L = \bigotimes_{q=1}^{n-1} H_q$  is very ample, by the graph embedding and Segre embedding. On a lifted line over  $G^\circ$ ,  $\deg H_q = q$ , since its  $q$ -minors have no common zero. The Hilbert polynomial with respect to  $L$  is therefore  $\binom{n}{2}m + 1$ . The lifted family defines a morphism

$$G^\circ \longrightarrow \text{Hilb}^{\binom{n}{2}m+1}(\text{CQ}(V)).$$

Take the reduced closure of its graph in  $G$  times this projective Hilbert scheme, and then its normalization; this is  $\hat{G}$ . Normalization is finite here, so the result is projective and normal. The graph is already closed over  $G^\circ$  and that open is smooth, so  $\rho$  is an isomorphism there. Equivariance of the family and the uniqueness of normalization give the stated group action.

Pull back the universal Hilbert family to obtain  $\mathcal{C}$ ; it is flat. To verify the asserted scheme-theoretic incidence, pull back the linear equations for the universal pencil line to  $\mathcal{C}$ . They vanish over the dense open  $G^\circ$ . On an affine open of the integral base  $\hat{G}$ , flatness makes the coordinate modules of  $\mathcal{C}$  torsion free over the base. A section which becomes zero at the generic point is therefore zero. Applied locally to these linear equations, this proves the incidence on the whole family. The power maps are everywhere defined on  $\text{CQ}(V)$  by Theorem Q.1, so their restrictions exist. Finally pull back the finite locally free algebra of Corollary I.7.5 along  $\rho$ ; arbitrary base change preserves its rank and multiplication.  $\square$

This is an incidence compactification over the Grassmannian, with its native congruence action, not a properness assertion for the quotient stack by that action. A limiting Hilbert fibre may have extra or nonreduced components; no classification of all such fibres is used. The construction is independent of  $h$ : its target, polarization and Hilbert polynomial were specified using the  $h$ -independent complete quadric graph. The unmarked inverse in the companion reconstruction paper recovers the underlying pencil, so the incidence construction is compatible with that inverse rather than a replacement of it.

#### APPENDIX AA. THE SIMPLE CORANK-TWO INCIDENCE MODIFICATION

The ordinary-collision calculation describes specified families of limits. We now identify the entire normalized Hilbert incidence space on a natural open of the pencil Grassmannian. The result holds in every dimension. It does not require a semisimple collision or a squarefree residual pencil.

Let  $n \geq 3$ ,  $M = \text{Sym}^2 V$ ,  $P = \mathbb{P}(M)$ , and  $G = \text{Gr}(2, M)$ . Write  $Z_j \subset P$  for the symmetric rank-at-most- $j$  scheme, and put  $Z_0 = \emptyset$  when  $n = 3$ . Let  $\mathcal{U} \subset G$  consist of regular pencils  $R$  such that their lines  $L_R = \mathbb{P}(R)$  avoid  $Z_{n-3}$  and their scheme-theoretic intersections with  $Z_{n-2}$  are either empty or a single reduced point. Write  $\mathcal{P}$  for the universal line over  $\mathcal{U}$ . We use projective spaces of lines, so that the exceptional fibre of a smooth blow-up is the projectivization of the normal space.

**Lemma AA.1** (The incidence centre and its normal space). *The locus  $\mathcal{U}$  is open and contains  $G^\circ$  of Proposition Z.1. Its incidence scheme*

$$\mathcal{D} = \mathcal{P} \times_P Z_{n-2} \longrightarrow \mathcal{U}$$

*is a closed immersion with smooth image  $\Sigma_2$  of codimension two. At  $R \in \Sigma_2$ , let  $p = L_R \cap Z_{n-2}$  and let  $a_R \subset N_{Z_{n-2}/P, p}$  be the nonzero image of  $T_p L_R$ . There is a canonical isomorphism*

$$N_{\Sigma_2/\mathcal{U}, R} \simeq N_{Z_{n-2}/P, p}/a_R. \quad (\text{AA.1})$$

*Near  $(p, R)$ , after an étale change of base and coordinates, the ideal of  $\mathcal{D}$  in  $\mathcal{P}$  is*

$$I_{n-1}(A) = (x, u, v), \quad I_{\Sigma_2} = (u, v), \quad (\text{AA.2})$$

*where  $x$  is a relative line coordinate and  $u, v$  are part of a regular system of parameters on the base. The generators may be changed by an invertible matrix; the asserted equality is an equality of ideals.*

*Proof.* On  $P \setminus Z_{n-3}$  the scheme  $Z_{n-2}$  is smooth of codimension three. A nonsingular  $(n-2)$ -block reduces its equations to the three entries of a symmetric  $2 \times 2$  Schur complement. In particular these are regular local equations for its given determinantal scheme structure.

The universal line evaluation  $\mathcal{P}_G \rightarrow P$  is smooth: over a point of  $P$  its fibre is the projective space of lines through that point. Hence its inverse image of the smooth rank locus is smooth and has dimension  $\dim G - 2$ . Lines contained in the rank locus, or meeting the lower rank locus, form closed exceptional sets after the usual proper incidence projection. On the complementary open the incidence projection is finite. The condition that its fibre have length at most one is open, by upper semicontinuity for a finite module. This proves the asserted openness. On this open the unit map  $\mathcal{O}_{\mathcal{U}} \rightarrow q_* \mathcal{O}_{\mathcal{D}}$  is surjective: on each geometric fibre its target is either zero or a one-dimensional algebra with unit, and Nakayama's lemma applies to its coherent cokernel. Thus  $q$  is a closed immersion. Its source is smooth by the smooth evaluation just noted; its codimension is two.

First-order motion of a line evaluates surjectively onto  $T_p P / T_p L_R$ : in the description  $T_R G = \text{Hom}(R, M/R)$  this is restriction to the one-dimensional subspace representing  $p$ . To remain incident to the rank locus, this evaluation must vanish in  $T_p P / (T_p Z_{n-2} + T_p L_R)$ . A reduced



intersection means that the normal image of  $T_p L_R$  is nonzero. The kernel is precisely  $T_R \Sigma_2$ , since the incidence point may move along both the line and the rank locus. The quotient is therefore (AA.1).

For the local assertion choose a rank-locus equation with nonzero relative derivative. Its zero hypersurface is etale over the base near  $(p, R)$ ; after an etale base change it gives a section. Use that equation as the coordinate  $x$ . Subtract multiples of  $x$  from the other two equations. Their restrictions to the section are base functions  $u, v$ , and the ideal becomes  $(x, u, v)$ . The normal-space calculation proves that  $du, dv$  are independent. This proves (AA.2). The argument is the smooth-centre coordinate calculation, not an assertion that arbitrary graph closures commute with base change.  $\square$

The open is nonempty along its centre. For example, a diagonal pencil with two equal linear entries and all other entries having distinct zeros away from their common zero belongs to  $\Sigma_2$ . Although  $a_R \neq 0$ , its residual binary quadratic form may have rank one. This case is allowed throughout the following theorem.

**Theorem AA.2** (The full simple-incidence Hilbert modification). *For the normalized Hilbert incidence modification  $\rho : \hat{G} \rightarrow G$  of Proposition Z.1, there is a canonical isomorphism over  $\mathcal{U}$*

$$\hat{G} \times_G \mathcal{U} \simeq B := \text{Bl}_{\Sigma_2} \mathcal{U}. \quad (\text{AA.3})$$

*It is equivariant under source congruence. In particular this part of the incidence space is smooth. Its exceptional divisor  $F = \mathbb{P}(N_{\Sigma_2/\mathcal{U}})$  is a  $\mathbb{P}^1$ -bundle, and*

$$K_B = \beta^* K_{\mathcal{U}} + F, \quad \mathcal{O}_B(F)|_{F_R} = \mathcal{O}_{\mathbb{P}^1}(-1), \quad (\text{AA.4})$$

where  $\beta : B \rightarrow \mathcal{U}$  is the blow-up.

*For every  $R \in \Sigma_2$  the entire fibre of  $\rho$  is  $F_R \simeq \mathbb{P}^1$ . Its point  $[\bar{b}] \in \mathbb{P}(N_{Z_{n-2}/P,p}/a_R)$  represents the reduced nodal curve*

$$C_R \cup \ell_{[\bar{b}]}, \quad \ell_{[\bar{b}]} = \mathbb{P}(a_R + \mathbb{C}b) \subset \pi^{-1}(p) = \mathbb{P}(N_{Z_{n-2}/P,p}). \quad (\text{AA.5})$$

*Here  $C_R$  is the strict transform of  $L_R$  and  $b$  is any lift of  $\bar{b}$ . The tail attaches to  $C_R$  at  $[a_R]$ . Every line through  $[a_R]$  in this exceptional  $\mathbb{P}^2$  occurs, and no further components, embedded points, or nonreduced fibres occur over this open. The two components have exterior multidegrees*

$$\deg_{C_R} H_q = q - \mathbf{1}_{q=n-1}, \quad \deg_{\ell} H_q = \mathbf{1}_{q=n-1} \quad (1 \leq q < n). \quad (\text{AA.6})$$

*Their common Hilbert polynomial is  $\binom{n}{2}m + 1$  for  $\bigotimes_{q=1}^{n-1} H_q$ .*

*The universal curve over  $B$  is the simultaneous multiplication graph on  $\mathcal{P} \times_{\mathcal{U}} B$ . Near a node it is etale locally  $xy = t$ , where  $t = 0$  defines  $F$ . Thus it is a flat family of nodal curves, including along directions for which the residual determinant has a repeated zero. It commutes with base change as this specified flat family. The blow-up is characterized by the universal property that  $I_{\Sigma_2} \mathcal{O}_B$  is an invertible ideal; the corresponding factorization property holds for every  $T \rightarrow \mathcal{U}$  on which that image ideal is an effective Cartier ideal.*

*Proof.* On  $P \setminus Z_{n-3}$ , the complete-quadric morphism is exactly the blow-up of the smooth centre  $Z_{n-2}$ : all earlier rank blow-ups are isomorphisms there and the determinant divisor requires no blow-up. We use the scheme-level classical construction [34, Remark 2.5 and Construction 2.6], originating in [33, Theorem 6.3]. Nothing in this step declares the space of complete quadrics itself new.

Pull the universal line back to  $B$ . In a base blow-up chart  $u = t, v = tw$ , the ideal (AA.2) becomes  $(x, t)$ . Its blow-up has the two charts

$$\mathcal{O}_B[x, y]/(xy - t), \quad \mathcal{O}_B[z], \quad x = tz.$$

They glue by the usual ratio change. Their total spaces are smooth, and they are flat over  $B$ . For example  $\mathcal{O}_B[x, y]/(xy - t)$  is free as an  $\mathcal{O}_B$ -module with basis  $1, x, x^2, \dots, y, y^2, \dots$ ; reduction by the monic relation  $xy - t$  gives this basis. Its fibres at  $t = 0$  have two reduced branches and one ordinary node. The other base chart has the same description. Outside the incidence section the pulled ideal is the unit ideal. Blowing up the globally defined pulled ideal therefore gives a projective flat family  $\mathcal{C}_B$  with these local charts. All smaller minor ideals are units on this open. The universal power identity makes the product of the nontrivial power ideals an  $h^2$ th power of  $I_{n-1}$ , up to an invertible determinant ideal. Thus this same blow-up is the simultaneous multiplication graph for every  $h$ , not only the exterior graph.

There is a closed immersion  $\mathcal{C}_B \hookrightarrow \text{CQ}(V) \times B$ . Indeed the base-changed Rees algebra of the rank-locus ideal surjects onto the Rees algebra of its image on the universal line; taking Proj gives a closed immersion into the base-changed complete-quadric graph. The universal line is already closed in  $P \times B$ . Composing these two closed immersions proves the assertion. Equivalently, use the simultaneous graph lemma on the integral total space. This argument does not identify a general base change of a blow-up with the blow-up of the image ideal without the indicated Rees quotient.

In normal coordinates at  $p$ , the exceptional map is  $[x : t] \mapsto [xa_R + tb]$ , with  $b$  representing the normal direction of the base motion. Formula (AA.1) implies that  $a_R, b$  are independent. This is a line embedded in  $\mathbb{P}(N_{Z_{n-2}/P, p})$ , not a multiple cover. The strict-transform branch meets it at  $[a_R]$ . This proves (AA.5), including the converse parametrization of all lines through that point. The local charts show the absence of embedded components and multiplicities.

The minor systems of orders  $q < n - 1$  have no base point on  $L_R$ . The  $(n - 1)$ -minors have one simple common zero, by (AA.2). Their degrees on  $C_R$  are therefore  $q$  and  $n - 2$ , respectively. On the exceptional  $\mathbb{P}^2$  only the last exterior system has positive degree, namely one. This proves (AA.6). The two projective lines have Euler characteristic  $2 - 1 = 1$ , since their intersection is one reduced point. Their total degree is  $\binom{n}{2}$ .

The flat embedded family defines a morphism from  $B$  to the reduced Hilbert graph closure over  $\mathcal{U}$ . It is proper because  $B$  is proper over  $\mathcal{U}$  and the Hilbert graph is separated over that base. Its image is exactly the graph closure: it is closed and contains the dense graph over  $\mathcal{U} \setminus \Sigma_2 = G^\circ$ , while  $B$  is integral. Over an incident pencil the displayed exceptional lines are pairwise distinct as embedded curves; over a nonincident pencil the fibre is a point. The morphism thus has finite geometric fibres and is finite by [Stacks Project, Tag 02LS]. It is birational, and  $B$  is normal because its centre is smooth. A finite birational map from a normal integral scheme identifies it with the normalization: on affine charts its ring is the integral closure in the common function field. Normalization commutes with restriction to the open  $\mathcal{U}$ . This proves (AA.3), without assuming that an arbitrary nondominant normal test scheme lifts to a normalization.

All maps are specified by the universal incidence and agree on the dense nonincident locus. Their uniqueness proves canonicity and congruence equivariance. For completeness, in coordinates  $v = uw$ ,  $du \wedge dv = u du \wedge dw$ , giving the discrepancy one in (AA.4). The exceptional normal bundle is tautological of degree  $-1$ . The universal property is the usual Rees-algebra factorization for an effective Cartier image ideal [Stacks Project, Tag 0806]; here the preceding argument identifies the actual Hilbert incidence modification with that universal object. Flat pullback preserves the explicitly constructed nodal family. No assertion about unrestricted graph-closure base change is used.  $\square$

#### AA.1. Every power on every exceptional direction.

**Proposition AA.3** (Complete exceptional power systems). *Fix  $R \in \Sigma_2$  and  $p = L_R \cap Z_{n-2}$ . For  $h \geq 1$  and  $1 \leq j \leq hn$ , define*

$$\delta_j = (j - h(n - 2))_+ - 2(j - h(n - 1))_+. \quad (\text{AA.7})$$

On the full exceptional plane  $\pi^{-1}(p) \simeq \mathbb{P}^2$ , the resolved  $j$ th multiplication map is the complete degree- $\delta_j$  Veronese map into its linear span, with degree zero interpreted as constant. Its restriction to every tail in (AA.5) is therefore the complete degree- $\delta_j$  system on  $\mathbb{P}^1$ . In particular  $j = h(n-2) + 1$  detects the tail as an embedded line. These assertions include tangent lines to the residual determinant conic and attachments on that conic.

*Proof.* A unit  $(n-2)$ -block leaves a residual two-dimensional space  $W$ . Lemma AC.2 identifies the first nonzero normal coefficient at degree  $j > h(n-2)$  with the power of order  $b = j - h(n-2)$  in  $B_h(W)$ , multiplied by a nonzero unit-block socle. For  $j \leq h(n-2)$  the order-zero coefficient is constant on the exceptional fibre. In the three-variable coordinate ring of  $\text{Sym}^2 W$ , put  $\mathfrak{a} = I_1$  and  $f = \det$ . The universal ideal formula gives

$$J_b = \mathfrak{a}^b \quad (b \leq h), \quad J_b = \mathfrak{a}^{2h-b} f^{b-h} \quad (h \leq b \leq 2h).$$

The generators all have degree  $b$ . Their span is the entire degree  $b$  part of the displayed ideal. After removal of the common factor  $f^{(b-h)_+}$  it is the full space of ternary forms of degree  $b - 2(b-h)_+ = \delta_j$ . Hence it defines the complete Veronese system on the exceptional plane. Restriction of all ternary forms to any line gives all binary forms of that degree, independently of its intersection multiplicities with the conic  $f = 0$ . This proves all assertions, including the endpoint  $j = hn$  where the map is constant.  $\square$

#### APPENDIX AB. A COMPUTED BOUNDARY OF THE PENCIL INCIDENCE SPACE

We give the incidence compactification a precise universal property and compute a family of its boundary fibres in every dimension. The new boundary is selected by one power base ideal, not by pulling an unrelated algebra back to a parameter space.

**Proposition AB.1** (The normal main-component flattening property). *Let  $G^\circ \subset G = \text{Gr}(2, \text{Sym}^2 V)$  and  $\widehat{G} \rightarrow G$  be as in Proposition Z.1. Among normal integral schemes  $T$  with a dominant morphism  $f : T \rightarrow G$  for which the closure of the lifted pencil family over  $f^{-1}(G^\circ)$  is an embedded flat family in  $\text{CQ}(V) \times T$  with Hilbert polynomial  $\binom{n}{2}m + 1$ , there is a unique morphism  $T \rightarrow \widehat{G}$  over  $G$ , and that family is the pullback of its universal family. Consequently  $\widehat{G}$  is the minimal normal dominant modification with this specified embedded-flat-closure property.*

*Proof.* The flat embedded family gives a morphism to the indicated Hilbert scheme. Together with  $f$  it agrees, over a dense open of  $T$ , with the graph defining the reduced graph closure. Since  $T$  is integral, its morphism factors through this closed graph scheme: each equation vanishing on the dense open vanishes on  $T$ . The induced morphism is dominant. A dominant morphism from a normal integral scheme to an integral scheme factors uniquely through the normalization: locally, an element integral over the target ring becomes integral over the source ring and lies in its fraction field, hence belongs to the normal source ring. Thus the morphism factors uniquely through  $\widehat{G}$ . The Hilbert-scheme representing property identifies the families. Conversely its pulled-back universal family is flat; flatness over an integral base makes it the schematic closure of its restriction to the dense open. Uniqueness follows also from separatedness.  $\square$

The hypothesis of dominance is part of this property; this proposition does not assert that normalization commutes with arbitrary base change. The property is a standard consequence of the Hilbert functor and normalization. Its usefulness here is that the following calculation identifies a nontrivial fibre of the resulting modification.

Assume  $n \geq 3$ . Choose distinct nonzero numbers  $a_3, \dots, a_n$  and put  $D = \text{Spec } \mathbb{C}[[\tau]]$ . On  $\mathbb{P}_D^1$ , consider the symmetric pencil

$$A_\tau(s, t) = \text{diag}(s, s + \tau t, t - a_3 s, \dots, t - a_n s). \quad (\text{AB.1})$$

For  $n = 3$  there is just one number  $a_3$ . Write  $X = \mathbb{P}_D^1$  and let  $o$  be the point  $s = \tau = 0$  in the special fibre. Its generic pencil has  $n$  distinct rank- $(n - 1)$  spectral points, so defines a point of  $G^\circ$ .

**Theorem AB.2** (A transverse corank-two tail and its power maps). *The closure of the generic lifted curve of (AB.1) in  $\text{CQ}(V) \times D$  is*

$$Z = \text{Bl}_o X. \quad (\text{AB.2})$$

*It is flat over  $D$ , and its special fibre is the reduced nodal union  $C \cup F$ , where  $C \simeq \mathbb{P}^1$  is the lifted special pencil and  $F \simeq \mathbb{P}^1$  is an exceptional tail. There are no embedded components. If  $H_q$  is the exterior-power hyperplane bundle of order  $q$ , then*

$$(\deg_C H_q, \deg_F H_q) = \begin{cases} (q, 0), & 1 \leq q \leq n - 2, \\ (n - 2, 1), & q = n - 1. \end{cases} \quad (\text{AB.3})$$

*Thus the special fibre has the same Hilbert polynomial  $\binom{n}{2}m + 1$  as the generic lifted line.*

*The tail lies in  $E_{n-2}$ , its two intersections with  $E_{n-1}$  are distinct and reduced, and its attachment to  $C$  is neither of those two points. For any  $h \geq 1$ , let  $L_p$  be the hyperplane bundle of the resolved  $p$ th power map. Then*

$$\deg_F L_p = \begin{cases} 0, & p \leq h(n - 2), \\ \min\{b, 2h - b\}, & p = h(n - 2) + b, \quad 0 \leq b \leq 2h. \end{cases} \quad (\text{AB.4})$$

*When this degree is positive, the restricted map on  $F$  is the complete Veronese map of that degree into its linear span. Most directly, in the local coordinate  $z = s/t$  at  $o$ ,*

$$J_{h(n-2)+1}(A_\tau) = (z, \tau). \quad (\text{AB.5})$$

*Hence a single multiplication power ideal selects the blow-up centre and its exceptional component.*

*Proof.* The rank is at least  $n - 2$  on  $X$ . In a neighbourhood of  $o$  the last  $n - 2$  diagonal entries are units; all other spectral points have rank  $n - 1$  and are pairwise distinct. Consequently the extended minor ideals  $I_q$  are the unit ideal for  $q \leq n - 2$ , while

$$I_{n-1} = (z, z + \tau) = (z, \tau), \quad I_n = (uz(z + \tau))$$

near  $o$ , with  $u$  a unit. Away from  $o$ ,  $I_{n-1}$  is a unit ideal;  $I_n$  is everywhere invertible. The simultaneous exterior-power graph on the integral surface  $X$  is therefore the blow-up of  $(z, \tau)$ . This is an application of the graph-by-generators lemma, not a claim that arbitrary base change of a blow-up is a blow-up. Since the universal family of pencil lines embeds  $X$  into  $\mathbb{P}(\text{Sym}^2 V) \times D$ , this graph embeds into  $\text{CQ}(V) \times D$  and is the required scheme-theoretic closure.

On the chart with  $\tau = zv$ , the special fibre has equation  $zv = 0$ ; on the other chart,  $z = \tau w$ , it has equation  $\tau = 0$ . These charts prove flatness over the DVR, multiplicity one of both components, their transverse intersection, and absence of embedded components. The map on  $C$  is the unique lift of the special pencil. The  $q$ -minors of that pencil have degree  $q$ . They have no common factor for  $q \leq n - 2$ , and precisely one factor  $s$  for  $q = n - 1$ . This proves their degrees on  $C$ . On  $F$ , all lower maps are constant, and the  $(n - 1)$ st exterior map is linear. This gives (AB.3). For  $L = \bigotimes H_q$ , the degrees on  $C$  and  $F$  add to  $\binom{n}{2}$ . The normalization sequence of  $C \cup F$  subtracts one point, so its Euler characteristic is one and its Hilbert polynomial is as stated. Thus it is the Hilbert limit used by  $\hat{G}$ ; equivalently the generic morphism from  $D$  extends to  $\hat{G}$  by properness, with this universal fibre.

Use homogeneous coordinates  $[\alpha : \beta] = [z : \tau]$  on  $F$ . After removing the exceptional factor, the residual quadratic form on the two-dimensional kernel is  $\text{diag}(\alpha, \alpha + \beta)$ . This is a line in  $\text{CQ}(\mathbb{C}^2) = \mathbb{P}(\text{Sym}^2 \mathbb{C}^2)$ , and gives the claimed image of  $F$  in the fibre of  $E_{n-2}$ . Its determinant

$\alpha(\alpha + \beta)$  has two simple zeros  $[0 : 1]$  and  $[-1 : 1]$ , which are its intersections with  $E_{n-1}$ . The main component attaches at  $[1 : 0]$ , where this residual form is nondegenerate.

For clarity, the whole list of extended power ideals near  $o$  is

$$J_p = \begin{cases} \mathcal{O}, & p \leq h(n-2), \\ (z, \tau)^b, & p = h(n-2) + b, \ 0 \leq b \leq h, \\ (z, \tau)^{2h-b} [z(z + \tau)]^{b-h}, & p = h(n-2) + b, \ h \leq b \leq 2h, \end{cases} \quad (\text{AB.6})$$

up to an invertible factor. These are actual ideals, by Theorem P.1. In particular  $b = 1$  gives (AB.5) for every  $h$ .

It remains to check the linear system on the tail, not only its degree. On diagonal matrices the coefficient polynomials of a  $p$ th power span exactly the monomials of total degree  $p$  in the diagonal entries with each exponent at most  $h$ . For containment, use the torus-weight bound in Lemma P.2; for the reverse inclusion, take the coefficients of the independent diagonal entries of  $X$  in  $\det(X + uA)^h$ . Thus, for  $p = h(n-2) + b$ , the leading normal coefficients use exponent  $h$  in each of the  $n-2$  invertible entries and span

$$\alpha^j(\alpha + \beta)^{b-j}, \quad \max\{0, b-h\} \leq j \leq \min\{b, h\}.$$

If  $b \leq h$  these span every binary form of degree  $b$ . If  $b > h$ , remove the common factor  $[\alpha(\alpha + \beta)]^{b-h}$ ; the remaining monomials span every binary form of degree  $2h - b$ . Lower powers have nonzero constant normal term and are constant on  $F$ . This proves the complete-linear-system assertion and (AB.4).  $\square$

*Remark AB.3* (What the failure family contributes). Along (AB.1), the original finite failure algebras form the finite locally free family already constructed in Corollary I.7.5. Its pullback to  $F$  is constant; we do not count that pullback as a new interaction. Instead, its first relation canonically recovers the source and the pencil line, and the source-normalized envelope has the total-base ideal (AB.5). Resolving that multiplication ideal creates  $F$ , and the other products prescribe all the linear systems (AB.4) on it. A degeneration arc, not only its closed failure algebra, supplies this total-base ideal. Thus the calculation distinguishes information in the family from information in the special point and supplies an explicit modular Hilbert limit.

## APPENDIX AC. ORDINARY COLLISION STRATA AND THEIR HILBERT FIBRES

We compute simultaneous collisions of arbitrary multiplicities, and allow higher-order perturbations of the total family. Besides describing its reduced components, we determine every resolved power system and exhibit positive-dimensional families in a fibre of the pencil incidence modification. The calculation uses the ordinary power ideals on the two-dimensional total base.

### AC.1. A local ideal calculation.

**Lemma AC.1** (Distinct tangent factors). *Let  $\ell_1, \dots, \ell_c$  be pairwise nonproportional linear forms in two variables. For  $0 \leq j < c$ , the products of  $j$  distinct  $\ell_i$  span all binary forms of degree  $j$ . Consequently, if  $R$  is a regular local complex surface ring with maximal ideal  $\mathfrak{n} = (z, \tau)$ , and a symmetric  $c \times c$  matrix  $S$  satisfies*

$$S = zC + \tau D + O(\mathfrak{n}^2), \quad C \text{ nonsingular}, \quad \det(\alpha C + \beta D) \text{ squarefree},$$

*then  $I_j(S) = \mathfrak{n}^j$  for  $j < c$ , while  $I_c(S) = (\det S)$ .*

*Proof.* Choose  $j+1$  of the linear forms. Their products obtained by omitting one factor are linearly independent: evaluation at the zero of the omitted factor kills all other products and not that one. Their number is  $j+1$ , the dimension of the required binary-form space. For  $j = 0$  the assertion is immediate.

The endomorphism  $C^{-1}D$  is self-adjoint for  $C$  and has distinct eigenvalues. Its eigenspaces are orthogonal and none is isotropic, since otherwise  $C$  would be degenerate. A constant congruence therefore makes  $\alpha C + \beta D$  diagonal with distinct linear entries. Its  $j$ -minors span  $\text{Sym}^j(\mathfrak{n}/\mathfrak{n}^2)$ . The entries of  $S$  lie in  $\mathfrak{n}$ , so  $I_j(S) \subset \mathfrak{n}^j$  and  $I_j(S) + \mathfrak{n}^{j+1} = \mathfrak{n}^j$ . Nakayama's lemma applied to  $\mathfrak{n}^j/I_j(S)$  proves equality. The argument may be made after completion and descended by faithful flatness. The determinant assertion is its definition.  $\square$

**Lemma AC.2** (Block products and the first normal term). *For  $V = U \oplus W$ , multiplication gives an injection*

$$B_h(U) \otimes B_h(W) \longrightarrow B_h(V).$$

*If  $\dim U = r$ ,  $Q \in \text{Sym}^2 U$  is nonsingular, and  $S \in \mathfrak{n} \otimes \text{Sym}^2 W$  has a generically nonsingular linear term  $S_1$ , the first nonzero normal term of  $(Q + S)^p$ , for  $hr < p \leq h \dim V$ , is*

$$\binom{p}{hr} Q^{hr} S_1^{p-hr}.$$

*For  $p \leq hr$  its order-zero term  $Q^p$  is nonzero. These statements remain valid with a nonsingular formal unit block and after a formal congruence, with the corresponding invertible changes of coefficients.*

*Proof.* The Cartan relations on each summand vanish in  $B_h(V)$ , so the algebra map is defined. Its source is Artin Gorenstein. Its socle is the product of the two socles, and that product is nonzero in  $B_h(V)$ : in determinant apolarity this is the nonzero mixed top derivative of  $\det(X_U)^h \det(X_W)^h$ , obtained by restricting to block diagonal matrices. A nonzero ideal in an Artin local Gorenstein algebra meets its socle: successive multiplication of a nonzero ideal by the maximal ideal ends in a nonzero subspace annihilated by that maximal ideal. The kernel is therefore zero.

Expand  $(Q + S)^p$  in the injected tensor product. Powers of  $Q$  above  $hr$  vanish, whereas  $Q^{hr}$  is a nonzero socle generator. For  $p > hr$  the term with the fewest factors of  $S$  consequently has precisely  $p - hr$  such factors and is the displayed term. The unit block and a formal change of frame specialize to invertible changes on the associated graded coefficient space. This proves the last assertion as well.  $\square$

**AC.2. The simultaneous collision theorem.** Let  $D_0 = \text{Spec } \mathbb{C}[[\tau]]$ , let  $\mathcal{R}$  be a rank-two pencil subbundle of  $\text{Sym}^2 V \otimes \mathcal{O}_{D_0}$ , and put  $X = \mathbb{P}(\mathcal{R})$ . Assume its generic pencil has  $n$  distinct spectral points. Its special pencil is assumed regular. Apart from simple spectral points, suppose it has distinct points  $x_1, \dots, x_k$  of coranks  $c_a$ , where  $2 \leq c_a \leq n - 1$ . Write  $r_a = n - c_a$ .

At  $x_a$ , choose a parameter  $z_a$  on the special pencil and a unit  $r_a \times r_a$  block. Its symmetric Schur complement is required to satisfy

$$S_a = z_a C_a + \tau D_a + O((z_a, \tau)^2), \quad C_a \text{ nonsingular}, \quad \det(\alpha C_a + \beta D_a) \text{ has } c_a \text{ simple zeros.} \quad (\text{AC.1})$$

We call this an ordinary transverse cluster. Intrinsically,  $C_a, D_a \in \text{Sym}^2 W_a$ , where  $W_a = (\ker(A(x_a) : V^* \rightarrow V))^*$ . Their pencil is the first derivative of the form restricted to its radical. Taking a Schur complement does not alter that first derivative. Replacing the parameter changes the basis of this residual pencil and preserves its marked member  $[C_a]$ . Thus the hypothesis does not depend on the chosen complement or frame.

**Theorem AC.3** (Simultaneous ordinary collision fibres). *Under (AC.1), the schematic closure of the generic lifted pencil in  $\text{CQ}(V) \times D_0$  is*

$$Z = \text{Bl}_{\{x_1, \dots, x_k\}} X. \quad (\text{AC.2})$$

*It is regular and flat over  $D_0$ . Its special fibre is the reduced nodal tree*

$$C \cup F_1 \cup \dots \cup F_k,$$

where  $C$  is the lift of the special pencil and each  $F_a \simeq \mathbb{P}^1$  meets  $C$  in one point. There are no embedded components and the tails are pairwise disjoint. The tail  $F_a$  is the complete-quadric lift of the pointed residual pencil  $([\alpha C_a + \beta D_a], [C_a])$  on  $W_a$ . It lies in  $E_{r_a}$ , meets  $E_{n-1}$  in  $c_a$  distinct reduced points, meets no other boundary divisor, and attaches at the nonsingular member  $[C_a]$ .

Let  $H_q$  be the hyperplane bundle of the resolved  $q$ th exterior map,  $1 \leq q \leq n-1$ . Then

$$\deg_{F_a} H_q = (q - r_a)_+, \quad \deg_C H_q = q - \sum_a (q - r_a)_+. \quad (\text{AC.3})$$

For the product polarization  $\bigotimes_{q=1}^{n-1} H_q$  the Hilbert polynomial is  $\binom{n}{2}m + 1$ . In particular  $\deg F_a = \binom{c_a}{2}$  and  $\deg C = \binom{n}{2} - \sum_a \binom{c_a}{2}$ .

For every  $h \geq 1$  and  $1 \leq p \leq hn$ , set

$$b_a = (p - hr_a)_+, \quad s_p = (p - h(n-1))_+, \quad \delta_{a,p} = b_a - c_a s_p. \quad (\text{AC.4})$$

These integers are nonnegative. The resolved  $p$ th power restricts to the complete linear system of degree  $\delta_{a,p}$  on  $F_a$ . It is constant if that degree is zero, and otherwise the complete Veronese map into its linear span. Near  $x_a$ , with  $\mathbf{n}_a = (z_a, \tau)$  and  $f_a = \det S_a$ , the exact ideals are

$$J_p(A) = \mathbf{n}_a^{\delta_{a,p}} (f_a)^{s_p}, \quad J_{hr_a+1}(A) = \mathbf{n}_a. \quad (\text{AC.5})$$

Equalities are up to multiplication by a unit after a local frame is fixed, not up to radical or integral closure. Consequently the multiplication powers select the graph modification and every tail linear system. This description depends only on the first normal jet in (AC.1); with the special pencil fixed, higher-order terms do not change its embedded special fibre or its power maps.

*Proof.* Near  $x_a$  the unit block and its Schur complement give  $I_q(A) = \mathcal{O}$  for  $q \leq r_a$  and  $I_{r_a+j}(A) = I_j(S_a)$ . Lemma AC.1 therefore gives

$$I_{r_a+j}(A) = \mathbf{n}_a^j \quad (1 \leq j < c_a), \quad I_n(A) = (u_a f_a), \quad u_a \in \mathcal{O}^\times.$$

At every other point the rank is at least  $n-1$ , so all minor ideals of order at most  $n-1$  are units. The determinant ideal is invertible. The simultaneous exterior graph is the blow-up of the product of its base ideals by Lemma O.2. Near  $x_a$  this product is  $\mathbf{n}_a^{\binom{c_a}{2}}$ . A positive power has the same blow-up as the ideal itself, and the points are disjoint. These local identifications glue uniquely over  $X$  to (AC.2). This computes the graph on the total surface; it is not an assertion that graph closure commutes with arbitrary base change.

The tautological family of genuine pencil lines embeds  $X$  into  $\mathbb{P}(\text{Sym}^2 V) \times D_0$ . Hence the simultaneous graph embedding places  $Z$  in  $\text{CQ}(V) \times D_0$ , with exactly the prescribed generic curve. The two charts  $\tau = z_a v$  and  $z_a = \tau w$  of each point blow-up show that  $Z$  is regular. In the first chart its special fibre is  $z_a v = 0$ . Both branches have multiplicity one and intersect transversally. In the second chart only the exceptional branch occurs. The parameter  $\tau$  is a nonzerodivisor, so  $Z$  is flat over the DVR. The charts also rule out embedded components.

On  $F_a$ , divide the Schur block by the exceptional parameter. Its value is  $\alpha C_a + \beta D_a$ . All earlier unit-block data specialize to their values at  $x_a$ , independently of the direction. The complete-quadric chart thus identifies  $F_a$  with the lifted residual pencil in the fibre over  $[A(x_a)]$ . Its only residual rank drops have corank one and occur at the  $c_a$  simple determinant zeros. This proves the boundary assertions, including the attachment at  $[\alpha : \beta] = [1 : 0]$ . For  $q = r_a + j < n$ , the leading  $q$ -minors are the  $j$ -minors of this residual pencil. They have no common zero and have degree  $j$ . The lower maps are constant. On  $C$ , the common factor in the  $q$ -minors has order  $(q - r_a)_+$  at  $x_a$  and no other zeros. This proves (AC.3). The sum of the component degrees is  $\binom{n}{2}$ . The normalization sequence for a tree of  $k+1$  projective lines subtracts its  $k$  nodes, so its Euler characteristic is one. This proves the Hilbert polynomial and identifies the curve as the fibre in the Hilbert incidence space.

For the power ideals apply Theorem P.1. If  $p \leq hr_a$ , the ideal is the unit ideal. Otherwise write  $p = hr_a + hj + s$  with  $0 \leq s < h$ . For  $j < c_a - 1$ , the two minor ideals are  $\mathfrak{n}_a^j$  and  $\mathfrak{n}_a^{j+1}$ , so the power ideal is  $\mathfrak{n}_a^{hj+s}$ . For  $j = c_a - 1$  it is

$$\mathfrak{n}_a^{(c_a-1)(h-s)}(f_a)^s.$$

At  $p = hn$  it is  $(f_a)^h$ . These cases are exactly (AC.5). The value  $p = hr_a + 1$  gives the point ideal for every  $h$ .

It remains to identify the complete coefficient space on  $F_a$ . By Lemma AC.2, its first normal coefficients, for  $p > hr_a$ , are exactly those of the  $b_a$ th power of the residual pencil, multiplied by a nonzero unit-block socle vector. In the polynomial ring  $\mathbb{C}[\alpha, \beta]$ , Lemma AC.1 gives  $I_j = (\alpha, \beta)^j$  for  $j < c_a$ . The universal power identity gives the homogeneous ideal

$$(\alpha, \beta)^{\delta_{a,p}} \det(\alpha C_a + \beta D_a)^{sp}.$$

All its original power generators have degree  $b_a$ ; therefore their span is its entire degree- $b_a$  part. Remove the common determinant factor. The remaining space is exactly  $H^0(\mathbb{P}^1, \mathcal{O}(\delta_{a,p}))$ , proving the assertion about the linear system. This argument determines the map as well as its degree. All surviving leading coefficients use only the first normal jet; the higher-order terms vanish on the exceptional divisor.  $\square$

**AC.3. A modular family over a fixed pencil.** Let  $R_0$  be a regular pencil whose spectral elementary divisors all have exponent one, with repeated spectral multiplicities  $c_1, \dots, c_k \geq 2$ . Thus these are semisimple collisions, not nontrivial Jordan blocks. At  $x_a$  keep the residual space  $W_a$  and the nonsingular tangent member  $[C_a]$ . Put

$$U_a = \{\ell \text{ a line through } [C_a] \text{ in } \mathbb{P}(\text{Sym}^2 W_a) : \ell \text{ meets the determinant in } c_a \text{ simple points}\}. \quad (\text{AC.6})$$

This is a nonempty open in  $\mathbb{P}(\text{Sym}^2 W_a / \langle C_a \rangle)$ , of dimension  $\binom{c_a+1}{2} - 2$ .

**Theorem AC.4** (Residual-pencil moduli in an incidence fibre). *The product  $U = \prod_a U_a$  parametrizes an algebraic flat family of distinct embedded Hilbert limits over the fixed pencil  $R_0$ . Each is the fixed main curve with the tails prescribed by its residual lines, and each is realized by an ordinary transverse one-parameter smoothing of  $R_0$ . Hence, for the normal incidence modification  $\rho : \hat{G} \rightarrow G$ , one has*

$$\dim \rho^{-1}(R_0) \geq \sum_a \left( \binom{c_a+1}{2} - 2 \right). \quad (\text{AC.7})$$

*Distinctness is detected by the resolved multiplication power  $J_{h(n-c_a)+1}$  on the  $a$ th tail. The statement describes a specified family in the fibre, not the entirety of the fibre or its normalization.*

*Proof.* Since  $C_a$  is nonsingular, lines through  $[C_a]$  are parametrized by the displayed projective quotient. The condition on determinant zeros is open and nonempty, as is seen by choosing  $D_a$  diagonal with distinct eigenvalues relative to  $C_a$ .

The complete-quadric morphism is an isomorphism over forms of rank at least  $c_a - 1$ . Each line in  $U_a$  stays in this open, and thus its universal line has an algebraic lift. The fibre of  $\text{CQ}(V) \rightarrow \mathbb{P}(\text{Sym}^2 V)$  over  $[A(x_a)]$  contains the residual  $\text{CQ}(W_a)$ : the nondegenerate first quotient form is fixed, and its radical carries the remaining complete form. This follows also by taking the unit block in the Schur charts used in the preceding proof. Thus the universal lifted lines give an algebraic family of embedded tails, each passing through the fixed attachment corresponding to  $[C_a]$ .

Adjoin the fixed main curve. The scheme-theoretic intersection of each tail family with that main curve is just this reduced section. Indeed the main curve has contact order one with  $E_{r_a}$ ,



since its local nonzero Smith exponents are 0 and 1, whereas the tail lies inside that divisor. The tails lie over different points, hence do not intersect each other. The union exact sequence

$$0 \longrightarrow \mathcal{O}_{C \cup \bigcup F_a} \longrightarrow \mathcal{O}_{C \times U} \oplus \bigoplus_a \mathcal{O}_{F_a} \longrightarrow \bigoplus_a \mathcal{O}_{\{\text{attachment}_a\} \times U} \longrightarrow 0$$

proves flatness over  $U$ : the two terms on the right are flat, so its kernel is flat. The degrees from (AC.3) give the required fixed Hilbert polynomial. We therefore obtain a morphism  $U$  to the Hilbert scheme. The embedded fibre over  $[A(x_a)]$  recovers its residual tail. Its first nonconstant exterior map, equivalently the power of index  $h(n - c_a) + 1$ , has degree one and recovers the line  $\ell$  itself. Distinct parameters give distinct embedded curves.

To see that these Hilbert points belong to the graph closure defining  $\widehat{G}$ , choose a nonsingular member of  $R_0$ . Semisimplicity makes its spectral eigenspaces orthogonal, so in a congruence frame the pencil is a direct sum of blocks  $(z - x_a)C_a$  and simple blocks. Choose any representative  $D_a$  of each residual line and replace these blocks by  $(z - x_a)C_a + \tau D_a$ . Each cluster splits into  $c_a$  simple spectral points and distinct clusters stay separated. Theorem AC.3 identifies the Hilbert limit with the constructed curve. This realizes every parameter by a smoothing, independently on all the blocks.

Let  $\overline{G}$  denote the reduced Hilbert graph before normalization. The morphism from the smooth, reduced scheme  $U$  factors through its fibre over  $R_0$ : the defining equations vanish at every closed point by the smoothing construction, hence vanish on  $U$ . Its geometric fibres are zero-dimensional by the distinctness just proved; its image consequently has dimension  $\dim U$ . Normalization  $\widehat{G} \rightarrow \overline{G}$  is finite and surjective, and the preimage of this image has the same dimension. This gives (AC.7). We use this finite preimage, rather than asserting that a nondominant map from a normal scheme must lift to a normalization. Such an assertion would not be a valid normalization universal property.  $\square$

**Example AC.5** (Triple collision). For one semisimple triple collision,  $c = 3$ , the tail has exterior multidegree  $(0, \dots, 0, 1, 2)$  and total degree three. At  $h = 2$  its successive residual powers  $b = 0, \dots, 6$  have degrees

$$0, 1, 2, 3, 4, 2, 0.$$

The residual lines through a fixed nonsingular ternary quadratic form form an open in  $\mathbb{P}^4$ , and give a four-dimensional family of distinct Hilbert limits over the same special pencil. Every member has three distinct spectral points on its tail and one nonspectral attachment. This is a family of embedded limits, not just a count of branches of a normalization fibre.

The special finite failure algebra remains  $A_{R_0}$  throughout this family of limits. What varies is the total-family first-relation geometry: its intrinsic envelope produces the maximal-ideal centre, its successive products give the tail systems, and their degree-one member recovers the residual line. Thus the family supplies information not contained in its closed algebra alone. The construction is equivariant under source congruence and under reparametrization of the pencil, but the dimension bound concerns the embedded incidence space, not a quotient by its stabilizer.

#### APPENDIX AD. SINGULARITIES OF THE UNIVERSAL POWER-ZERO SCHEMES

There is an explicit further consequence for all the universal ideals, which we record separately from the geometry of the full reciprocal normalization fibre. Multiplier ideals are taken on the affine smooth space  $M$ , and  $I_{r+1}^{(a)}$  denotes symbolic power, with exponent zero interpreted as the unit ideal.

**Corollary AD.1** (Multiplier ideals and threshold). *For  $1 \leq p \leq nh$ , put  $w_r = (p - hr)_+$  and  $c_r = \binom{n-r+1}{2}$  for  $0 \leq r < n$ . For every real  $b \geq 0$ ,*

$$\mathcal{J}(M, b \cdot J_p) = \bigcap_{r=0}^{n-1} I_{r+1}^{(\max\{0, \lfloor bw_r \rfloor + 1 - c_r\})}, \quad (\text{AD.1})$$

$$\text{lct}_M(J_p) = \min_{\substack{0 \leq r < n \\ p > hr}} \frac{\binom{n-r+1}{2}}{p - hr}. \quad (\text{AD.2})$$

*Proof.* Blow up the affine origin and then the strict rank loci. This is the affine complete-quadratic resolution, whose charts are (Q.4) with one additional scalar coordinate. Let  $E_0$  be the divisor over the origin. The same minor calculation now includes  $r = 0$ , and the total transform of  $J_p$  is  $\sum_{r=0}^{n-1} w_r E_r$ . The discrepancy along  $E_r$  is  $c_r - 1$ : over the generic rank- $r$  point the earlier blow-ups are isomorphisms, and the smooth rank- $r$  locus has codimension  $c_r$ ; for  $r = n - 1$  this is the strict determinant divisor and the discrepancy is zero. All boundary divisors have normal crossings, so this is a log resolution.

The defining pushforward formula for the multiplier ideal requires  $\text{ord}_{E_r}(f) \geq \lfloor bw_r \rfloor + 1 - c_r$ . At the generic rank- $r$  point this valuation is the order in the regular local ring with maximal ideal  $I_{r+1}$ , so the condition on polynomials is exactly membership in the indicated symbolic power. Intersecting the conditions proves the first formula; the first value at which one becomes nontrivial proves the second. Equivalently this is the specialization, via Theorem P.1, of the classical symmetric-determinantal multiplier-ideal calculation [32, §2.4 and Theorem 4.8].  $\square$

The ideal formula, graph identification and contact reconstruction are statements about the entire universal coefficient space. They do not supply the conductor of every multivariate common-divisor image or the Hilbert function of every full  $F_{g,k}(W)$ . Those objects remain in the preserved incidence theory; the present results connect their exact quadratic sections to a native global boundary and its spectral schemes.

#### APPENDIX AE. SPECTRAL DIVISOR FIBRES AND THE RECOVERY OF DEGENERATION TYPE

The preceding completion concerns the native congruence action. To retain degeneration type before passage to a coarse quotient, we associate a tower of binary divisor incidences to a regular pencil. This construction is intrinsic after the unmarked inverse and is unchanged by a simultaneous change of the pencil and source frames.

Let  $\Lambda = \mathbb{P}(R)$  be the pencil line and let  $\mathcal{T}_R$  be the cokernel of the universal symmetric map on  $\Lambda$ . A regular pencil means that its determinant is not identically zero. Thus  $\mathcal{T}_R$  is a torsion sheaf. Define  $D_j(R)$  by

$$\mathcal{I}_{D_j(R)} = \text{Fitt}_j(\mathcal{T}_R), \quad 0 \leq j \leq n, \quad (\text{AE.1})$$

where  $D_n = \emptyset$ . These are effective divisors on the smooth curve  $\Lambda$ . Write  $s_j = \deg D_j$ . If  $s_j > 0$ , let  $G_j$  be its binary equation and put

$$J_j = G_j H^0(\Lambda, \mathcal{O}_\Lambda(1)) \subset H^0(\Lambda, \mathcal{O}_\Lambda(s_j + 1)).$$

Changing the scalar representative of  $G_j$  does not change  $J_j$ . Let  $\mathcal{F}_j(R)$  be the whole scheme fibre at  $J_j$  of the degree-one divisor normalization  $\nu_1$ , with  $r = 2$  and  $D = s_j + 1$ . For  $s_j = 0$  put  $\mathcal{F}_j = \emptyset$ .

**Theorem AE.1** (A fibre-theoretic Segre invariant). *There is a natural identification of finite schemes over  $\Lambda$*

$$\mathcal{F}_j(R) \simeq D_j(R). \quad (\text{AE.2})$$

At a spectral point  $x$ , write the positive elementary divisors of  $\mathcal{T}_R$  as  $\lambda_{x,1} \geq \cdots \geq \lambda_{x,r_x} > 0$ , and extend this sequence by zeroes. Then

$$(\mathcal{F}_j(R))_x = \text{Spec } \mathbb{C}[\epsilon]/(\epsilon^{b_{x,j}}), \quad b_{x,j} = \sum_{i>j} \lambda_{x,i}, \quad (\text{AE.3})$$

with the empty-scheme convention when  $b_{x,j} = 0$ . Consequently

$$\lambda_{x,j+1} = b_{x,j} - b_{x,j+1}. \quad (\text{AE.4})$$

The tower, with its maps to the same intrinsic spectral line, therefore determines the full Segre symbol. Conversely, the elementary-divisor partitions together with their spectral support on that line determine the tower. The tower retains the projective configuration of the support as well as the Segre symbol. It is an invariant of the unmarked finite failure algebra.

The construction is compatible with base change on every family stratum where the Fitting ideals in (AE.1) define relative effective Cartier divisors of fixed degrees. On such a stratum  $\mathcal{F}_j$  is finite locally free of degree  $s_j$ . These hypotheses are part of the relative statement; no assertion of flatness across a jump of  $s_j$  is intended.

*Proof.* Over the discrete valuation ring  $\mathcal{O}_{\Lambda,x}$ , Smith normal form gives diagonal entries with exponents  $\lambda_{x,r_x}, \dots, \lambda_{x,1}$  and  $n - r_x$  zero exponents. The minors of size  $n - j$  have greatest common divisor of order  $\sum_{i>j} \lambda_{x,i}$ . This proves the formula for  $D_j$ , including its nonreduced structure. It is the usual Fitting-ideal description of elementary divisors, used here on the intrinsic spectral line.

The residual two-space  $H^0(\mathcal{O}_{\Lambda}(1))$  is gcd-free. Hence Theorem AI.1 identifies the degree-one normalization fibre at  $J_j$  with the scheme of linear factors of  $G_j$ . On an affine chart write a monic linear factor as  $z - a$ . Monic division gives  $G_j = (z - a)H + G_j(a)$ , so the entire factor scheme is  $G_j(a) = 0$ , not its reduction. These descriptions agree on overlaps of the projective line. They give (AE.2) and the local algebra (AE.3). Taking successive differences proves (AE.4). The common map to  $\Lambda$  is essential: forgetting it separately in each member of the tower would discard the alignment of the spectral supports.

Pencil frame changes act on  $\Lambda$  by projective transformations; source congruence changes give isomorphic cokernels and preserve all Fitting ideals. The intrinsic inverse of Theorem I.8.3 therefore defines this invariant from the unmarked algebra. The equivalence with Segre data uses the classical elementary-divisor convention of [4, §5].

For a family with the stated Cartier and degree conditions, the same monic division argument works over any base algebra, including one with nilpotents. It represents the divisor's functor of points, so it commutes with base change. A relative effective divisor of fixed degree on a projective-line bundle is finite locally free of that degree. Local choices of  $G_j$  differ by units and glue. Formation of Fitting ideals itself commutes with base change; requiring them to remain flat Cartier divisors is precisely what permits the additional finite-flat conclusion.  $\square$

**Example AE.2** (Equal discriminant, different fibre towers). At a root of multiplicity four, the partitions  $(3, 1)$  and  $(2, 2)$  have the same zeroth fibre  $\mathbb{C}[\epsilon]/(\epsilon^4)$ . Their first fibres are respectively  $\mathbb{C}$  and  $\mathbb{C}[\epsilon]/(\epsilon^2)$ ; all higher fibres are empty there. Thus the discriminant alone does not detect the distinction, whereas the divisor-fibre tower does. Both are realized by regular symmetric pencils: for a block of size  $l$ , take the reversal matrix  $H_l$  and the symmetric pair  $(H_l, H_l J_l(\alpha))$ , where  $J_l(\alpha)$  is a Jordan block. Direct sums realize every indicated partition. Additional simple blocks may be appended. The example is a statement on the regular-pencil stack, not a claim that distinct Segre strata always remain distinct after strictly semistable GIT identification.

*Remark AE.3* (The two divisor constructions). The primitive first relation of an actual pencil has exact common factor  $(\det T)^{n-1}$ , by Lemma I.M.1. Its common-factor degree does not jump

when the pencil changes Segre type. The divisors  $D_j$  above instead lie on the recovered spectral line and define auxiliary binary first-relation subspaces  $J_j$ . Thus (AE.4) relates pencil data to intrinsically induced divisor ramification, not to a nonexistent change in the conductor of the original exact-gcd locus. The identification supplies a moduli consequence of the all-binary-fibre theorem while respecting this distinction.

#### APPENDIX AF. COMMON-DIVISOR STRATA AND INTRINSIC FAMILIES

The coefficient condition in Theorem I.J.1 is not imposed in this section. We instead retain the common divisor of a space of first relations. This geometric condition gives a larger smooth parameter space, includes relations without either factor symmetry, and has a scheme-theoretic inverse. The distinction between a stratum and the condition on its geometric points is essential when the parameter base is nonreduced.

Write  $S_j(E) = \text{Sym}^j E$ , where  $E$  is a complex vector space. For  $r \geq 2$ , denote by

$$X_{h,r}(E) \subset \text{Gr}(r, S_h(E))$$

the locus of subspaces with greatest common divisor equal to 1, up to a scalar. Empty Grassmannians are omitted throughout. When  $h = 0$  this notation can also be used for  $r = 1$ .

**Lemma AF.1** (Universal common divisor). *Let  $D = g + h$ . The spaces of  $r$  degree- $D$  forms whose exact greatest common divisor has degree  $g$  form a locally closed reduced subscheme  $Z_g$  of  $\text{Gr}(r, S_D(E))$ . Multiplication induces an isomorphism*

$$\mathbb{P}(S_g(E)) \times X_{h,r}(E) \xrightarrow{\sim} Z_g, \quad ([f], J) \mapsto fJ. \quad (\text{AF.1})$$

*In particular, the common-divisor line and the residual subbundle of a family with classifying map in  $Z_g$  are recovered uniquely and commute with arbitrary complex base change.*

*Proof.* For each  $a$ , multiplication from  $\mathbb{P}(S_a(E)) \times \text{Gr}(r, S_{D-a}(E))$  has closed image: its domain is projective. The union for  $a > g$  is closed. Remove this union from the image for  $a = g$ , and give the resulting locally closed set its reduced structure. Its points are exactly those in the statement. The inverse image of this set under multiplication is  $\mathbb{P}(S_g(E)) \times X_{h,r}(E)$ . The same closed-image argument with  $D = h$  shows that  $X_{h,r}(E)$  is open.

The restricted multiplication map is proper, by base change. It is injective on geometric points over every algebraically closed extension of  $\mathbb{C}$ , since a greatest common divisor is unique up to scalar in a polynomial ring. Its differential is injective as well. Indeed, a tangent vector in its kernel satisfies

$$\dot{f}j + f\dot{j} \in fJ \quad (j \in J).$$

Consequently  $f \mid \dot{f}j$  for every  $j \in J$ . For each irreducible factor of  $f$ , some element of  $J$  is not divisible by that factor. Comparison of its valuation gives  $f \mid \dot{f}$ . Equality of degrees makes  $\dot{f}$  a scalar multiple of  $f$ ; then  $\dot{J} = 0$  in  $\text{Hom}(J, S_h(E)/J)$ .

In characteristic zero the geometric-point injectivity and this differential calculation imply that the map is universally injective and unramified. A proper, universally injective, unramified morphism is a closed immersion [12, Lemma 41.7.2]. Here its image is all of the reduced target. The defining ideal of a closed immersion with this support is zero in a reduced scheme. This proves the isomorphism, including its assertion for nonreduced test schemes. Equivalently, one can check properness on the base change of the original projective multiplication map; that base change has the same underlying open preimage. Its source is reduced because it is this open subscheme of the original smooth product.  $\square$

*Remark AF.2.* A map from a nonreduced scheme is required to factor through the locally closed scheme  $Z_g$ , not merely to have geometric points in its underlying set. These are different requirements. For example, over  $\mathbb{C}[\epsilon]/(\epsilon^2)$  the span of  $x^2 + \epsilon y^2, xy$  specializes to a pair with

common divisor  $x$ , but has no factor  $x + \epsilon(ax + by)$ . Comparison of the  $y^2$  coefficient in the first form proves this. The stratum in Lemma AF.1 excludes this transverse infinitesimal family. No use of a fibrewise radical can replace its scheme structure.

Now let  $\dim V = \dim U = n \geq 3$ ,  $E = V^* \otimes U$ , and  $\delta = \det T$ . Denote by  $G \subset \mathrm{GL}(E)$  the stabilizer of the line  $\mathbb{C}\delta$ . By Lemma I.4.1,

$$G^\circ = (\mathrm{GL}(V) \times \mathrm{GL}(U))/\mathbb{G}_m, \quad G/G^\circ = \mathbf{Z}/2\mathbf{Z}. \quad (\text{AF.2})$$

The second component exchanges the two tensor factors; the diagonal scalars in the displayed product act trivially on the matrix space. Fix  $s \geq 1$  and  $D = sn + h$ . Inside  $Z_{sn}$ , impose the condition that the common-divisor line is the  $s$ -th power of a determinant form after a linear change of the  $n^2$  variables. Call the resulting locally closed scheme  $Z_{s,h,r}^{\det}$ .

**Theorem AF.3** (Intrinsic determinant-divisor stratum). *There are canonical isomorphisms of schemes and algebraic stacks*

$$Z_{s,h,r}^{\det} \simeq \mathrm{GL}(E) \times^G X_{h,r}(E), \quad [Z_{s,h,r}^{\det} / \mathrm{GL}(E)] \simeq [X_{h,r}(E) / G]. \quad (\text{AF.3})$$

*They hold over arbitrary complex parameter schemes. The stack is smooth. At a residual relation  $J$ , its tangent complex, in degrees  $-1, 0$ , is*

$$[\mathrm{Lie}(G) \longrightarrow \mathrm{Hom}(J, S_h(E)/J)], \quad X \longmapsto (j \mapsto Xj \bmod J). \quad (\text{AF.4})$$

*No invariance of  $J$  under either full factor group is assumed. The equivalence identifies stabilizers, infinitesimal families, and specializations whose classifying maps remain in this stratum.*

*Proof.* The power map  $\mathbb{P}(S_n(E)) \rightarrow \mathbb{P}(S_{sn}(E))$ ,  $[f] \mapsto [f^s]$ , is a closed immersion in characteristic zero. It is proper and geometrically injective, and its differential has kernel given by  $sf^{s-1}\dot{f} \in \mathbb{C}f^s$ , hence by  $\dot{f} \in \mathbb{C}f$ . The same closed-immersion criterion as above applies. The orbit of  $[\delta]$  is the smooth locally closed homogeneous space  $\mathrm{GL}(E)/G$ . The preimage of its power orbit under the first projection in (AF.1) is therefore a locally closed subscheme. It is exactly  $Z_{s,h,r}^{\det}$ .

Pulling back along the  $G$ -torsor  $\mathrm{GL}(E) \rightarrow \mathrm{GL}(E)/G$  trivializes the common-divisor form, leaving the unrestricted residual relation  $J \in X_{h,r}(E)$ . Descent gives the first isomorphism in (AF.3); descent of equivariant maps and morphisms gives the second. This proof uses morphisms of schemes, not a comparison only on complex points. It also supplies local lifts of all isomorphisms over an fppf cover.

The space  $X_{h,r}(E)$  is open in a Grassmannian and  $G$  is smooth. The quotient is algebraic and smooth [13, Theorem 97.17.2]; see also [14, Lemma 101.33.7]. For completeness, write a first-order residual subspace as the graph of  $\epsilon\phi : J \rightarrow S_h(E)/J$ . An infinitesimal change  $1 + \epsilon X$  replaces  $\phi$  by  $\phi + X|_J \bmod J$ . This proves the tangent complex. Its kernel records infinitesimal automorphisms and its cokernel records first-order isomorphism classes. The same torsor description proves compatibility with specialization. No assertion is made about limits outside the specified common-divisor stratum.  $\square$

The next step connects these parameter spaces with ungraded finite algebras. It is useful to state the family condition intrinsically, since an absolute nilradical is unsuitable over a nonreduced base. Put  $e = \dim E$  and  $\ell = \binom{e+D}{D} - r$ . Consider finite locally free commutative unital algebras  $\mathcal{A}$  of rank  $\ell$  on a complex scheme  $T_0$ , subject to the following conditions. The map  $\epsilon(a) = \ell^{-1} \mathrm{tr}(M_a)$  is multiplicative. For  $\mathcal{N} = \ker \epsilon$ , one has  $\mathcal{N}^{D+1} = 0$ ; all quotients of its power filtration are locally free;  $\mathcal{E} = \mathcal{N}/\mathcal{N}^2$  has rank  $e$ ; and multiplication gives isomorphisms  $\mathrm{Sym}^j \mathcal{E} \simeq \mathcal{N}^j / \mathcal{N}^{j+1}$  for  $j < D$ . In degree  $D$  its kernel has rank  $r$ . These conditions specify the maximal-lower-Hilbert-function stratum; they do not assert that arbitrary smoothings of a local algebra remain in it.

**Proposition AF.4** (Relative first-relation algebra). *Every such algebra is locally isomorphic, over its base, to*

$$\mathcal{A}(\mathcal{K}) = \mathrm{Sym} \mathcal{E} / ((\mathcal{K}) + \mathfrak{m}^{D+1}), \quad \mathcal{K} = \ker(\mathrm{Sym}^D \mathcal{E} \longrightarrow \mathcal{N}^D).$$

*Its trace augmentation,  $\mathcal{E}$ , and  $\mathcal{K}$  are intrinsic and compatible with arbitrary base change in this stratum. For two such algebras, taking the linear part gives a locally surjective map of isomorphism sheaves to the sheaf of isomorphisms of  $(\mathcal{E}, \mathcal{K})$ . Its kernel is the smooth substitution group whose underlying scheme in a local presentation is  $\mathrm{Hom}(\mathcal{E}, \mathcal{N}^2)$ .*

*After quotienting morphisms by this tangent-identity kernel and fppf stackifying, the algebras with first-relation map in  $Z_{s,h,r}^{\det}$  have stack  $[X_{h,r}(E)/G]$ .*

*Proof.* Locally lift a basis of  $\mathcal{E}$  to  $\mathcal{N}$ . Nilpotence and multiplication on the successive quotients make the induced map from the truncated symmetric algebra surjective. A relation with a nonzero term of least degree  $j < D$  would give a nonzero kernel in degree  $j$  of the associated graded algebra, which is impossible. Its entire kernel therefore lies in degree  $D$  and is precisely  $\mathcal{K}$ . Changing a lift by an element of  $\mathcal{N}^2$  does not alter a product of  $D$  generators, since the difference has degree at least  $D + 1$ . This also proves the asserted uniqueness of the first relation.

In this presentation multiplication by a positive-degree element is strictly filtration-increasing. Its trace is zero over any base ring. Multiplication by a scalar has trace  $\ell$  times that scalar, proving that the presentation's augmentation is the specified intrinsic trace augmentation. All filtration sequences are locally split sequences of vector bundles, so their kernels and quotients commute with base change.

An isomorphism must preserve trace and induces a linear map carrying  $\mathcal{K}$  to  $\mathcal{K}'$ . Conversely such a map lifts locally to an isomorphism by the homogeneous presentations. All other lifts are obtained by substituting generators plus arbitrary elements of  $\mathcal{N}^2$ ; the inverse is constructed degree by degree in a finite filtration. This gives the stated kernel and its smoothness, without replacing its substitution law by addition. After identifying morphisms with the same linear part, frames of  $\mathcal{E}$  give the quotient  $[Z_{s,h,r}^{\det} / \mathrm{GL}(E)]$ . Theorem AF.3 completes the proof.  $\square$

*Remark AF.5.* The original algebra stack is not this quotient stack: it has the additional nonlinear inertia just computed. The phrase *effective first-relation stack* will refer to the specified quotient on morphisms followed by stackification. The construction retains the full automorphism group before taking that quotient. It does not identify algebras with the same geometric fibres, nor does it discard nilpotent parameters. In particular, its relation to a stratum of first relations is stronger than a bijection of isomorphism classes over  $\mathbb{C}$ .

## APPENDIX AG. THE COMMON-DIVISOR BOUNDARY AND ITS NORMALIZATION

The exact-common-divisor stratum has a canonical completion which need not be normal. Its normalization records a divisor, and its failure to be an isomorphism is controlled by two different phenomena: choices of that divisor and overlap with the residual common factor. We establish these assertions for arbitrary spaces of forms. No determinant, Cauchy, or Pluecker equation is imposed in this section.

Let  $E$  have dimension  $e \geq 2$ , put  $S_j = \mathrm{Sym}^j E$  and  $s_j = \dim S_j = \binom{e+j-1}{j}$ , and fix

$$g, h \geq 1, \quad D = g + h, \quad 2 \leq r \leq s_h.$$

Write  $\mathcal{G} = \mathrm{Gr}(r, S_D)$ . A *degree- $g$  divisor incidence* over a complex scheme  $T_0$  consists of a line subbundle  $\mathcal{L} \subset S_g \otimes \mathcal{O}_{T_0}$  and a rank- $r$  subbundle  $\mathcal{K} \subset S_D \otimes \mathcal{O}_{T_0}$  such that

$$\mathcal{K} \longrightarrow (S_D \otimes \mathcal{O}_{T_0}) / (\mathcal{L} \cdot S_h) = 0. \tag{AG.1}$$

Subbundle always means that the quotient is locally free. In particular, the form generating  $\mathcal{L}$  is nonzero on every geometric fibre. Condition (AG.1) is the vanishing of a bundle map, not a condition imposed only on closed points.

**Proposition AG.1** (Canonical divisor incidence). *The incidence functor is represented, with its full scheme structure, by*

$$\mathcal{X}_g = \mathbb{P}(S_g) \times \mathrm{Gr}(r, S_h).$$

*Its forgetful morphism is multiplication*

$$\nu_g : \mathcal{X}_g \longrightarrow \mathcal{G}, \quad ([f], J) \longmapsto fJ. \quad (\text{AG.2})$$

*This morphism is finite. Its scheme-theoretic image  $Y_g$  is an integral projective scheme. If  $K$  is a geometric point of  $Y_g$  and  $F_K = \gcd(K)$ , the underlying set of  $\nu_g^{-1}(K)$  is*

$$\{[f] : f \mid F_K, \deg f = g\}, \quad (\text{AG.3})$$

*with residual space  $J = K/f$ . The incidence construction and its scheme-theoretic fibres commute with arbitrary complex base change.*

*Proof.* On  $\mathbb{P}(S_g)$ , multiplication from the tautological line tensored with  $S_h$  to  $S_D$  has constant rank  $s_h$ . Indeed, multiplication by a nonzero polynomial over a field is injective. A nonzero maximal minor locally splits this map, so its image and cokernel are vector bundles, also after a nonreduced base change. The closed incidence (AG.1) is therefore the relative Grassmannian of rank- $r$  subbundles of  $\mathcal{L} \otimes S_h$ . Tensoring by  $\mathcal{L}^{-1}$  identifies it canonically with the stated product. In more elementary terms,  $\mathcal{K}$  has a unique inverse image in  $\mathcal{L} \otimes S_h$ ; its quotient there is locally free because the quotient of  $S_D/\mathcal{K}$  onto  $S_D/(\mathcal{L} \cdot S_h)$  is a surjection of vector bundles. This proves the functorial assertion, including its scheme structure.

The source of  $\nu_g$  is projective. Unique factorization in the polynomial ring over an algebraically closed field gives (AG.3). A fixed polynomial has only finitely many divisors of a fixed degree, up to scalar. The morphism is thus proper with finite fibres and is finite [15, Tag 02LS]. Define  $Y_g$  by the kernel of  $\mathcal{O}_{\mathcal{G}} \rightarrow (\nu_g)_* \mathcal{O}_{\mathcal{X}_g}$ , as in [16, Tag 01R5]. Its integrality follows from the integrality of  $\mathcal{X}_g$ : on an affine open meeting the image, the coordinate ring of the image injects into the domain which is the coordinate ring of its finite inverse image. Reducedness is therefore a consequence of this incidence construction, not an extra choice. Representability of incidence gives the base-change assertion.  $\square$

*Remark AG.2.* The condition defining  $Y_g$  is the existence of a divisor of *degree*  $g$ . In several variables it is not the condition  $\deg \gcd(K) \geq g$ : an irreducible polynomial of larger degree need not have a degree- $g$  factor. Nor do we assert that formation of the scheme-theoretic image, or normalization itself, commutes with arbitrary nonflat base change. The assertion for arbitrary bases in Proposition AG.1 concerns the incidence functor and the base change of its representing morphism.

We next retain the nilpotents of a fibre. This will distinguish a ramified divisor choice from several distinct divisor choices.

**Lemma AG.3** (Fibre equations and the overlap space). *Let  $K = fJ$ , set  $H = \gcd(J)$ , and put  $c = \gcd(f, H)$  and  $k = \deg c$ . Then the tangent space of the scheme-theoretic fibre at  $([f], J)$  is naturally, after the displayed representatives have been chosen,*

$$\ker(d\nu_g)_{([f], J)} \simeq S_k/\mathbb{C}c, \quad \dim \ker(d\nu_g) = s_k - 1. \quad (\text{AG.4})$$

*More explicitly, choose a projective coefficient chart  $f(u) = f + \sum u_\alpha m_\alpha$ , with one coefficient normalized to its value in  $f$ . Let  $M(f(u))$  be the matrix of multiplication  $S_h \rightarrow S_D$ . Choose a*

set  $A_0$  of  $s_h$  rows for which  $M_{A_0}(f)$  is invertible, and let  $B_0$  denote the remaining rows. For a matrix  $C$  whose columns form a basis of  $K$ , the full fibre on this chart is defined by

$$C_{B_0} - M_{B_0}(f(u))M_{A_0}(f(u))^{-1}C_{A_0} = 0, \quad \det M_{A_0}(f(u)) \neq 0. \quad (\text{AG.5})$$

In particular its completed local ring is the power-series ring in the  $s_g - 1$  variables  $u_\alpha$ , modulo the entries in (AG.5); no radical is taken.

*Proof.* For each column of  $C$ , its possible quotient by  $f(u)$  is forced by the  $A_0$  rows to be  $M_{A_0}^{-1}C_{A_0}$ . The remaining rows are exactly the condition that this quotient works. These quotients span a subbundle wherever the equations hold, by the incidence argument above. Thus elimination has introduced neither additional solutions nor discarded nilpotents, proving the fibre presentation over arbitrary coefficient rings in the chart.

A tangent vector is a pair  $(\dot{f} \bmod \mathbb{C}f, \dot{J})$ . It lies in the fibre precisely when

$$\dot{f}j + f\dot{j} \in fJ \quad \text{for all } j \in J.$$

It is necessary and sufficient that  $f \mid \dot{f}j$  for every  $j \in J$ ; the class of  $\dot{j}$  modulo  $J$  is then forced. Taking the minimum valuation over a basis of  $J$ , for each irreducible polynomial, makes this equivalent to  $f \mid \dot{f}H$ . Writing  $f = cf_1$ ,  $H = cH_1$ , with  $\gcd(f_1, H_1) = 1$ , gives  $\dot{f} = f_1w$  with  $w \in S_k$ . The scalar tangent direction  $\dot{f} \in \mathbb{C}f$  is exactly  $w \in \mathbb{C}c$ . This proves (AG.4) and its dimension assertion.  $\square$

**Theorem AG.4** (Normalization and the exact normal locus). *The finite map*

$$\nu_g : \mathbb{P}(S_g) \times \text{Gr}(r, S_h) \longrightarrow Y_g$$

*is the normalization of the natural degree- $g$  divisor locus. Its dimension is*

$$\dim Y_g = s_g - 1 + r(s_h - r). \quad (\text{AG.6})$$

*For a geometric point  $K$ , the scheme  $Y_g$  is normal at  $K$  if and only if  $F_K$  has exactly one degree- $g$  divisor  $[f]$  and*

$$\gcd(f, F_K/f) = 1. \quad (\text{AG.7})$$

*Normality at a point is also equivalent to smoothness there. Thus the two precise obstructions are more than one divisor choice, or overlap between the selected divisor and the residual common factor. The latter gives the fibre tangent dimension in (AG.4), and (AG.5) determines its higher nilpotents.*

*Proof.* The locus of gcd-free  $r$ -planes in  $S_h$  is nonempty: a space containing  $x_1^h, x_2^h$  is one example, and it can be extended to every allowed rank  $r$ . It is open, since the loci possessing a positive-degree divisor are the finitely many closed images of projective multiplication maps. It is therefore dense in the irreducible Grassmannian. Its product with  $\mathbb{P}(S_g)$  maps to the locus where the exact gcd has degree  $g$ .

This last locus is open in  $Y_g$ : remove the finitely many closed images for divisor degrees  $a > g$ . Call it  $U_g$ . Every fibre over a geometric point of  $U_g$  has one point and, by Lemma AG.3 with  $H = 1$ , zero tangent space. A finite local algebra over an algebraically closed field with zero cotangent space is the field itself, by Nakayama's lemma. The fibre consequently has length one, not merely one point.

Here is the local argument which also settles the scheme structure. At a closed point of  $U_g$ , write  $A$  for the local ring of  $Y_g$ , and  $B$  for its finite inverse-image algebra. The map  $A \rightarrow B$  is injective, and  $B/\mathfrak{m}_A B = \mathbb{C}$ , generated by 1. Nakayama's lemma, applied to the finite module  $B/A$ , gives  $B = A$ . The non-isomorphism locus of a finite morphism is closed, so the same conclusion holds throughout  $U_g$ . Hence  $\nu_g$  is birational. The source is smooth and integral, thus normal. On affine opens, a finite birational extension  $A \subset B$  with  $B$  integrally closed is the



integral closure of  $A$  in the common fraction field: every element integral over  $A$  is integral over  $B$  and hence belongs to  $B$ . This proves the normalization assertion; compare [17, Tag 035E]. The dimension follows from the product description.

If  $Y_g$  is normal at  $K$ , its normalization is an isomorphism there, so its fibre is one reduced point. Formula (AG.3) gives uniqueness of  $[f]$ , and (AG.4) gives  $s_k - 1 = 0$ , equivalent to  $k = 0$  because  $e \geq 2$ . Conversely uniqueness and  $k = 0$  again make the entire fibre a single reduced point. The same finite local Nakayama argument shows that the normalization is an isomorphism near  $K$ . Its source is normal, so  $Y_g$  is normal at  $K$ . The source is smooth, so every normal point of the target is smooth; the converse holds for every complex variety. This proves both directions without replacing a nonreduced fibre by its set of points.  $\square$

**Corollary AG.5** (Universality of the exact stratum). *The open subscheme  $U_g \subset Y_g$  represents the unmarked relation families in the divisor incidence whose residual space is gcd-free on every geometric fibre. Over it the divisor line and residual subbundle are unique and commute with arbitrary base change. Moreover  $U_g$ , as a locally closed subscheme of  $\mathcal{G}$ , is the reduced exact-gcd stratum of Lemma AF.1.*

*Proof.* The inverse image of  $U_g$  is  $\mathbb{P}(S_g) \times X_{h,r}(E)$ , and the proof above shows that its finite map to  $U_g$  is an isomorphism. Its source is reduced, so  $U_g$  has the reduced induced structure on its locally closed support. The incidence functor supplies the asserted universal property. A family in the ambient Grassmannian which only has the right geometric gcd need not factor through  $Y_g$ . In particular, the transverse dual-number family of Remark AF.2 is not admitted by this corollary. The corollary is a scheme-theoretic identification, not a claim that fibrewise gcd tests detect nilpotent equations.  $\square$

**Example AG.6** (A nonreduced fibre at a divisor collision). Take  $E = \langle x, y, z \rangle$ ,  $g = 1, h = 2, r = 2$ , and  $K = \langle x^2y, x^2z \rangle$ . Its gcd is  $x^2$ , so the only point of the normalization fibre has divisor  $x$ . In the chart  $f = x + ay + bz$ , division by this monic linear form amounts to substitution  $x = -ay - bz$ . The two remainders are

$$(ay + bz)^2y, \quad (ay + bz)^2z.$$

Their coefficient ideal is  $(a^2, ab, b^2)$ . Thus the full fibre is

$$\operatorname{Spec} \mathbb{C}[a, b]/(a, b)^2, \tag{AG.8}$$

of length three and tangent dimension two. It is not a reduced point, despite uniqueness of the geometric divisor. The family

$$K_t = x \langle xy + ty^2, xz + tz^2 \rangle$$

has independent generators for every  $t$ , exact gcd  $x$  when  $t \neq 0$ , and the displayed collision at  $t = 0$ . Consequently this nonreduced fibre occurs on the boundary of the exact stratum, not in an unrelated example.

**Example AG.7** (Several branches and a normal gcd jump). For the same  $g, h, r, E$ , let  $K = xy \langle x, z \rangle$ . The divisors  $x$  and  $y$  give two points of the normalization fibre. At both points the overlap is one, so the fibre consists of two reduced points. The point of  $Y_1$  is nonnormal. By contrast, put  $q = x^2 + yz$  and  $K = xq \langle y, z \rangle$ , now with  $h = 3$ . The polynomial  $q$  is irreducible and is not divisible by  $x$ . There is exactly one linear divisor of gcd  $K = xq$ , and its overlap with the residual gcd  $q$  is one. This point of  $Y_1$  is normal although its gcd degree is three. These two examples show why neither gcd degree nor the number of geometric lifts alone is a sufficient normality criterion.

**Corollary AG.8** (An intrinsic boundary for first-relation algebras). *On the maximal-lower-Hilbert-function stratum of Proposition AF.4, the algebras whose intrinsic degree- $D$  kernel has*

classifying map in  $Y_g$  have, after removing only tangent-identity substitutions, stack  $[Y_g/\mathrm{GL}(E)]$ . Its normalization is the finite representable morphism

$$[\mathcal{X}_g/\mathrm{GL}(E)] \longrightarrow [Y_g/\mathrm{GL}(E)]. \quad (\text{AG.9})$$

All members of the universal relation family give finite locally free algebras of rank  $\binom{e+D}{D} - r$ . This construction imposes no pencil image equations and permits the divisor and residual common factor to collide.

*Proof.* For every subbundle  $\mathcal{K}$  classified by the Grassmannian,  $\mathrm{Sym} E/((\mathcal{K}) + \mathfrak{m}^{D+1})$  is, as a module, the sum of  $S_j$  for  $j < D$  and  $S_D/\mathcal{K}$ . This proves finite local freeness and the rank. The first-relation proposition identifies the quotient after the specified nonlinear inertia is removed. Incidence and multiplication are  $\mathrm{GL}(E)$ -equivariant. Pulling (AG.9) back along the smooth frame atlas  $Y_g \rightarrow [Y_g/\mathrm{GL}(E)]$  gives precisely  $\nu_g$ . Finiteness, representability, and the normalization identification follow by descent; equivalently normality and integral closure may be checked on these smooth atlases. This quotient is not claimed to classify smoothings outside the specified Hilbert stratum.  $\square$

For the pencil first relations one can take  $e = n^2$ ,  $g = n(n-1)$ ,  $h = 3n-4$ , and  $r = \binom{n+1}{2}$ . Every pencil then lies in the exact-gcd open of this canonical boundary. The boundary theorem concerns the much larger space of all degree- $g$  divisorial first relations, rather than a closure defined by recovering a pencil and reimposing its Pluecker equations. It determines its normalization and normal locus, but does not assert a classification of all its linear orbits or of all finite-flat algebra deformations.

#### APPENDIX AH. FORMAL IMAGES AND MARKED DIVISOR GERMS

All rings and schemes in the following sections are over  $\mathbb{C}$ . Write  $S_j = \mathrm{Sym}^j E$ , and let  $Y_g \subset \mathrm{Gr}(r, S_D)$  be the scheme-theoretic image of

$$\nu_g : \mathbb{P}(S_g) \times \mathrm{Gr}(r, S_{D-g}) \longrightarrow \mathrm{Gr}(r, S_D), \quad (f, J) \longmapsto fJ,$$

where  $\dim E \geq 2$ ,  $g \geq 1$ , and  $2 \leq r \leq \dim S_{D-g}$ . The source is smooth, and  $\nu_g$  is its finite normalization: a fibre records the degree- $g$  divisors of the common divisor, and on the exact-degree- $g$  locus there is one reduced lift. We shall retain the scheme structures of these fibres and of the image.

**Lemma AH.1** (Completed finite images). *Let  $X \rightarrow M$  be a finite morphism of complex finite-type schemes, with  $X$  reduced and  $M$  smooth, and let  $Y$  be its scheme-theoretic image. Fix a closed point  $y \in Y$ , put  $R = \widehat{\mathcal{O}}_{M,y}$ , and write  $B_i = \widehat{\mathcal{O}}_{X,x_i}$  for the points over  $y$ . Then*

$$\widehat{\mathcal{O}}_{Y,y} = R / \bigcap_i \ker(R \rightarrow B_i).$$

*This completed image is reduced. If the cotangent map from  $R$  to  $B_i$  is surjective, then  $R \rightarrow B_i$  is surjective. If there are two such points, writing  $I_i = \ker(R \rightarrow B_i)$  gives*

$$\widehat{\mathcal{O}}_{Y,y} = (R/I_1) \times_{R/(I_1+I_2)} (R/I_2).$$

*The formal schemes at the different  $x_i$  are disjoint. A local Artin point of the finite inverse image belongs to exactly one of them.*

*Proof.* On an affine neighbourhood the image ring is the image of the ambient ring in a finite algebra. Exactness of completion on finite modules preserves this kernel. The completed finite algebra is the product of the  $B_i$ : the orthogonal idempotents of its Artin closed fibre lift uniquely through the powers of the maximal ideal. Reduced finite-type complex rings are excellent and analytically unramified, so their completions are reduced; alternatively this applies directly to the

reduced image ring. For cotangent surjectivity, lift a basis of the maximal ideal of  $B_i$  modulo its square to  $R$ . The resulting lifts generate every successive associated-graded piece. Subtracting successive homogeneous lifts and taking their convergent sum expresses any element of  $B_i$  as the image of an element of  $R$ . The finite map factors through the completed image, so this is also a closed immersion into that image, not only into  $M$ . Finally the elementary exact sequence

$$0 \longrightarrow R/(I_1 \cap I_2) \longrightarrow R/I_1 \oplus R/I_2 \longrightarrow R/(I_1 + I_2) \longrightarrow 0$$

proves the fibre-product assertion. The idempotent decomposition gives the last assertion, including over nonreduced Artin bases.  $\square$

**Lemma AH.2** (Coprime lifts and the split codimension). *If homogeneous forms  $a, b$  have coprime reductions over a local Artin base  $A$ , then  $(a) \cap (b) = (ab)$  in  $A[E]$ . The relevant graded quotients and Koszul kernels are  $A$ -flat. In the split-divisor model, with  $s_j = \binom{e+j-1}{j}$ ,  $e \geq 2$ ,  $h > g \geq 1$ , and  $r \geq 2$ ,*

$$c = r(s_h - s_{h-g}) - (s_g - 1) \geq (r-1)(s_g - 1) \geq 1.$$

*Proof.* Over the residue field the two forms form a regular sequence. In each degree their Koszul complex consists of finite free modules and has its prescribed ranks. The acyclic-complex criterion over an Artin local ring, proved by lifting splittings successively along its maximal ideal, gives exactness and flatness of the lifted cokernels. Thus  $b$  is a nonzerodivisor modulo  $a$ , proving the intersection identity. For the inequality, express  $s_h - s_{h-g}$  as the sum of  $g$  consecutive first differences of the sequence  $s_j$ . These differences are nondecreasing for  $e \geq 2$ . Their sum is at least  $s_g - s_0 = s_g - 1$ .  $\square$

In particular the two formal marked lifts in a coprime split model are separated before their images are compared. Their intersection functor is precisely  $K_A = a_A b_A L_A$ . Division gives a subbundle: multiplication by a fibrewise nonzero form is locally split, and the quotient of the ambient bundle by  $K_A$  is locally free. Combining this Artin-functor identification with Lemma AH.1 justifies the completed-image fibre product without any assumption that a tangent-space comparison determines an intersection scheme.

## APPENDIX AI. A GLOBAL FACTORIZATION THEOREM FOR BINARY NORMALIZATION FIBRES

For this section  $E$  has dimension two. Set  $N_0 = r(D+1-r)$  and  $m = r-1$ . Let

$$\mathcal{T}_s = Y_s \setminus Y_{s+1}, \quad 1 \leq s \leq D+1-r,$$

with the reduced locally closed structure; an inadmissible  $Y_j$  is empty. Binary forms split into linear factors, so the  $Y_j$  are nested. The multiplication map identifies

$$\mathcal{T}_s \simeq \mathbb{P}(S_s) \times X_{D-s,r},$$

where  $X_{D-s,r}$  is the open Grassmannian of gcd-free residual spaces. The first projection is the algebraic common-divisor morphism, not a choice made independently on individual fibres.

For nonnegative  $a, b$  define  $H_{a,b} = \mathbb{C}$  when  $ab = 0$ . Otherwise, put  $k = \min(a, b)$  and  $l = \max(a, b)$ , and set

$$H_{a,b} = \mathbb{C}[q_1, \dots, q_k] / (c_{l+1}, \dots, c_{l+k}), \quad \sum_{j \geq 0} c_j z^j = (1 + q_1 z + \dots + q_k z^k)^{-1}. \quad (\text{AI.1})$$

The weight of  $q_i$  is  $i$ . This grading is not the maximal-ideal adic grading when  $k > 1$ .

**Theorem AI.1** (Stratified finite flatness and all binary fibres). *For  $g \leq s$ , the restriction of  $\nu_g$  over  $\mathcal{T}_s$  is the base change, along the common-divisor morphism, of*

$$\mathbb{P}(S_g) \times \mathbb{P}(S_{s-g}) \longrightarrow \mathbb{P}(S_s), \quad (f, H) \mapsto fH.$$

It is finite locally free of degree  $\binom{s}{g}$ . If  $\gcd(K) = \prod_{i=1}^b \ell_i^{s_i}$ , with distinct  $\ell_i$  and  $\sum_i s_i = s$ , its entire normalization fibre is

$$\prod_{\substack{0 \leq a_i \leq s_i \\ \sum_i a_i = g}} \bigotimes_{i=1}^b H_{a_i, s_i - a_i}. \quad (\text{AI.2})$$

Every local factor is an Artin complete intersection and is Gorenstein. Its length, weighted Hilbert series, and socle weight are, respectively,

$$\prod_i \binom{s_i}{a_i}, \quad \prod_i \prod_{j=1}^{\min(a_i, s_i - a_i)} \frac{1 - z^{\max(a_i, s_i - a_i) + j}}{1 - z^j}, \quad \sum_i a_i (s_i - a_i).$$

The total length is  $\binom{s}{g}$ , including at arbitrary simultaneous root collisions. Over fixed  $s$ , the closure order of common-root partition strata is the order generated by merging parts.

*Proof.* Over  $\mathcal{T}_s$  the common divisor  $G$  and residual subbundle  $L$  are defined scheme-theoretically by the inverse to the exact-divisor incidence. Locally on any base change, choose a member of  $L$  coprime to  $G$  on each relevant geometric fibre. There is such a member over  $\mathbb{C}$ , since only finitely many roots must be avoided. Regular-sequence cancellation, or monic division and its flat quotient, gives  $f \mid GL$  if and only if  $f \mid G$ . This equivalence is an equality of incidence functors over nonreduced bases. The residual subspace is then forced. This proves the base-change description.

The displayed multiplication map is finite by properness and the finiteness of the divisors of a fixed binary form. Its source and target are smooth of dimension  $s$ . A finite map between these smooth spaces is flat: a regular system of parameters in the target is a system of parameters in each source local ring, hence a regular sequence; the local flatness criterion applies. Its degree can be computed at a squarefree form, where the  $\binom{s}{g}$  divisors are reduced and distinct.

Coprime root clusters split uniquely over Artin bases by Hensel factorization. On the cluster with polynomial  $x^{s_i}$ , choosing a degree- $a_i$  factor is the equation  $f_i H_i = x^{s_i}$  with both factors monic. Eliminating one factor by reciprocal expansion gives (AI.1). This proves the full product decomposition, not only its closed points. Its equations have their sole common zero at the origin: a monic factor of  $x^{a+b}$  over  $\mathbb{C}$  is a power of  $x$ . Thus its  $k$  equations in  $k$  variables are a homogeneous system of parameters and a regular sequence. The Hilbert series is the quotient of the corresponding weighted polynomial Hilbert series. Taking its value at one gives  $\binom{a+b}{a}$ ; its top weight is  $ab$ . A zero-dimensional complete intersection has a one-dimensional socle. Equivalently  $q_k^l$ , up to a nonzero scalar, represents the rectangular Schur class. The identification with the usual Grassmannian cohomology presentation is classical; a purely algebraic Schur-basis statement is [24, Theorem 2.7]. Tensoring the local factors proves all three formulas. The total length identity is the coefficient of  $z^g$  in  $\prod_i (1+z)^{s_i}$ . Finally the exact-gcd stratum is the product with  $X_{D-s,r}$  above. In the symmetric power of the projective line, root clusters specialize by coalescence, and every merging is realized by moving the corresponding roots together. This gives precisely the stated closure order within fixed gcd degree.  $\square$

*Remark AI.2.* The finite flat statement is on the exact-gcd stratum with its indicated scheme structure. It does not assert that normalization or formation of scheme-theoretic images commutes with an arbitrary nonflat base change of the ambient Grassmannian. It identifies a specific base change of the already constructed incidence morphism.

## APPENDIX AJ. CONDUCTORS AND ALL BRANCH INTERSECTIONS ON THE SQUAREFREE LOCUS

Let  $\mathcal{U} \subset \text{Gr}(r, S_D)$  be the open set on which the common divisor is squarefree. Its complement is the proper image of the incidence  $K = \ell^2 L$ . This condition concerns the common divisor, not every member of the subspace.

**Theorem AJ.1** (Squarefree conductor stratification). *Let  $K \in Y_g \cap \mathcal{U}$ , let  $s = \deg \gcd(K)$ , and set*

$$t = s - g, \quad m = r - 1, \quad q = N_0 - ms.$$

*Introduce  $s$  disjoint blocks  $u_i = (u_{i1}, \dots, u_{im})$ , and let  $J_j$  be the ideal generated by products of one variable from each of  $j$  distinct blocks. Put  $J_0 = (1)$  and  $J_{s+1} = 0$ . Then*

$$\hat{\mathcal{O}}_{Y_g, K} \simeq \mathbb{C}[[z_1, \dots, z_q, u_{11}, \dots, u_{sm}]]/J_{t+1}. \quad (\text{AJ.1})$$

*The branches are indexed by the subsets  $A \subset \{1, \dots, s\}$  of cardinality  $g$ . The branch  $A$  has ideal  $I_A = (u_{ij} : i \in A)$  in the ambient power-series ring. For any collection of branches, their full scheme-theoretic intersection has ideal  $I_{\cup A}$ ; it is smooth of dimension  $q + m(s - |\cup A|)$ .*

*The conductor in (AJ.1) is  $J_t/J_{t+1}$ . Its restriction to branch  $A$  is  $\prod_{i \notin A} (u_{i1}, \dots, u_{im})$ . Consequently the global conductor sheaf on  $Y_g \cap \mathcal{U}$  is exactly*

$$\mathfrak{c}_{Y_g}|_{\mathcal{U}} = \mathcal{I}_{Y_{g+1}/Y_g}|_{\mathcal{U}}. \quad (\text{AJ.2})$$

*In particular it is radical there. These local rings are seminormal. Their dimension, depth and multiplicity are*

$$q + mt, \quad q + t, \quad \binom{s}{g}.$$

*They are Cohen–Macaulay exactly when  $s = g$  or  $r = 2$ . The Hilbert series of the nonsmooth factor in its standard grading is*

$$\sum_{j=0}^t \binom{s}{j} ((1-z)^{-m} - 1)^j. \quad (\text{AJ.3})$$

*The whole collection of branch intersections and conductor strata is invariant under permutation of the roots and descends from any ordered-root etale chart.*

*Proof.* Choose  $F_0 \in K$  with a simple zero at each of the  $s$  common roots, and complete it to a frame  $F_0, \dots, F_m$ . Hensel factorization gives moving roots  $x_1, \dots, x_s$  in the completion. Use as coordinates the  $s$  root parameters and the  $ms$  evaluations  $u_{ij} = F_j(x_i)$ , with harmless invertible normalizing factors. These are part of a regular parameter system: the evaluation map  $S_D/K \rightarrow \mathbb{C}^s$  is surjective because  $K$  vanishes on the common-root scheme, and  $D \geq s - 1$ . Thus  $\text{Hom}(K, S_D/K)$  allows independent variation of all  $rs$  jets. The passage to the displayed coordinates is triangular with invertible diagonal, since  $F_0$  has simple zeros. The other  $N_0 - rs$  parameters and the  $s$  root parameters give exactly  $q = N_0 - ms$  smooth parameters.

A marked degree- $g$  divisor near this fibre chooses a subset  $A$  of these roots. Its incidence equations are exactly  $u_{ij} = 0$  for  $i \in A$ , with no nilpotents suppressed. There are no other divisor lifts. Lemma AH.1 therefore identifies the completed image ideal with  $\bigcap_{|A|=g} I_A$ . A monomial belongs to this intersection exactly when its block support meets every  $g$ -element subset, or equivalently has at least  $s - g + 1 = t + 1$  blocks. This proves (AJ.1). Sums of the coordinate ideals give all intersections, including intersections of three or more branches.

For completeness compute the conductor inside the product of the branch rings. Its component on branch  $A$  consists of functions whose extension by zero on every other branch belongs to the image ring. The image of  $\bigcap_{B \neq A} I_B$  modulo  $I_A$  is precisely  $\prod_{i \notin A} (u_i)$ . Indeed a surviving monomial has support outside  $A$ , and must meet every  $g$ -subset other than  $A$ . Replacing one

element of  $A$  by each  $i \notin A$  forces all those blocks to occur. Conversely their product meets every such subset. Hence a conductor monomial has exactly  $t$  surviving blocks; monomials on more blocks are zero. This is  $J_t/J_{t+1}$ . The same coordinates identify  $Y_{g+1}$  with  $J_t$ , proving the sheaf equality by faithful flatness of completion. The case  $t = 0$  means that the conductor is the unit ideal and the next stratum is absent locally.

The ring is the ring of tuples on its coordinate branches that agree on every pairwise intersection. To verify this directly, compare the coefficient of each monomial on all branches where it survives; the pairwise agreements give a unique common coefficient. If an element  $b$  of the total fraction ring has  $b^2, b^3$  in the image, it first belongs to the normalization, by integrality. Its two restrictions on any intersection have equal squares and cubes, hence are equal in that reduced ring. Thus  $b$  lies in the image. This proves seminormality by the square-and-cube criterion.

Formula (AJ.3) follows by sorting nonzero monomials by their block support. The top-dimensional contributions come from the  $\binom{s}{t}$  supports of size  $t$ , giving dimension  $mt$  and multiplicity  $\binom{s}{t}$ . We supply the homological step for the depth, since branch counting alone does not give it. The simplicial complex of the squarefree monomial ideal  $J_{t+1}$  consists of sets using at most  $t$  blocks. Any induced subcomplex meeting  $b$  blocks is homotopy equivalent to the  $(t-1)$ -skeleton of the simplex on those  $b$  blocks: collapse each block to a representative vertex. The two composites are contiguous to the identity, because adjoining the representatives never increases the number of blocks. If  $b \leq t$  the complex is a simplex; otherwise its only reduced homology is in degree  $t-1$ , with rank  $\binom{b-1}{t}$ . This last assertion follows also by the elementary induction that adds the last vertex to a simplex skeleton.

The squarefree multidegree- $W$  part of the Koszul complex on all  $sm$  variables is the augmented simplicial cochain complex of this induced subcomplex, with degree shifted by  $|W| - 1$ . Non-squarefree multidegrees are contractible in the relevant summands, by pairing faces using a variable whose exponent exceeds one. Consequently a nonzero positive Betti number has homological degree  $|W| - t$ . The full vertex set gives the nonzero degree  $sm - t$ , since  $s > t$  when  $g > 0$ . The projective dimension is therefore  $sm - t$ , and the Auslander–Buchsbaum formula gives depth  $t$ . When  $t = 0$  the quotient is simply  $\mathbb{C}$ . Adjoining  $q$  power-series parameters gives depth  $q + t$  in every case, and comparison with  $q + mt$  proves the Cohen–Macaulay criterion. All the constructions commute with relabelling roots, proving descent.  $\square$

**Corollary AJ.2** (Binary divisor-image intersections). *For admissible binary divisor degrees  $a_1, \dots, a_v$ , their scheme-theoretic intersection is  $Y_{\max a_i}$ . On the squarefree locus all multiple intersections of normalization branches, rather than only the degree images, are given by the union-of-subsets rule in Theorem AJ.1.*

*Proof.* The reduced binary images are nested as closed subschemes: a binary form of degree  $b$  has a divisor of each degree  $a \leq b$ , and reduced closed subschemes are determined by their closed points. Their ideals are thus nested, so their sum is the largest ideal. The assertion about branches has already been proved as an identity of coordinate ideals, without taking radicals.  $\square$

#### APPENDIX AK. AN INCIDENCE ATLAS FOR ALL BINARY ROOT COLLISIONS

The squarefree theorem has a uniform replacement when any number of root clusters collide. This replacement specifies the full completed image, not merely a transverse family in it.

**Theorem AK.1** (Simultaneous collision atlas). *Let  $K \in Y_g$  be binary with exact common divisor  $\prod_{i=1}^b \ell_i^{s_i}$  and  $s = \sum_i s_i$ . There are formal ambient coordinates consisting of the coefficients of monic polynomials  $P_i$  of degree  $s_i$ , polynomials  $R_{ij}$  of degree less than  $s_i$  ( $1 \leq j \leq m$ ), and  $N_0 - rs$  smooth parameters  $z$ . They are centred at  $P_i = x_i^{s_i}$  and  $R_{ij} = 0$ . For every  $\alpha = (a_i)$  with  $0 \leq a_i \leq s_i$  and  $\sum a_i = g$ , let  $B_\alpha$  be the regular power-series ring in the coefficients of monic*

$A_i, B'_i$  of degrees  $a_i, s_i - a_i$ , polynomials  $U_{ij}$  of degree less than  $s_i - a_i$ , and the same  $z$ . The completed normalization and image are exactly

$$T \longrightarrow \prod_{\alpha} B_{\alpha}, \quad P_i \mapsto A_i B'_i, \quad R_{ij} \mapsto A_i U_{ij}, \quad \widehat{\mathcal{O}}_{Y_g, K} = T / \bigcap_{\alpha} \ker(T \rightarrow B_{\alpha}). \quad (\text{AK.1})$$

Here  $T = \mathbb{C}[[\text{coeff}(P_i), \text{coeff}(R_{ij}), z]]$ . Degree-zero monic factors are one, and negative-degree remainder spaces are zero. The formula is independent, up to isomorphism of completed incidence diagrams, of the frame, the selected  $F_0$ , and the local root coordinates. It covers every higher and simultaneous binary collision.

*Proof.* Choose  $F_0 \in K$  having exact order  $s_i$  at every common root. The required nonvanishing of its residual value excludes finitely many hyperplanes in  $K$ , so a choice exists. Complete it to a frame. Weierstrass preparation at the distinct roots produces  $P_i$ ; division of the remaining frame members produces  $R_{ij}$ . The map

$$S_D/K \longrightarrow \bigoplus_i \mathbb{C}[x_i]/(x_i^{s_i})$$

is surjective: its kernel contains  $K$ , and the usual interpolation map from  $S_D$  is surjective. Thus the Grassmannian tangent space permits independent variation of all  $rs$  jets. Division and preparation change these jets by an invertible triangular matrix, whose diagonal contains the nonzero residual values of  $F_0$ . Their differentials are independent. Formal inverse coordinates supply the remaining  $N_0 - rs$  parameters; this integer is nonnegative since  $r \leq D - s + 1$ .

A marked divisor near a chosen normalization point splits uniquely among the coprime root clusters. In cluster  $i$  its monic representative has degree  $a_i$  and divides both  $P_i$  and all  $R_{ij}$ . Monic division is equivalent, over every local Artin base, to the unique equations  $P_i = A_i B'_i$  and  $R_{ij} = A_i U_{ij}$ . Conversely these equations give the marked divisor and the residual subbundle. Thus the normalized incidence functor is represented by the displayed regular ring. It is finite over  $T$ : monic factor coefficients are integral over the coefficients of  $P_i$ , while the remaining factors and quotients are uniquely determined by division. Lemma AH.1 now gives the exact completed image and the intersection of kernels.

Changing a frame or a coordinate changes only a presentation of the same functor of a subbundle with a marked divisor. The mutually inverse changes of incidence data therefore induce isomorphisms of its complete representing rings and of the resulting finite-image diagrams. This also proves that the calculation describes the entire completed image. A point of that image over an Artin ring need not lift to its normalization; no such lifting assertion is used or implied.  $\square$

#### APPENDIX AL. THE SINGULAR FITTING IDEAL OF EVERY DOUBLE-ROOT MODEL

The atlas specializes for  $s = 2, g = 1$  to the following ring, after completion of the square and suppression of smooth parameters:

$$A_m = R \oplus It \subset B = \mathbb{C}[[u_1, \dots, u_m, t]], \quad R = \mathbb{C}[[u_1, \dots, u_m, \Delta]], \quad I = (u_1, \dots, u_m), \quad \Delta = t^2.$$

Put  $w_i = u_i t$ . The defining equations are  $u_i w_j - u_j w_i$  and  $w_i w_j - \Delta u_i u_j$ .

**Theorem AL.1** (Closed singular-ideal formula). *For every  $m \geq 1$ , the intrinsic singular Fitting ideal is*

$$\text{Fitt}_{m+1} \Omega_{A_m/\mathbb{C}}^{\text{cont}} = (I^{m+1} + \Delta I^m) \oplus I^m t = I^{m-1} (I^2, (w_1, \dots, w_m), \Delta I). \quad (\text{AL.1})$$

The first equality is a decomposition as an  $R$ -submodule of  $B$ ; the right side is an ideal of  $A_m$ . It is not merely an ideal of Jacobian minors left unevaluated. For  $m = 1$  it is  $(u^2, w, \Delta u)$ , and its radical is the conductor  $(u, w)$  for all  $m$ .

*Proof.* There are  $2m + 1$  ambient variables and dimension  $m + 1$ , so the Fitting ideal consists of the  $m$ -minors of the Jacobian of the displayed relations. Evaluate this matrix in  $B$  using  $w_i = u_i t$  and  $\Delta = t^2$ . A minor not using the  $\Delta$  column has  $u$ -degree  $m$ . It vanishes at  $t = 0$ : the only remaining rows are the Koszul rows in the  $w$  columns, of rank at most  $m - 1$ . Hence its terms have a positive power of  $t$ , and belong to  $\Delta I^m \oplus I^{m+1}t$ . A minor using the  $\Delta$  column has  $u$ -degree  $m + 1$ , and hence lies in  $I^{m+1} \oplus I^{m+1}t$ . This proves the first inclusion.

Use the  $m$  rows

$$u_1 w_j - u_j w_1 \quad (2 \leq j \leq m), \quad w_1^2 - \Delta u_1^2.$$

The columns  $(w_1, w_2, \dots, w_m)$ ,  $(u_1, w_2, \dots, w_m)$ , and  $(\Delta, w_2, \dots, w_m)$  give, up to sign and nonzero scalar,  $u_1^m t$ ,  $\Delta u_1^m$ , and  $u_1^{m+1}$ . The ideal is invariant under simultaneous linear change of  $u$  and  $w$ . In characteristic zero the powers of linear forms span each symmetric power, so these three minors generate the three indicated spaces of all  $u$ -monomials. They generate the claimed  $R$ -module; it is an  $A_m$ -ideal since multiplication by  $w_i = u_i t$  preserves it. This proves equality. The formula also immediately gives its radical, without discarding the nonreduced Fitting structure in the assertion itself.  $\square$

#### APPENDIX AM. A STRUCTURAL DICHOTOMY FOR PURE-POWER DIVISOR FIBRES

Let  $E = \mathbb{C}a \oplus W$ ,  $d = \dim W \geq 1$ , and  $g, k \geq 1$ . The factorization fibre of  $a^{g+k}$  records monic factors of degrees  $g$  and  $k$ . Interchanging these factors identifies its two descriptions; we may and shall take  $g \geq k$ . Write

$$Q_i \in \text{Sym}^i W \quad (1 \leq i \leq k), \quad C_0 = 1, \quad C_j = -\sum_{i=1}^k Q_i C_{j-i}, \quad C_j = 0 \quad (j < 0).$$

Let  $P = \text{Sym}(\bigoplus_{i=1}^k (\text{Sym}^i W)^*)$ , with its generators of weight  $i$ , and let  $\mathfrak{n}$  be its variable ideal. Put

$$F_{g,k}(W) = P/\mathcal{I}, \quad \mathcal{I} = (\text{coeff } C_{g+1}, \dots, \text{coeff } C_{g+k}). \quad (\text{AM.1})$$

It is an Artin local algebra. Indeed its complex points are exactly the factorizations of  $a^{g+k}$ , hence the single origin by unique factorization. The polynomial and completed presentations coincide. If  $K = a^{g+k}L$  with  $\gcd(L) = 1$ , cancellation by a member of  $L$  not divisible by  $a$ , as in Lemma AH.2, identifies this with the full degree- $g$  normalization fibre at  $K$ .

**Theorem AM.1** (Minimal equations and complete-intersection classification). *The embedding dimension and minimal number of equations of  $F_{g,k}(W)$  in a minimal regular local presentation are*

$$\text{edim } F_{g,k}(W) = \binom{d+k}{k} - 1, \quad (\text{AM.2})$$

$$\mu(\mathcal{I}) = \sum_{j=g+1}^{g+k} \binom{d+j-1}{j} = \binom{d+g+k}{d} - \binom{d+g}{d}. \quad (\text{AM.3})$$

More precisely the minimal relation module has the equivariant weighted decomposition

$$\mathcal{I}/\mathfrak{n}\mathcal{I} \simeq \bigoplus_{j=g+1}^{g+k} (\text{Sym}^j W)^*. \quad (\text{AM.4})$$

Consequently this fibre is a complete intersection if and only if  $d = 1$ . For  $d = 1$ , it has length  $\binom{g+k}{k}$ , weighted Hilbert series  $\prod_{i=1}^k (1 - z^{g+i})/(1 - z^i)$ , and one-dimensional socle generated by



$Q_k^g$ , of weight  $gk$ . For  $k = 1$  and arbitrary  $d$ ,

$$F_{g,1}(W) = \text{Sym}(W^*)/\mathfrak{n}^{g+1}, \quad \text{length } F_{g,1}(W) = \binom{d+g}{g}, \quad \text{Soc } F_{g,1}(W) = \text{Sym}^g(W^*). \quad (\text{AM.5})$$

In particular the last algebra is Gorenstein precisely when  $d = 1$ . No Gorenstein classification for  $d > 1, k > 1$  is needed for the theorem.

*Proof.* Since  $g \geq k$ , each defining form has no linear term in the coefficient variables. Thus the presentation is minimal and (AM.2) follows by summing  $\dim \text{Sym}^i W$ . For  $d = 1$ , the regular-sequence argument in the proof of Theorem AI.1 applies to the  $k$  reciprocal relations. In particular none of them belongs to  $\mathfrak{n}\mathcal{I}$ . It gives the stated Hilbert series, length, and the classical rectangular socle class.

For general  $W$ , taking coefficients of  $C_j$  gives an equivariant map  $(\text{Sym}^j W)^* \rightarrow \mathcal{I}/\mathfrak{n}\mathcal{I}$ . Its source is irreducible. This map is nonzero: choose one coordinate  $x$  of  $W$ , keep only the coefficients of  $x^i$  in every  $Q_i$ , and set every other coefficient to zero. The coefficient of  $x^j$  maps to the  $j$ th binary reciprocal relation. Were it in  $\mathfrak{n}\mathcal{I}$  before specialization, its specialization would be in the binary  $\mathfrak{n}\mathcal{I}$ , contradicting the preceding minimality. Irreducibility now makes the entire coefficient map injective. The different  $j$  have different weights, so their images are independent. They generate the relation module by the chosen presentation, proving (AM.4) and (AM.3).

If  $d \geq 2$ , the sequence  $\binom{d+j-1}{j}$  strictly increases in  $j$ . Comparing its terms at  $j = g+i$  and  $j = i$  shows that the minimal number of relations is strictly greater than the embedding dimension. A zero-dimensional complete intersection in a minimal regular local presentation has exactly that many minimal relations as variables; hence these fibres are not complete intersections. This proves both directions of the classification. Finally when  $k = 1$  the sole form is  $C_{g+1} = (-Q_1)^{g+1}$ . Its coefficients are, with nonzero multinomial factors, every monomial of degree  $g+1$  in the  $d$  variables. This gives (AM.5), including the socle and length.  $\square$

**Corollary AM.2** (The ternary-pencil ramification fibre). *At the common limiting relation algebra with  $e = 9$ ,  $g = 6$  and  $k = 2$ , the normalization fibre has embedding dimension 44 and exactly*

$$\binom{14}{7} + \binom{15}{8} = 9867$$

*minimal scalar equations. Their two weighted pieces have dimensions 3432 and 6435, in weights seven and eight. In particular this actual pencil-orbit boundary fibre is not a complete intersection.*

*Proof.* Here  $d = e - 1 = 8$ . Apply (AM.2) and (AM.4). These are minimal equation counts, not the number of coefficient variables or the length of the fibre.  $\square$

The binary factorization rings in this section are classical. The additional point of the multivariate argument is that polarization preserves every full irreducible coefficient space in the minimal relation module. Neither the Grassmannian identification nor a finite calculation in a few variables is being used as a claim of priority or as a substitute for that argument.

## APPENDIX AN. HOMOLOGICAL INVARIANTS OF THE DIVISOR FIBRES

We first extract all graded Betti numbers from the squarefree local model and then identify a determinant-apolar quotient of the multivariate fibres. A separate exact calculation treats an entire multivariate fibre, rather than only a quotient or its minimal relations. The grading conventions will be specified in each case.

**Theorem AN.1** (Betti numbers of the squarefree block model). *Let  $S = \mathbb{C}[u_{ij} : 1 \leq i \leq s, 1 \leq j \leq m]$  with its standard multigrading, let  $0 \leq t < s$ , and put  $A = S/J_{t+1}$ , with the block-support*

ideal of Theorem AJ.1. For a nonempty squarefree multidegree  $F$  let  $b(F)$  be the number of blocks it meets. Besides  $\beta_{0,0} = 1$ , the only nonzero multigraded Betti numbers are

$$\beta_{i,F}^S(A) = \binom{b(F)-1}{t} \quad \text{if } b(F) > t \text{ and } i = |F| - t. \quad (\text{AN.1})$$

For  $t = 0$  this is the Koszul resolution of  $\mathbb{C}$ . For all  $t$ , the singly graded numbers are

$$\beta_{i,i+t}^S(A) = \sum_{b=t+1}^{\min(s,i+t)} \binom{s}{b} \binom{b-1}{t} [z^{i+t}]((1+z)^m - 1)^b \quad (i \geq 1). \quad (\text{AN.2})$$

The ideal  $J_{t+1}$  has a  $(t+1)$ -linear resolution. The projective dimension of  $A$  is  $sm - t$ . In the Cohen–Macaulay case  $m = 1$ , its type is  $\binom{s-1}{t}$ ; it is Gorenstein exactly when  $t = 0$  or  $t = s - 1$ . The same Betti numbers over the completed ambient regular ring apply to the local model after adjoining its smooth parameters.

*Proof.* The Stanley–Reisner complex consists of sets meeting at most  $t$  blocks. If  $t \geq 1$ , its induced subcomplex on  $F$  has, in each nonempty block, a simplex all of whose vertices have identical external links. Contract that simplex to one of its vertices: the map and the inclusion are contiguous, because adjoining the chosen vertex to any face does not change its block support. Applying this one block at a time gives a homotopy equivalence with the  $(t-1)$ -skeleton of the simplex on  $b(F)$  vertices. If  $b(F) \leq t$  this simplex is contractible. Otherwise its reduced homology is concentrated in degree  $t-1$ , with dimension  $\binom{b(F)-1}{t}$ . The latter assertion follows by attaching its top faces successively, or from the augmented simplicial chain complex and the binomial identity for its Euler characteristic. Hochster’s multigraded formula now gives (AN.1); see [26, Chapter 5]. If  $t = 0$ , every variable is in the ideal and the Koszul formula gives the same result directly.

For a support of size  $b$ , the coefficient in (AN.2) counts choices of  $i+t$  vertices meeting each selected block. Summing proves the formula. All ideal syzygies therefore have the indicated linear degrees. Taking  $F$  to be all  $sm$  vertices gives the nonzero last Betti number  $\binom{s-1}{t}$  in homological degree  $sm - t$ ; no larger degree occurs. When  $m = 1$ , Auslander–Buchsbaum gives depth  $t$ , equal to dimension, and the last Betti number is the Cohen–Macaulay type. Completion and adjoining independent regular parameters are flat operations preserving the minimal free resolution. These are classical squarefree-monomial methods. The assertion here is their full specialization to the normalization image model, not a claim of priority for Hochster’s formula or skeleton resolutions.  $\square$

For the remaining results use the weighted reciprocal algebra  $F_{g,k}(W)$  of (AM.1), where  $d = \dim W$  and  $g \geq k \geq 2$ . Set all  $Q_i$  except  $Q_2$  equal to zero and, in addition, impose every coefficient of  $Q_2^2$ . Denote the resulting algebra by

$$B_d = \text{Sym}((\text{Sym}^2 W)^*) / (\text{coeff } Q_2^2). \quad (\text{AN.3})$$

Its coefficient variables have reciprocal weight two. When referring to its ordinary polynomial grading, we instead give them degree one.

**Theorem AN.2** (A determinant-apolar quotient of every higher fibre). *There is a  $\text{GL}(W)$ -equivariant graded surjection  $F_{g,k}(W) \twoheadrightarrow B_d$  for every  $g \geq k \geq 2$ . After fixing the standard coefficient/differential pairing,  $B_d$  is the apolar algebra of the determinant of a generic symmetric  $d \times d$  matrix. Its polynomial-degree Hilbert function is*

$$\dim(B_d)_p = \frac{1}{d+1} \binom{d+1}{p} \binom{d+1}{p+1} \quad (0 \leq p \leq d). \quad (\text{AN.4})$$

It is Gorenstein, its reciprocal-weight socle is  $2d$ , and

$$\text{length } F_{g,k}(W) \geq \text{length } B_d = \frac{1}{d+2} \binom{2d+2}{d+1}. \quad (\text{AN.5})$$

For  $k = 2$  and  $g = 2$  or  $3$ , the quotient by the coefficients of  $Q_1$  alone is already  $B_d$ . In particular the actual ternary-pencil fibre  $F_{6,2}(\mathbb{C}^8)$  has length at least 4862, independently of its embedding dimension 44 and its 9867 minimal equations.

*Proof.* Under the stated substitution, the reciprocal series becomes  $(1 + Q_2 z^2)^{-1}$ . Its odd coefficients vanish and its even coefficients are powers of  $Q_2$ . Since  $g + 1 \geq 3$ , imposing  $Q_2^2 = 0$  annihilates all defining coefficients. This proves the surjection, functorially for linear changes of  $W$ . When  $k = 2$  and  $g = 2$  or  $3$ , the first even defining coefficient is  $Q_2^2$  and there are no further independent equations after  $Q_1 = 0$ .

Choose coordinates  $x_1, \dots, x_d$  on  $W$  and write  $Q_2 = \sum_i y_{ii} x_i^2 + \sum_{i < j} y_{ij} x_i x_j$ . Let  $X = (X_{ij})$  be symmetric, with independent entries for  $i \leq j$ , and interpret  $y_{ij}$  as  $\partial/\partial X_{ij}$ . Then  $Q_2(x)$  acts as directional differentiation in the rank-one symmetric direction  $xx^t$ . The determinant of  $X + zxx^t$  is linear in  $z$ , so every coefficient of  $Q_2(x)^2$  annihilates  $\det X$ . These coefficients are, up to nonzero scalars,  $y_{ii}^2$ ,  $y_{ii}y_{ij}$ ,  $y_{ij}^2 + 2y_{ii}y_{jj}$ ,  $y_{ii}y_{jl} + y_{ij}y_{il}$  for distinct  $i, j, l$ , and  $y_{ij}y_{kl} + y_{ik}y_{jl} + y_{il}y_{jk}$  for four distinct indices. They are exactly the quadratic generators in [27, Proposition 3.1 and Theorem 3.10]. That theorem proves equality with the full apolar ideal, not merely containment. Its Corollary 2.7 gives (AN.4) and the Catalan length. Alternatively, that Hilbert function counts the independent minors obtained by differentiating  $\det X$ . Apolarity to one homogeneous form gives a perfect pairing between complementary degrees and a one-dimensional socle in degree  $d$ . Replacing polynomial degree by twice that degree proves the weight statement. The length inequality follows from the surjection. For  $d = 8$  the Catalan number is 4862.

The apolar Hilbert function and its Gorenstein property are classical results of the cited determinant theory. The new identification in this setting is the functorial quotient of the entire family of normalization fibres. A quotient's Gorenstein property is not being attributed to the original fibre.  $\square$

**Proposition AN.3** (The complete first multivariate two-factor fibre). *Let  $A = F_{2,2}(\mathbb{C}^2)$  and write  $Q_1 = ax + by$ ,  $Q_2 = cx^2 + dxy + ey^2$ . Then*

$$H_A^{\text{wt}}(z) = 1 + 2z + 6z^2 + 6z^3 + 7z^4, \quad \text{length } A = 22, \quad \dim \text{Soc } A = 7, \quad (\text{AN.6})$$

$$H_{\text{gr}_m A}(z) = 1 + 5z + 6z^2 + 5z^3 + 5z^4. \quad (\text{AN.7})$$

The weight grading gives  $a, b$  weight one and  $c, d, e$  weight two. In the minimal weighted free resolution over  $S = \mathbb{C}[a, b, c, d, e]$ , the shift polynomials  $B_i(z) = \sum_j \beta_{i,j}^S(A) z^j$  are

$$\begin{aligned} B_0 &= 1, & B_1 &= 4z^3 + 5z^4, \\ B_2 &= 4z^5 + 12z^6 + 12z^7, & B_3 &= 21z^8 + 20z^9 + z^{10}, \\ B_4 &= 21z^{10} + 8z^{11}, & B_5 &= 7z^{12}. \end{aligned}$$

In particular the total Betti numbers are  $(1, 9, 28, 42, 29, 7)$ . This fibre is not Gorenstein. The assertion concerns this whole five-variable fibre, not the determinant-apolar quotient.

*Proof.* The nine initial equations are the coefficients of  $-Q_1^3 + 2Q_1Q_2$  and  $Q_1^4 - 3Q_1^2Q_2 + Q_2^2$ . For graded reverse lexicographic order  $e > d > c > b > a$ , a Groebner basis is the following list (a

scalar multiple does not change a member):

$$\begin{aligned}
& (c, d, e)^3, \quad b(c, d, e)^2, \quad a(c, d, e)^2, \\
& b^2e - e^2, \quad b^2d - de, \quad b^2c - a^2e, \quad b^3 - 2be, \\
& 2abe - de, \quad 2a^2e + 2abd - 2ce - d^2, \quad 2abc - cd, \\
& 3ab^2 - 2ae - 2bd, \quad a^2d - cd, \quad a^2c - c^2, \\
& 3a^2b - 2ad - 2bc, \quad a^3 - 2ac.
\end{aligned}$$

Here a monomial ideal in the list means all its minimal monomial generators. There are  $10 + 6 + 6 + 12 = 34$  polynomials. Polynomial division gives both equality with the ideal of the nine equations and zero remainders for all their basis S-pairs. This is a finite exact arithmetic verification; the explicit input and all reduction and rank tests are supplied in the accompanying calculation. The standard monomials, grouped by weight, are

$$\begin{array}{c|l}
0 & 1 \\
1 & a, b \\
2 & a^2, ab, b^2, c, d, e \\
3 & ac, bc, ad, bd, ae, be \\
4 & c^2, cd, d^2, a^2e, ce, de, e^2.
\end{array} \tag{AN.8}$$

Reduction of products of these monomials gives  $\dim A/\mathfrak{m} = 1$  and  $\dim(\mathfrak{m}^i) = (21, 16, 10, 5, 0)$  for  $1 \leq i \leq 5$ . The common kernel of multiplication by  $a, b, c, d, e$  is exactly the last row of (AN.8). This proves both distinct Hilbert series and the type.

For precision, the resolution calculation uses no numerical rank thresholds. In weight  $w$  and homological degree  $i$ , take basis pairs  $(J, m)$  with  $J \subset \{a, b, c, d, e\}$  of size  $i$ ,  $m$  a standard monomial, and  $\text{wt}(J) + \text{wt}(m) = w$ . The differential is

$$d(J, m) = \sum_{v \in J} (-1)^{\text{pos}(v, J)-1} (J \setminus \{v\}, \text{NF}(vm)). \tag{AN.9}$$

All entries are rational, explicitly determined by the displayed 34 polynomials. Exact row reduction and  $\beta_{i,w} = \dim C_{i,w} - \text{rank } d_{i,w} - \text{rank } d_{i+1,w}$  give the six shift polynomials above. The calculation is fully specified by (AN.8)–(AN.9); the archived rank table makes each finite elimination independently repeatable. As additional checks, their alternating sum is  $H_A^{\text{wt}}(z)(1 - z)^2(1 - z^2)^3$  and the last Betti space is the socle shifted by the total variable weight eight. These checks do not replace the exact rank computations.  $\square$

#### APPENDIX AO. SPECTRAL FITTING SCHEMES AS POWER-ZERO LOCI IN ONE RECIPROCAL SECTION

The algebra just calculated supplies a single geometric object in which all spectral Fitting schemes can be read simultaneously. This avoids constructing a separate binary form after each Fitting ideal has already been recovered.

Let  $V$  have dimension  $n$  and  $R \subset \text{Sym}^2 V$  be any pencil. Fix  $h \geq 1$  and put  $\tilde{E} = \mathbb{C}a \oplus V^*$ . Consider the divisor-incidence point

$$K_h = a^{2h+2} \tilde{E} \subset \text{Sym}^{2h+3} \tilde{E}. \tag{AO.1}$$

The entire degree- $2h$  normalization fibre over  $K_h$  is  $F_{2h,2}(V^*)$ , because  $\tilde{E}$  is gcd-free. Its section by the linear-factor coefficients is  $B_{n,h}$  and  $(B_{n,h})_1 = \text{Sym}^2 V$ . Thus the given pencil defines a line  $\Lambda = \mathbb{P}(R)$  in the projectivized degree-one space of this actual reciprocal section. This construction uses a new formal line  $\mathbb{C}a$  and is functorial for isomorphisms of  $V$ .

For  $0 \leq p \leq hn$  let  $\mathcal{J}_p$  be the ideal sheaf on  $\Lambda$  locally generated by the coordinates of

$$\ell^p \in (B_{n,h})_p \otimes \mathcal{O}_\Lambda(p), \quad (\text{AO.2})$$

where  $\ell$  is the universal pencil member. Multiplying a local representative of  $\ell$  by a unit does not change this ideal. We set  $\mathcal{J}_0 = \mathcal{O}_\Lambda$ . Let  $\mathcal{T}_R$  be the cokernel of the universal symmetric matrix  $V^* \otimes \mathcal{O}_\Lambda(-1) \rightarrow V \otimes \mathcal{O}_\Lambda$ . This definition includes singular pencils.

**Lemma AO.1** (The local power-ideal formula). *Let  $A(t)$  be a symmetric matrix over  $\mathbb{C}[[t]]$  of rank  $r$  over  $\mathbb{C}((t))$ . Write its finite elementary exponents as  $0 \leq e_1 \leq \dots \leq e_r$  and put  $e_i = +\infty$  for  $i > r$ . For  $1 \leq p \leq hn$ , write  $p = hq + s$ ,  $0 \leq s < h$ . The ideal generated by the coordinates of  $\ell_{A(t)}^p$  in  $(B_{n,h})_p \otimes \mathbb{C}[[t]]$  is*

$$\left( t^{h \sum_{i=1}^q e_i + s e_{q+1}} \right). \quad (\text{AO.3})$$

An empty sum is zero, the last term is omitted if  $s = 0$ , and  $t^{+\infty}$  means the zero ideal.

*Proof.* A symmetric matrix over this discrete valuation ring is congruent to

$$\text{diag}(t^{e_1}, \dots, t^{e_r}, 0, \dots, 0).$$

Here is the required elementary reduction. Divide by the least valuation of an entry. A diagonal unit is a pivot; if only an off-diagonal entry is a unit, the value of the form on a sum of two basis vectors supplies a unit because 2 is invertible. Split off the resulting rank-one summand by elimination and continue. Every remaining unit has a square root in  $\mathbb{C}[[t]]$ , so the units can be removed from the diagonal. The finite exponents are the Smith exponents, as congruence is in particular equivalence.

Congruence acts on every graded piece of  $B_{n,h}$  by an invertible matrix over  $\mathbb{C}[[t]]$ . It therefore preserves the ideal in question. The apolar embedding

$$(B_{n,h})_p \longrightarrow \mathbb{C}[X_{ij}]_{hn-p}, \quad D \longmapsto D(\det X)^h$$

is injective and splits as a map of complex vector spaces. After tensoring with  $\mathbb{C}[[t]]$ , the coordinates of a vector and of its image generate the same ideal. We may therefore calculate the coefficient ideal of  $[z^p] \det(X + zA(t))^h$ .

For diagonal  $A(t)$ , expand each of the  $h$  determinant factors in its chosen diagonal entries. Every term is divisible by  $t^{\sum c_i e_i}$  with  $0 \leq c_i \leq h$  and  $\sum c_i = p$ . The least such exponent is obtained by filling the smallest  $e_i$  to capacity  $h$ , giving (AO.3). There is no cancellation at this least exponent: specialize  $X$  to  $\text{diag}(x_1, \dots, x_n)$ . Then

$$\det(X + zA(t))^h = \prod_i (x_i + zt^{e_i})^h$$

(with  $x_i^h$  for a zero diagonal entry of  $A$ ). The monomial  $\prod x_i^{h-c_i}$  has coefficient  $\prod \binom{h}{c_i} t^{\sum c_i e_i}$ , which is nonzero and has exactly the required valuation. If  $p > hr$ , every term is zero. This proves all the conventions, including the zero-ideal case.  $\square$

**Theorem AO.2** (A reciprocal multiplication model for every spectral Fitting ideal). *For every complex quadratic pencil, regular or singular, and for  $0 \leq j \leq n$ , there is an equality of coherent ideal sheaves on its intrinsic line*

$$\mathcal{J}_{h(n-j)} = \text{Fitt}_j(\mathcal{T}_R)^h. \quad (\text{AO.4})$$

*In particular, when  $h = 1$ , the complete spectral Fitting schemes, including their multiplicities and any whole-line components, are exactly the power-zero schemes inside the one quadratic section of the normalization fibre at  $K_1$ .*

For a regular pencil and a spectral point  $x$ , set  $b_{x,j} = \text{length } \mathcal{O}_{\Lambda,x} / \text{Fitt}_j(\mathcal{T}_R)_x$ . The local length of the power-zero scheme is  $hb_{x,j}$ , and, for  $0 \leq j < n$ ,

$$\lambda_{x,j+1} = \frac{1}{h} \left( \text{length}(\mathcal{O}_{\Lambda,x} / \mathcal{J}_{h(n-j),x}) - \text{length}(\mathcal{O}_{\Lambda,x} / \mathcal{J}_{h(n-j-1),x}) \right). \quad (\text{AO.5})$$

Thus its multiplication, together with the embedded pencil line, recovers the full Segre data and their common projective support. For  $n \geq 3$ , the unmarked inverse makes this an invariant of the finite failure algebra. For a singular pencil, the construction recovers its normal rank and all its spectral Fitting ideals; it does not assert that these ideals alone recover the singular minimal indices.

*Proof.* At a point of the smooth curve  $\Lambda$ , pass to the completed discrete valuation ring. The ideal of  $(n-j)$ -minors of a matrix with the stated Smith exponents is  $(t^{e_1+\dots+e_{n-j}})$ , with the zero-ideal convention. Apply Lemma AO.1 with  $p = h(n-j)$ . The equality of completed ideals descends by faithful flatness. This proves (AO.4) at every stalk and hence globally. This proof does not replace an ideal by its radical.

For a regular pencil the positive Smith exponents at  $x$ , in reverse order, are  $\lambda_{x,1} \geq \lambda_{x,2} \geq \dots$ . Consequently  $b_{x,j} = \sum_{i>j} \lambda_{x,i}$ . Equation (AO.4) multiplies these lengths by  $h$ , so taking successive differences proves (AO.5). The powers that vanish at the generic point similarly determine the normal rank in the singular case. Frame changes act on  $\Lambda$  by projectivities and congruences act equivariantly on the reciprocal construction. Theorem I.8.3 recovers precisely the needed pencil up to these changes, proving the intrinsic assertion.  $\square$

*Remark AO.3* (Families and the scope of the ideal identity). For a vector bundle and pencil subbundle over any complex base, the quadratic-section algebra is obtained by applying the fixed  $\text{GL}(V)$ -equivariant construction to local frames. Its graded pieces are vector bundles, and the coefficient ideal of each power map commutes with arbitrary base change: it is the image of the dual of a map between vector bundles. No flatness of its zero scheme is needed for that assertion. Formula (AO.4) has been proved on each complex pencil line (and on a regular one-dimensional base by the same local argument). We do not infer an equality of those relative ideals on an arbitrary nonreduced parameter space from equality on its geometric fibres. Nor are the power-zero schemes asserted to be flat across a change in their length.

**Corollary AO.4** (The native specialization order in reciprocal coordinates). *On the fixed-discriminant, fixed-corank locus  $X_{m,c}(A)$  of (I.T.1), a specialization from the labelled partition tuple  $\lambda$  to  $\mu$  exists if and only if, at every labelled root and every  $j$ , the local length of the corresponding power-zero scheme does not decrease. Equivalently the special power-zero divisor contains the generic one, using the common labelled spectral line. The reduced closure of a full  $O(A)$ -orbit within this locus is therefore specified by these length inequalities. Each inequality can be written as the vanishing of the requisite jets of the coordinates of the power map (AO.2). Every strict transition is realized by a flat family of the original finite failure neighbourhoods and is first distinguished at order  $n^2 + 2n - 4$ .*

*Proof.* For fixed multiplicity  $m_x$ , dominance  $\mu_x \preceq \lambda_x$  is equivalent to  $\sum_{i>j} \mu_{x,i} \geq \sum_{i>j} \lambda_{x,i}$  for every  $j$ . Use (AO.5) and Theorem I.T.2. In a local coordinate  $t$  at a labelled root, a coefficient ideal has colength at least  $b$  exactly when each of its generators is divisible by  $t^b$ ; this is the asserted finite jet condition. The closure order and the flat realization are the inherited full-orthogonal results, ultimately using Ohta's theorem. The new assertion is their expression inside one reciprocal multiplication algebra. No reducedness assertion is made for the scheme cut out by the jet equations before taking its reduction.  $\square$

**Example AO.5** (A rank profile with a genuinely different power-zero scheme). At a root of multiplicity four the partitions  $(3,1)$  and  $(2,2)$  have equal determinant and corank. In  $B_{4,h}$  their

sixth powers when  $h = 2$  (in general their  $3h$ th powers) cut out local schemes of lengths  $h$  and  $2h$ , respectively. Their  $4h$ th powers have length  $4h$  in both cases. Thus the nonreduced power-zero scheme, not just the set of non-Lefschetz elements, distinguishes them. The specialization  $(3, 1) \rightsquigarrow (2, 2)$  increases the first of these lengths. Additional simple spectral blocks give the same distinction in all larger dimensions, with power indices  $h(n - 1)$  and  $hn$ .

The representation theory and strong Lefschetz property used above are classical inputs, explicitly credited to Nagaoka–Wachi and, for  $h = 1$ , Shafiei. The geometric statements here are the exact section of the whole reciprocal family, its size and Loewy-depth consequences for actual first-relation boundary fibres, and the power-ideal identity on pencil lines. The last construction is an intrinsically induced reciprocal model; it is not a claim that the common extreme  $\mathrm{GL}(E)$  boundary point remembers the starting pencil. Its embedded pencil line is essential.

The preceding relative caution concerns the valuation-ring proof. The following polynomial identity establishes the equality over arbitrary complex base algebras.

#### APPENDIX AP. INCIDENCE DESCENT AND THE LOCAL GRAPH CALCULATIONS

We record the scheme and deformation details used by the reduced and nonreduced boundary theorems. These also fix the relation between the local coordinates and their invariant meanings.

**Proposition AP.1** (The reduced open and its tangent sequence). *In the regular-pencil locus remove lines meeting rank at most  $n - 3$ . The universal corank-two incidence  $D \rightarrow U$  is finite. Its open locus of reduced geometric fibres includes empty fibres and is precisely*

$$U_{\mathrm{red}} = U \setminus q(\mathrm{Supp} \Omega_{D/U}).$$

At  $(p, R)$  in this open, with  $Z = Z_{n-2} \subset P = \mathbb{P}(M)$ , there is an exact tangent-normal sequence

$$0 \longrightarrow T_R \Sigma_p \longrightarrow T_R G \xrightarrow{e_p} N_{Z/P, p} / \overline{T_p L_R} \longrightarrow 0. \quad (\mathrm{AP}.1)$$

For distinct incidence points the direct sum of the maps  $e_p$  is surjective, without a semisimplicity assumption. Each  $\Sigma_p$  is smooth of codimension two. The local labelling of these centres is étale; permutations on overlaps give the unlabelled boundary and its Fitting ideal.

*Proof.* The determinant is a nonzero polynomial on a regular pencil, so its corank-two incidence is finite on every fibre. The incidence projection is proper, and is therefore finite by the proper quasi-finite theorem [Stacks Project, Tag 02LS]. The rank- $(n - 2)$  locus on this open is smooth of codimension three, by its three Schur-complement entries. A finite complex algebra has zero Kahler differentials if and only if it is reduced: after decomposition into local Artin factors, a nontrivial maximal ideal has nonzero cotangent space by Nakayama; a residue field  $\mathbb{C}$  contributes zero. The same holds geometrically after extension of the residue field in characteristic zero. Formation of relative differentials commutes with that extension. Their support has closed finite image. Its complement is the asserted open, and the zero algebra, representing an empty fibre, is allowed. Accordingly  $\mathrm{Fitt}_0(0) = \mathcal{O}$ .

Let  $\ell \subset R \subset M$  represent  $p$ . Then  $T_R G = \mathrm{Hom}(R, M/R)$  and  $T_p P / T_p L_R = \mathrm{Hom}(\ell, M/R)$ . Restriction from  $R$  to  $\ell$  is surjective. Compose it with the quotient to  $T_p P / (T_p Z + T_p L_R)$ ; this is exactly  $e_p$  in (AP.1). Moving the incidence point along the line and inside  $Z$  shows that its kernel is  $T_R \Sigma_p$ . For the convention  $A : V^* \rightarrow V$ , put  $K = \ker A \subset V^*$ . Restriction of the represented bilinear form to  $K$  identifies the normal space with  $\mathrm{Hom}(\ell, \mathrm{Sym}^2 K^*) = \ell^* \otimes \mathrm{Sym}^2 K^*$ . Thus the projective factor  $\ell^*$  occurs on both sides of the restriction map; it is not silently discarded. A frame of  $\ell$  identifies the map with restriction of the symmetric variation to the radical. Changing that frame multiplies both maps by a unit. A reduced intersection makes the normal image of  $T_p L_R$  a line, so the final quotient has dimension two.

For several points choose a nonsingular member and its self-adjoint operator. Radicals at distinct eigenvalues have direct sum, even when Jordan blocks are present. A symmetric form can prescribe its restrictions on this direct sum independently. Choose a pencil coefficient combination nonzero at every one of the finitely many points; varying that coefficient realizes all restrictions. Variations in the pencil itself land in their respective tangent-normal lines, so the joint map descends to  $\text{Hom}(R, M/R)$  and is still surjective. Smooth evaluation of the universal line to  $P$  also gives the incidence dimension  $\dim G + 1 - 3 = \dim G - 2$ .

Finally strict henselization of the base splits the finite algebra into its finitely many local idempotent factors. Every idempotent and its finitely many equations descend to an étale neighbourhood, since the strict henselization is a filtered limit of such neighbourhoods. Each factor has one-dimensional closed fibre, so its unit map has zero cokernel after tensoring with that residue field. Nakayama, followed by shrinking, makes the unit map surjective. It is therefore a closed immersion, not just a labelled point set. The tangent sequence identifies its two local equations, and joint surjectivity makes all equation pairs independent. Labels on étale overlaps are permuted. This proves the claimed descent description.  $\square$

**Lemma AP.2** (Independent-centre Proj and normalization). *Suppose étale locally a smooth base has independent parameter pairs  $I_i = (u_i, v_i)$ , with any number of complementary parameters. Then*

$$\text{Bl}_{\prod_i I_i} U = \prod_U \text{Bl}_{I_i} U.$$

*The multi-Rees algebra has the relations  $v_i U_i - u_i V_i$ , and its diagonal subalgebra is standard graded and is the ordinary Rees algebra of  $\prod_i I_i$ . Its ordinary Proj is the displayed scheme; the full multigraded algebra uses MultiProj.*

*Proof.* In polynomial coordinates tensor the one-pair Rees injections over  $\mathbb{C}$ ; all variables are independent. The same relations persist under étale flat extension. On a chart choose for each  $i$  either  $u_i$  or  $v_i$  as denominator. Localizing by the product of the chosen generators and taking diagonal degree zero adjoins exactly the individual ratios, with equations  $v_i = u_i w_i$  or their opposites. These are the fibre-product blow-up charts. The diagonal degree-one space consists of all products of one generator from every pair; its  $d$ th products give precisely  $(\prod I_i)^d$ . This proves the standard-grading and ordinary-Proj assertions. The equality of intersection and product of the ideals is checked on disjoint monomial variable blocks and descends by faithful flatness.

In its Hilbert application the reduced graph closure is integral and has the same function field as the pencil base, since both contain the same dense nonincident open. Over a fixed pencil distinct choices of exceptional lines produce distinct embedded subschemes: restriction to each fixed exceptional plane recovers its chosen line. Hilbert points do not identify them by an ambient congruence. Thus the proper map from the smooth product blow-up has finite geometric fibres, is quasi-finite, and is finite by [Stacks Project, Tag 02LS]. Only after that finiteness step does normality identify the source with the normalization of the integral graph in their common function field.  $\square$

**Lemma AP.3** (Jordan constants and normal thick charts). *For  $L_m = E_m(xI - N_m)$  the determinant of its complementary  $(m-1)$ -block and that of  $E_m$  are both  $(-1)^{m(m-1)/2}$ . Thus its scalar Schur complement is exactly  $x^m$ . In the two-equal-block family (V.1), the determinant constant  $c_m$  is one. The homogenized pencil at infinity is  $E_m \oplus E_m$  and is nonsingular. Its thick graph charts are flat and normal; their special fibres have only their displayed component primes.*

*Proof.* The complementary block is anti-triangular of size  $m-1$  with antidiagonal  $-1$ . Its sign is  $(-1)^{m-1+(m-1)(m-2)/2} = (-1)^{m(m-1)/2}$ . The determinant ratio proves the claim, and the two signs multiply to one. Homogenization gives  $E_m \oplus E_m$  when the finite-coordinate homogenizing variable is zero.



In the chart  $x^m = tz$ , the coordinate ring is free of rank  $m$  over the base polynomial ring in  $z$ , with basis  $1, x, \dots, x^{m-1}$ . In the other chart  $t = x^m y$ , the hypersurface equation is a nonzerodivisor in every geometric fibre of its smooth ambient relative affine plane. The fibrewise Cartier criterion [Stacks Project, Tag 062Y, part (3)] proves its flatness. The chart  $t = x^m y$  is smooth after eliminating  $t$ . The chart  $x^m = tz$  is a domain and a hypersurface, hence Cohen–Macaulay. Its singular locus  $x = t = z = 0$  has codimension two in the total space. It satisfies  $S_2$  and  $R_1$ , hence is normal by Serre’s criterion. At  $t = 0$  the  $t = x^m y$  chart has the irredundant primary decomposition

$$(x^m y) = (x^m) \cap (y) \quad \text{in } \mathbb{C}[x, y].$$

Its associated primes are  $(x)$  and  $(y)$ . The other chart  $(x^m)$  has the sole associated prime  $(x)$ . Polynomial smooth parameters do not change these conclusions. This gives the exact absence of embedded points, not just the support of the fibre.  $\square$

**Proposition AP.4** (Power coefficient modules on Artin tails). *Let  $A = \mathbb{C}[x]/(d)$  for a monic polynomial of degree  $r$ , and let  $T = \mathbb{P}_A^1$  be a primitive line tail in the corank-two normal plane. For  $1 \leq j \leq hn$ , put*

$$\delta_j = (j - h(n - 2))_+ - 2(j - h(n - 1))_+.$$

*After removal of the common Cartier factors, the actual coefficient map of the  $j$ th power surjects onto  $H^0(T, \mathcal{O}_T(\delta_j)) = \text{Sym}^{\delta_j}(A^2)^*$ . In particular it is the full relative system, including nilpotent coefficients; its cycle degree is  $r\delta_j$ .*

*Proof.* The unit  $(n - 2)$ -block supplies a socle factor with a unit coordinate, so its multiplication inclusion on the residual coefficient module is split over  $A$ . Put  $b = j - h(n - 2) > 0$  when the normal power is nonconstant. On the residual symmetric two-space its degree- $b$  coefficient span is, by the universal identity,

$$A[\alpha, \beta, \gamma]_b \quad (b \leq h), \quad (\alpha\gamma - \beta^2)^{b-h} A[\alpha, \beta, \gamma]_{2h-b} \quad (b \geq h).$$

Remove the displayed determinant factor. A primitive line is a split rank-two submodule of the three-coordinate module over the semilocal Artin algebra  $A$ . Restriction of all degree- $\delta_j$  ternary forms therefore surjects onto all binary forms of that degree. Composition with the split unit-block map is the required explicit coefficient-module surjection. For  $j \leq h(n - 2)$  the normal system is constant, and for  $j = hn$  the same is true after removing the full determinant factor. Invertible Gaussian changes act invertibly on these modules. The original  $\mathcal{O}(j)$  twist on the pencil line is trivial over the affine Artin support; this trivialization is separate from the graph line bundle  $\mathcal{O}_T(\delta_j)$ . Thus no reduced-point argument or ignored twist occurs.  $\square$

**Proposition AP.5** (The transverse surface and ramified normalization). *For the intersection calculation in Proposition V.3, take the specified normal surface  $S_m = \text{Bl}_{(x^m, t)} \mathbb{A}_{x, t}^2$  and its proper exceptional curve  $F$ . Then  $mF$  is Cartier,  $F^2 = -1/m$ , and the minimal resolution chain starts at  $F$  with multiplicities  $m - 1, m - 2, \dots, 1$ . The main component meets  $F$  at the other, nonsingular end point of its parameter line; it does not meet the far end of that exceptional chain.*

*After  $t = s^m$ , the normalization is  $\text{Bl}_{(x, s)} \mathbb{A}^2$ . Its reduced tail maps to the original residual line by  $[X : S] \mapsto [X^m : S^m]$ . With its attachment marked at  $[1 : 0]$ , its automorphism group over that fixed map is  $\mu_m$ ; it is a stable nonconstant component. No other marked points are required for this assertion.*

*Proof.* On  $S_m$  the exceptional Cartier ideal is  $\mathcal{O}(-mF)$  and restricts to  $\mathcal{O}_F(1)$  on the proper curve  $F \simeq \mathbb{P}^1$ . Intersection with that proper curve is well-defined even though the surface is not proper. Hence  $(mF) \cdot F = -1$ . In the singular chart  $x^m = tz$  the quotient parametrization is  $t = A^m, z = B^m, x = AB$ . The intermediate primitive rays of the  $A_{m-1}$  resolution have orders of  $t$  equal to  $m - 1, \dots, 1$  in order away from the ray  $A = 0$ , which is  $F$ . The main component

meets  $F$  at  $z = \infty$  in the other smooth chart. The far end of the resolution chain faces the strict transform of  $z = 0$ , which is horizontal for the map to the  $t$ -line, not the main special-fibre component.

After ramification the two rings are  $\mathbb{C}[x, s, y]/(s^m - x^m y)$  and  $\mathbb{C}[x, s, z]/(x^m - s^m z)$ . Their normalizations are obtained by adjoining respectively  $s/x$  and  $x/s$ . These elements are integral, with  $m$ th powers  $y$  and  $z$ , and belong to the common function field  $\mathbb{C}(x, s)$ . The resulting rings  $\mathbb{C}[x, s/x]$  and  $\mathbb{C}[s, x/s]$  are regular and finite over the displayed rings, and glue to the ordinary point blow-up. This proves the normalization claim with its common function field. The graph coordinates are the  $m$ th powers of the exceptional coordinates, as stated. Its deck transformations multiply  $X/S$  by an  $m$ th root of unity; they fix the attachment and form a finite group. The local cotangent module  $T^1$  computed previously refers only to abstract total-surface hypersurface deformations. Neither this normalization nor that calculation identifies embedded, fixed-base relative, or raw Artin-algebra deformations with those abstract deformations.  $\square$

#### APPENDIX AQ. DESCENT AND COMPLETE-LOCAL DETAILS

The following statements supply the local-to-global steps used in Sections AH–AM. They also specify which assertions concern a completed image, a normalization fibre, or a family on an exact-divisor stratum.

**Lemma AQ.1** (Complete-local surjectivity and reduced completion). *In the cotangent-surjectivity assertion of Lemma AH.1, the rings are Noetherian complete local  $\mathbb{C}$ -algebras with residue field  $\mathbb{C}$ , and the homomorphism is local, continuous, and the identity on that residue field. Under these hypotheses cotangent surjectivity implies ring surjectivity. A reduced local ring essentially of finite type over  $\mathbb{C}$  has reduced completion.*

*Proof.* Choose preimages in the source maximal ideal of generators of the target cotangent space. Their degree- $q$  products span every quotient  $\mathfrak{m}_B^q/\mathfrak{m}_B^{q+1}$ . Given an element of  $B$ , first lift its residue, then its error successively in degrees  $q = 1, 2, \dots$ . The correction in degree  $q$  lies in  $\mathfrak{m}_B^q$ . Their sum converges in  $R$ , and continuity and separatedness imply that its image is the desired element of  $B$ . This is not a statement about arbitrary noncomplete local homomorphisms.

Finite-type complex rings are excellent. Quasi-excellent local rings are Nagata and have geometrically reduced formal fibres; see [29, Tags 07QV and 0BJ0]. For a reduced local ring  $A$ , the injection into the product of its finitely many minimal-prime quotients remains injective after flat completion. Each quotient is a Nagata domain and its completion is reduced. Thus  $\hat{A}$  is a subring of a product of reduced rings. This justifies the reduced completed-image step of the earlier lemma.  $\square$

**Lemma AQ.2** (Coprime cluster products over Artin bases). *Let  $A$  be a local Artin  $\mathbb{C}$ -algebra. If the reduction of a monic polynomial has a factorization into pairwise coprime monic factors, there is a unique factorization into monic lifts of the same degrees. In a factorization fibre, the functor of divisors with prescribed degrees in the distinct residual clusters is the product of the cluster divisor functors. Its coordinate algebra is therefore the tensor product of their Artin coordinate algebras.*

*Proof.* For two coprime factors  $p, q$ , the linear correction map

$$(\dot{p}, \dot{q}) \longmapsto q\dot{p} + p\dot{q}, \quad \deg \dot{p} < \deg p, \quad \deg \dot{q} < \deg q,$$

is an isomorphism on coefficient spaces over the residue field. Its kernel is zero by coprimeness, and source and target have equal dimension. It is therefore an isomorphism over  $A$  as well. Across a square-zero extension of Artin rings it uniquely corrects a lifted product to the given polynomial. Induction on the nilpotence order and then on the number of clusters proves existence

and uniqueness. Apply the same argument to both the marked divisor and its complement. The cluster product and decomposition constructions are mutually inverse on every Artin base; Yoneda gives the asserted product of finite schemes. Passing to inverse limits gives the corresponding assertion on complete local bases.  $\square$

**Lemma AQ.3** (The local choice and descent in the binary stratum). *The base-change identification of Theorem AI.1 holds as an identification of incidence functors, not only on closed fibres. The required choice of a member of the residual space coprime to the common divisor can be made on a Zariski open cover of the exact-gcd stratum; the choices are auxiliary and the resulting identifications glue uniquely.*

*Proof.* Write a point of the stratum as  $(G, L)$  in  $\mathbb{P}(S_s) \times X_{D-s,r}$ . The vanishing of a member of  $L$  at one of the finitely many roots of  $G$  is a proper linear condition on that member. Over  $\mathbb{C}$  a finite union of these proper hyperplanes does not cover  $L$ . Choose a linear combination avoiding them, lift its coefficients on a bundle-trivializing neighbourhood, and invert its resultant with  $G$ . This gives the required Zariski neighbourhood. These neighbourhoods cover the stratum and may be pulled back by any base change.

On the marked-divisor incidence, pass locally to a monic coordinate for the proposed divisor  $f$ . The quotient by  $f$  is finite free. On every geometric fibre  $f$  divides  $G$ , so the chosen residual member is invertible in that quotient, by coprimeness. Its multiplication matrix has invertible determinant locally, hence it is invertible also over a nonreduced base. Thus  $f \mid GL$  implies  $f \mid G$  as an equation on that base, and the converse is immediate. Monic division forces the remaining subspace. The identification is independent of which coprime residual member established cancellation, so it glues on overlaps. No lifting of arbitrary image points to the normalization is asserted.

The finite factorization map used there is flat by miracle flatness: its target local rings are regular, its source local rings are Cohen–Macaulay, it is finite, and dimensions agree. These are exactly the hypotheses of [29, Tag 00R4]. This names the flatness input separately from the incidence-functor argument.  $\square$

**Lemma AQ.4** (The coordinate equalizer and coherent conductor). *For the block ideals  $I_A$  of Theorem AJ.1, the completed image ring is the equalizer of the branch rings with their restrictions to all pairwise intersections. This assertion uses the coordinate monomial structure. The completed conductor calculation there implies equality of the corresponding coherent ideal sheaves on the squarefree open.*

*Proof.* A monomial has a coefficient on each branch where its support survives. If it survives on two branches, it also survives on their coordinate intersection. Pairwise agreement therefore assigns it one well-defined coefficient. Collecting these coefficients gives one formal series modulo the intersection of the  $I_A$ ; conversely such a series has compatible restrictions. Formal convergence causes no issue, since there are finitely many monomials in each degree. The claim would be false for general configurations of ideals without this coordinate argument.

Write  $A \subset B$  for the local finite normalization algebra. The conductor is  $\text{Ann}_A(B/A)$ , a coherent ideal. Flat completion preserves the exact finite-module sequence and its annihilator: compute the latter as the intersection of the kernels of multiplication on a finite generating set. The completed finite algebra is the product of the normal branch rings. Thus the conductor and the ideal of  $Y_{g+1}$  have equal completed stalks at every squarefree closed point. Faithful flatness gives equal stalks, and coherence on a finite-type complex scheme gives equality of the ideal sheaves.  $\square$

**Proposition AQ.5** (Finiteness and uniqueness of the collision diagram). *Every branch map in Theorem AK.1 is finite, and the completed incidence diagram is determined, up to the unique*

*isomorphism respecting its incidence data, by its marked-divisor functor. It is independent of the chosen frame and root coordinates.*

*Proof.* For each cluster  $P_i = A_i B'_i$ , the coefficients of the monic  $A_i$  are integral over the coefficients of  $P_i$ . This follows from the finite factorization morphism, or by expressing them as elementary symmetric functions of a selected subset of the roots in a finite splitting algebra. Division by a monic polynomial expresses every coefficient of  $B'_i$  polynomially in those of  $A_i$  and  $P_i$ . Likewise  $R_{ij} = A_i U_{ij}$  expresses all quotient coefficients in terms of  $A_i$  and  $R_{ij}$ . Consequently the branch algebra is generated over the ambient coefficient algebra by finitely many integral coefficients of the  $A_i$ , so it is finite. Completing preserves this finite-module assertion.

The equations represent the functor of a subbundle with a marked divisor and its uniquely determined quotient. A change of frame or local coordinate gives inverse natural transformations of this same functor and of its forgetful map to the ambient Grassmannian. The complete representing rings and their maps are uniquely isomorphic by Yoneda. Their scheme-theoretic images consequently agree as completed diagrams. This proves coordinate independence without identifying arbitrary Artin points of an image with lifted Artin points of its normalization.  $\square$

**Partition closures.** For a fixed partition  $(s_1, \dots, s_b)$ , the proper map  $(\mathbb{P}^1)^b \rightarrow \mathbb{P}(S_s)$  sending  $(\ell_i)$  to  $\prod \ell_i^{s_i}$  has closed image. Its open set of distinct  $\ell_i$  is dense. At the boundary precisely some of these points coincide, and the associated parts add. Every such coincidence can be approached from distinct points. This proves exactly the merging closure order in the fixed-degree symmetric product; compare the coincident-root constructions of [19, 20]. It is not a closure classification for all analytic singularities of the completed collision atlas.

**Equivariance of the singular Fitting calculation.** The group used in Theorem AL.1 is  $\mathrm{GL}_m(\mathbb{C})$ , acting simultaneously on the columns  $u$  and  $w$  and fixing  $\Delta$ . It preserves the presentation and therefore its intrinsic Fitting ideal. In the normalization the three displayed minors span, under this group, respectively  $\mathrm{Sym}^m(\mathbb{C}^m)t$ ,  $\Delta \mathrm{Sym}^m(\mathbb{C}^m)$ , and  $\mathrm{Sym}^{m+1}(\mathbb{C}^m)$ , with the coordinate/dual convention determined by the action on  $u$ . To see spanning directly, their translates are nonzero scalar multiples of the corresponding powers of all linear forms; polarization in characteristic zero spans each symmetric power. Multiplication by  $R$  gives exactly the modules asserted in (AL.1). This is stronger than invoking invariance without identifying the representations.

**Specialization of minimal relations.** For the surjective coefficient specialization  $\pi : P \rightarrow P'$  to one coordinate in Theorem AM.1, one has  $\pi(\mathbf{n}) = \mathbf{n}'$  and  $\pi(\mathcal{I}) = \mathcal{I}'$ . Consequently  $\pi(\mathbf{n}\mathcal{I}) = \mathbf{n}'\mathcal{I}'$ . A coefficient whose image is a minimal binary relation cannot lie in  $\mathbf{n}\mathcal{I}$ . This explicitly justifies the nonzero irreducible coefficient map used in that proof. All reciprocal series and socle weights in these arguments use the weight of a  $Q_i$  coefficient equal to  $i$ . They do not claim ordinary maximal-ideal Hilbert functions; (AN.7) gives one such function separately.

## APPENDIX AR. CONDUCTORS AND LOCAL MODELS OF DIVISOR COLLISIONS

We retain the notation of Section AG. All completed local rings below are completed at closed complex points. For a reduced local ring  $A$  with finite normalization  $B$ , its *conductor* is  $\mathfrak{c} = \mathrm{Ann}_A(B/A)$ , regarded also as an ideal of  $B$ . We use the intrinsic singular subscheme defined by  $\mathrm{Fitt}_{\dim A} \Omega_{A/\mathbb{C}}$ ; on completions, differentials mean continuous differentials. The conductor and this Fitting ideal need not agree. Our first result determines both on a multivariate stratum. The second identifies the way two branches merge when a common binary root becomes double.

The formal-image and separated-germ steps below are isolated in Lemma AH.1. Lemma AH.2 gives the Artin cancellation and a direct positive lower bound for the split codimension. Theorem AK.1 supplies the full Weierstrass incidence functor in every collision order; Theorem AL.1 evaluates the singular ideal for all ranks.

**AR.1. Two coprime divisor choices.**

**Theorem AR.1** (The split-divisor stratum). *Suppose  $h > g$ ,  $2 \leq r \leq s_{h-g}$ , and*

$$K = abL, \quad [a], [b] \in \mathbb{P}(S_g), \quad L \in \text{Gr}(r, S_{h-g}), \quad \gcd(L) = 1.$$

*Assume that  $a, b$  are coprime and that the only degree- $g$  divisors of  $ab$ , up to scalar, are  $a$  and  $b$ . Put*

$$\begin{aligned} t &= 2(s_g - 1) + r(s_{h-g} - r), \\ c &= r(s_h - s_{h-g}) - (s_g - 1), \quad d_g = t + c. \end{aligned}$$

*Then  $c \geq 1$ , and there are formal coordinates in which*

$$\hat{\mathcal{O}}_{Y_g, K} = \mathbb{C}[[z_1, \dots, z_t, x_1, \dots, x_c, y_1, \dots, y_c]] / (x_i y_j : 1 \leq i, j \leq c). \quad (\text{AR.1})$$

*Its normalization is  $\mathbb{C}[[z, x]] \times \mathbb{C}[[z, y]]$ . The conductor is*

$$\mathfrak{c} = (x_1, \dots, x_c, y_1, \dots, y_c), \quad \hat{\mathcal{O}}_{Y_g, K} / \mathfrak{c} = \mathbb{C}[[z]], \quad (\text{AR.2})$$

*and the intrinsic singular subscheme has ideal  $\mathfrak{c}^c$ . The two smooth branches meet scheme-theoretically in a smooth  $t$ -dimensional subscheme. The local ring has multiplicity two and depth  $t + 1$ ; it is Cohen–Macaulay exactly when  $c = 1$ .*

*Proof.* We first determine the intersection as a scheme, not by comparing closed points or tangent dimensions. The normalization has precisely two points above  $K$ , namely  $(a, bL)$  and  $(b, aL)$ . At either point, the overlap in Lemma AG.3 is zero. The map from the completed local ring of the ambient Grassmannian to each completed source ring is consequently surjective: its cotangent map is surjective, and completeness and Nakayama’s lemma lift successive homogeneous terms. Thus each source completion is a closed smooth formal branch of the image.

Here is the required assertion over infinitesimal bases. Let  $A_0$  be a local Artin  $\mathbb{C}$ -algebra. A point on both formal branches carries unique marked lifts  $a_{A_0}, b_{A_0}$  near  $a, b$ . If homogeneous polynomials have coprime reductions, those reductions form a regular sequence in the polynomial ring over  $\mathbb{C}$ . Its graded Koszul complex is exact. In each degree the polynomial modules are finite free; lifting a splitting of the complex, or applying the local criterion for flatness successively, shows that its lift over  $A_0$  is exact and that the quotient is  $A_0$ -flat. In particular

$$(a_{A_0}) \cap (b_{A_0}) = (a_{A_0} b_{A_0}). \quad (\text{AR.3})$$

This also follows by noting that  $b_{A_0}$  is a nonzerodivisor modulo  $a_{A_0}$  and solving a relation between them. Applying (AR.3) to the columns of the relation subbundle gives, uniquely,

$$K_{A_0} = a_{A_0} b_{A_0} L_{A_0}.$$

The residual module is a subbundle: multiplication by  $a_{A_0} b_{A_0}$  is a split injection in degree  $D$ , and the quotient of the ambient free module by  $K_{A_0}$  is free. The kernel of the induced surjection onto the complementary quotient is therefore free. This proves that the formal intersection is represented by the product of the two divisor charts and the Grassmannian chart for  $L$ , with no nilpotents omitted.

The differential of this product parametrization is injective. Indeed, an infinitesimally constant product gives

$$(\dot{a}b + a\dot{b})l + ab\dot{l} \in abL \quad (l \in L).$$

Taking valuations at the factors of  $a$ , and using  $\gcd(a, b) = \gcd(L) = 1$ , forces  $a \mid \dot{a}$ ; similarly  $b \mid \dot{b}$ . These are the two projective scalar directions, and then  $\dot{L} = 0$ . The intersection is thus a smooth formal closed subscheme of dimension  $t$ .

Let  $R_0$  be the ambient completed local ring and  $I_a, I_b$  the ideals of these branches. Finiteness of the normalization, exactness of completion on finite modules, and the definition of the scheme-theoretic image give

$$\widehat{\mathcal{O}}_{Y_g, K} = R_0 / (I_a \cap I_b) = (R_0 / I_a) \times_{R_0 / (I_a + I_b)} (R_0 / I_b).$$

This is the elementary conductor-square construction; see also [21, Lemma 1.3 and §1.3.2]. All three quotient rings are regular. Choosing compatible regular parameters for the two surjections gives (AR.1). The branch dimension is  $d_g = s_g - 1 + r(s_h - r)$ , so its codimension along the intersection is the asserted  $c$ . It is positive: if  $c = 0$ , both branch ideals would coincide locally with the intersection ideal, contradicting the two distinct normalization branches of a reduced local ring.

In the fibre-product description, an element belongs to the conductor exactly when both its restrictions to the intersection vanish. This gives (AR.2). The Jacobian of the relations  $x_i y_j$  has nonzero entries only among the  $x_i$  and  $y_j$ . The Fitting ideal in question is its ideal of  $c$ -minors. Every such minor belongs to  $\mathfrak{c}^c$ . Conversely, choosing the  $c$  columns indexed by the  $y_j$  and one row  $(i_j, j)$  for each  $j$  gives every monomial  $x_{i_1} \cdots x_{i_c}$ . The symmetric choice gives every degree- $c$  monomial in the  $y$ 's. Mixed monomials vanish in the ring. Thus the ideal is exactly  $\mathfrak{c}^c$ , including its nonreduced structure.

Finally use

$$0 \longrightarrow \widehat{\mathcal{O}}_{Y_g, K} \longrightarrow \mathbb{C}[[z, x]] \oplus \mathbb{C}[[z, y]] \longrightarrow \mathbb{C}[[z]] \longrightarrow 0,$$

where the last arrow is the difference of restrictions. The depth lemma gives depth  $t + 1$ ; for  $c = 1$  this equals the dimension. For  $c > 1$  it is strictly smaller. Additivity of leading Hilbert coefficients gives multiplicity two, since the last module has smaller dimension than the two regular branches.  $\square$

*Remark AR.2.* The hypotheses are nonempty in every number of variables: for  $e = 2$  one may take  $g = 1$ , and for  $e \geq 3$  one may take two distinct irreducible forms of degree  $g$ . The theorem treats an explicitly specified stratum, not all pairs of overlapping degree- $g$  divisors. In particular, it does not substitute a binary factorization rule for multivariate divisibility. Its high-codimension branch intersections need not be normal-crossing divisors.

**AR.2. A double common root.** For  $m \geq 1$ , put  $R_m = \mathbb{C}[[u_1, \dots, u_m, \Delta]]$  and  $I = (u_1, \dots, u_m) \subset R_m$ . Define

$$A_m = R_m \oplus It \subset B_m = \mathbb{C}[[u_1, \dots, u_m, t]], \quad \Delta = t^2. \quad (\text{AR.4})$$

The right side of (AR.4) is a subring, not a square-zero extension:  $(It)^2 = \Delta I^2$ . For  $m = 1$  it is the ordinary pinch-point ring.

**Theorem AR.3** (Collision of two binary divisor choices). *Let  $e = 2$ ,  $g = 1$ ,  $D = h + 1$ , and  $2 \leq r \leq D - 1$ . At a point  $K_0 = \ell^2 L$  with  $L \in \text{Gr}(r, S_{D-2})$  and  $\gcd(L) = 1$ , set*

$$m = r - 1, \quad q = r(D - 1 - r) + 1.$$

Then

$$\widehat{\mathcal{O}}_{Y_1, K_0} \simeq A_m[[z_1, \dots, z_q]]. \quad (\text{AR.5})$$

Writing  $w_i = u_i t$ , a complete presentation of  $A_m$  is

$$\mathbb{C}[[u_1, \dots, u_m, \Delta, w_1, \dots, w_m]] / (u_i w_j - u_j w_i, w_i w_j - \Delta u_i u_j), \quad (\text{AR.6})$$

where the first relations have  $i < j$  and the second have  $i \leq j$ . Its normalization is  $B_m$ , and its conductor is

$$\mathfrak{c} = (u_1, \dots, u_m, w_1, \dots, w_m), \quad \mathfrak{c} B_m = I B_m.$$

The induced map on conductor quotients is

$$\mathbb{C}[[\Delta]] \longrightarrow \mathbb{C}[[t]], \quad \Delta \longmapsto t^2. \quad (\text{AR.7})$$

Thus the two reduced divisor lifts merge through a ramified double cover of the conductor. The completed local ring has dimension  $q + m + 1$  and depth  $q + 2$ . It is Cohen–Macaulay if and only if  $r = 2$ . For  $r = 2$ , its intrinsic singular ideal, with the smooth variables suppressed, is  $(w, \Delta u, u^2)$ , whereas its conductor is  $(u, w)$ .

*Proof.* Choose an affine coordinate  $x$  with the root of  $\ell$  at zero. Some member  $F_0 \in K_0$  has multiplicity exactly two there, since  $\gcd(L) = 1$ . Extend it to a frame  $F_0, \dots, F_{r-1}$ . In the completed local ring of the ambient Grassmannian the formal Weierstrass theorem writes the universal  $F_0$  as a unit times

$$P(x) = x^2 + ax + b.$$

Divide each  $F_i$ ,  $i > 0$ , by this monic polynomial. Its remainder is  $u_i x + v_i$ . The  $2r$  functions  $a, b, u_1, v_1, \dots, u_m, v_m$  are part of a regular parameter system. To see this rather than assume versality, note that  $K_0$  has zero two-jet at the root, while  $S_D/K_0 \rightarrow \mathbb{C}[x]/(x^2)$  is surjective. The Grassmannian tangent space  $\text{Hom}(K_0, S_D/K_0)$  therefore allows independent variations of both jets in every frame column. Passing from these jets to the displayed remainders is an invertible triangular operation, with leading coefficient  $F_0/x^2(0) \neq 0$ . Their differentials are consequently independent.

Set

$$t = x + a/2, \quad \Delta = a^2/4 - b, \quad w_i = au_i/2 - v_i.$$

The incidence of a common root near zero is now exactly

$$\Delta = t^2, \quad w_i = u_i t.$$

There are no other normalization points above  $K_0$ , because its exact common divisor is  $\ell^2$ . Consequently the image of this completed incidence is the completed image defining  $Y_1$ . The ambient dimension is  $N_0 = r(D + 1 - r)$ . Of the  $2r$  parameters above,  $a$  is free, as are the other  $N_0 - 2r$  parameters. This accounts for  $q = N_0 - 2r + 1$  smooth variables in (AR.5).

It remains to justify elimination scheme-theoretically. The relations in (AR.6) reduce any series to  $p + \sum_i w_i p_i$  with  $p, p_i \in R_m$ . If its image in  $B_m$  is zero, the decomposition  $B_m = R_m \oplus R_m t$  gives  $p = 0$  and  $\sum_i u_i p_i = 0$ . The first syzygies of the regular sequence  $u_1, \dots, u_m$  are generated by the Koszul syzygies. They are precisely the relations  $u_i w_j - u_j w_i$ . This proves the presentation without taking a radical. The same argument works first over polynomial rings and then by exact completion.

The extension  $A_m \subset B_m$  is finite, since  $t^2 = \Delta$ , and birational, since  $t = w_1/u_1$  in the fraction field. The regular ring  $B_m$  is thus the normalization. For  $p + vt \in R_m \oplus It$ , multiplication by  $t$  stays in  $A_m$  if and only if  $p \in I$ . As  $1, t$  generate  $B_m$ , the conductor is  $I \oplus It$ , proving the quotient map (AR.7). Equivalently,

$$A_m = B_m \times_{\mathbb{C}[[t]]} \mathbb{C}[[\Delta]], \quad (\text{AR.8})$$

where the first map sets  $u = 0$  and the second sends  $\Delta$  to  $t^2$ . This is a specific conductor square, not an assumption that the normalization descends through its reduced fibres.

As an  $R_m$ -module,  $A_m = R_m \oplus I$ . For  $m = 1$ , the ideal  $I$  is free. For  $m > 1$ , the exact sequence  $0 \rightarrow I \rightarrow R_m \rightarrow \mathbb{C}[[\Delta]] \rightarrow 0$  gives  $\text{depth}_{R_m} I = 2$ . Hence  $A_m$  has depth two for every  $m \geq 1$ . The maximal ideal of  $R_m$  generates an ideal with maximal radical in  $A_m$ , so this is also its local ring depth. Adding the  $q$  smooth parameters gives the assertion. When  $m = 1$ , the equation is  $w^2 - \Delta u^2 = 0$ ; its partial derivatives generate  $(w, \Delta u, u^2)$  in characteristic zero.  $\square$

**Corollary AR.4** (The split and ramified parts of the same model). *On the binary locus where the common divisor has degree exactly two, there are two strata. Over two distinct roots the formal local ring has the split form (AR.1), with  $t = q + 1$  and  $c = r - 1$ . Over a double root it has (AR.5). The support of the singular locus in the latter model is the conductor, and the normalization fibre at a collision is  $\text{Spec } \mathbb{C}[t]/(t^2)$ . For arbitrary  $m$ , its singular subscheme is given without radicalization by the  $m$ -minors of the Jacobian of the displayed  $m^2$  relations in (AR.6).*

*Proof.* The assertion at distinct roots is Theorem AR.1. In the collision model the normalization is an isomorphism wherever some  $u_i$  is invertible. On  $u = w = 0$ , a nonzero value of  $\Delta$  gives two distinct points after an étale square-root extension, while  $\Delta = 0$  gives the stated length-two fibre. The exact normality criterion proves the assertion about singular support. There are  $2m + 1$  nonsmooth coordinates and dimension  $m + 1$ ; the cotangent presentation therefore gives the  $m$ -minors as claimed. This formula specifies the whole singular scheme, even where it is thicker than its conductor support.  $\square$

*Remark AR.5* (Descent to intrinsic first relations). On the maximal-lower-Hilbert-function stratum of Corollary AG.8, these statements compute the completed smooth-atlas singularities of the intrinsic first-relation algebra stack after tangent-identity rigidification. The conductor is equivariant, and formation of its finite-module annihilator and of the Fitting ideals commutes with flat base change. Thus the calculations descend through the frame atlas. This does not identify these rings with completed local rings of a coarse orbit space, where stabilizers can introduce further singularities.

#### APPENDIX AS. INTERSECTIONS OF DIVISOR-DEGREE IMAGES

Here the total relation degree  $D$  is fixed. Write  $Y_a \subset \text{Gr}(r, S_D)$  for the image admitting a divisor of degree  $a$ ; empty Grassmannians are omitted. In several variables these images are not generally nested. Their reduced intersections nevertheless have a finite description by common parts of two marked divisors.

**Proposition AS.1** (The common-part formula). *For  $1 \leq a, b < D$  and  $0 \leq j \leq \min(a, b)$ , let  $M_j$  be the scheme-theoretic image of*

$$\begin{aligned} \mathbb{P}(S_j) \times \mathbb{P}(S_{a-j}) \times \mathbb{P}(S_{b-j}) \times \text{Gr}(r, S_{D-a-b+j}) &\longrightarrow \text{Gr}(r, S_D), \\ (c, u, v, L) &\longmapsto cuvL. \end{aligned} \tag{AS.1}$$

*A negative last degree or insufficient dimension gives the empty image, and  $\mathbb{P}(S_0)$  is a point. Then*

$$(Y_a \cap Y_b)_{\text{red}} = \left( \bigcup_j M_j \right)_{\text{red}}. \tag{AS.2}$$

*On the locus of two marked divisors  $f_a, f_b$  with exact common part of degree  $j$ , the corresponding factorization is  $c = \gcd(f_a, f_b)$ ,  $f_a = cu$ ,  $f_b = cv$ , with  $\gcd(u, v) = 1$ . In particular, at a relation  $K = a_0^m L$  with  $a_0$  a nonzero linear form and  $\gcd(L) = 1$ , the set of positive divisor degrees is exactly  $\{1, \dots, m\}$ . For the boundary point in Theorem AT.1,  $m = D - q_n$ .*

*Proof.* Each map in (AS.1) is defined on a projective scheme: multiplication by a nonzero form has constant rank, so the subspace construction is a morphism. Its image is closed and admits the two divisors  $cu$  and  $cv$ , proving one inclusion. Conversely, if  $K$  has divisors  $f_a, f_b$ , unique factorization gives their common part  $c$  of some degree  $j$ . Their least common multiple is  $cuv$  and divides every member of  $K$ . Division yields  $L$  of degree  $D - a - b + j$ , of the same rank. This proves equality on geometric points. Over  $\mathbb{C}$ , a reduced closed subscheme of a finite-type



scheme is determined by its closed points, giving (AS.2). The last assertion follows because every homogeneous divisor of a power of one linear form is a power of that form.  $\square$

The reduction in (AS.2) is essential to the statement of this global formula. It is not a prescription to reduce the individual  $Y_a$ , their normalization fibres, or their singular subschemes. The full intersection of the two formal branches on the coprime stratum was computed over Artin bases in Theorem AR.1; Theorem AR.3 computes the thickening when those branches collide. Thus the degree-poset description and the local scheme calculations have distinct, explicit roles.

#### APPENDIX AT. A COMMON RAMIFIED BOUNDARY OF PENCIL FAILURE ORBITS

The fixed tensor presentation of a pencil has exact common divisor  $\delta^{n-1}$ . Varying the pencil alone therefore stays in the exact-divisor locus. The intrinsic finite algebra permits a larger, linear coordinate-change action: changing the basis of its entire cotangent space. We now describe a boundary point produced by such changes. The generic members are genuine pencil failure algebras. The central member has a larger common divisor, and its normalization fibre is nonreduced. Thus the boundary of Section AG occurs inside the orbit closures of the motivating algebras themselves.

Let  $n = \dim V = \dim U \geq 3$ ,  $E = V^* \otimes U$ , and put

$$e = n^2, \quad N = \binom{n+1}{2}, \quad r = \binom{N}{2}, \quad g = n(n-1), \quad h = 3n-4, \quad D = g+h.$$

For a pencil  $R$ , write  $K_R = \delta^{n-1} C_R \subset \text{Sym}^D E$  as in Lemma I.M.1. Define the projective orbit closure

$$Z_R = \overline{\text{GL}(E) \cdot K_R} \subset \text{Gr}(r, \text{Sym}^D E).$$

It is independent of a chosen basis of the cotangent space. Since  $Y_g$  is closed and invariant,  $Z_R \subset Y_g$ .

The group in  $Z_R$  is the full  $\text{GL}(E)$ , not the  $\text{PGL}(V)$  congruence group of the pencil. Fixing the standard flag, scalar direction and complement specifies the displayed common point; no choice-independent distinguished moduli point is asserted.

**Theorem AT.1** (A common point in full first-relation orbit closures). *Set*

$$q_n = \begin{cases} 3, & n = 3, \\ 4, & n \geq 4, \end{cases} \quad k_n = h - q_n.$$

*There is an explicitly constructed point  $K_\infty$  belonging to  $Z_R$  for every quadratic pencil  $R$ , with*

$$K_\infty = a^{g+k_n} L_n, \quad L_n \subset \text{Sym}^{q_n} E, \quad \text{rank } L_n = r, \quad \gcd(L_n) = 1, \quad (\text{AT.1})$$

*where  $a \in E$  is a nonzero linear form. The flag pencil  $R_* = \langle x_1^2, x_1 x_2 \rangle$  admits a finite-flat family of first-relation algebras over  $\mathbb{A}^1$ , with special relation space  $K_\infty$  and every nonzero fibre isomorphic to its failure algebra. Together with Theorem I.O.1, this gives a chain of two explicit finite-flat specializations from the failure algebra of every  $R$  to the same boundary algebra.*

*The fibre of  $\nu_g$  over  $K_\infty$  has a single geometric point, marked by  $a^g$ , and is nonreduced. Its tangent dimension is*

$$\binom{e + k_n - 1}{k_n} - 1. \quad (\text{AT.2})$$

*In particular  $K_\infty$  lies in the singular locus of  $Y_g$  and outside the exact-degree- $g$  locus. For  $n = 3$  this tangent dimension is 44.*

The construction requires the exact osculating order of one elementary flag embedding. We include the calculation to distinguish the theorem from a computationally suggested degeneration.

**Lemma AT.2** (The flag coefficient filtration). *Write a matrix of independent linear coordinates as*

$$T = aI_n + B, \quad B_{11} = 0.$$

*For  $C_* = C_{R_*}$ , the filtration by vanishing order at  $T = I_n$  on the chart  $T_{11} = 1$  has last nonzero graded piece in degree  $q_n$ . It has a nonzero degree-zero piece. Hence an adapted basis  $c_1, \dots, c_r$  has orders  $0 \leq d_j \leq q_n$ , attaining both endpoints, and*

$$\lim_{\tau \rightarrow 0} \tau^{-d_j} c_j(aI_n + \tau B) = a^{h-d_j} p_j(B), \quad (\text{AT.3})$$

*where the displayed initial forms are linearly independent.*

*Proof.* The complementary-minor identity for  $\text{Sym}^2 T$  identifies the vector of primitive coefficients, up to a fixed invertible coordinate change, with

$$\delta(T)^3(v^2 \wedge vw), \quad v = T^{-1}e_1, \quad w = T^{-1}e_2. \quad (\text{AT.4})$$

Indeed  $\det(\text{Sym}^2 T) = \delta^{n+1}$ , the complementary minors have order two, and division of  $\delta I_{N-2}$  by  $\delta^{n-1}$  leaves  $\delta^3$ . This identity is initially on  $\delta \neq 0$ ; both sides are the same polynomial vector of degree  $h$ .

A standard affine flag chart has representatives

$$v = e_1 + \sum_{i=2}^n z_i e_i, \quad w = e_2 + \sum_{i=3}^n y_i e_i.$$

Every coordinate of  $v^2 \wedge vw$  is a polynomial of degree at most four. The map from  $\text{GL}_n$  to this flag chart is smooth, and restricting to  $T_{11} = 1$  remains smooth: the scalar subgroup is contained in each flag stabilizer and is transverse to this chart. Normalizing  $v, w$  only multiplies the whole wedge by a common unit. Smooth pullback preserves the order of a nonzero function, as does multiplication by a unit. The coordinates in (AT.4) are linearly independent by Lemma I.6.2 and the nonzero matrix-coefficient inclusion in (I.6.6). Thus no nonzero linear combination can have order exceeding four. For  $n \geq 4$ , the coordinate indexed by  $e_3^2 \wedge e_3 e_4$  is

$$z_3^2(z_3 y_4 - z_4 y_3),$$

which is nonzero and has order exactly four.

For  $n = 3$ , put  $z = z_2$ ,  $w = z_3$ ,  $t = y_3$ . In the ordered basis  $e_1^2, e_1 e_2, e_1 e_3, e_2^2, e_2 e_3, e_3^2$ , the two factors are

$$(1, 2z, 2w, z^2, 2zw, w^2), \quad (0, 1, t, z, zt + w, wt).$$

Their fifteen wedge coordinates have the same linear span as

$$\begin{aligned} 1, z, w, t, z^2, zw, w^2, zt, wt, z^2 t, zwt, w^2 t, \\ z^2 w - z^3 t, \quad zw^2 - z^2 wt, \quad w^3 - zw^2 t. \end{aligned} \quad (\text{AT.5})$$

This is obtained by subtracting the two coordinates with a common quadratic term and rescaling by 2. Their lowest homogeneous terms are distinct monomials, all of degrees at most three, and some have degree three. Thus the third jet is injective on this coefficient space and the second is not. This proves  $q_3 = 3$ . The coordinate  $e_1^2 \wedge e_1 e_2$  is 1 in every dimension, proving the order-zero assertion. Finally, homogeneity of degree  $h$  gives (AT.3); an adapted basis is by definition a basis on each associated-graded piece, so its initial forms are independent.  $\square$

*Proof of Theorem AT.1.* The linear substitution  $T \mapsto aI_n + \tau B$  is invertible for  $\tau \neq 0$ . Divide the adapted coefficient  $c_j$  by its exact power  $\tau^{d_j}$  after substitution. There are no negative powers: all its terms have  $B$ -degree at least  $d_j$ . The resulting columns span a rank- $r$  subbundle  $J_\tau \subset \text{Sym}^h E \otimes \mathbb{C}[\tau]$ . Independence at zero follows from the lemma, and away from zero from the invertible substitution. The ideal of its maximal minors in  $\mathbb{C}[\tau]$  has no common closed zero, and

is therefore the unit ideal. On each maximal-minor open a corresponding square submatrix is invertible and splits the image. These opens cover  $\text{Spec } \mathbb{C}[\tau]$ , proving the subbundle assertion over the whole line, including all scheme points. Set

$$K_\tau = \det(aI_n + \tau B)^{n-1} J_\tau.$$

Multiplication by the nonzero homogeneous polynomial in every fibre again has constant rank. Consequently  $K_\tau$  is a subbundle, and

$$\mathcal{A}_\tau = \text{Sym } E \otimes \mathbb{C}[\tau] / (K_\tau, \mathfrak{m}^{D+1})$$

is finite locally free of rank  $\binom{e+D}{D} - r$ . For  $\tau \neq 0$ , its relation space is a linear transform of  $K_{R_*}$ .

At zero the marked factor becomes  $a^g$ , while  $J_0 = \langle a^{h-d_j} p_j(B) \rangle$ . It contains a nonzero multiple of  $a^h$ , so its gcd is a power of  $a$ . It also contains a polynomial with exact  $a$ -adic order  $h - q_n$ , and every column is divisible by that power. Therefore  $\gcd(J_0) = a^{k_n}$ , proving (AT.1). Notice that  $0 < k_n < g$  for every  $n \geq 3$ .

The closed-flag specialization of Theorem I.O.1 and its failure-algebra lift imply  $K_{R_*} \in Z_R$ . A closed invariant subset containing a point contains that point's orbit closure. Hence the just-constructed  $K_\infty \in Z_{R_*}$  belongs to every  $Z_R$ . This proves the assertion about the two-stage specialization without replacing it by an unproved assertion about one fixed one-parameter subgroup acting on every initial pencil.

The sole degree- $g$  divisor of  $a^{g+k_n}$  is  $a^g$ . Its overlap with the residual gcd is  $a^{k_n}$ . Lemma AG.3 gives (AT.2); the fibre is supported at one point and has positive tangent space. The normalization criterion then proves singularity. With  $n = 3$ ,  $e = 9$  and  $k_n = 2$ , the dimension is  $\binom{10}{2} - 1 = 44$ .  $\square$

**AT.1. The complete normalization fibre at the limiting algebra.** The last theorem is not restricted to the tangent space of its fibre. The full local algebra can be given uniformly.

**Proposition AT.3** (Pure-power divisor fibre). *Let  $E = \mathbb{C}a \oplus W$ ,  $g \geq k \geq 1$ , and  $K = a^{g+k}L$  with  $\gcd(L) = 1$ . Introduce coefficient variables for  $q_i \in \text{Sym}^i W$ ,  $1 \leq i \leq k$ , and define forms  $c_j$  recursively by*

$$c_0 = 1, \quad c_j = - \sum_{i=1}^k q_i c_{j-i}, \quad c_j = 0 \ (j < 0). \quad (\text{AT.6})$$

*The entire fibre of  $\nu_g$  over  $K$ , supported at its unique point, is the Artin algebra*

$$\mathbb{C}[[\text{coefficients of } q_1, \dots, q_k]] / (\text{coefficients of } c_{g+1}, \dots, c_{g+k}). \quad (\text{AT.7})$$

*Its tangent dimension is  $s_k(E) - 1$ .*

*Proof.* Choose  $p \in L$  not divisible by  $a$ . Over any local Artin base, a divisor  $f$  near  $a^g$  is monic in  $a$ , and the quotient by  $f$  is flat over the base. The reductions of  $f, p$  are coprime. The regular-sequence argument used in (AR.3) makes  $p$  a nonzerodivisor modulo  $f$ . Thus  $f \mid a^{g+k}p$  is equivalent to  $f \mid a^{g+k}$ , functorially on these bases. Once this condition holds, it holds for every member of  $K$ .

Polynomial division gives a unique monic residual factor  $H = a^k + q_1 a^{k-1} + \dots + q_k$ . The equation  $fH = a^{g+k}$  determines the coefficients of  $f$  successively as  $c_1, \dots, c_g$  in (AT.6). The remaining  $k$  product coefficients vanish if and only if  $c_{g+1}, \dots, c_{g+k}$  vanish. This follows successively by comparing the product with the reciprocal series  $(1 + \sum q_i a^{-i})^{-1}$ . Hence (AT.7) represents the full infinitesimal fibre, without reduction. Finiteness of  $\nu_g$  makes it Artin; there is no other geometric divisor, so completion loses no part of the fibre. Since  $g \geq k$ , these equations have no linear terms in the coefficient variables. The number of coefficient variables is  $\sum_{i=1}^k \dim \text{Sym}^i W = \dim \text{Sym}^k E - 1$ , as asserted.  $\square$

**Example AT.4** (The nine-generator limit). For  $n = 3$ , the construction can be written with four of the nine linear coordinates, denoted  $a, u, v, w$ . The other five coordinates remain generators of the algebra. Up to nonzero scalar changes of the relation basis,  $K_\infty = a^8 L_3$ , where

$$L_3 = \langle a^3, a^2u, a^2v, a^2w, au^2, auv, av^2, auw, avw, \\ u^2v, u^2w, uv^2, uvw, v^3, v^2w \rangle. \quad (\text{AT.8})$$

Here one may take  $u = B_{21}$ ,  $v = B_{31}$ ,  $w = B_{32}$ ; the monomials are the lowest terms in (AT.5). The relation space has dimension 15 and exact gcd  $a^8$ . The selected divisor has degree  $g = 6$ , and its fibre has 44 coefficient variables: eight coefficients of a linear form  $Q_1$  and thirty-six of a quadratic form  $Q_2$  in  $W$ . Its defining ideal consists of all coefficients of the two forms

$$c_7 = -Q_1^7 + 6Q_1^5Q_2 - 10Q_1^3Q_2^2 + 4Q_1Q_2^3, \\ c_8 = Q_1^8 - 7Q_1^6Q_2 + 15Q_1^4Q_2^2 - 10Q_1^2Q_2^3 + Q_2^4.$$

This is a full presentation of the normalization fibre. Its embedding dimension 44 is not being identified with its length, nor is this fibre algebra being identified with the completed local ring of  $Y_g$ .

*Remark AT.5* (Relation to pencil invariants). The action defining  $Z_R$  is the intrinsic  $\text{GL}(E)$  action on first relations, not merely the congruence action on  $V$ . The flag pencil is singular: every member has rank at most two. It is therefore outside the regular-pencil open set treated by [4, Theorem 1.1 and §5], where Segre symbols organize classification and degenerations. The theorem above does not purport to classify that poset or all orbits in  $Z_R$ . Its conclusion is instead a uniform common boundary and an exact ramification algebra for genuine finite failure algebras. The increase from  $g$  to  $g + k_n$  in the intrinsic gcd is precisely what is invisible in a fixed-determinant pencil specialization.

#### ACKNOWLEDGMENT OF ASSISTANCE

AI assistance was used in developing and drafting the arguments and finite verification scripts. All structural proofs are supplied for independent mathematical scrutiny; computational receipts are not formal proof certificates. No AI system is listed as an author.

For the present revision, GPT-6 Astra Pro assisted specifically with the incidence normalization and overlap criterion in Section AG, the universal-property proof in Section I.N, the explicit specialization in Section I.O, and the associated exact tests and exposition. These arguments require independent human verification. No statement here represents a completed journal review or a formal proof-assistant certification.

Additional disclosure for the present revision. GPT-6 Astra Pro assisted specifically with the conductor and collision models, the scalar-matrix degeneration, the flag-jet calculation, the reciprocal-polynomial fibre presentation, literature comparisons, and exact computational checks. The proofs above state the complete mathematical arguments; finite computations are consistency checks, not universal proof certificates. No journal decision or historical priority determination is inferred from the build and source audits.

#### ASSISTANCE IN THE PRESENT REVISION

An AI assistant helped formulate and write the binary conductor stratification, the simultaneous incidence atlas, the evaluated singular Fitting ideal, and the minimal-relation-module argument for pure-power fibres. It also prepared the source-preserving reorganization and finite exact consistency checks. The written proofs, rather than those finite checks, are the mathematical basis of the statements and are submitted to independent scrutiny.

Revision 154. AI assistance was used for the polarized moduli compatibility, spectral divisor-fibre construction, homological calculations and proof expansions. The fixed five-variable Betti table uses exact computer-assisted rational elimination with its complete finite inputs recorded. The general theorems rely on the written arguments and the explicitly attributed classical results, not on regression counts or source-preservation receipts.

Revision 155. AI assistance was used to formulate and draft the exact quadratic coefficient-section theorem and the reciprocal power-ideal description of spectral Fitting schemes, to check their finite examples, and to prepare the companion-paper organization. Classical apolar and orthogonal-orbit results are explicitly attributed. The general claims rely on the written proofs; finite exact tests and source preservation are not formal proof certificates.

Revision 157. AI assistance was used in deriving and drafting the universal power-ideal formula, its complete-quadric graph interpretation, the contact-divisor and resolved-pencil statements, and their exact finite checks. Classical determinantal, apolar and complete-quadric results are attributed at their points of use. The proofs, rather than the computations, support the general statements. No formal proof certification or journal endorsement is claimed.

Scope of the new intrinsic construction. The source-normalized envelope is an algebra bundle with an intrinsic projective power diagram. It is not a claimed preferred subquotient of the sharp failure algebra. Singular-pencil classification is read through the classical complex symmetric Kronecker form. The explicit Hilbert tail is a boundary calculation for the stated transverse family, not a classification of every fibre of the incidence compactification. Exact computations accompanying the source check finite instances; they do not certify the general proofs.

Revision 169. AI assistance was used in deriving and drafting the all-order row factorization, monic chart equations, associated-point and genus-correction calculations, equivariant chart gluing, Quot comparison, and finite exact checks. The written arguments are offered for independent mathematical scrutiny. No external independent audit of the sharp inverse, formal proof-assistant certification, or journal endorsement is represented as completed by this revision.

## REFERENCES

- [1] E. Ballico, On the failure locus of higher order properties of embeddings in projective spaces, *Math. Nachr.* **163** (1993), 5–13. doi:10.1002/mana.19931630102.
- [2] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–313.
- [3] S.-J. Cheng and W. Wang, Howe duality for Lie superalgebras, *Compositio Math.* **128** (2001), 55–94; arXiv:math/0008093v1, §2 and Theorem 3.5.
- [4] C. Fevola, Y. Mandelshtam, and B. Sturmfels, Pencils of quadrics: old and new, *Le Matematiche* **76** (2021), no. 2, 319–335. doi:10.4418/2021.76.2.2; arXiv:2009.04334v2, Theorem 1.1, Corollary 2.1, Remark 2.3, Conjecture 4.5, and Theorem 5.1.
- [5] T. Ohta, The singularities of the closures of nilpotent orbits in certain symmetric pairs, *Tohoku Math. J. (2)* **38** (1986), 441–468. doi:10.2748/tmj/1178228456.
- [6] I. Smith, Floer cohomology and pencils of quadrics, *Invent. Math.* **189** (2012), 149–250. doi:10.1007/s00222-011-0364-1; arXiv:1006.1099, §4.2.
- [7] J. Elias and M. E. Rossi, Isomorphism classes of short Gorenstein local rings via Macaulay’s inverse system, *Trans. Amer. Math. Soc.* **364** (2012), no. 9, 4589–4604. doi:10.1090/S0002-9947-2012-05430-4; arXiv:0911.3565v1, Theorem 3.3 and Corollaries 3.4–3.5.
- [8] J. Elias and M. E. Rossi, Analytic isomorphisms of compressed local algebras, *Proc. Amer. Math. Soc.* **143** (2015), no. 3, 973–987. doi:10.1090/S0002-9939-2014-12313-6; arXiv:1207.6919v1, Theorem 3.1 and Examples 3.3–3.5.
- [9] M. Marcus and B. N. Moyls, Transformations on tensor product spaces, *Pacific J. Math.* **9** (1959), no. 4, 1215–1221. doi:10.2140/pjm.1959.9.1215, Theorem 1.
- [10] The Stacks Project Authors, *The Stacks Project*, Section 39.8, Lemma 39.8.2, <https://stacks.math.columbia.edu/tag/0BF6>, accessed September 24, 2026.
- [11] J. S. Milne, *Algebraic Groups: The Theory of Group Schemes of Finite Type over a Field*, Cambridge Studies in Advanced Mathematics **170**, Cambridge University Press, Cambridge, 2017.

- [12] The Stacks Project Authors, *The Stacks Project*, Lemma 41.7.2 (Tag 04XV), <https://stacks.math.columbia.edu/tag/04XV>, accessed September 24, 2026.
- [13] The Stacks Project Authors, *The Stacks Project*, Theorem 97.17.2 and Section 97.18, <https://stacks.math.columbia.edu/tag/06PI>, accessed September 24, 2026.
- [14] The Stacks Project Authors, *The Stacks Project*, Lemma 101.33.7 (Tag 0DLS), <https://stacks.math.columbia.edu/tag/0DLS>, accessed September 24, 2026.
- [15] The Stacks Project Authors, *The Stacks Project*, Lemma 37.44.1 (Tag 02LS), <https://stacks.math.columbia.edu/tag/02LS>, accessed September 24, 2026.
- [16] The Stacks Project Authors, *The Stacks Project*, Section 29.6, Scheme theoretic image (Tag 01R5), <https://stacks.math.columbia.edu/tag/01R5>, accessed September 24, 2026.
- [17] The Stacks Project Authors, *The Stacks Project*, Section 29.55, Normalization (Tag 035E), <https://stacks.math.columbia.edu/tag/035E>, accessed September 24, 2026.
- [18] The Stacks Project Authors, *The Stacks Project*, Subsection 112.5.7, Rigidification (Tag 04V2), including the discussion of noncentral normal inertia subgroups, <https://stacks.math.columbia.edu/tag/04V2>, accessed September 24, 2026.
- [19] J. V. Chipalkatti, On equations defining coincident root loci, *J. Algebra* **267** (2003), 246–271. doi:10.1016/S0021-8693(03)00336-3; arXiv:math/0110224v1, Proposition 5.1, Theorem 5.4, and Corollary 5.8.
- [20] S. Kurmann, Some remarks on equations defining coincident root loci, *J. Algebra* **352** (2012), 223–231. doi:10.1016/j.jalgebra.2011.10.045; arXiv:1108.4532v1, Proposition 1.3, Theorem 1.5, and Proposition 2.1.
- [21] D. Ferrand, Conducteur, descente et pincement, *Bull. Soc. Math. France* **131** (2003), no. 4, 553–585. doi:10.24033/bsmf.2455.
- [22] D. Abramovich, M. Olsson, and A. Vistoli, Tame stacks in positive characteristic, *Ann. Inst. Fourier (Grenoble)* **58** (2008), no. 4, 1057–1091. doi:10.5802/aif.2378; Appendix A, Theorem A.1.
- [23] Y. Hu, J. Lin, and Y. Shao, A compactification of the space of algebraic maps from  $\mathbb{P}^1$  to  $\mathbb{P}^n$ , arXiv:math/0701255v1 (2007), Theorem 1.1, §2.2, and Corollary 3.6.
- [24] D. Grinberg, *A basis for a quotient of symmetric polynomials*, arXiv:1910.00207, version of September 24, 2021, Theorem 2.7.
- [25] D. Mumford, J. Fogarty, and F. Kirwan, *Geometric Invariant Theory*, third edition, Ergebnisse der Mathematik und ihrer Grenzgebiete (2), vol. 34, Springer-Verlag, Berlin, 1994.
- [26] E. Miller and B. Sturmfels, *Combinatorial Commutative Algebra*, Graduate Texts in Mathematics, vol. 227, Springer, New York, 2005. Chapter 5 (Stanley–Reisner rings and Hochster’s formula).
- [27] S. Shafiei, *Apolarity for determinants and permanents of generic symmetric matrices*, arXiv:1303.1860v3 (2021), Proposition 3.1, Theorem 3.10, and Corollary 2.7.
- [28] A. Krishna, *A universal discriminant formula for pencils of quadrics*, arXiv:2607.05340v2 (2026), Theorem 1.1 and Lemma 3.1.
- [29] The Stacks Project Authors, *The Stacks Project*, Tags 00R4 (miracle flatness), 07QV (quasi-excellent rings are Nagata), and 0BJ0 (geometrically reduced formal fibres), accessed September 25, 2026.
- [30] T. Nagaoka and A. Wachi, *The strong Lefschetz property of Gorenstein algebras generated by relative invariants*, arXiv:2403.05492v1 (2024), Theorem 4, Example 5, Example 9, Proposition 10, Remark 11, Lemma 13 and Proposition 16. Published version: doi:10.1090/proc/16870.
- [31] W. Bruns and A. Conca, *Gröbner bases and determinantal ideals*, arXiv:math/0302058v3 (2003), Lemma 2.5 and Theorem 2.4.
- [32] I. Henriques and M. Varbaro, *Test, multiplier and invariant ideals*, Adv. Math. **287** (2016), 704–732, doi:10.1016/j.aim.2015.09.028; arXiv:1407.4324.
- [33] I. Vainsencher, *Schubert calculus for complete quadrics*, in Enumerative Geometry and Classical Algebraic Geometry (Nice, 1981), Progress in Mathematics **24**, Birkhäuser (1982), 199–235.
- [34] A. Massarenti, *On the birational geometry of spaces of complete forms I: collineations and quadrics*, Proc. Lond. Math. Soc. **121** (2020), 1579–1618, doi:10.1112/plms.12377; arXiv:1803.09161v2, §2.
- [35] A. Casarotti, E. Corniani and A. Massarenti, *Complete singular collineations and quadrics*, International Mathematics Research Notices **2023** (2023), no. 18, 15370–15407. DOI: 10.1093/imrn/rnac271. The cited construction is also in arXiv:2111.02940, Definition 2.5, Remark 2.6 and Theorem 2.14.
- [36] R. C. Thompson, *Pencils of complex and real symmetric and skew matrices*, Linear Algebra and its Applications **147** (1991), 323–371. DOI: 10.1016/0024-3795(91)90238-R.
- [Stacks Project] The Stacks Project Authors, *The Stacks Project*, Tags 0806 (universal property of blowing up), 02LS (proper quasi-finite morphisms), and 0H1G (smooth centres and their blow-ups), <https://stacks.math.columbia.edu>, accessed September 25, 2026.
- [Li(2009)] L. Li, Wonderful compactification of an arrangement of subvarieties, *Michigan Math. J.* **58** (2009), 535–563; <https://arxiv.org/abs/math/0611412>.

- [De Teran–Dmytryshyn–Dopico(2018)] F. De Terán, A. Dmytryshyn and F. M. Dopico, *Generic symmetric matrix pencils with bounded rank*, arXiv:1808.03118 (2018), Theorem 2.1; <https://arxiv.org/abs/1808.03118>.
- [Gathmann(1998)] A. Gathmann, *Gromov–Witten invariants of blow-ups*, arXiv:math/9804043 (1998), Section 7; <https://arxiv.org/abs/math/9804043>.
- [Ballico(1996)] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–314.
- [Stacks Project] The Stacks Project Authors, *The Stacks Project*, Section 29.55 (normalization), Tags 080C (strict transform), 0806 (universal property of blowing up), and 02LS (finite morphisms), <https://stacks.math.columbia.edu>, accessed September 25, 2026.
- [Dmytryshyn(2011/2018)] A. Dmytryshyn, *Miniversal deformations of pairs of symmetric matrices under congruence*, arXiv:1104.2530, version 2 (2018), Theorem 2.1, Corollary 2.1 and Lemma 3.1; <https://arxiv.org/abs/1104.2530>.
- [Hu–Lin–Shao(2011)] Y. Hu, J. Lin and Y. Shao, A compactification of the space of algebraic maps from  $\mathbb{P}^1$  to  $\mathbb{P}^n$ , *Comm. Anal. Geom.* **19** (2011), 1–30; arXiv:math/0701255.
- [Chung–Hong–Kiem] K. Chung, J. Hong and Y.-H. Kiem, *Compactified moduli spaces of rational curves in projective homogeneous varieties*, arXiv:1010.0068; <https://arxiv.org/abs/1010.0068>.
- [Stacks Project] The Stacks Project Authors, *The Stacks Project*, Tags 062Y (relative effective Cartier divisors), 02LS (finite morphisms), and 0806 (the blow-up universal property), <https://stacks.math.columbia.edu>, accessed September 25, 2026.
- [Bhatt(2014)] B. Bhatt, *Algebraization and Tannaka duality*, arXiv:1404.7483, 2014, Theorem 1.4 and §5.
- [Marian–Oprea–Pandharipande(2011)] A. Marian, D. Oprea, and R. Pandharipande, *The moduli space of stable quotients*, *Geom. Topol.* **15** (2011), 1651–1706; arXiv:0904.2992.
- [Ciocan-Fontanine–Kim–Maulik(2011)] I. Ciocan-Fontanine, B. Kim, and D. Maulik, *Stable quasimaps to GIT quotients*, arXiv:1106.3724, 2011.
- [Manolache(2013)] C. Manolache, *Stable maps and stable quotients*, arXiv:1301.4393, 2013.
- [Lahoz–Lehn–Macrì–Stellari(2016)] M. Lahoz, M. Lehn, E. Macrì, and P. Stellari, *Generalized twisted cubics on a cubic fourfold as a moduli space of stable objects*, arXiv:1609.04573, 2016.
- [Stacks Project] The Stacks Project Authors, *The Stacks Project*, Lemma 37.53.6 (connected fibres of a proper birational morphism) and Lemma 41.11.3 (completed local criterion for etaleness), <https://stacks.math.columbia.edu>, accessed September 25, 2026.
- [Stacks Project] The Stacks Project Authors, *The Stacks Project*, Tag 0539 (torsion-free modules over valuation rings), <https://stacks.math.columbia.edu/tag/0539>, accessed September 26, 2026.
- [Chung and Moon(2017)] K. Chung and H.-B. Moon, *Mori’s program for the moduli space of conics in Grassmannian*, *Taiwanese J. Math.* **21** (2017), 621–652; arXiv:1608.00181.
- [Grothendieck(1961)] A. Grothendieck, *Techniques de construction et théorèmes d’existence en géométrie algébrique IV: les schémas de Hilbert*, Séminaire Bourbaki, exp. 221 (1960/61), 249–276; with the published erratum to exp. 221.
- [Stacks Project] The Stacks Project Authors, *The Stacks Project*, Tags 00MI (fibrewise exactness and flatness), 0BJ0 (geometrically reduced formal fibres), 05PS (universal flattening), and 08S3 (square-zero extensions and the naive cotangent complex), <https://stacks.math.columbia.edu>, accessed September 26, 2026.
- [Li(2001)] J. Li, *Stable morphisms to singular schemes and relative stable morphisms*, *J. Differential Geom.* **57** (2001), 509–578; arXiv:math/0009097.
- [Li and Wu(2011)] J. Li and B. Wu, *Good degeneration of Quot-schemes and coherent systems*, arXiv:1110.0390, 2011.
- [Gotzmann(1978)] G. Gotzmann, *Eine Bedingung für die Flachheit und das Hilbertpolynom eines graduierten Ringes*, *Math. Z.* **158** (1978), 61–70, doi:10.1007/BF01214566.
- [Bayer–Morrison(1988)] D. Bayer and I. Morrison, *Standard bases and geometric invariant theory. I. Initial ideals and state polytopes*, *J. Symbolic Comput.* **6** (1988), no. 2–3, 209–217, doi:10.1016/S0747-7171(88)80043-9.
- [Stacks Project] The Stacks Project Authors, *The Stacks Project*, Section 31.33, “Blowing up”, Tag 010F, <https://stacks.math.columbia.edu/tag/010F>.
- [Speyer–Sturmfels(2004)] D. Speyer and B. Sturmfels, *The tropical Grassmannian*, *Adv. Geom.* **4** (2004), 389–411; arXiv:math/0304218.
- [Manubens–Montes(2006)] M. Manubens and A. Montes, *Improving DISPGB algorithm using the discriminant ideal*, *J. Symbolic Comput.* **41** (2006), 1245–1263; arXiv:math/0601763.
- [Marian–Oprea–Pandharipande(2011)] A. Marian, D. Oprea, and R. Pandharipande, *The moduli space of stable quotients*, *Geom. Topol.* **15** (2011), 1651–1706; arXiv:0904.2992.
- [Manolache(2014)] C. Manolache, *Stable maps and stable quotients*, *Compos. Math.* **150** (2014), 1457–1481; arXiv:1301.4393.